

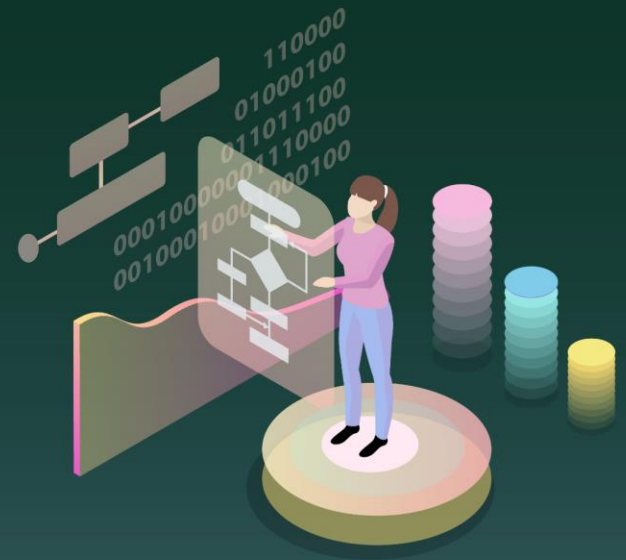
Unified Modeling Language (UML)

Romi Satria Wahono

romi@romisatriawahono.net

http://romisatriawahono.net

08118228331



Romi Satria Wahono

- **SMA Taruna Nusantara** Magelang (1993)
- **B.Eng, M.Eng** and **Ph.D** in Software Engineering
Saitama University Japan (1994-2004)
Universiti Teknikal Malaysia Melaka (2014)
- Core Competency in **Enterprise Architecture**,
Software Engineering and **Machine Learning**
- **LIPI** Researcher (2004-2007)
- Founder and **CEO**:
 - PT **Brainmatics** Cipta Informatika (2005)
 - PT IlmuKomputerCom **Braindevs** Sistema (2014)
- Professional **Member** of IEEE, ACM and PMI
- IT and Research **Award Winners** from WSIS (United Nations),
Kemdikbud, Ristekdikti, LIPI, etc
- SCOPUS/ISI Indexed **Q1 Journal Reviewer**: **Information and Software Technology**, **Journal of Systems and Software**, **Software: Practice and Experience**, **Empirical Software Engineering**, etc
- Industrial **IT Certifications**: TOGAF, ITIL, CCAI, CCNA, etc
- **Enterprise Architecture Consultant**: KPK, RistekDikti, INSW, BPPT, Kemsos
Kemenkeu (Itjend, DJBC, DJPK), Telkom, FIF, PLN, PJB, Pertamina EP, etc





BAGAIMANA MELAKUKAN PENELITIAN YANG BAIK?

Pada artikel ini, saya mencoba merangkumkan bahan-bahan melakukan penelitian yang baik. Artikel ini adalah bagian dari seri 'How to do Good Research' yang akan saya lanjutkan dalam artikel-artikel berikutnya.

RomiSatriaWahono.Net

Lecture Notes in Software Engineering, Computing Research and Technopreneurship

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di universitas di Indonesia. Seluruh materi kuliah bisa diunduh dan materi mata kuliah memuat course description, standard competency, dan book yang digunakan.

Research Methodology (updated January 2015)
Data Mining (updated January 2015)
Theory of Computation (updated March 2015)

Java Fundamentals (updated October 2013)
Java Enterprise Edition

Systems Analysis and Design (updated January 2015)
Business Process Model and Notation (updated January 2015)
Software Engineering
Software Testing
Software Quality Assurance
Project Management

TOGAF 9.1 Fundamental
TOGAF 9.1 Foundation
TOGAF 9.1 Certified

1 Hour Online 10 Mit



Romi
Resea
TOGA
PJB, F
Inform

📅 **Senin, 4**

🕒 **09.00 -**

Terbatas 500 Pe
Minggu, 3 Mei 2
mitosse1hour.br

#TrainingD

Contact Us: 📞 📧

Irma 0811822888
Annisa 0811822888
Vina 0811822888
Lina 0811822888
Rachma 0811822888

1 Hour Online Unified M



Romi Satri
Researcher &
TOGAF & ITIL
PJB, FIF, INSV
Information &

📅 **Rabu, 6 Mei**

🕒 **09.00 - 10.00**

Terbatas 500 Peserta
Selasa, 5 Mei 2020, 10.00
um1hour.brainmatics

#TrainingDirur

Contact Us: 📞 📧

Irma 08118228881
Annisa 08118228882
Vina 08118228883
Lina 08118228884
Rachma 08118228885

Training

1 Hour Online Training

Software Engineering Research Trends



Romi Satria Wahono, Ph.D

Researcher & Technopreneur. Founder & CEO PT Braindevs & PT Brainmatics. TOGAF & ITIL Certified, Enterprise Architect di KPK, Pertamina EP, BPPT, PLN, PJB, FIF, INSW, RistekDikti, Kemenkeu. SCOPUS Q1 Journal Reviewer: Information & Software Technology, Journal of System & Software, etc.

📅 **Jum'at, 8 Mei 2020**

🕒 **09.00 - 10.00 WIB**

Terbatas 500 Peserta, Registrasi Sebelum
Kamis, 7 Mei 2020, Pukul 17:00 Melalui
sert1hour.brainmatics.com

#TrainingDirumahAja

Contact Us: 📞 📧

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Lina 08118228884
Rachma 08118228885



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Jl. Gatot Subroto Kav 71-73 Jakarta 12870
info@brainmatics.com http://brainmatics.com

📱 📧 📞 📧 BrainmaticsID

Course Outline

1. System Planning

- 1.1 Siklus dan Metodologi Pengembangan Software
- 1.2 System Request dan Feasibility Analysis

2. Systems Analysis

- 2.1 Requirement Gathering
- 2.2 Identifikasi Proses Bisnis dengan Use Case Diagram
- 2.3 Pemodelan Proses Bisnis dengan AD atau BPMN
- 2.4 Realisasi Proses Bisnis dengan Sequence Diagram

3. Systems Design

- 3.1 Perancangan Class Diagram
- 3.3 Perancangan User Interface Design
- 3.4 Perancangan Data Model
- 3.5 Perancangan Deployment Diagram



1. Systems Planning



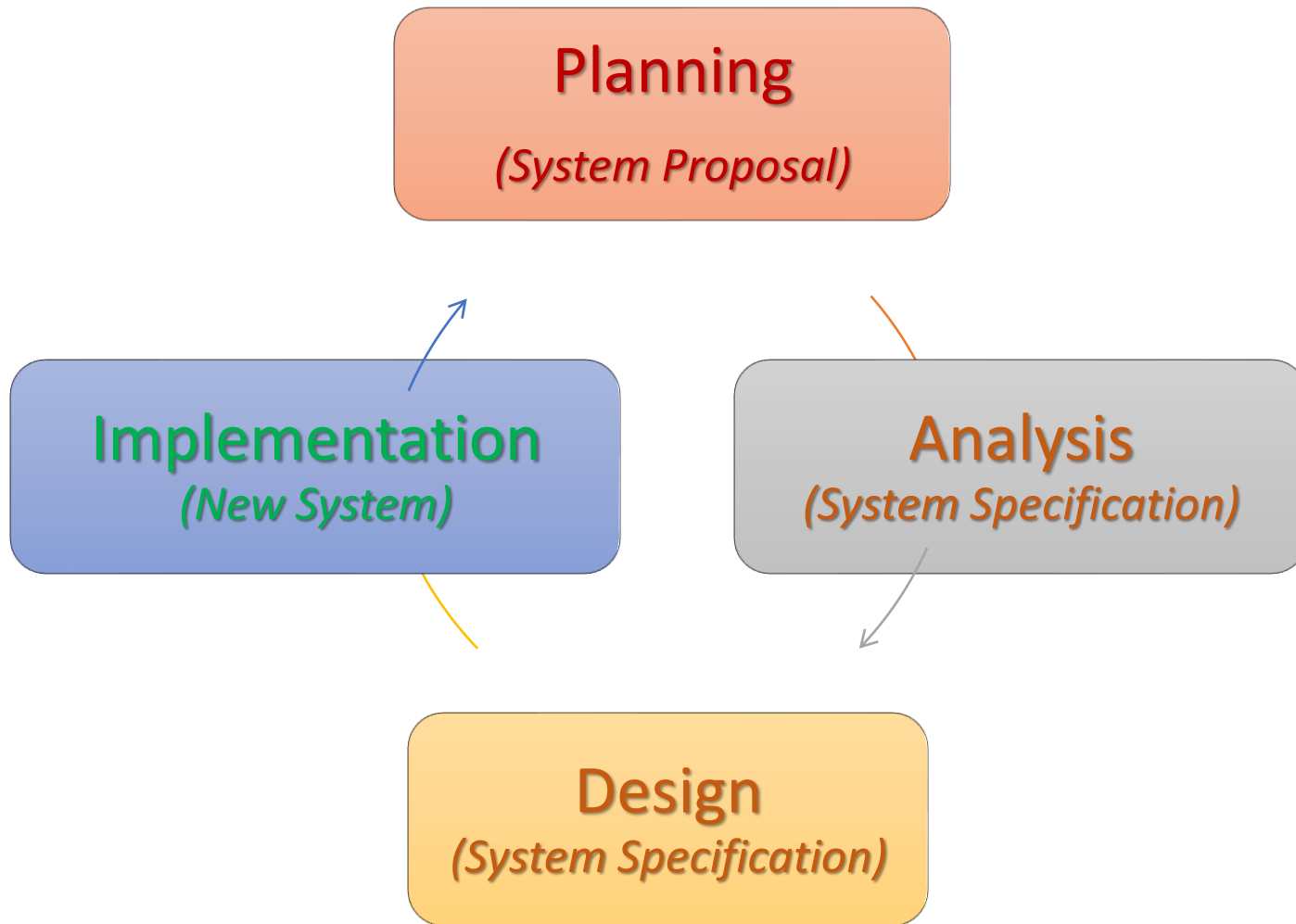
1.1 Siklus dan Metodologi Pengembangan Software

Siklus Pengembangan Software:

Alur, Peran, dan Tahapan (*Deliverable*) (Tilley, 2012) (Dennis, 2016) (Valacich, 2017)

1. **User/Product Owner** membawa permintaan kebutuhan (perubahan) software (**System Request**) ke System Analyst
 2. **System Analyst** membuat analisis kelayakan (**Feasibility Analysis**) dari System Request tersebut
 3. Setelah dinyatakan layak, System Analyst melakukan analysis dan design, dan hasilnya adalah **System Specification**
 - **Business Analyst** membantu System Analyst memahami proses bisnis dari software yang akan dibangun
 4. System Specification diserahkan oleh System Analyst ke **Programmer** untuk dilakukan **Konstruksi (Coding)**
 5. Hasil Konstruksi berupa **Kode Program** diserahkan ke **Software Tester** untuk dilakukan **Pengujian (Unit, Integration, System, User Acceptance Testing)**
 6. **Instalasi (delivery)** software dan **manajemen perubahan**
 - **Software** = Kode Program + Dokumentasi (Pengembangan dan Penggunaan)
 7. Siklus kembali ke 1 apabila ada permintaan perubahan (**Permintaan Perubahan Software**)
- Planning**
(*System Proposal*)
- Analysis and Design**
(*System Specification*)
- Implementation**
(*Software*)
- Maintenance**
(*Updated Software*)

Siklus Pengembangan Software



(Tilley, 2012)

(Dennis, 2016)

(Valacich, 2017)

Metodologi Pengembangan Software (Model Process)

- A formalized **approach to implementing** the Software Development Life Cycle (*Dennis, 2012*)
- A **simplified representation** of a software process (*Sommerville, 2015*)
- A distinct **set of activities**, actions, tasks, milestones, and work products required to engineer high quality software (*Pressman, 2015*)

Metodologi Pengembangan Software

1. Structured Design

(Prescriptive) (1967-)

- Waterfall method
- Parallel development

2. Rapid Application Development

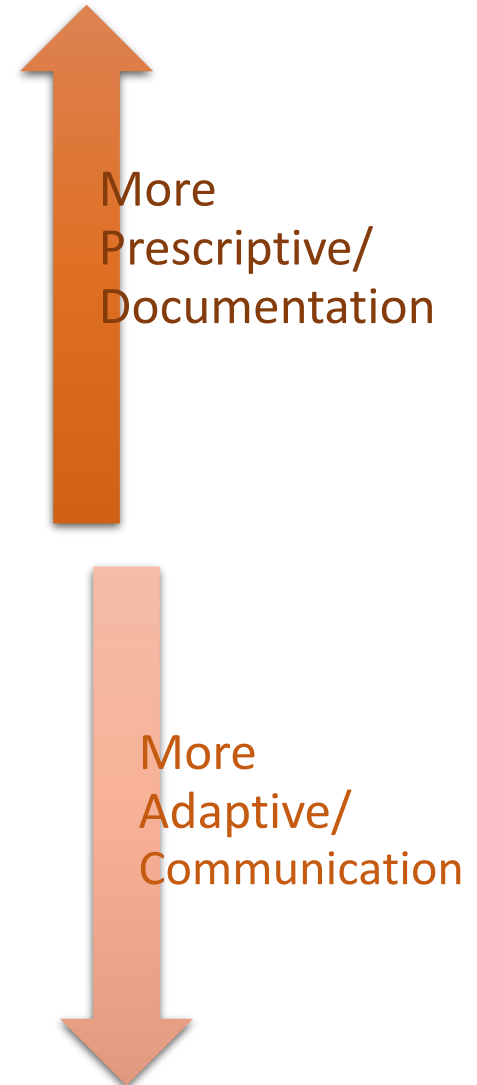
(Iterative) (1985-)

- Phased Development
- Prototyping

3. Agile Development

(Adaptive) (1995-)

- Extreme Programming (XP)
- Scrum



Compiled from (Dennis, Wixom and Tegarden, 2016)

DRAINMATIC'S

System Request: Sistem Penjualan Musik Online

Project Sponsor: Margaret Mooney, Vice President of Marketing

Business Needs: Project ini dibangun untuk:

1. Mendapatkan pelanggan baru lewat Internet

Studi Kelayakan Sistem Penjualan Musik Online

Business Requirement: Margaret Mooney dan Alec Adams membuat studi kelayakan untuk pengembangan Sistem yang mendukung Sistem Penjualan Musik Online

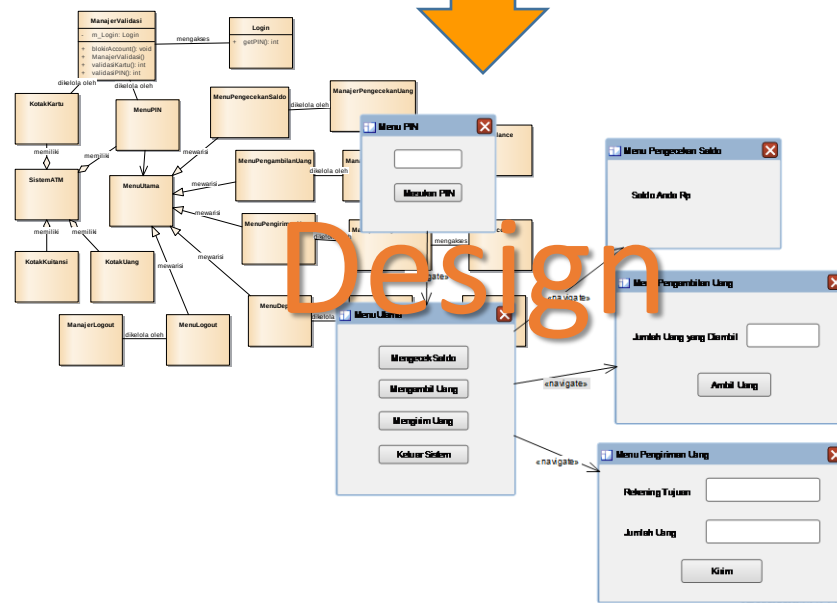
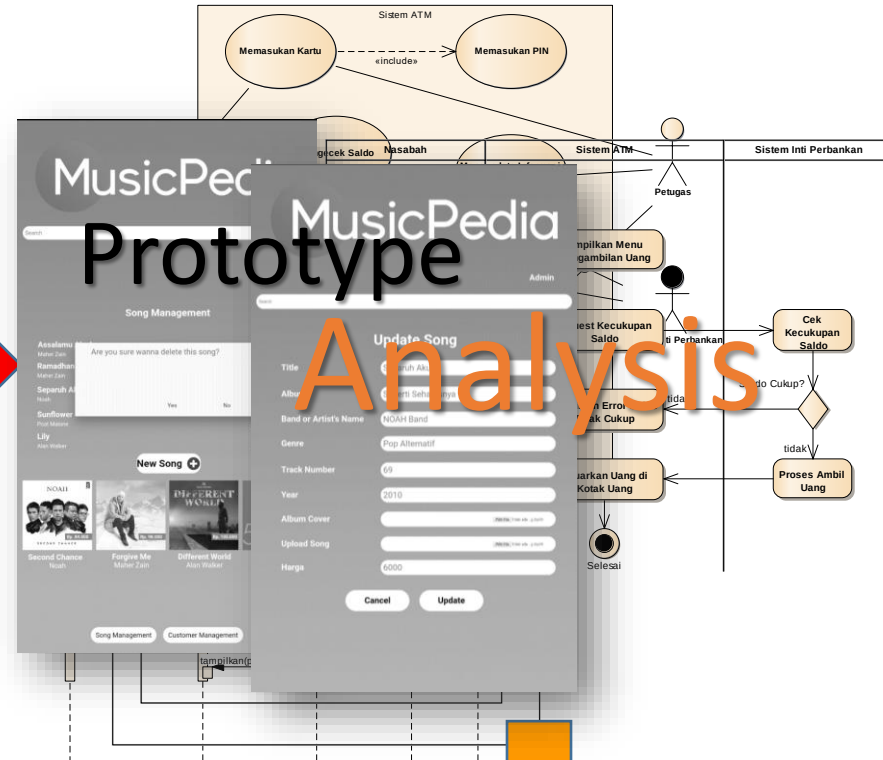
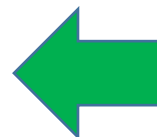
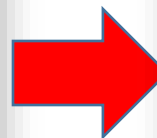
Kelayakan Teknis

1. Fitur Pencarian Produk
2. Fitur Pencarian Toko
3. Fitur Pemetaan Produk

Sistem penjualan musik online layak secara teknis, meskipun memiliki beberapa risiko.

	2016	2017	2018
Peningkatan penjualan dari pelanggan baru	0	400,000,000	500,000,000
Peningkatan penjualan dari pelanggan lama	0	600,000,000	700,000,000
Pengurangan biaya operasional dan telepon	0	100,000,000	100,000,000
Total Benefits:	0	1,100,000,000	1,300,000,000
PV of Benefits:	8,991,084	9,091,084	9,101,112
PV of All Benefits:	8,991,084	9,091,084	9,101,112
Honor Tim (Planning, Analysis, Design, and Implementation)	160,000,000	0	0
Honor Konsultan Infrastruktur Internet	10,000,000	0	0
Total Development Costs:	170,000,000	0	0
Honor Pengelola Web	60,000,000	70,000,000	80,000,000
Biaya Lisensi Software	50,000,000	60,000,000	70,000,000
Hardware upgrades	100,000,000	100,000,000	100,000,000
Biaya Komunikasi	20,000,000	30,000,000	40,000,000
Biaya Marketing	100,000,000	200,000,000	300,000,000
Total Operational Costs:	330,000,000	460,000,000	590,000,000
Total Costs:	780,000,000	460,000,000	590,000,000
PV of Costs:	735,849,057	409,398,362	495,375,377
PV of all Costs:	735,849,057	1,145,247,419	1,640,622,796
Total Project Costs Less Benefits:	-780,000,000	640,000,000	710,000,000
Yearly NPV:	-735,849,057	569,507,722	596,129,691
Cumulative NPV:	-735,849,057	-166,251,335	429,878,356
Return on Investment (ROI) di Tahun 3:	26.2%		
Break-even Point (BEP):	2.28 tahun		

Implementation



Design

System Request: Sistem Penjualan Musik Online

Project Sponsor: Margaret Mooney, Vice President of Marketing

Business Needs: Project ini dibangun untuk:

1. Mendapatkan pelanggan baru lewat Internet

Studi Kelayakan Sistem Penjualan Musik Online

Business Requirement: Margaret Mooney dan Alec Adams membuat studi kelayakan untuk pengembangan Sistem yang mendukung Sistem Penjualan Musik Online

Kelayakan Teknis

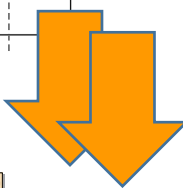
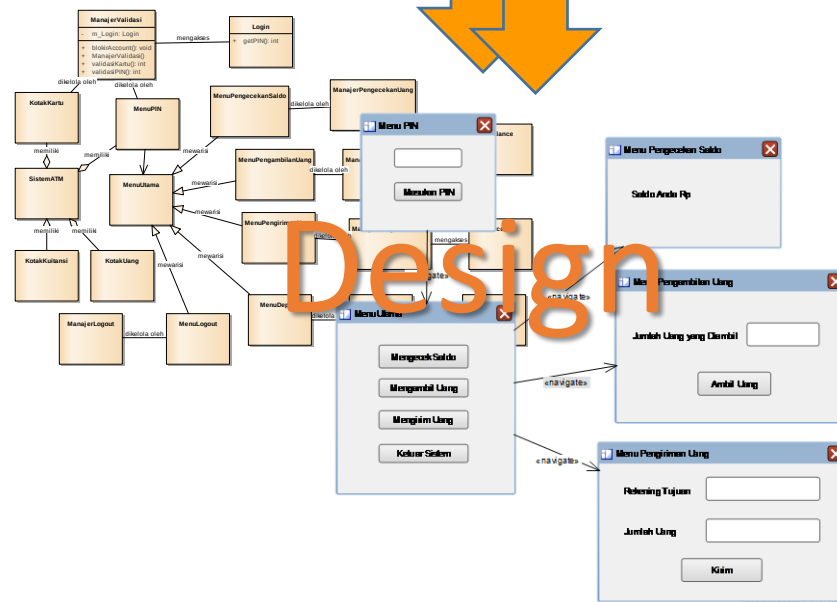
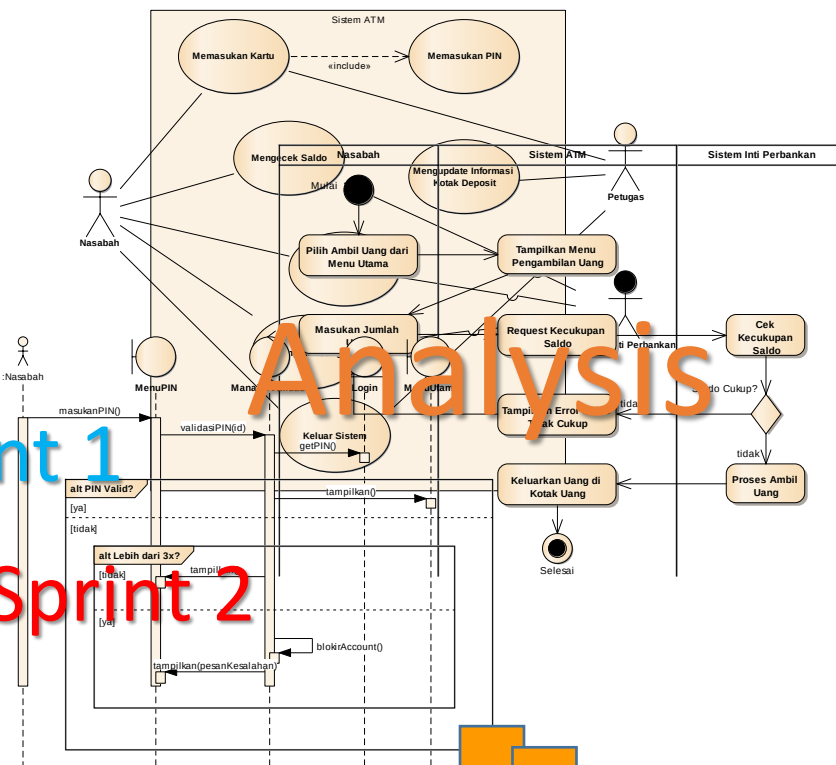
1. Fitur Pencarian Produk
2. Fitur Pencarian Toko
3. Fitur Pemetaan Produk
4. Fitur Rekomendasi Produk

	2016	2017	2018
Peningkatan penjualan dari pelanggan baru	0	400,000,000	500,000,000
Peningkatan penjualan dari pelanggan lama	0	600,000,000	700,000,000
Pengurangan biaya operasional dan telepon	0	100,000,000	100,000,000
Total Benefits:	0	1,100,000,000	1,300,000,000
PV of Benefits:	8,990,084	9,091,084	9,091,084
PV of All Benefits:	9,091,084	9,091,084	9,091,084
Honor Tim (Planning, Analysis, Design, and Implementation)	160,000,000	160,000,000	160,000,000
Honor Konsultan Infrastruktur Internet	10,000,000	10,000,000	10,000,000
Total Development Costs:	450,000,000	0	0
Honor Pengelola Web	60,000,000	70,000,000	80,000,000
Biaya Lisensi Software	50,000,000	60,000,000	70,000,000
Hardware upgrades	100,000,000	100,000,000	100,000,000
Biaya Komunikasi	20,000,000	30,000,000	40,000,000
Biaya Marketing	100,000,000	200,000,000	300,000,000
Total Operational Costs:	330,000,000	460,000,000	590,000,000
Total Costs:	780,000,000	460,000,000	590,000,000
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PV of all Costs:	735,849,057	1,145,247,419	1,640,622,796
Total Project Costs Less Benefits:	-780,000,000	640,000,000	710,000,000
Yearly NPV:	-735,849,057	569,507,722	596,129,691
Cumulative NPV:	-735,849,057	-166,251,335	429,878,356
Return on Investment (ROI) di Tahun 3:	26.2%	0.262021445	0.262021445
Break-even Point (BEP): 2.28 tahun	2 + (596,129,691 / 429,878,356)	2.278884507	2.278884507



Sprint 1

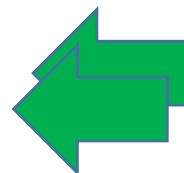
Sprint 2



Hasil Sprint 1

Hasil Sprint 2

Implementation





1.2 System Request dan Feasibility Analysis

Siklus Pengembangan Software

1. **Planning:** Mengapa Software harus Dikembangkan?

- System request
- Feasibility Analysis (project size estimation)



2. **Analysis:** Siapa Pengguna dan Alur Kerja Software?

- Requirement gathering
- Business process modeling



3. **Design:** Bagaimana Komposisi dari Software?

- Program design
- User interface design
- Data design



4. **Implementation:** Konstruksi dan Penyerahan Software

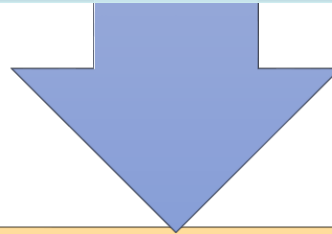
- System construction
- Testing
- Documentation
- Installation

System Request (Business Value Identification)

Lower Cost

*Increase
Productivity*

Increase Profit



Feasibility Analysis

*Technical
(Capabilities)*

*Economic
(ROI, BEP)*

*Organizational
(Goals, Core Business)*

System Request

Elemen	Deskripsi	Contoh
Business Need	- The business-related reason for initiating the software development project Reason prompting the project, and why the project should be funded?	- Meningkatkan penjualan - Mengurangi biaya operasional - Meningkatkan produktifitas pegawai - Meningkatkan kualitas layanan - Mengurangi kebocoran/kecurangan - Mengurangi cacat produksi - Meningkatkan efisiensi kerja
Business Value	- The benefits that the software will create for the organization Tangible value (a quantiable value) and intangible value (intuitive believe)	- Peningkatan penjualan 3% - Pengurangan biaya operasional 10% - Peningkatan produktifitas pegawai 10% (dihitung rasio pekerjaan dan gaji) - Pengurangan cacat produksi 20% - Peningkatan efisiensi kerja 20%
Business Requirements	- The business capabilities that software will provide - Can be replaced by Use Case Diagram	- Fitur registrasi, login, dan logout - Fitur pengelolaan data pengguna - Fitur pengiriman notifikasi otomatis - Fitur cetak laporan bulanan dan tahunan

Software Berkualitas?

Software quality is (IEEE, 1991):

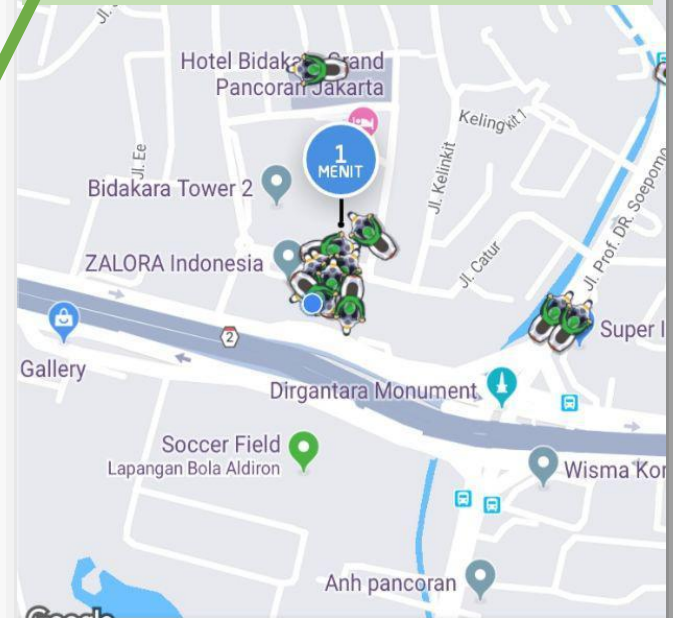
1. The degree to which a system, component, or

Sesuai Kebutuhan

2. The degree to which a system, component, or process meets customer

Ada Keuntungan

Business Requirements



Set lokasi jemput



Jl. Gatot Subroto No.177a

Jl. Gatot Subroto No.177a, RT.8/RW.8, Menteng Dalam, Tebet, Kota Jakarta Selatan, Daerah Khusus Ibukota Jakarta 12870, Indonesia

Business Value

System Request: Sistem Penjualan Musik Online

Project Sponsor: Margaret Mooney, Vice President of Marketing

Business Needs: Project ini dibangun untuk:

1. Mendapatkan pelanggan baru lewat Internet

2. Meningkatkan efisiensi penanganan masalah pelanggan melalui internet

Business Requirements:

Sistem yang mendukung penjualan musik secara online. Fitur-fitur yang harus ada:

1. Fitur Pencarian Produk
2. Fitur Pencarian Toko yang Menyediakan Stok Produk
3. Fitur Pemesanan Produk Melalui Toko yang Menyediakan
4. Fitur Pembayaran dengan Berbagai Pilihan Pembayaran

Business Value:

Intangible Value:

- Meningkatkan kenyamanan dan **kepuasan pelanggan**
- Meningkatkan **brand recognition** tentang perusahaan di dunia Internet

Tangible Value:

1. Meningkatkan penjualan dari pelanggan baru lewat Internet:

- Rp 400 juta **peningkatan penjualan** dari pelanggan baru dan Rp 600 juta dari pelanggan lama

2. Mengurangi biaya operasional untuk menangani komplain dari pelanggan

- Rp 100 juta **pengurangan** tahunan biaya telepon untuk menangani pelanggan

Studi Kelayakan Sistem Penjualan Musik Online

Margaret Mooney dan Alec Adams membuat studi kelayakan untuk pengembangan Sistem Penjualan Musik Online

Kelayakan Teknis

Sistem penjualan musik online layak secara teknis, meskipun memiliki beberapa risiko.

Risiko Berhubungan dengan **Kefamiliaran dengan Aplikasi**: Risiko **Tinggi**

- Divisi Marketing **tidak memiliki pengalaman** menggunakan sistem penjualan online
- Divisi IT memiliki pemahaman yang baik tentang sistem penjualan offline, akan tetapi **tidak berpengalaman** mengembangkan sistem penjualan musik online

Risiko Berhubungan dengan **Kefamiliaran dengan Teknologi**: Risiko **Sedang**

- Divisi IT tidak menguasai masalah infrastruktur dan ISP, tetapi akan menyewa konsultan
- Divisi IT cukup familier dengan framework dan IDE yang akan digunakan
- Divisi Marketing tidak memiliki pengalaman menggunakan teknologi Web

Risiko berhubungan dengan **Ukuran Project**: Risiko **Rendah**

- Perusahaan memiliki total **30 orang pengembang**
- Project dikerjakan oleh **5 orang pengembang** dengan estimasi waktu **6 bulan**

Kompatibilitas dengan sistem dan infrastruktur yang ada: Risiko **Rendah**

- Sistem pemesanan yang ada sekarang menggunakan *open standard*, jadi sangat **kompatibel** dengan sistem penjualan berbasis web yang akan dibangun

Kelayakan Ekonomi

Cost benefit analysis telah dilakukan. Sistem Penjualan musik online memiliki peluang yang baik untuk bisa **meningkatkan pendapatan perusahaan**.

- Return on Investment (ROI) setelah 3 tahun: **31%**
- Break-even point (BEP): **2.25 tahun**
- Total keuntungan setelah 3 tahun: **Rp. 503.559.986,-**

Kelayakan Organisasi

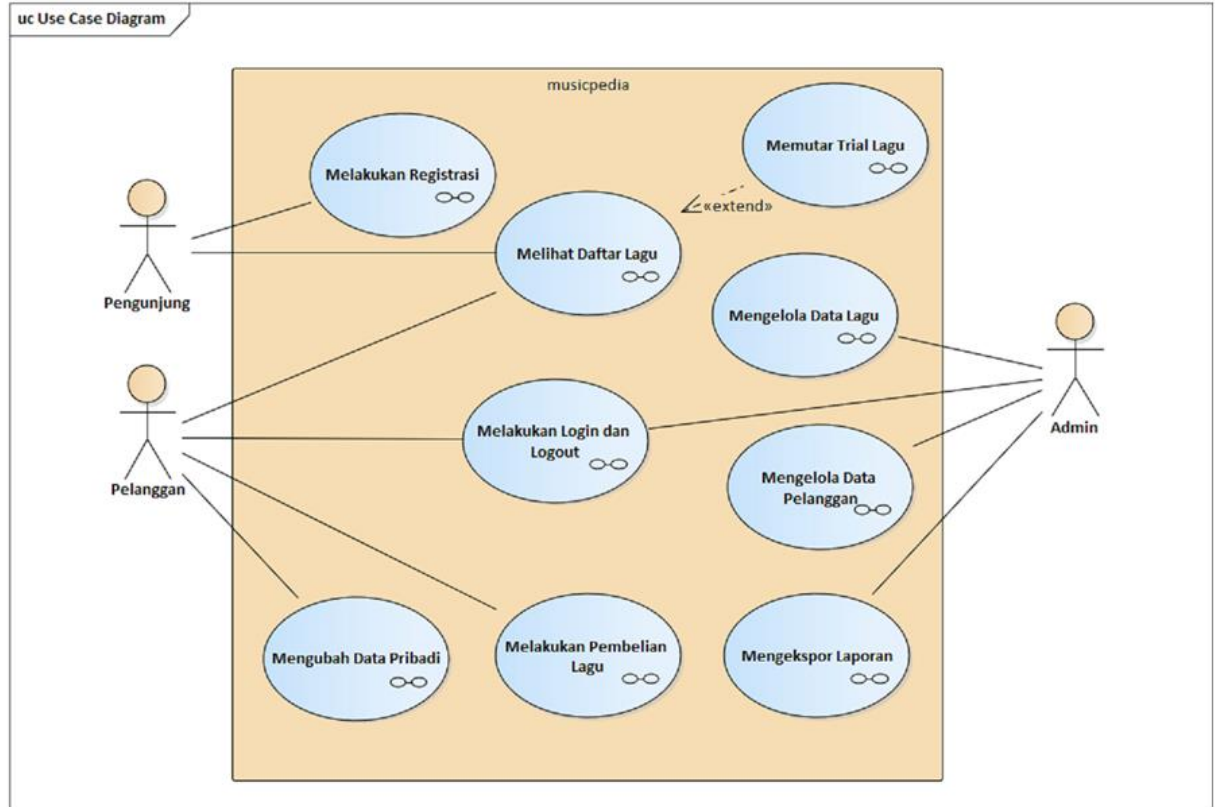
- Secara organisasi, **resikonya rendah**. Tujuan dari pengembangan sistem penjualan musik online adalah meningkatkan penjualan perusahaan. Dan ini selaras dengan KPI marketing yang ke arah peningkatan kuantitas penjualan
- Project champion dari pengembangan sistem penjualan musik online ini adalah Margaret Mooney, Vice President of Marketing

	2019	2020	2021
Peningkatan penjualan dari pelanggan baru	0	400,000,000	500,000,000
Peningkatan penjualan dari pelanggan lama	0	600,000,000	700,000,000
Pengurangan biaya operasional dan telepon	0	100,000,000	100,000,000
Total Benefits:	0	1,100,000,000	1,300,000,000
PV of Benefits:	0	978,996,084	1,091,505,068
PV of All Benefits:	0	978,996,084	2,070,501,152
Honor Tim (Analysis, Design and Implementation)	360,000,000	0	0
Honor Konsultan	90,000,000	0	0
Total Development Costs:	450,000,000	0	0
Honor Pengelola Web	60,000,000	70,000,000	80,000,000
Biaya Lisensi Software	50,000,000	60,000,000	70,000,000
Hardware upgrades	100,000,000	100,000,000	100,000,000
Biaya Komunikasi	20,000,000	30,000,000	40,000,000
Biaya Marketing	100,000,000	200,000,000	300,000,000
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PV of all Costs:	735,849,057	1,145,247,419	1,640,622,796
Total Project Costs Less Benefits:	-780,000,000	640,000,000	710,000,000
Yearly NPV:	-735,849,057	569,597,722	669,811,321
Cumulative NPV:	-735,849,057	-166,251,335	503,559,986
Return on Investment (ROI) di Tahun 3: 30.70%	-100.00%	-0.145166304	0.306932213
Break-even Point (BEP): 2.25 tahun	24		2.248206218

Software Request

musicpedia				
Date	26 Oktober 2018			
Description	Musicpedia adalah aplikasi layanan download musik & audio dimana saja dan kapan saja, menawarkan akses lengkap ke jutaan lagu dari semua artis papan atas di industri musik barat maupun musik lokal, mendengarkan musik baru dan top musik dunia dalam bentuk audio mp3 secara offline			
Project Sponsor	Wahyu Utomo, VP Business Development, PT Musika Indonesia			
Business Need	1. Tidak Setuju	2. Ragu-Ragu	3. Setuju	4. Sangat Setuju
Aplikasi yang dikembangkan mampu meningkatkan pendapatan perusahaan?	☐	☐	☐	☒
Aplikasi yang dikembangkan mampu mengurangi biaya operasional perusahaan?	☐	☐	☒	☐
Aplikasi yang dikembangkan mampu meningkatkan produktifitas kerja pegawai?	☐	☒	☐	☐
Aplikasi yang dikembangkan mampu meningkatkan nilai tambah perusahaan yang bersifat intangible?	☐	☐	☒	☐
Business Value	Intangible Value: a. Meningkatkan brand recognition perusahaan di dunia internet b. Meningkatkan produktivitas kerja pegawai dan mengurangi kuantitas pegawai Tangible Value: a. Mengurangi biaya operasional perusahaan: - Sewa ruangan: Rp120.000.000,- - Biaya komunikasi: Rp6.000.000,- b. Meningkatkan penjualan musik: Rp400.000.000,-			

Business Requirements



Technical Feasibility

musicpedia

Date: 26 Oktober 2018

Penjelasan isian	1. Sangat Kurang	2. Kurang	3. Baik	4. Sangat Baik
Kefamiliaran dengan Aplikasi				
	1	2	3	4
Pengguna familiar terhadap pengoperasian aplikasi ini.	☐	☐	☒	☐
Pengembang familiar terhadap pengembangan aplikasi ini.	☐	☐	☐	☒
Kefamiliaran dengan Teknologi				
	1	2	3	4
Pengguna familiar dengan teknologi pendukung aplikasi.	☐	☒	☐	☐
Pengembang familiar mengembangkan aplikasi dengan platform, bahasa pemrograman dan tool IDE yang dipilih.	☐	☐	☐	☒
Ukuran Proyek				
Jumlah pengembang yang dibutuhkan.	7 Man/Month			
Waktu yang dibutuhkan dalam mengembangkan aplikasi ini.	6 Month			
Kompatibilitas				
	1	2	3	4
Kebutuhan pengguna terhadap kompatibilitas aplikasi untuk terintegrasi dengan aplikasi lain.	☐	☐	☒	☐
Kompatibilitas aplikasi terhadap teknologi yang ada pada organisasi.	☐	☐	☐	☒
Secara analisi kelayakan teknis, apakah aplikasi layak dikembangkan sesuai kriteria di atas?				
			☒ Layak	☐ Tidak Layak

Technical Feasibility

Use Case Points

Tahap 1 - Menghitung Person Hours (PH)				
Use Case Points (UCP)		Person Hours Multiplier (PHM)	Person Hours (PH)	
51		20	1020	
51		28	1428	
Tahap 2 - Menghitung Person Month (PM)				
PHM	Person Hours (PH)	Lama Bekerja Perhari	Jumlah Bekerja Sebulan	Person Months (PM)
20	1020	8	22	5.80
	1020	10	26	3.92
28	1428	8	22	8.11
	1428	10	26	5.49
Tahap 3 - Menghitung Time (Month)				
PHM	Formula Penghitung Waktu		Jumlah Bekerja Sebulan	Waktu dalam Bulan (M)
20	3 * PM^(1/3)		22	5.39
			26	4.73
28			22	6.03
			26	5.29

Economic Feasibility

Cost-Benefit Analysis

Tahun	2019	2020	2021	2022
Peningkatan Pendapatan Penjualan Lagu		400,000,000	400,000,000	400,000,000
Pengurangan Biaya Sewa Ruangan		120,000,000	120,000,000	120,000,000
Pengurangan Biaya Komunikasi		6,000,000	6,000,000	6,000,000
Total Benefits	0	526,000,000	526,000,000	526,000,000
PV of Benefits	0	468,138,127	441,639,743	416,641,267
PV of All Benefits	0	468,138,127	909,777,870	884,779,394
Honor Tim (Analysis, Design and Implementation)	250,000,000	120,000,000	120,000,000	120,000,000
Total Development Costs	250,000,000	120,000,000	120,000,000	120,000,000
Honor Pengelola Web	72,000,000	72,000,000	72,000,000	72,000,000
Biaya Lisensi Software	10,000,000	10,000,000	10,000,000	10,000,000
Hardware upgrades	50,000,000	50,000,000	50,000,000	50,000,000
Biaya Komunikasi	1,000,000	1,000,000	1,000,000	1,000,000
Biaya Marketing	50,000,000	50,000,000	50,000,000	50,000,000
Total Operational Costs	183,000,000	183,000,000	183,000,000	183,000,000
Total Costs	433,000,000	303,000,000	303,000,000	303,000,000
PV of Costs	408,490,566	269,668,921	153,650,329	144,953,140
PV of all Costs	408,490,566	678,159,487	831,809,816	976,762,957
Total Project Costs Less Benefits	-433,000,000	223,000,000	223,000,000	223,000,000
Yearly NPV	-408,490,566	198,469,206	187,235,100	176,636,887
Cumulative NPV	-408,490,566	-210,021,360	-22,786,260	153,850,627
Return on Investment (ROI)	-100.00%	-0.309693168	-0.027393593	0.15751071
Break-even Point (BEP)				3.129000574

Organizational Feasibility

musicpedia		
Date	26 Oktober 2018	
Anggota Tim		
User/Product Owner	Wahyu Utomo	
Project Manager	Haris Dermawan	
System Analyst	Risa Dhani Horasman Purba	
Business Analyst	Mulyana	
Programmer	Achmad Fatkarrofiqi	
Tester	Januar Sapareza	
Apakah aplikasi ini mendukung visi dan misi organisasi?		
Ya		
Apakah aplikasi ini sesuai dengan tugas, fungsi dan KPI unit kerja anda?		
Ya		
Apakah aplikasi ini selaras dengan proses bisnis unit kerja anda?		
Ya		
Secara analisis kelayakan organisasi, apakah aplikasi layak dikembangkan sesuai kriteria di atas?	☒ Layak	☐ Tidak Layak

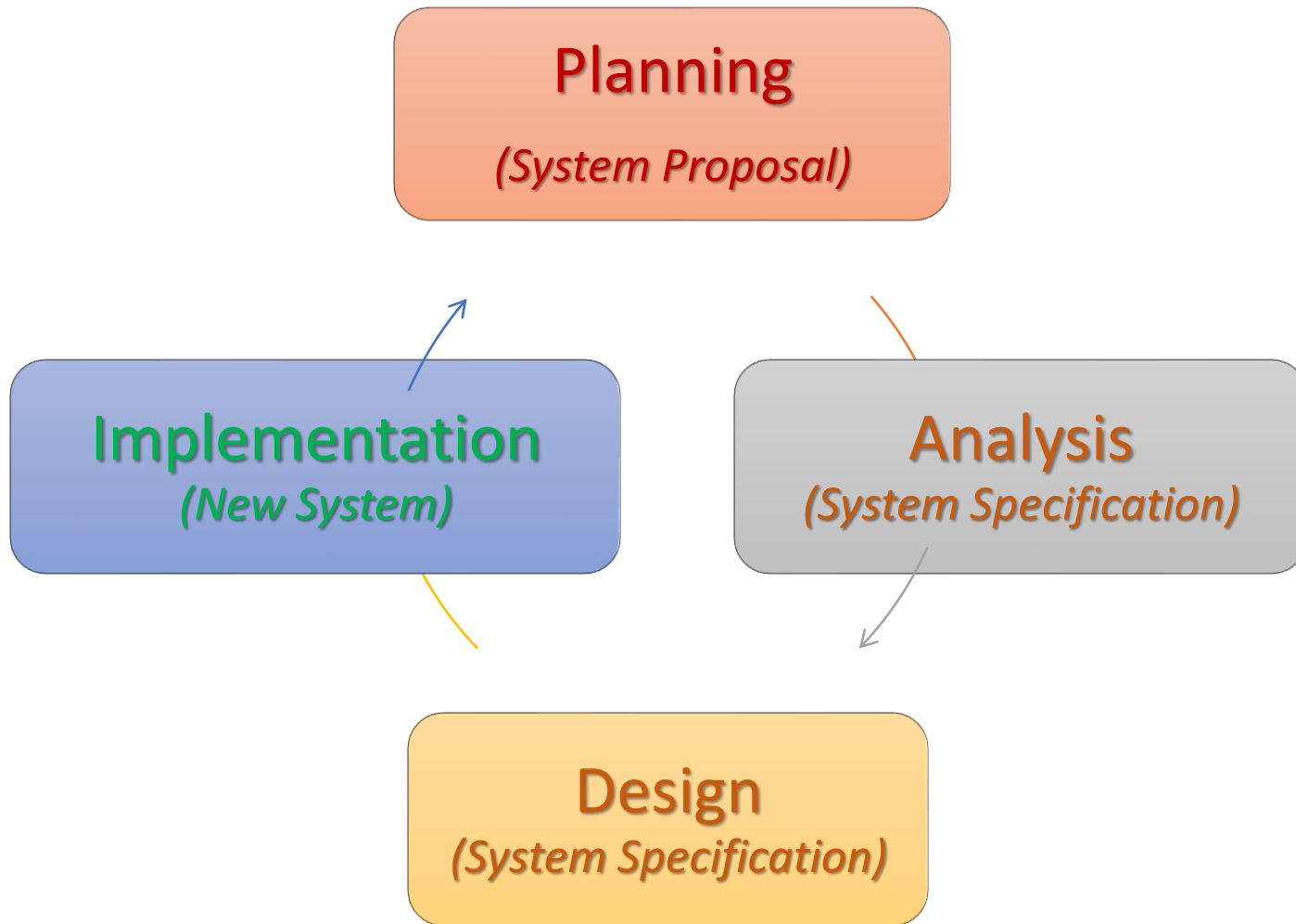


2. Systems Analysis



2.1 Requirement Gathering

Siklus Pengembangan Software



(Tilley, 2012)

(Dennis, 2016)

(Valacich, 2017)

Boehm's First Law

Errors are **most frequent during the requirements and design** activities, and are the more expensive the later they are removed

(Endres, 2003)

[L2]

What is a Requirements

- Business **Requirements**
 - Statement of **what the system must do**
 - Focus on what the system must do, **not how to do it**
- There are **2 kinds** of **requirements**

1. Functional Requirements

2. Nonfunctional Requirements

D. Nonfunctional Requirements

1. Operational Requirements

- 1.1. The system will operate in Windows and Macintosh environments
- 1.2. The system will be able to read and write Word documents, RTF, and HTML
- 1.3. The system will be able to import Gif, Jpeg, and BMP graphics files

2. Performance Requirements

- 2.1. Response times must be less than 7 seconds
- 2.2. The Inventory database must be updated in real time

3. Security Requirements

- 3.1. No special security requirements are anticipated

4. Cultural and Political Requirements

- 4.1. No special cultural and political requirements are anticipated

C. Functional Requirements

1. Printing

- 1.1. The user can select which pages to print
- 1.2. The user can view a preview of the pages before printing
- 1.3. The user can change the margins, paper size (e.g., letter, A4) and orientation on the page

2. Spell Checking

- 2.1. The user can check for spelling mistakes; the system can operate in one of two modes as selected by the users
 - 2.1.1. Mode 1 (Manual): The user will activate the spell checker and it will move the user to the next misspelled word
 - 2.1.2. Mode 2 (Automatic): As the user types, the spell checker will flag misspelled words so the user immediately see the misspelling
- 2.2. The user can add words to the dictionary
- 2.3. The user can mark words as not misspelled but not add them to the dictionary

Contoh Template System Specification (SRS, BRD, FSD)

1. System Planning

- 1.1 Project Scope
- 1.2 Project Schedule
- 1.3 Project Team

2. System Design

2.1 Functional Requirements

- 2.1.1 Actor
- 2.1.2 Use Case Diagram
- 2.1.3 Activity Diagram (BPMN)
- 2.1.4 Sequence Diagram
- 2.1.5 Class Diagram
- 2.1.6 Data Model
- 2.1.7 User Interface Design
- 2.1.8 Deployment Diagram
- 2.1.9 Relational Matrices
 - 2.1.9.1 Actor – Activity Diagram
 - 2.1.9.2 Actor – Sequence Diagram

2.2 Nonfunctional Requirements

- 2.2.1 Operational
- 2.2.2 Performance
- 2.2.3 Security
- 2.2.4 Hardware
- 2.2.5 Development Platform
- 2.2.6 Deadline

3. System Implementation

- 3.1 Testing Strategy
- 3.2 Installation Strategy
- 3.3 Change Management Strategy

1. Functional Requirement

- Kebutuhan tentang **fungsi software** secara menyeluruh
- Pemodelan dengan **UML**, ataupun penjelasan fitur-fitur dalam bentuk **problem statements**, adalah termasuk dalam **Functional Requirement**
 - **Diagrams:**
 - **Use Case** Diagrams
 - **Activity** Diagrams
 - **Problem Statements:**
 - Must **search** for inventory
 - Must **perform** these calculations
 - Must **produce** a specific report



2. Nonfunctional Requirements

1. **Operational** – Physical/technical environment
2. **Performance** – Speed and reliability
3. **Security** – Who can use the system
4. **Cultural & Political** – Company policies, legal issues

Nonfunctional Requirements

Jaringan Dokumentasi dan Informasi Hukum (JDIH)

Date: 23 Oktober 2018

Operational Requirements

1. Sistem dapat digunakan oleh semua perangkat lunak yang lainnya.
2. Sistem layanan harus memiliki tingkat ketersediaan 99%.
3. System harus go live sebelum tanggal 25 Januari 2018.

Performance Requirements

1. Waktu respon harus tidak lebih dari 2 detik.
2. Mampu diakses oleh lebih dari 50 pengguna secara bersamaan.
3. Server harus bersih dari virus.

Security Requirements

1. Akses ke dalam system tanpa otorisasi tidak dimungkinkan.
2. Data hanya dapat diubah oleh administrator sistem.
3. Seluruh data harus di-backup setiap 24jam serta hasil backup-nya disimpan di lokasi yang berbeda dengan sistem.
4. Seluruh komunikasi antara client-server harus dipastikan keamanannya.

Deadline Requirements

1. Pengembangan aplikasi tidak boleh melewati jangka waktu yang sudah ditentukan.

Metode Requirement Gathering

1. Document Analysis
2. Interviews
3. Joint Application Design (JAD)
4. Questionnaires
5. Observation

1. Document Analysis

- Provides clues about the "formal" existing **As-Is system**
- Typical **documents**
 - Forms
 - Reports
 - Policy manuals
- Look for user additions to forms
- Look for **unused form** elements
- Do document analysis before interviews

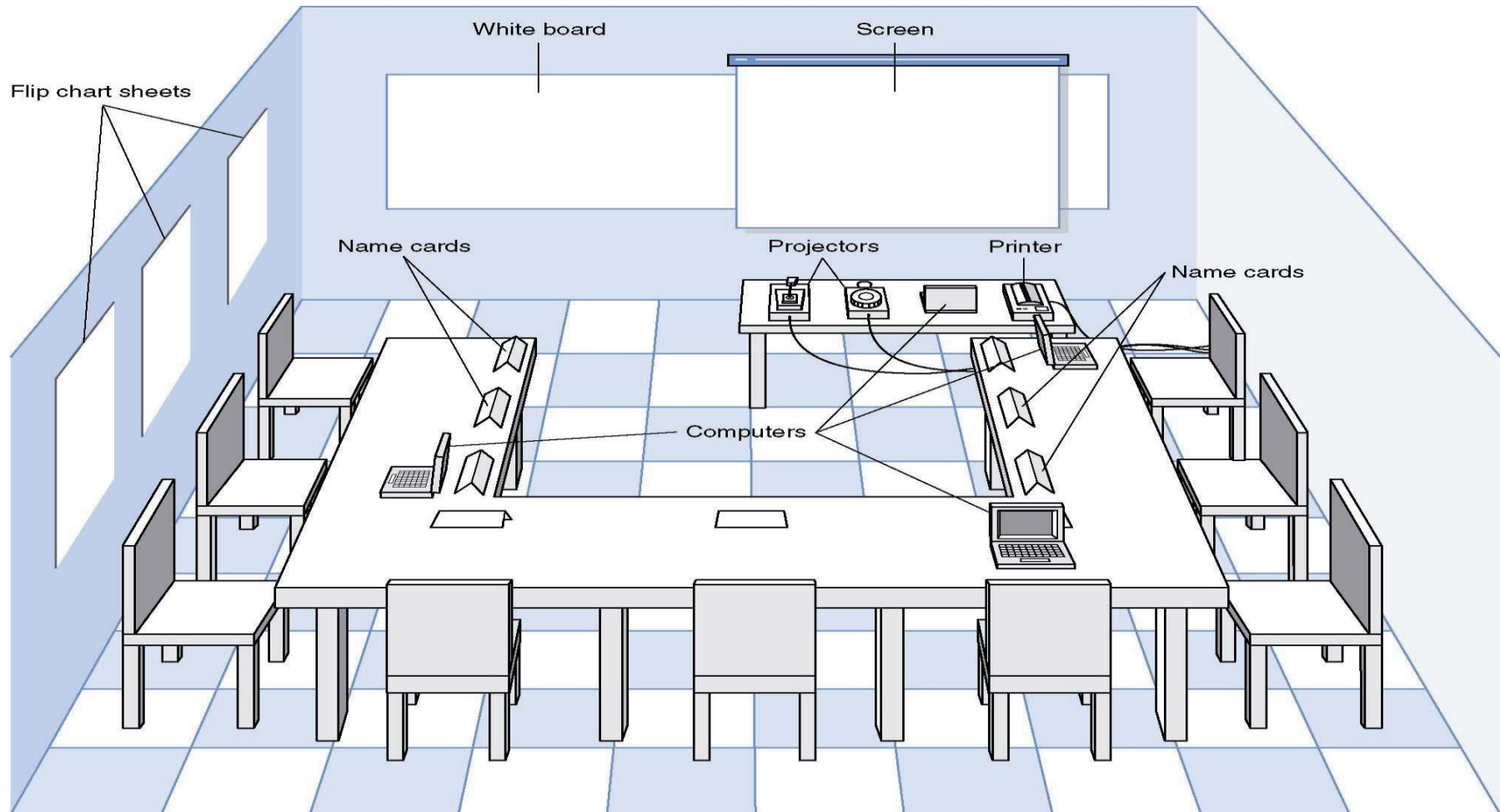
2. Interviews

- Most **commonly used technique** and very natural
- If you need to know something, you **ask someone**
- Five basic **steps**:
 1. Selecting interviewees
 2. Designing interview questions
 3. Preparing for the interview
 4. Conducting the interview
 5. Post-interview follow-up

3. Joint Application Design (JAD)

- Allows project managers, users, and developers to work together
- May reduce scope creep by 50%
- Avoids requirements being too specific or too vague
- Include 10 to 20 users
- Tend to last 5 to 10 days over a three-week period

JAD Meeting Room



4. Questionnaire

- **Selecting** participants
 - Using **samples** of the population
- **Designing the questionnaire**
 - More important than interview questions
 - Prioritize questions to grab attention
 - Distinguish between:
 1. **Fact-oriented questions** (specific answers)
 2. **Opinion questions** (agree – disagree scale)
- **Administering** the questionnaire
 - Explain its **importance** & **how it will be used**
 - Give **expected response date**
 - **Follow up** on late returns and have **supervisors** follow up
 - **Promise to report** results
- **Questionnaire follow-up**
 - Send results to participants

5. Observation

- Users/managers often **don't remember** everything they do
- **Validates info gathered** in other ways
- **Behaviors change** when people are **watched**
- **Keep low profile**, don't change the process
- Careful **not to ignore** periodic activities
 - Weekly ... Monthly ... Annual

Karakteristik Metode Requirement Gathering

	Interviews	JAD	Questionnaires	Document Analysis	Observation
Type of Information	As-Is Improve. To-Be	As-Is Improve. To-Be	As-Is Improve.	As-Is	As-Is
Depth of Information	High	High	Medium	Low	Low
Breadth of Information	Low	Medium	High	High	Low
Integration of Info.	Low	High	Low	Low	Low
User Involvement	Medium	High	Low	Low	Low
Cost	Medium	Low-Medium	Low	Low	Low-Medium

Business Process Analysis Strategies

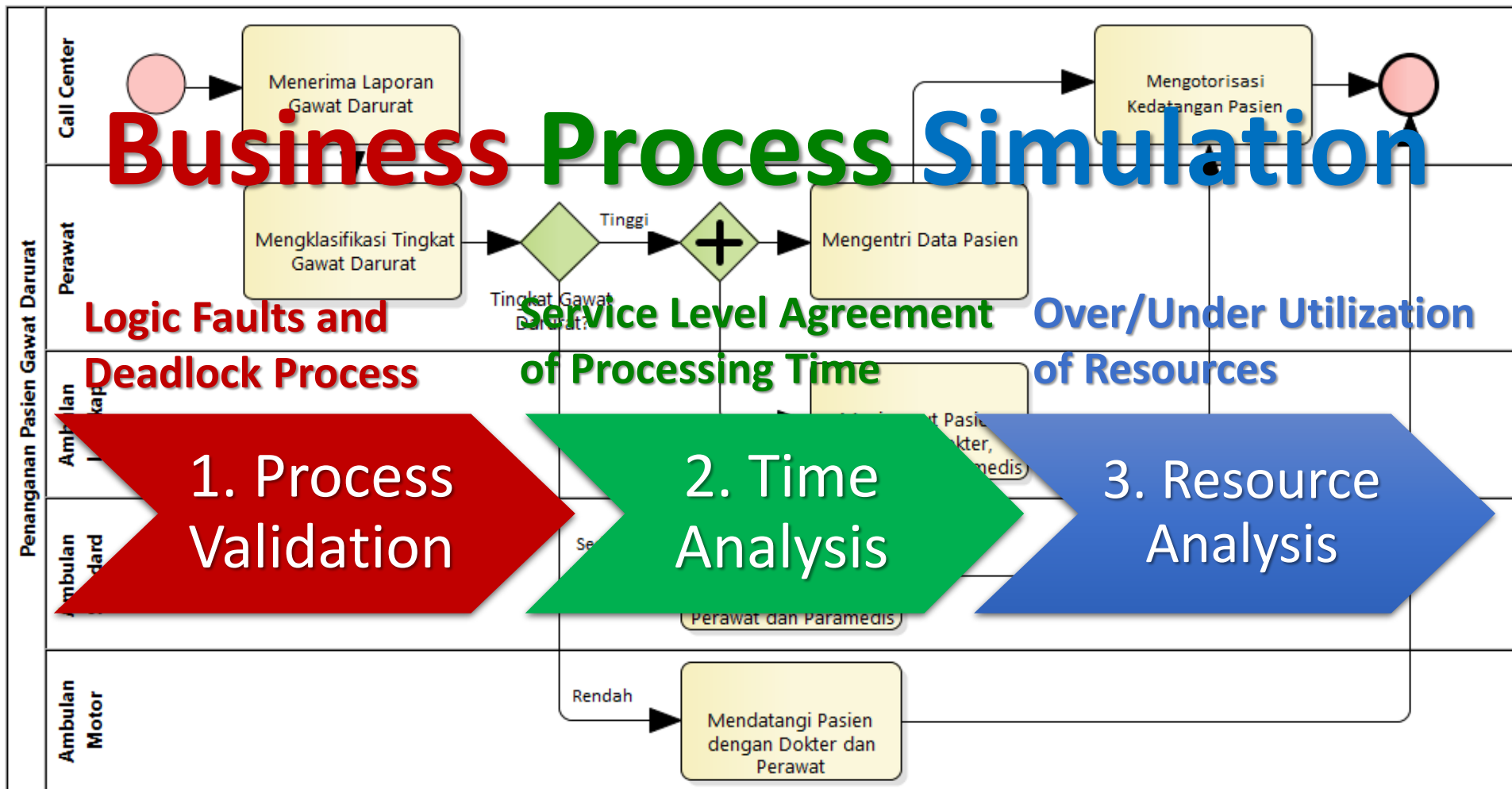
1. BPA (Business Process **Automation**)
 - Makes almost **no changes** to business processes, just makes them more efficient
2. BPI (Business Process **Improvement**)
 - **Change** what the users do, not just how efficiently they do it
3. BPR (Business Process **Reengineering**)
 - Throw away everything, **start with a blank** page

Business Process Analysis Strategies

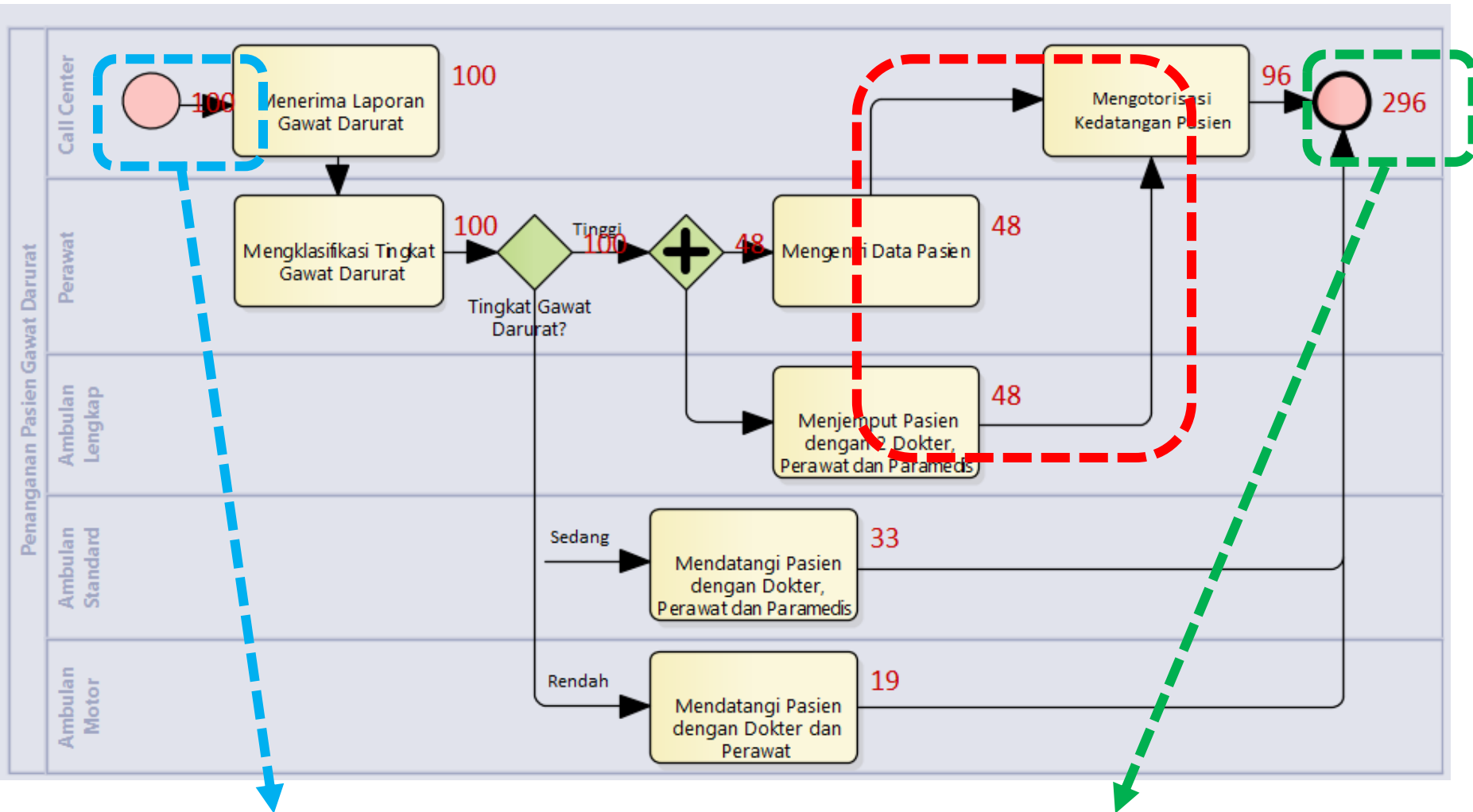
Comparison

	Business Process Automation	Business Process Improvement	Business Process Reengineering
Potential Business Value	Low-Moderate	Moderate	High
Project Cost	Low	Low-Moderate	High
Breadth of Analysis	Narrow	Narrow-Moderate	Very Broad
Risk	Low	Low-Moderate	Very High

Business Process Simulation: Penanganan Pasien Gawat Darurat



Validasi Proses Bisnis

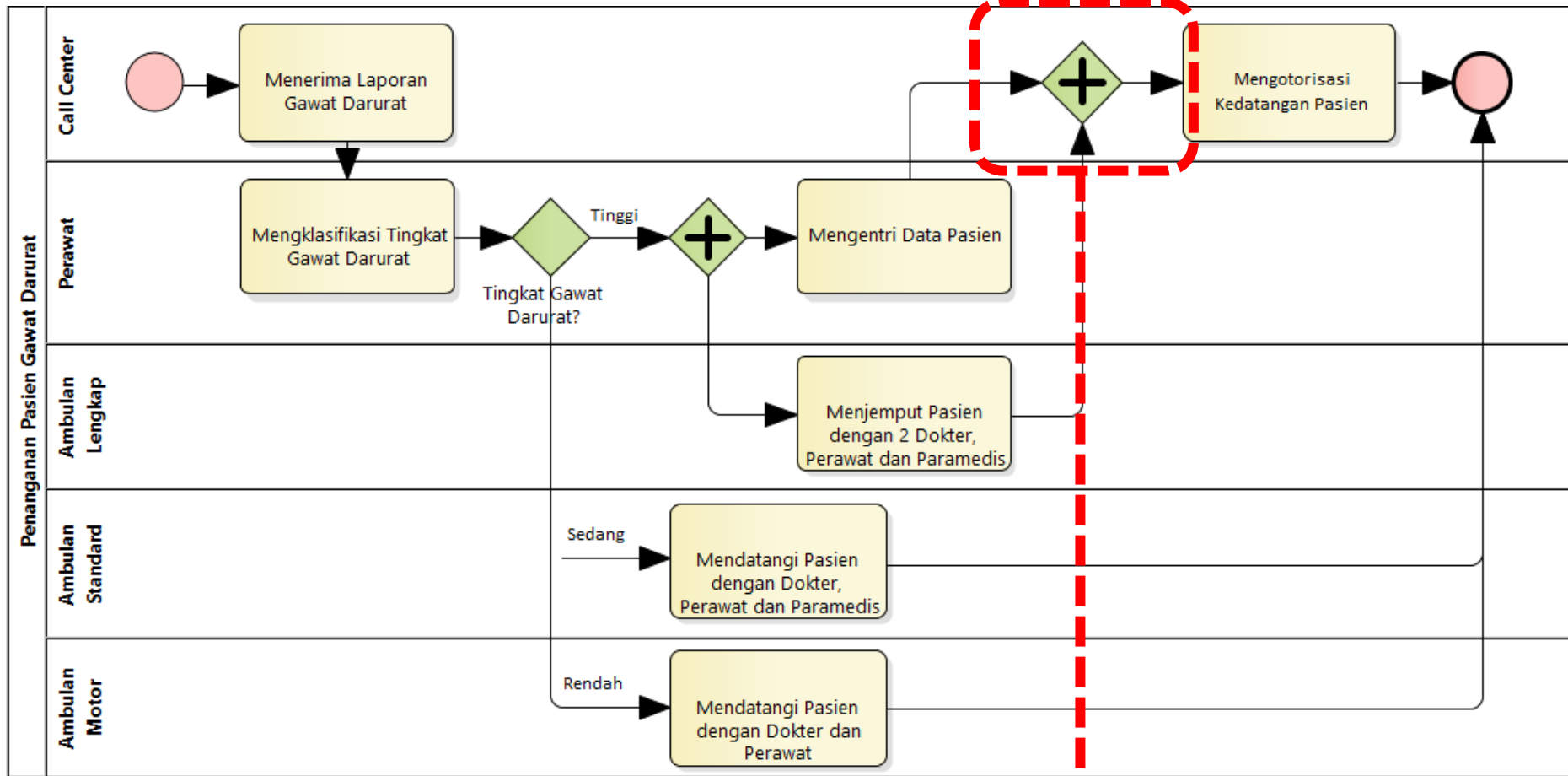


Menangani 100 Orang Pasien

Seperti Menangani 296 Orang Pasien

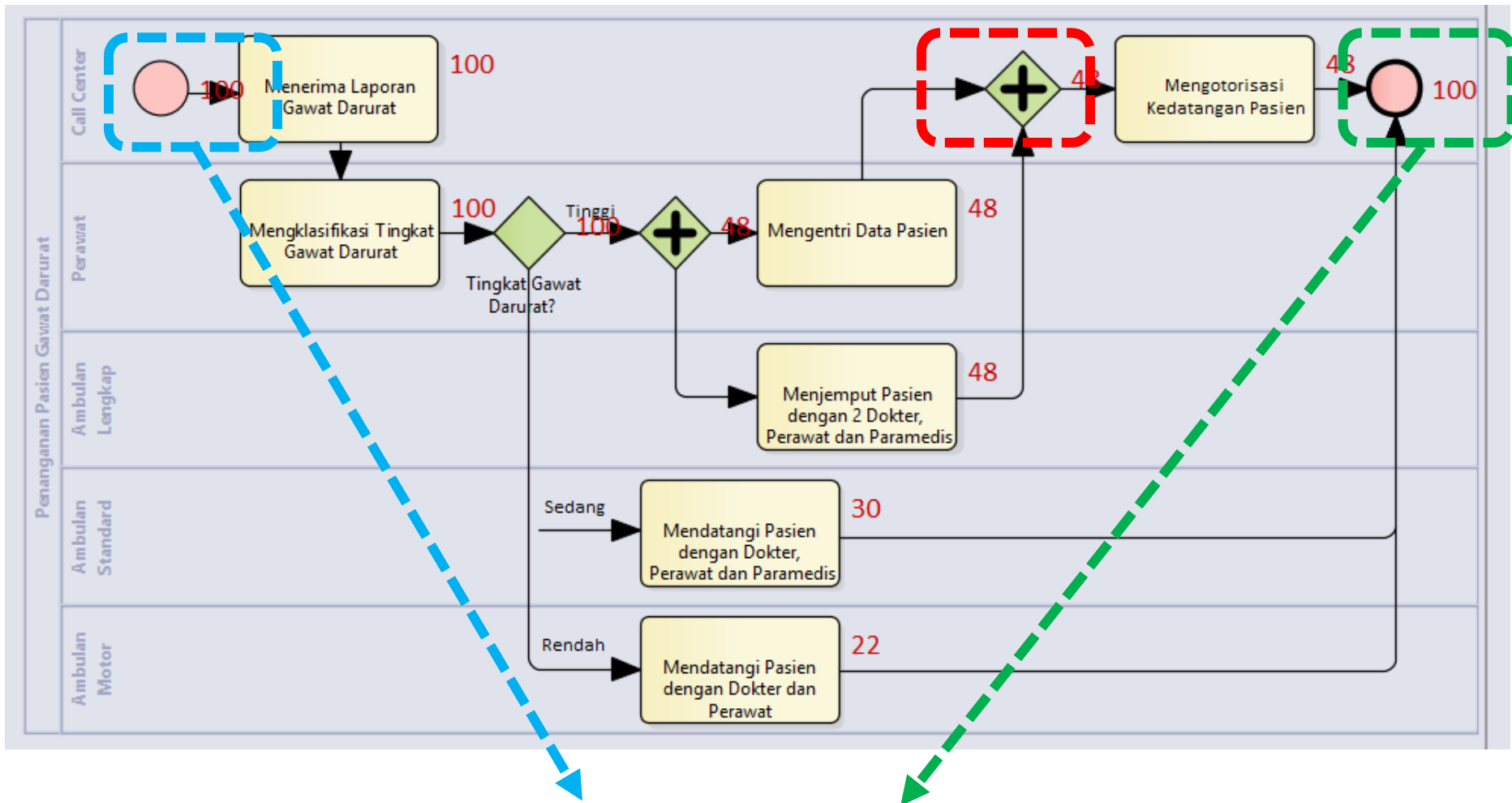
PROSES BISNIS INI TIDAK EFISIEN!

Validasi Proses Bisnis



**Perlu Sinkronisasi Pekerjaan,
Sebelum Otorisasi Kedatangan Dilakukan**

Validasi Proses Bisnis

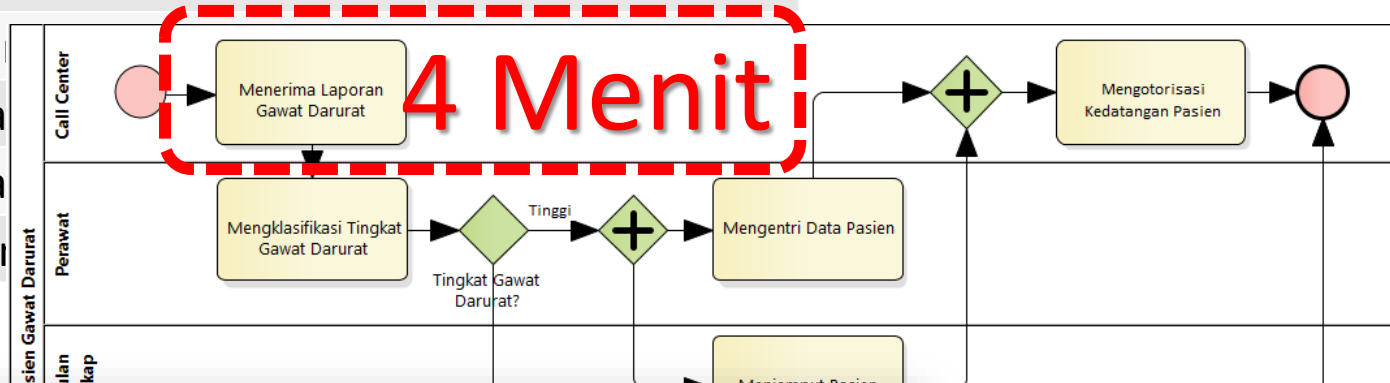


**Proses Bisnis Terbukti Valid dan
100 Pasien Tertangani dengan Efisien dan Efektif!**

Analisis Waktu

Aktifitas	Waktu (Menit)
Menerima Laporan Gawat Darurat	4
Mengklasifikasi Tingkat Gawat Darurat	5
Mengentri Data Pasien	11

Menjemput Pasien dengan
Mendatangi Pasien dengan
Mendatangi Pasien dengan
Mengotorisasi Kedatangan



Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
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Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat

Average Time

249.63

Mean Of Processing Time

17.73

**Perlu Waktu 17.73 Menit
untuk Menangani 1 Pasien**

Number Of Processes Started

100

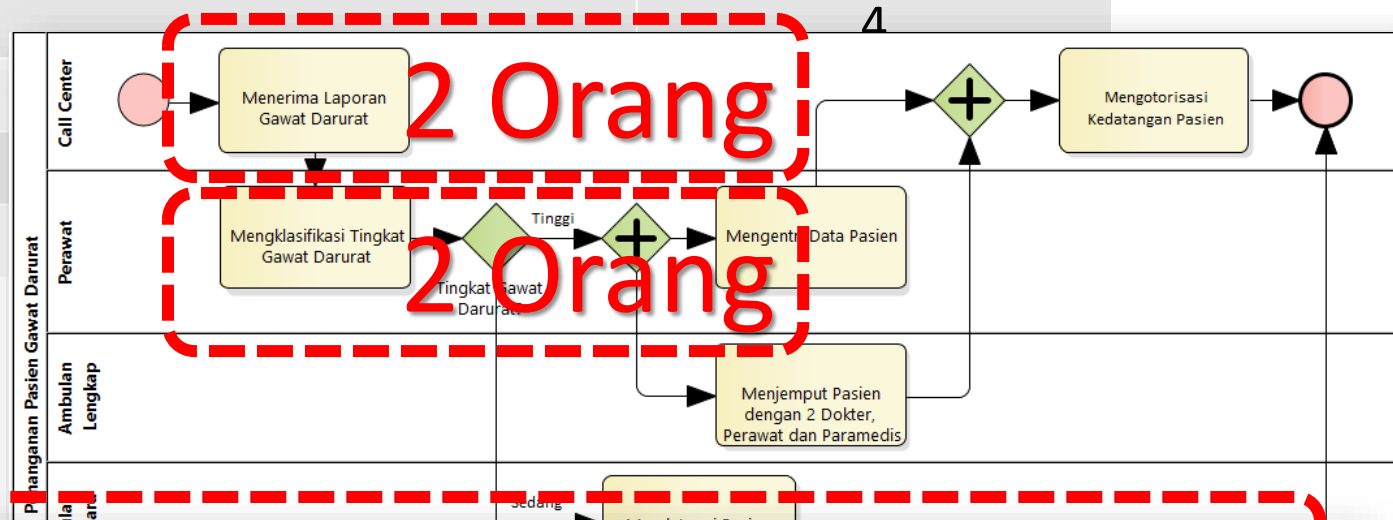
Standard Deviation Time

131.97

Analisis Sumber Daya Manusia

Sumber Daya	Kuantitas
Call center	2
Perawat	2
Ambulan Lengkap	4

Ambulan Standar
Ambulan Motor
Resepsionis

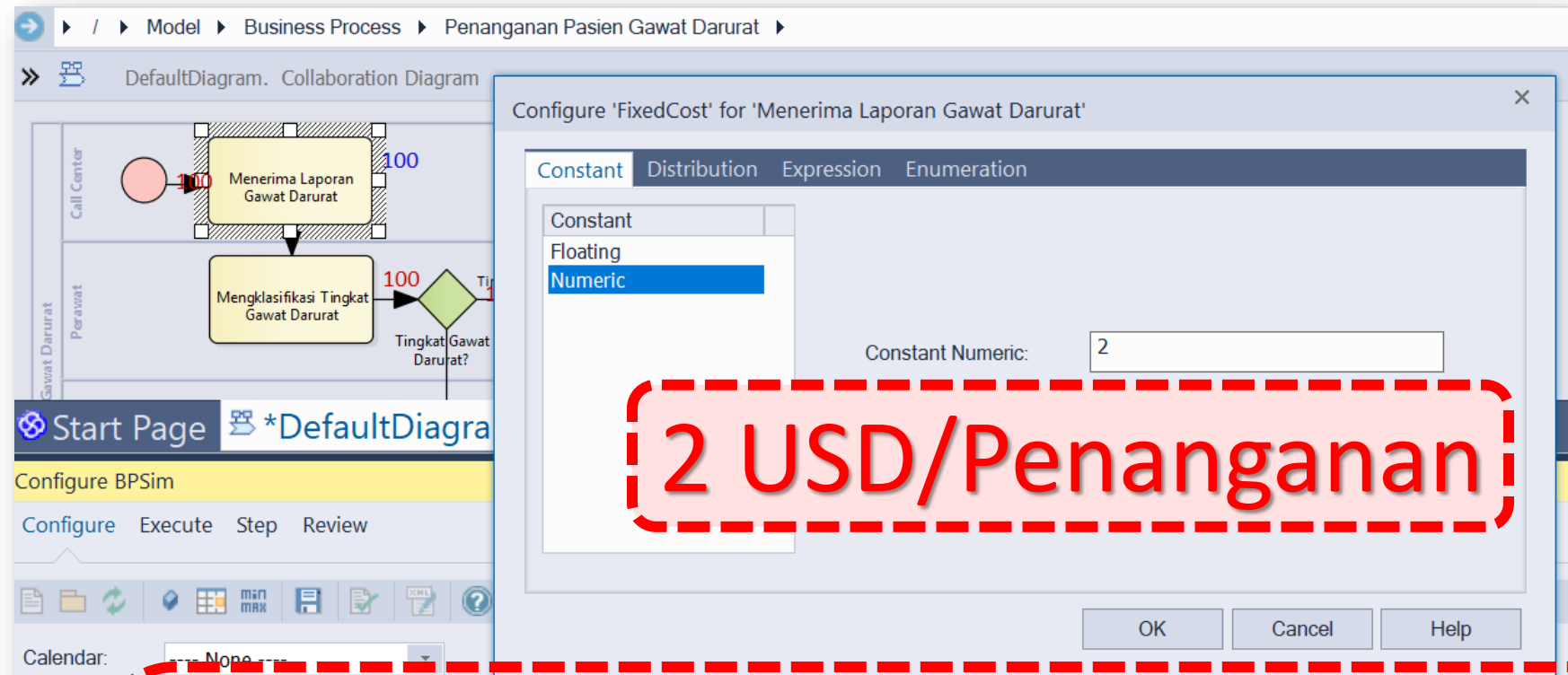


Perawat	Degree Of Utilisation	99%
Paramedis	Degree Of Utilisation	98.34%
Call Center	Degree Of Utilisation	65.56%
Dokter	Degree Of Utilisation	50.28%
Supir	Degree Of Utilisation	33.22%
Ambulan Standard	Degree Of Utilisation	33.22%
Ambulan Motor	Degree Of Utilisation	17.05%

Overutilization

Underutilization

Analisis Biaya



2 USD/Penanganan

Total Biaya Perhari

Menerima Laporan Gawat Darurat	Total Completion Cost	200
Mengentri Data Pasien	Total Completion Cost	48
Mengklasifikasi Tingkat Gawat Darurat	Total Completion Cost	100
Mengotorisasi Kedatangan Pasien	Total Completion Cost	48

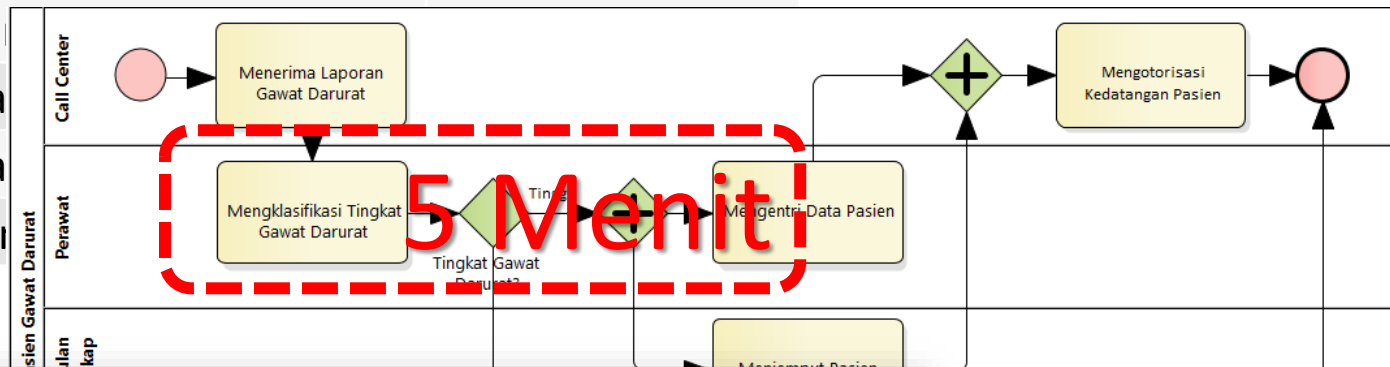
Business Process Automation

- Dikembangkan Sistem Cerdas yang membantu perawat dengan **Mengautomasi Klasifikasi Tingkat Gawat Darurat**
- Pekerjaan “Mengklasifikasi Tingkat Gawat Darurat” yang **sebelumnya perlu waktu 5 menit**
 - Dengan sistem cerdas menjadi **hanya 1 menit**

Business Process Simulation

Aktifitas	Waktu (Menit)
Menerima Laporan Gawat Darurat	4
Mengklasifikasi Tingkat Gawat Darurat	5
Mengentri Data Pasien	11

Menjemput Pasien dengan
Mendatangi Pasien dengan
Mendatangi Pasien dengan
Mengotorisasi Kedatangan



Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
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Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat

Average Time 249.63

Mean Of Processing Time 17.73

Number Of Processes Started 100

Standard Deviation Time 131.97

**Perlu Waktu 17.73 Menit
untuk Menangani 1 Pasien**

Business Process Automation

Aktifitas	Waktu (Menit)
Menerima Laporan Gawat Darurat	4
Mengklasifikasi Tingkat Gawat Darurat	1
Mengentri Data Pasien	11

Menjemput Pasien denga
Mendatangi Pasien denga
Mendatangi Pasien denga
Mengotorisasi Kedatanga



Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat
Penanganan Pasien Gawat Darurat

Average Time

249.63

Hanya Perlu Waktu 8 Menit
untuk Menangani 1 Pasien

Number Of Processes Completed

52

Number Of Processes Started

100

Standard Deviation Time

131.97

Tantangan di Requirement Gathering

1. The "Yes, But" syndrome

- Stems from **human nature** and the **users' inability** to experience the software as they might a physical device

2. The "Undiscovered Ruins"

- Searching for requirements is like **searching** for "Undiscovered Ruins"
- The **more you find**, the **more you know** remain

3. The "User and the Developer" syndrome

- Reflects the profound differences between these two, making **communication difficult**

■ ■ ■ The "Yes, But" Syndrome

For whatever reason, we always see **two** immediate, distinct, and separate **reactions** when the users see the system implementation for the first time

1. "**Wow, this is so cool**; we can really use this, what a neat job" and so on.
2. "Yes, **but, hmmm**, now that I see it, what about this ... ? Wouldn't it be nice if ... ? Whatever happened to ... ?"

The "Undiscovered Ruins" Syndrome

- In many ways, the search for requirements is **like a search for undiscovered ruins**
 - The more you find, the more you know remain
 - You never really feel that you have found them all, and perhaps you never will.
- Indeed, software development teams always **struggle to determine when they are done** with requirements elicitation. When have they found
 - all the requirements
 - or at least enough requirements?

The "User and the Developer" Syndrome

- **Communication gap** between the user and the developer
- **Users and developers** are
 - typically **from different worlds**,
 - may even **speak different languages**,
 - have **different backgrounds**,
 - have **different motivations**, and
 - have **different objectives**

■ ■ ■ The "User and the Developer" Syndrome

Reasons for this problem and some **suggested solutions**

Problem	Solution
Users do not know what they want, or they know what they want but cannot articulate it.	Recognize and appreciate the user as domain expert; try alternative communication and elicitation techniques.
Users think they know what they want until developers give them what they said they wanted.	Provide alternative elicitation techniques earlier: storyboarding, role playing, throwaway prototypes, and so on.
Analysts think they understand user problems better than users do.	Put the analyst in the user's place. Try role playing for an hour or a day.
Everybody believes everybody else is politically motivated.	Yes, its part of human nature, so let's get on with the program.

Evolusi Paradigma Analysis dan Design

	Paradigm	Diagrams
1	Process-oriented Paradigm	Flowchart
2	Data-oriented Paradigm	DFD
3	Object-oriented Paradigm (data + process)	UML

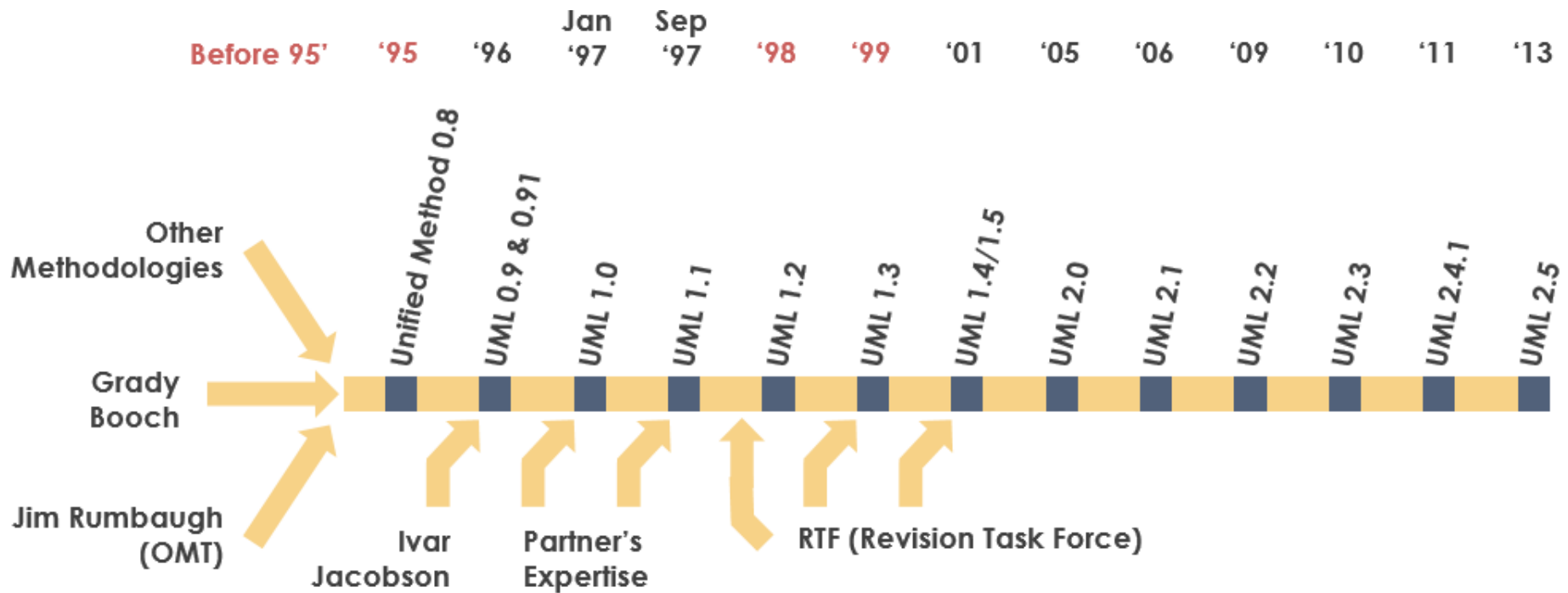
What is the UML?

- UML: **Unified Modeling Language**
- UML can be used for **modeling all processes in the development life cycle** and across different implementation technologies (technology and language independent)
- UML is the **standard language** for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system
- UML is a **communication tool** – for the team, and other stakeholders

Sejarah UML



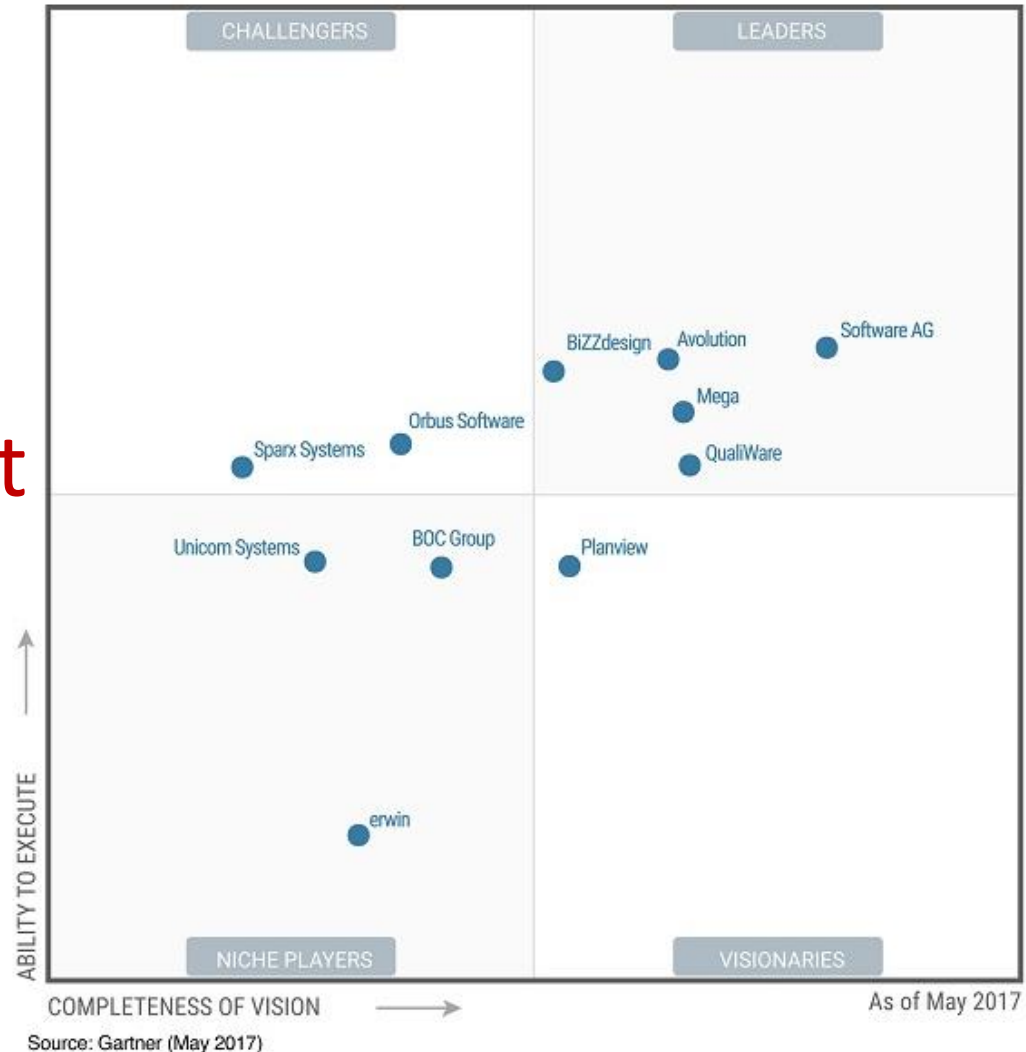
Booch, Jacobson, Rumbaugh



Before 95' - Fragmentation ► 95' - Unification ► 98' - Standardization ► 99' - Industrialization

State of the Art UML Tools

- Rational Rose
- Visual Paradigm
- Sparx Systems Enterprise Architect
- Microsoft Visio
- Star UML



UML 2.0 Diagrams

Diagram Name	Used to	Primary Phase
Structure Diagrams		
Class	Illustrate the relationships between classes modeled in the system.	Analysis, Design
Object	Illustrate the relationships between objects modeled in the system. Used when actual instances of the classes will better communicate the model.	Analysis, Design
Package	Group other UML elements together to form higher level constructs.	Analysis, Design, Implementation
Deployment	Show the physical architecture of the system. Can also be used to show software components being deployed onto the physical architecture.	Physical Design, Implementation
Component	illustrate the physical relationships among the software components.	Physical Design, Implementation
Composite Structure	Illustrate the internal structure of a class, i.e., the relationships among the parts of a class.	Analysis, Design
Behavioral Diagrams		
Activity	Illustrate business workflows independent of classes, the flow of activities in a use case, or detailed design of a method.	Analysis, Design
Sequence	Model the behavior of objects within a use case. Focuses on the time-based ordering of an activity.	Analysis, Design
Communication	Model the behavior of objects within a use case. Focuses on the communication among a set of collaborating objects of an activity.	Analysis, Design
Interaction Overview	Illustrate an overview of the flow of control of a process.	Analysis, Design
Timing	Illustrate the interaction that takes place among a set of objects and the state changes in which they go through along a time axis.	Analysis, Design
Behavioral State Machine	Examine the behavior of one class.	Analysis, Design
Protocol State Machine	Illustrates the dependencies among the different interfaces of a class.	Analysis, Design
Use-Case	Capture business requirements for the system and to illustrate the interaction between the system and its environment.	Analysis

UML Problems

1. UML is modeling notation, it is **not a development process** or a methodology
 - UML driven development process?
2. UML is **too complex, difficult to understand quickly**
 - Which UML diagrams should we use?

UML based Software Analysis and Design

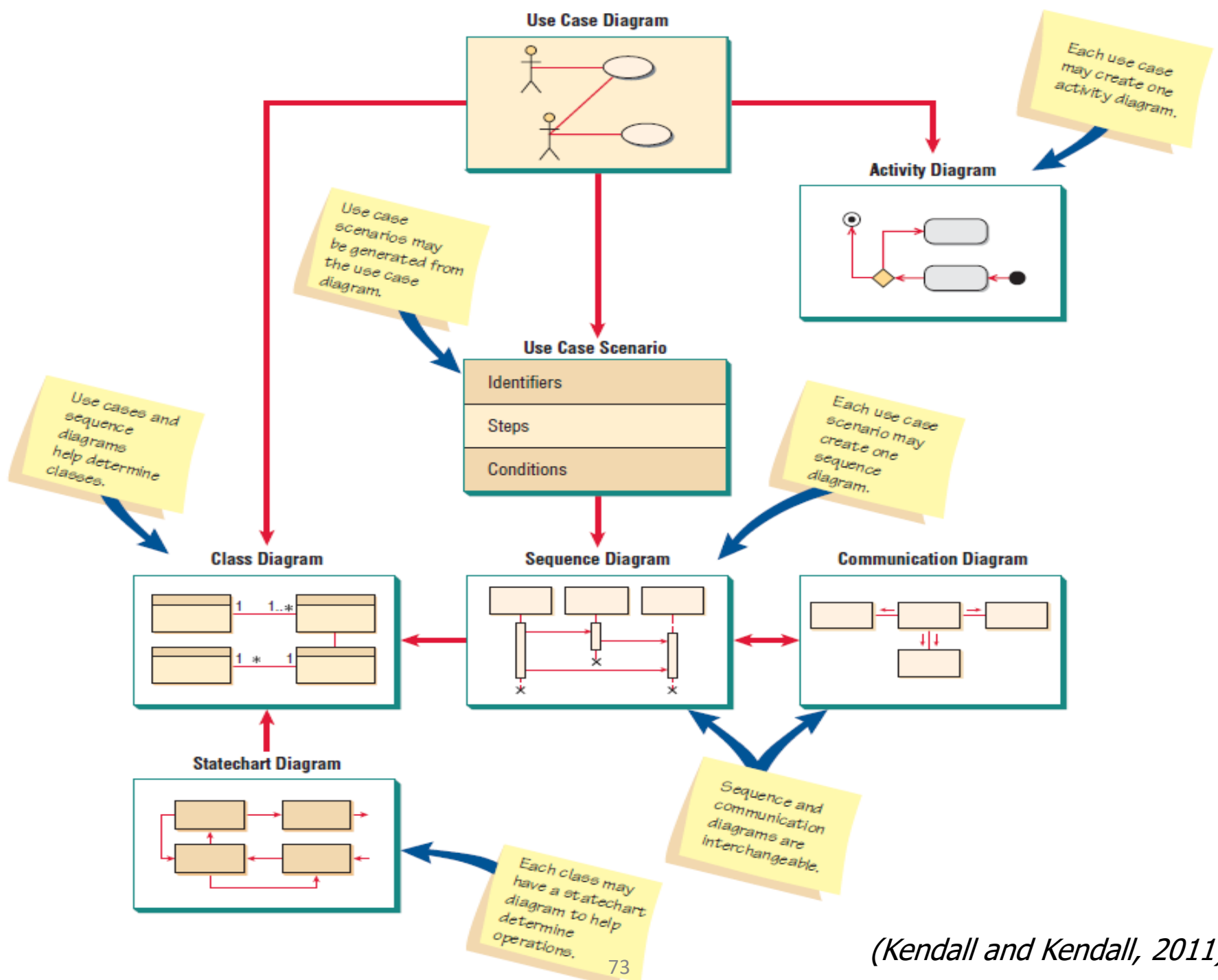
(Sparx Systems EA)

1. Display the boundary of a system and its major functions using **use cases and actors**
2. Model the organization's business process with **activity diagram**
3. Illustrate use case realizations with **sequence diagrams**
4. Represent a static structure of a system using **class diagrams**
5. Reveal the physical implementation architecture with **deployment diagrams**

UML based Software Analysis and Design

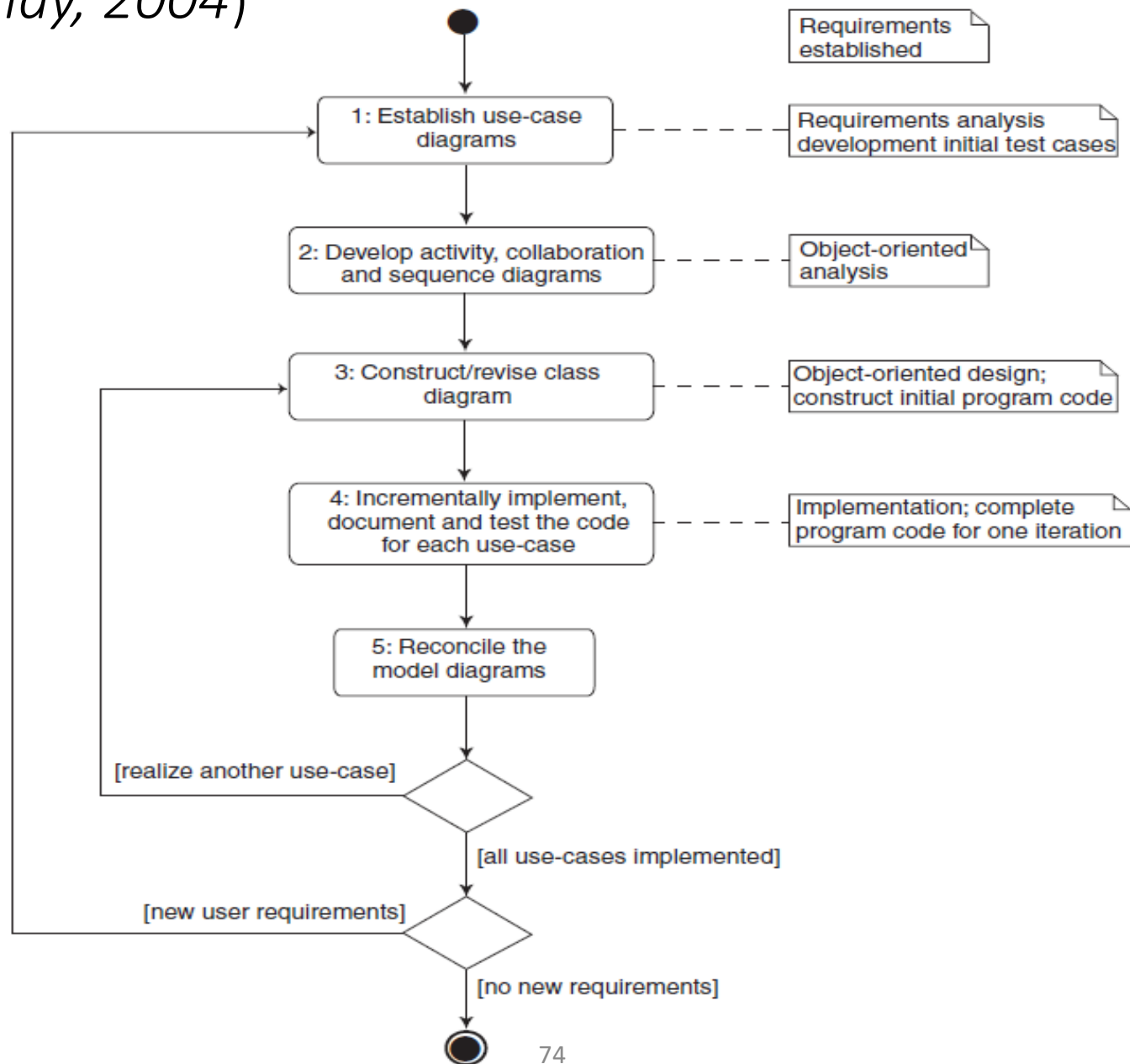
(Kendal, 2011)

1. A **use case diagram**, describing how the system is used. **Analysts start with a use case diagram**
2. An **activity diagram**, illustrating the overall flow of activities. **Each use case may create one activity diagram**
3. **Sequence diagrams**, showing the sequence of activities and class relationships. **Each use case may create one or more sequence diagrams**
4. **Class diagrams**, showing the classes and relationships. **Sequence diagrams are used to determine classes**
5. **Statechart diagrams**, showing the state transitions. **Each class may create a statechart diagram, which is useful for determining class methods**



UML based Software Analysis and Design

(Barclay, 2004)



UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

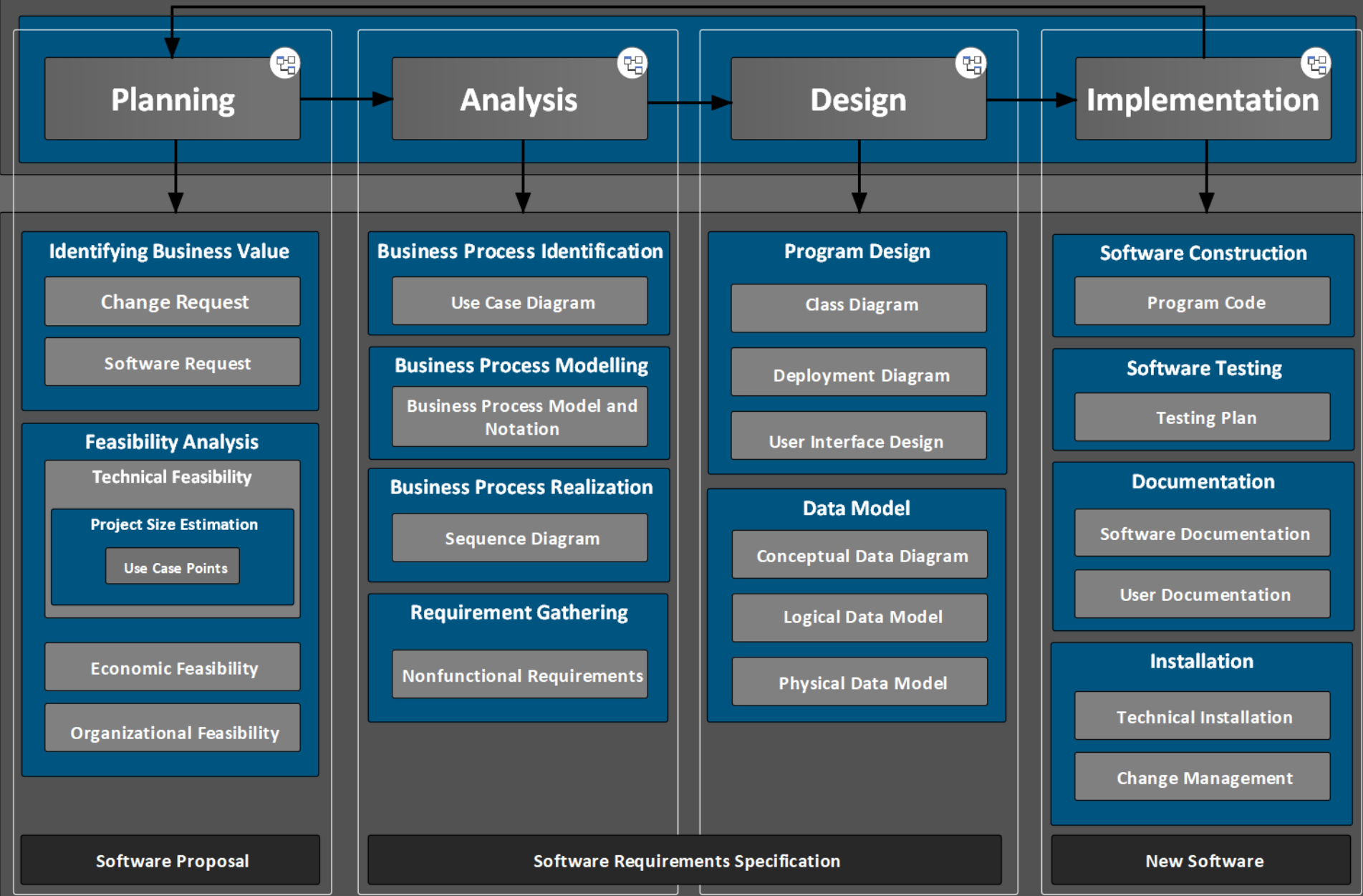
2.2 Pemodelan **User Interface Design**

2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

Application Development Governance

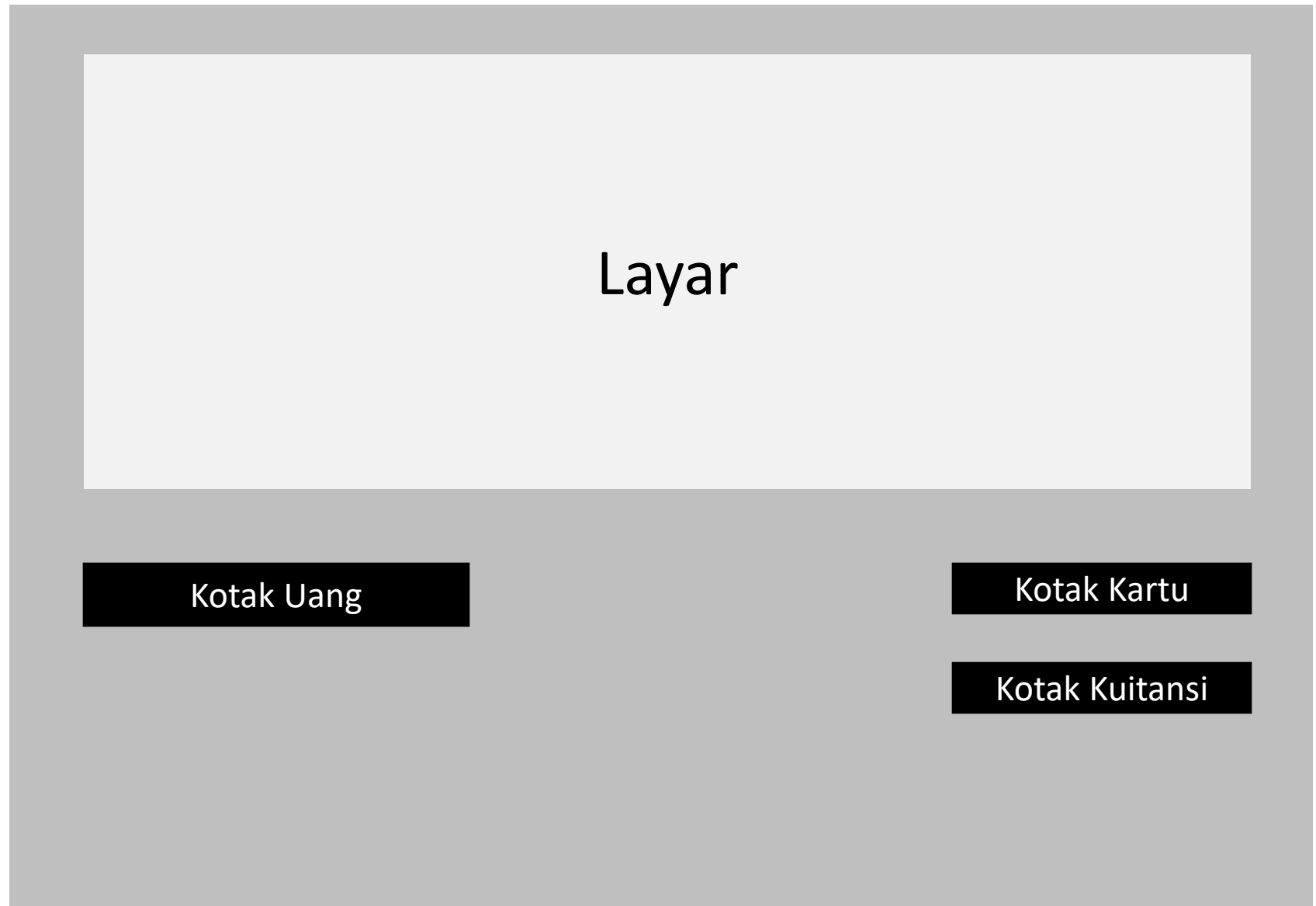
Software Development Life Cycle



Studi Kasus: ATM System



ATM System



Menu PIN

Masukkan PIN:

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Utama

1. Mengecek Saldo
2. Mentransfer Uang
3. Mengambil Uang
4. Logout

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengecekan Saldo

1. Saldo anda adalah ...

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengiriman Uang

1. No Account Penerima:

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengiriman Uang

1. Jumlah uang yang dikirim:

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengiriman Uang

1. Uang berhasil terkirim

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengambilan Uang

1. Jumlah uang yang diambil:

Kotak Uang

Kotak Kartu

Kotak Kuitansi

Menu Pengambilan Uang

Uang berhasil diambil

Kotak Uang

Kotak Kartu

Kotak Kuitansi



2.2 Identifikasi Proses Bisnis dengan Use Case Diagram

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

Use Case Diagram

- Summarized into a **single picture**
- All of the use cases for the part of the system being modeled
- Use case represents the discrete **activities performed by the user**
- Use Case Diagram tells **what the system will do**
- Good for **communicating with users**

Use Case Diagram Syntax

- **Actor**

- person or system that derives benefit from and is external to the subject

- **Use Case**

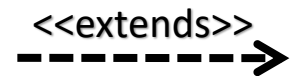
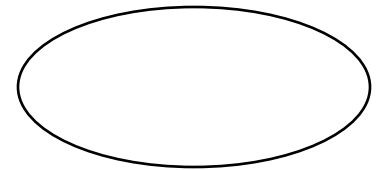
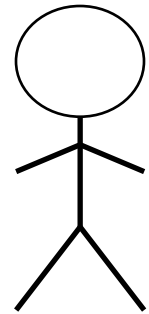
- Represents a major piece of system functionality

- **Association** Relationship

- **Include** Relationship

- **Extend** Relationship

- **Generalization** Relationship



Use Case

- A major piece of **system functionality**
- Can **extend** other Use Cases
- Placed inside system boundary
- Labeled with descriptive **verb - noun phrase**



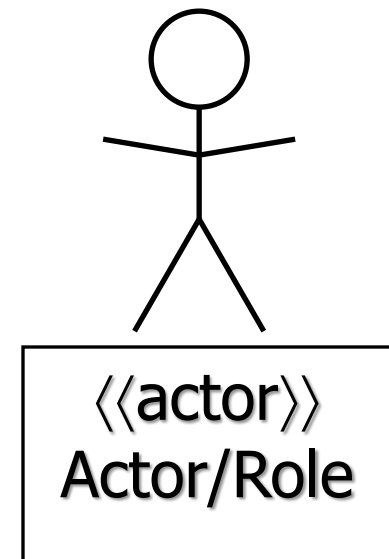
System Boundary

- Includes the **name of the system** inside or on top
- Represents the **scope of the system**
- **Actors are outside** the scope of the system

Boundary

Actor

- A **person** or **another system** that interacts with the current system
- A **role**, not a specific user
- **Provides input, receives output, or both**

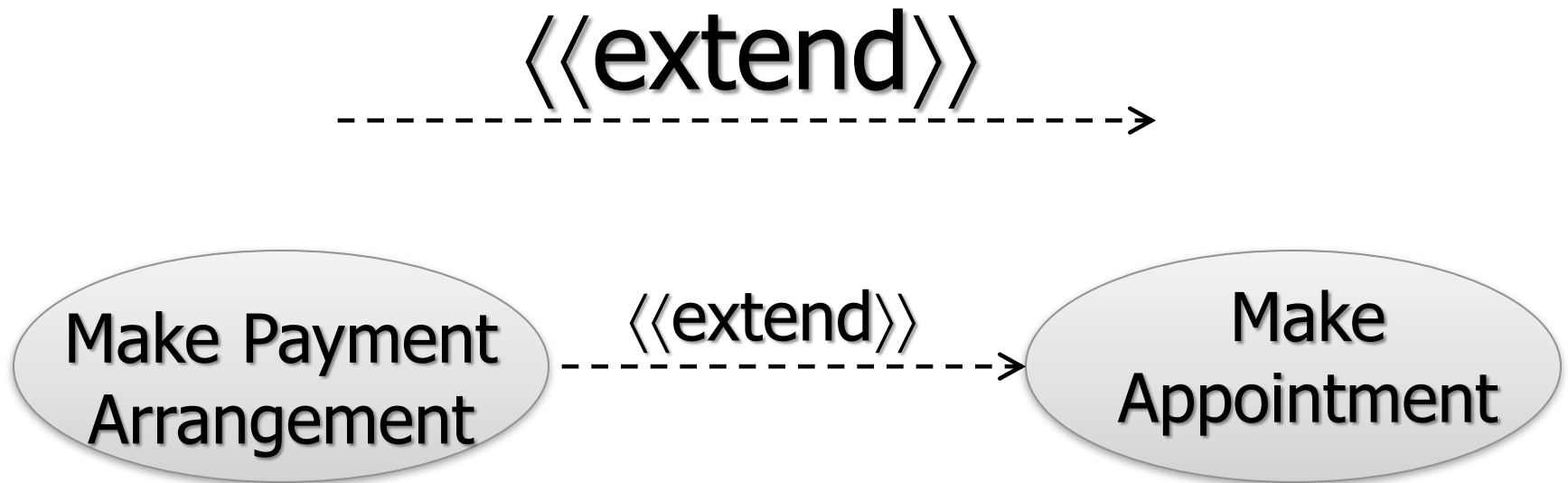


Association Relationship

- **Links** actor and the Use Case
- Shows **two-way communication**
 - If one-way, arrows are used

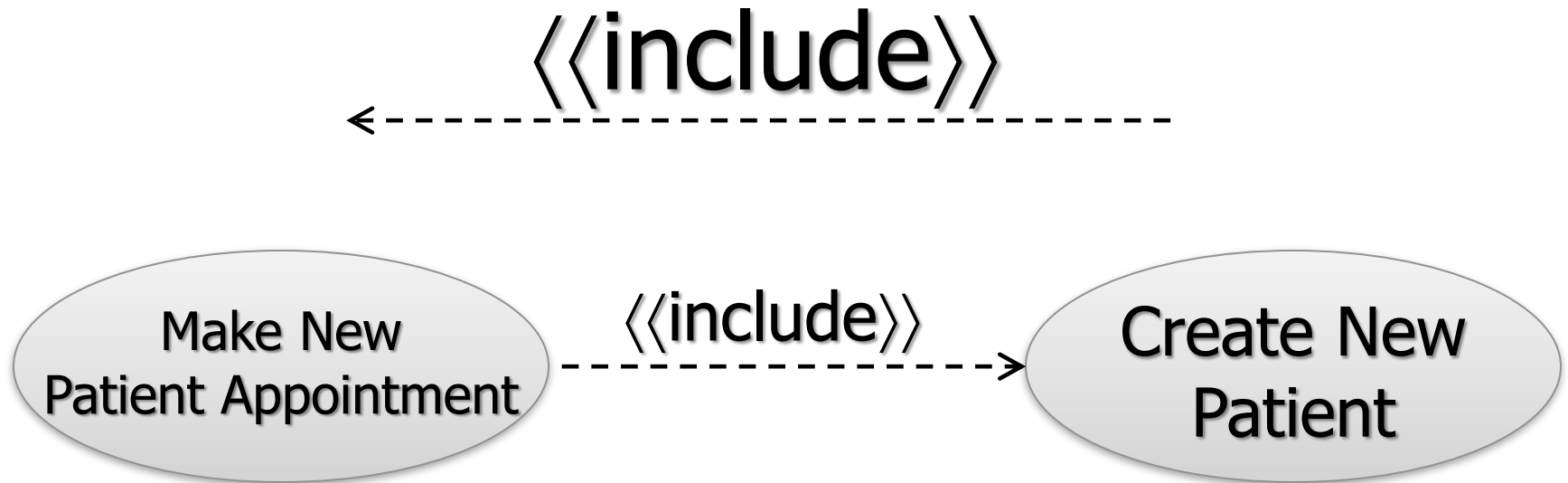
Extends Relationship

- **Extends** Use Case to include **Optional** behavior
- **Arrow points** from the extension Use Case **to** the base Use Case



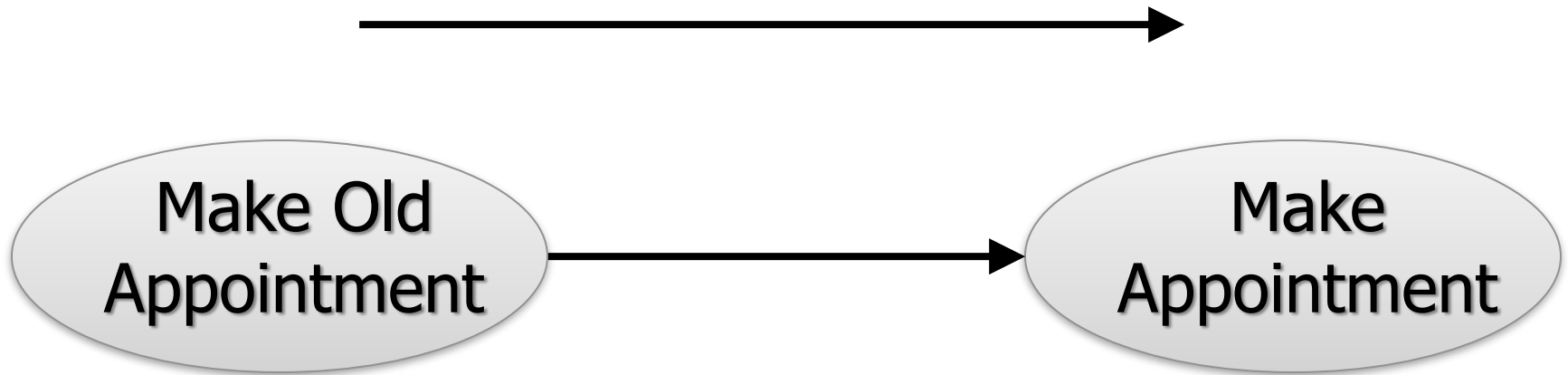
Include Relationship

- **Include** one Use Case from **within another**
- **Arrow points** from base Use Case **to the included Use Case**

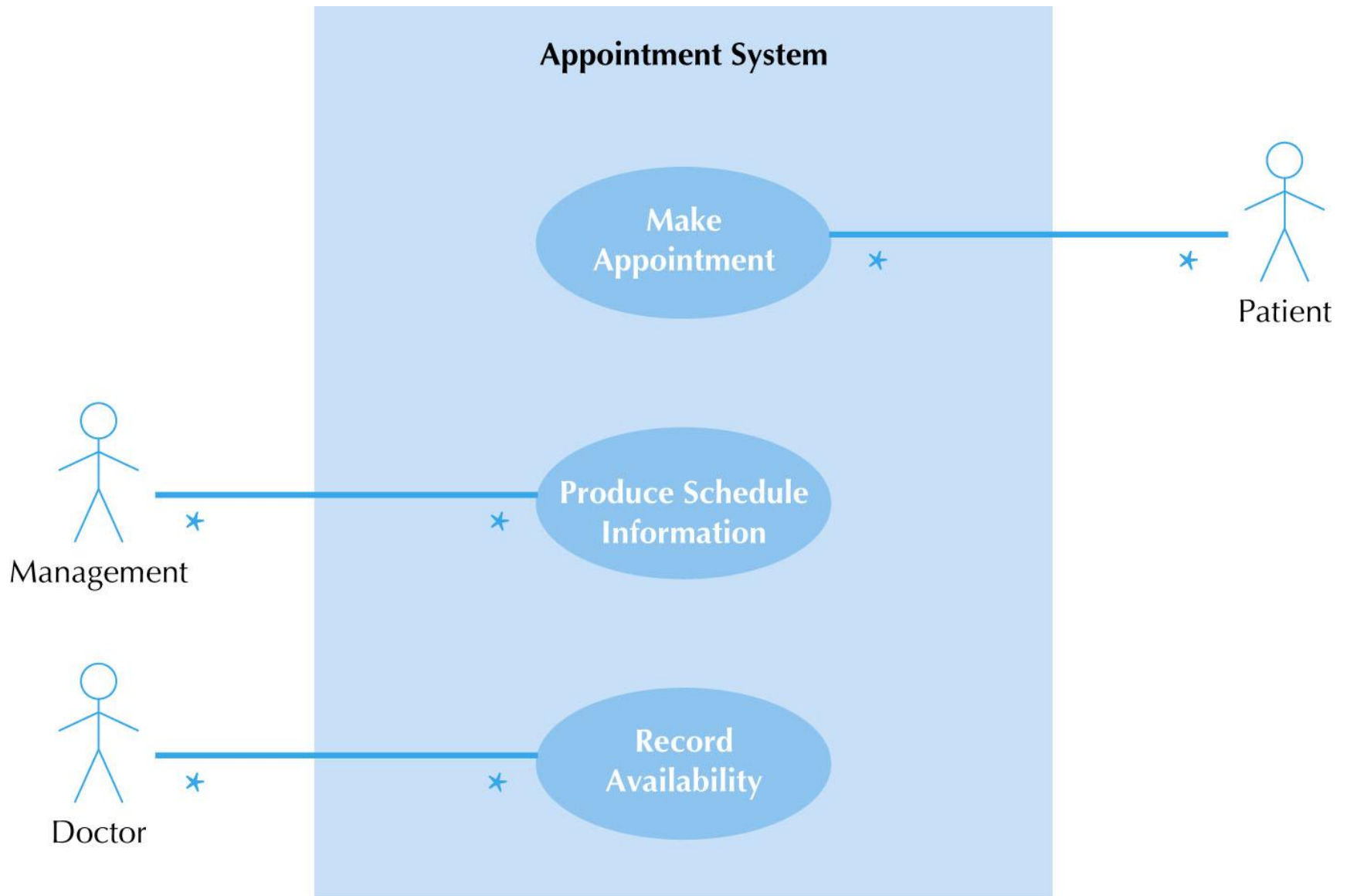


Generalization Relationship

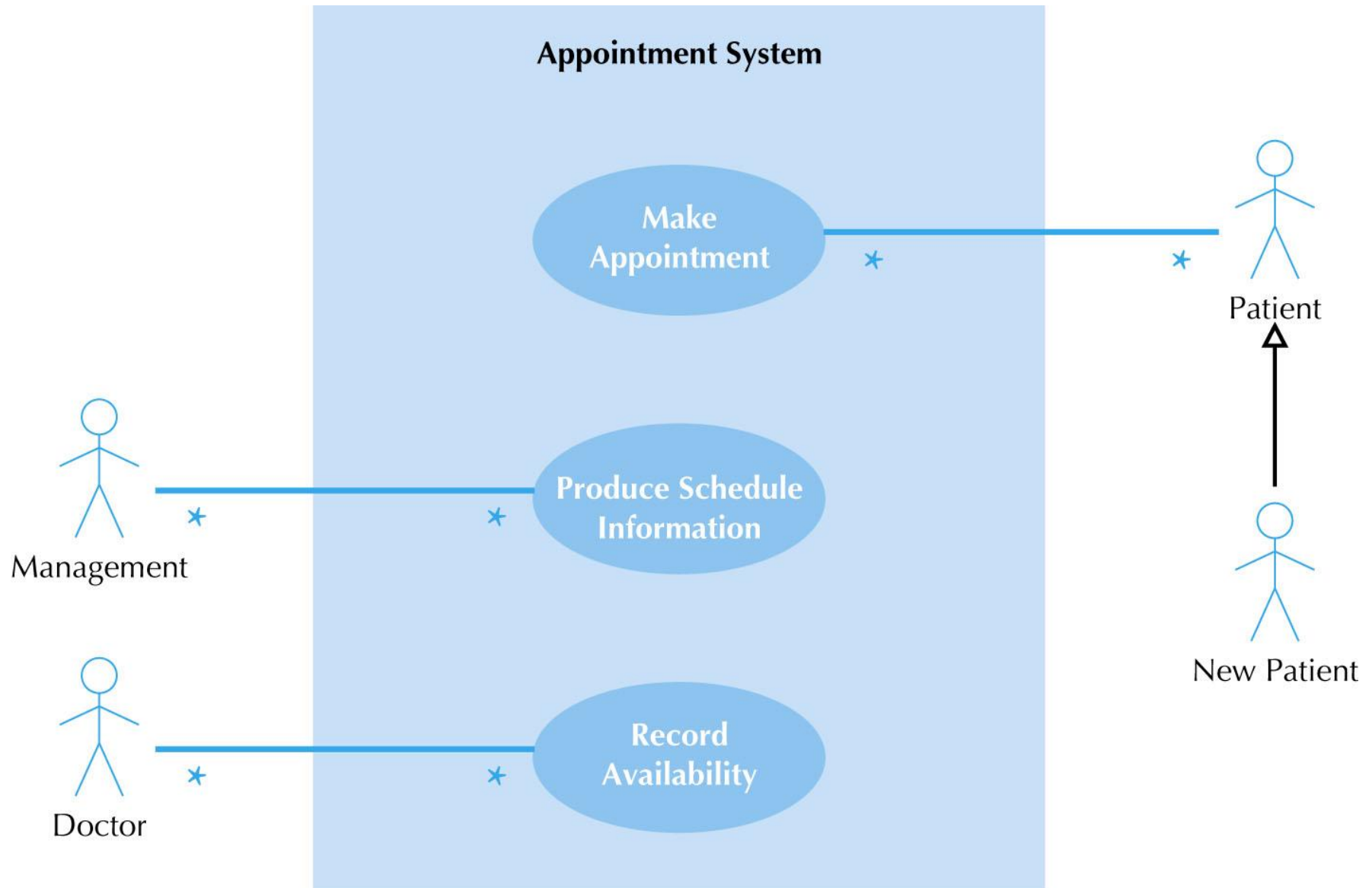
- A specialized Use Case to a more generalized Use Case
- **Arrow points** from specialized **to** general Use Case



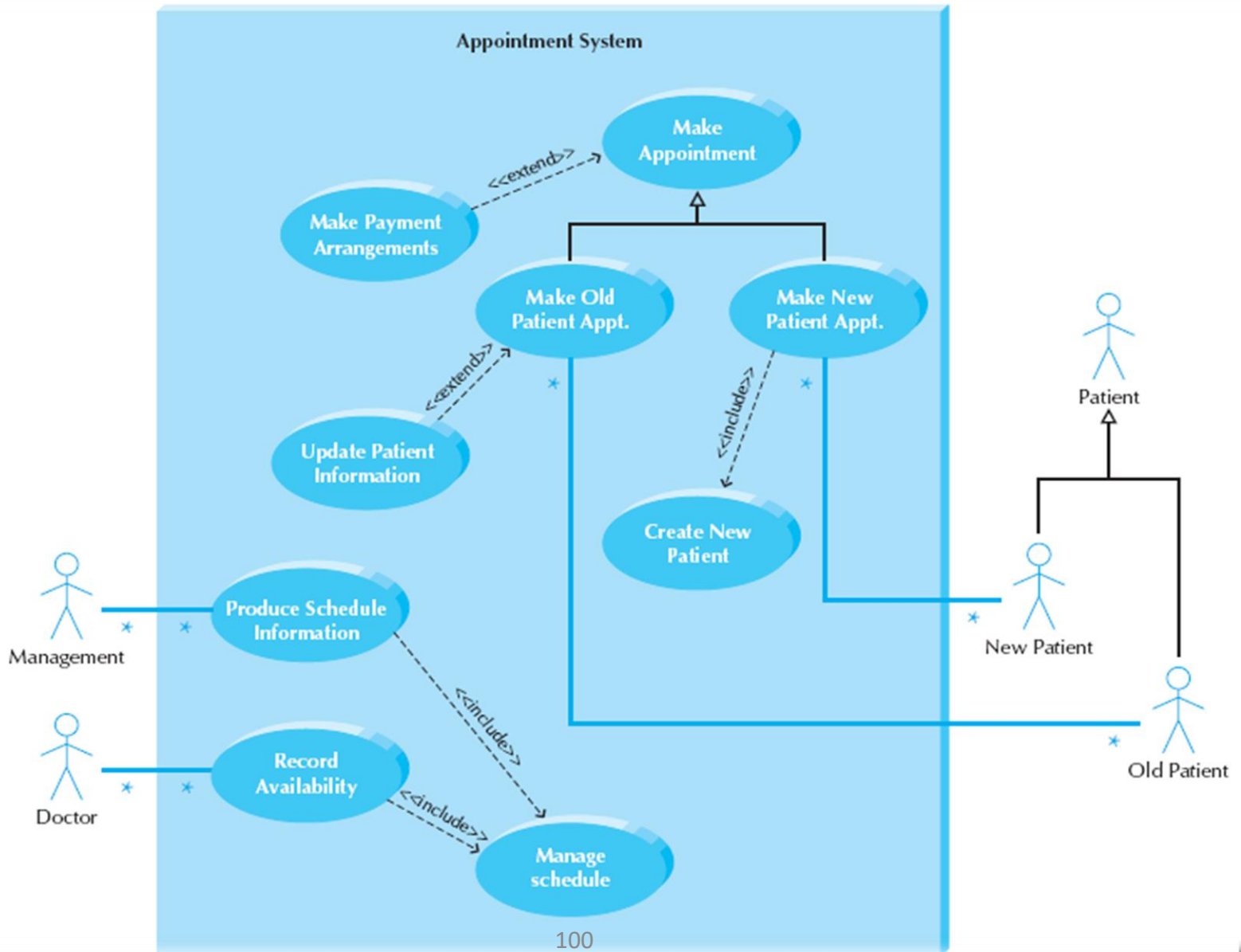
Use Case Diagram for Appointment System



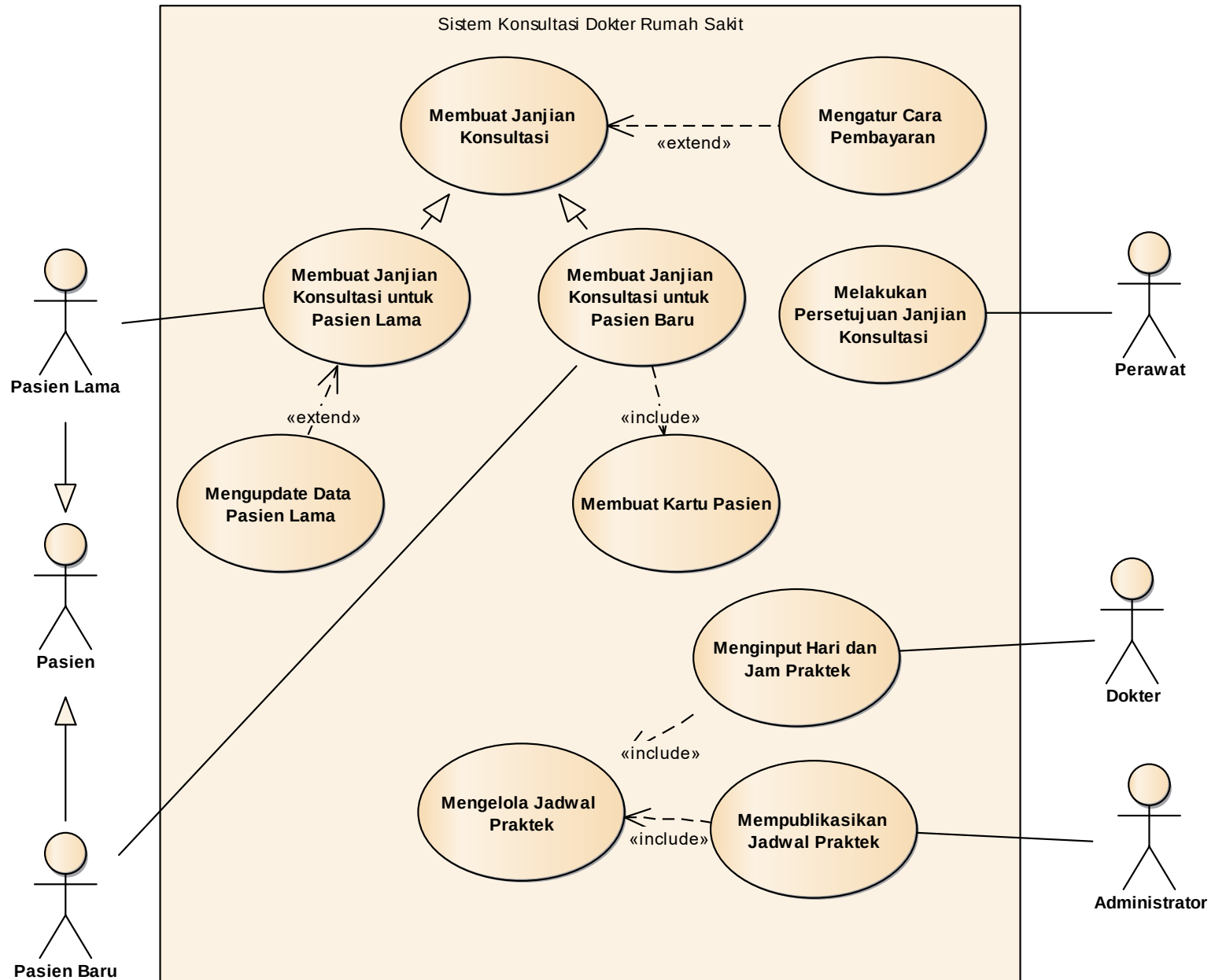
Use Case Diagram with Specialized Actor



Extend and Include Relationships



Sistem Konsultasi Dokter Rumah Sakit

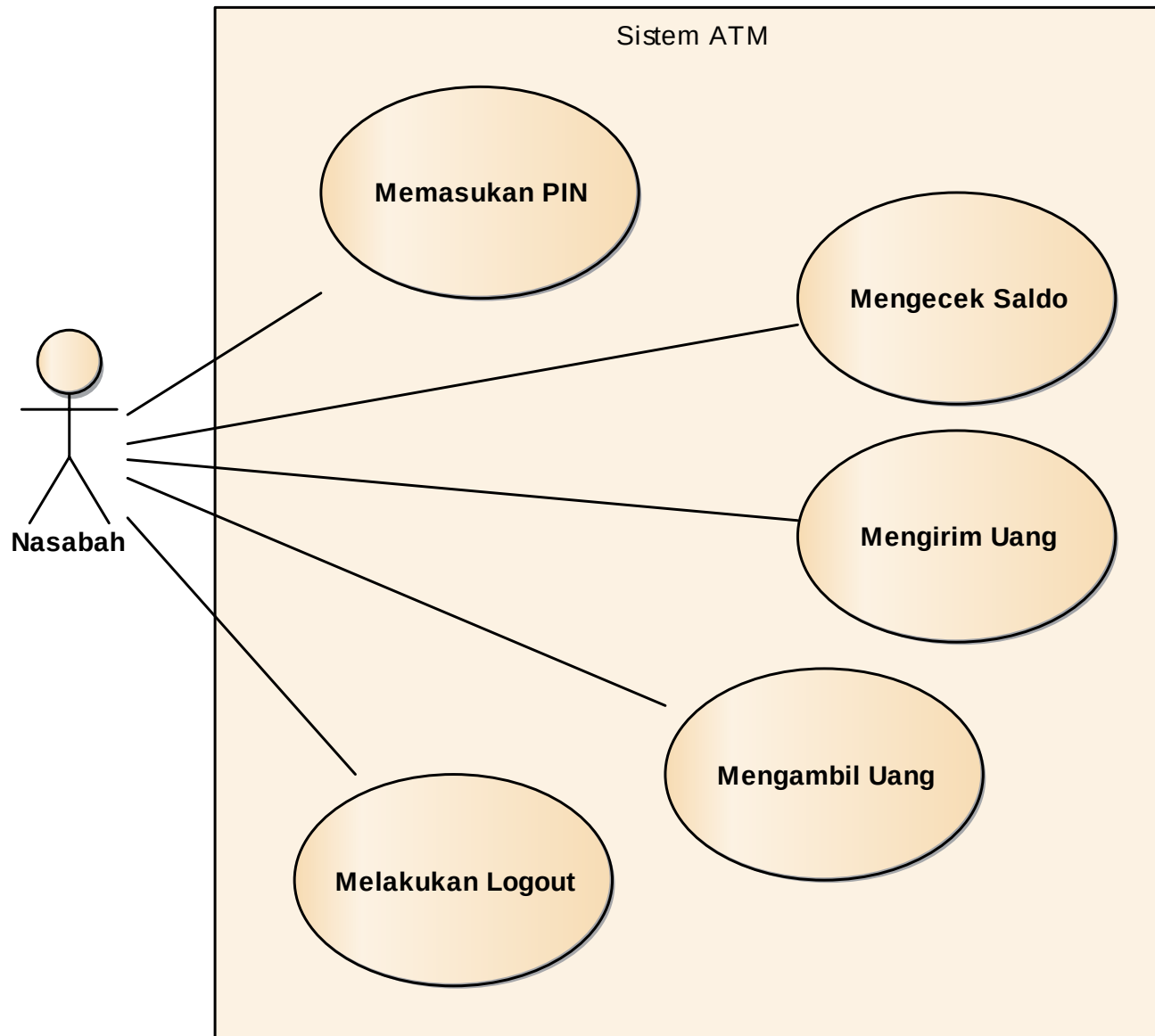




Studi Kasus: Use Case Diagram Sistem ATM

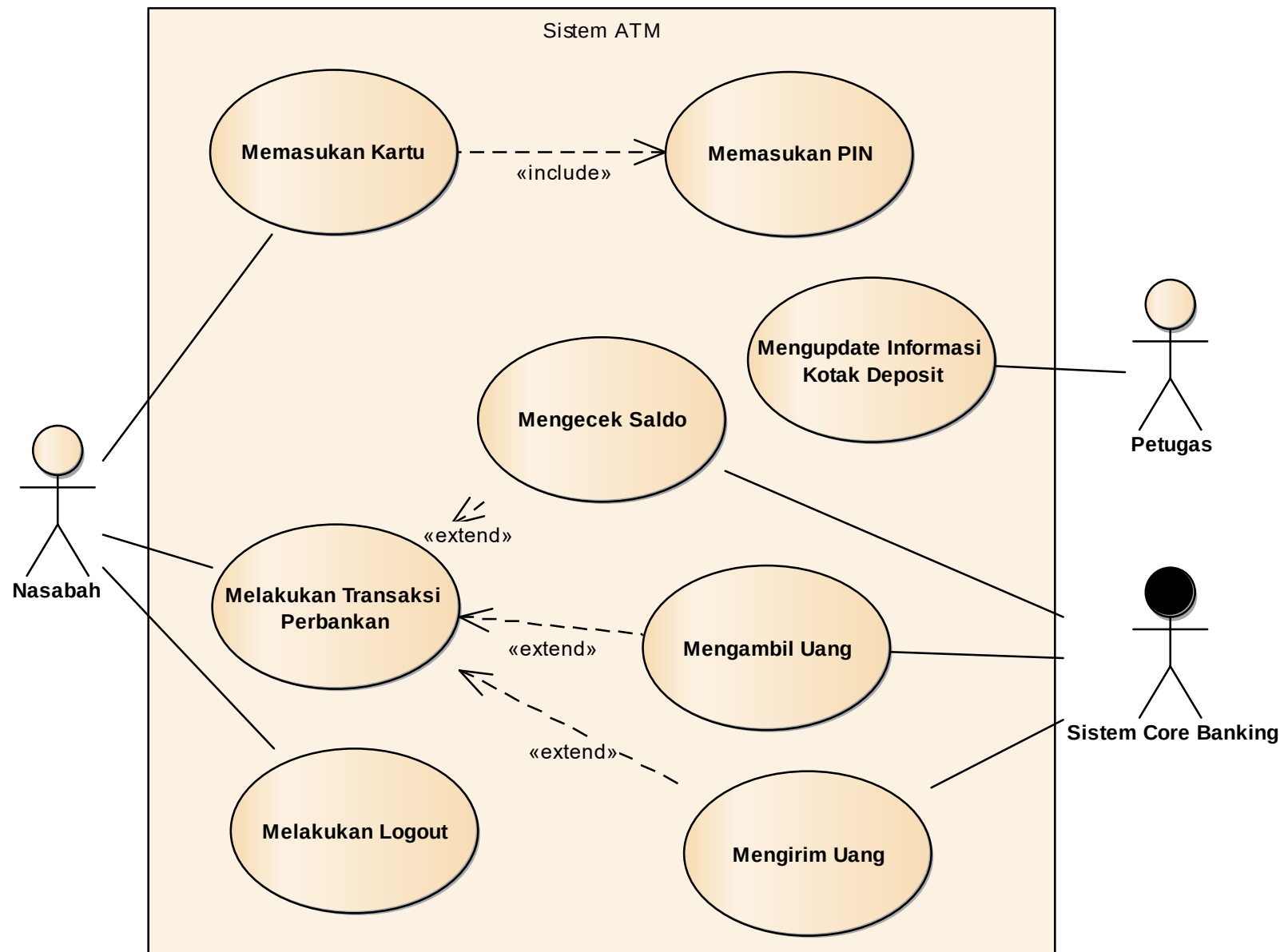
Use Case Diagram Sistem ATM

(versi Sederhana)



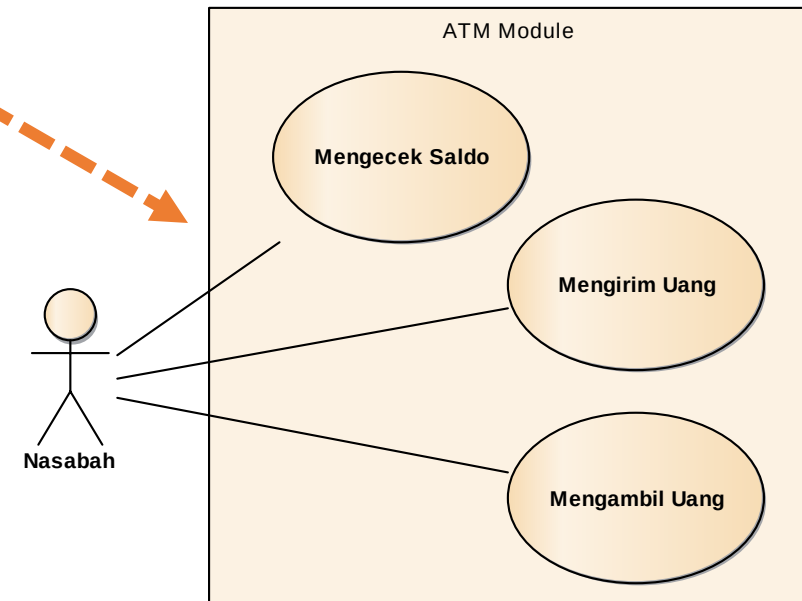
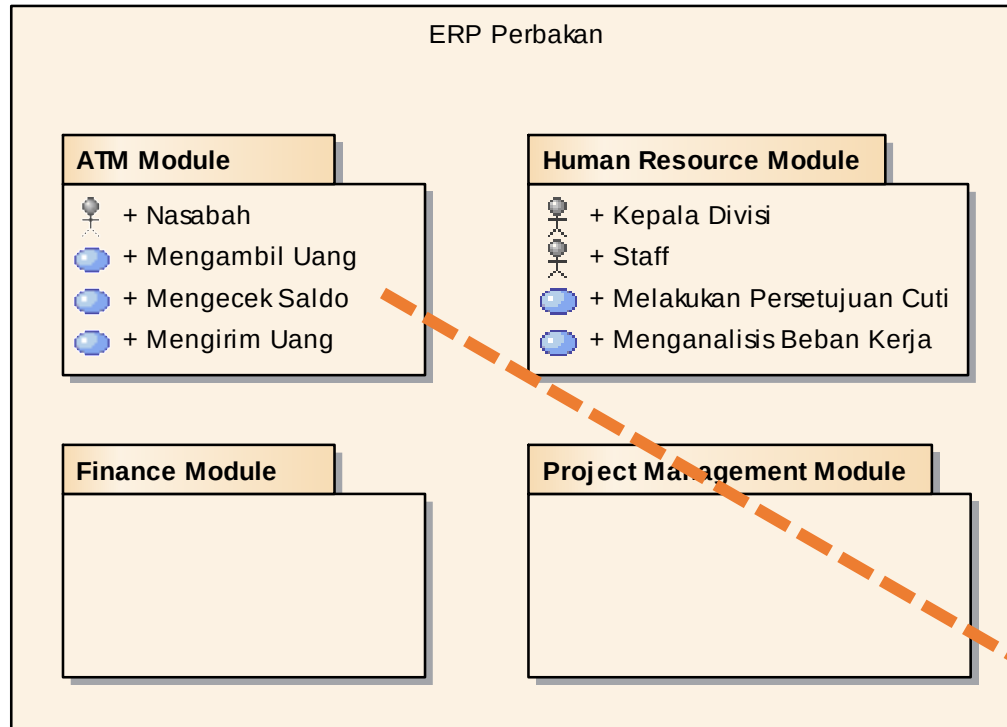
Use Case Diagram Sistem ATM

(Versi Include dan Extends)



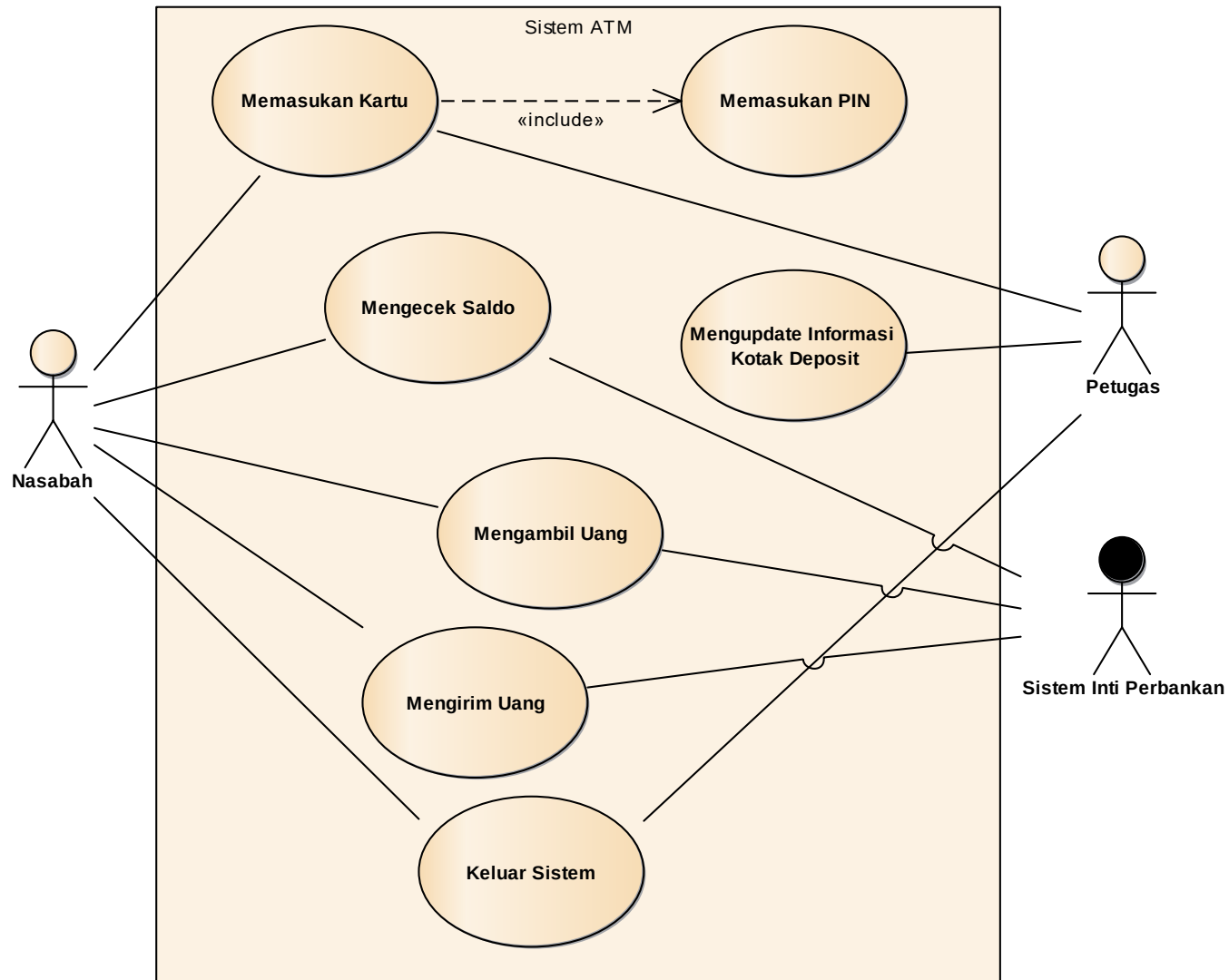
Use Case Diagram ERP Perbankan

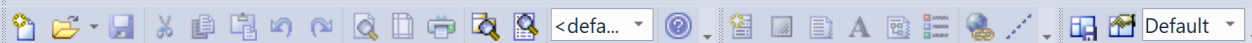
(Sistem Lebih Kompleks)



Use Case Diagram Sistem ATM

(Versi Normal)





Toolbox ▾

More tools

Common

Artifacts

Start Page

Start Page x

Project Browser ▾

Model

Model



Enterprise Architect

Version 12

Start

[New File](#)[Open File](#)[Server Connection](#)[Cloud Connection](#)

Recent

[SistemATM](#)[SistemATM](#)[test](#)[SistemATM](#)[SistemATM](#)[SistemABC](#)[njlk](#)[telkom](#)[test10](#)[test6](#)

New Package

Owner: Model

Name: Package1

Initial Content:

- ☐ Select and Apply Model Pattern
- ☒ Create Diagram
- ☐ Package Only

OK

Cancel

Help

SistemATM - Enterprise Architect

FILE EDIT VIEW PROJECT PACKAGE DIAGRAM ELEMENT TOOLS ANALYZER EXTENSIONS WINDOW HELP

Toolbox ▾ + ×
More tools
Common
Artifacts

Start Page
Enterprise Architect
Version 12

Start

New File
Open File
Server Connection
Cloud Connection

Recent
SistemATM
SistemATM
test
SistemATM
SistemATM
SistemABC
njlk
telkom
test10
test6

New Package

Owner: Model

Name: 1 Use Case Diagram

Initial Content:

Select and Apply Model Pattern

Create Diagram

Package Only

OK Cancel Help

Project Browser
Model

Project Br... Resources

CAP NUM SCRL CLOUD

17:22
16-Oct-2017

SistemATM - Enterprise Architect

FILE EDIT VIEW PROJECT PACKAGE DIAGRAM ELEMENT TOOLS ANALYZER EXTENSIONS WINDOW HELP

Toolbox ▾ + x Start Page

More tools
Common
Artifacts

Enterprise Architect
Version 12

Start

New File
Open File
Server Conn
Cloud Conn

Recent

SistemATM
SistemATM
test
SistemATM
SistemATM
SistemABC
njlk
telkom
test10
test6

New Diagram

Package: 1 Use Case Diagram

Diagram: UCD Sistem ATM

Type

Select From:

- UML Structural
- UML Behavioral
- Extended
- ArcGIS
- ArchiMate
- Archimate 2.0
- BPMN 1.0
- BPMN 1.1
- BPMN 2.0
- Code Engineering
- Data Flow Diagrams
- Entity Relationship Diagram

Diagram Types:

- Use Case
- Activity
- State Machine
- Communication
- Sequence
- Timing
- Interaction Overview

UML Diagrams for Behavioral Modeling.

OK Cancel Help

Project Browser

Model

1 Use Case Diagram

Package: 1 Use Case Diagram

17:22
16-Oct-2017

Actor : Actor1

Properties

General

Rules

Requirements

Constraints

Scenarios

Related

Files

Links

Nasabah

B I U A

Stereotype:

Status:

Proposed

Alias:

Keywords:

Author:

romis

Complexity:

Easy

Language:

<none>

Version:

1.0

Phase:

1.0

Package:

1 Use Case Diagram

Created:

16-Oct-2017 17:23:56

Modified:

16-Oct-2017 17:23:56

Main

Tags

OK

Ca...

Apply

Help

SistemATM - Enterprise Architect

FILE EDIT VIEW PROJECT PACKAGE DIAGRAM ELEMENT TOOLS ANALYZER EXTENSIONS WINDOW HELP

Toolbox

- Use Case
 - Actor
 - Use Case
 - Test Case
 - Collaboration
 - Collaboration Use
 - Boundary
 - Package
- Use Case Relationships
- Use Case Patterns
- Common
- Artifacts

Use case and Actor elements for Usecase diagrams

Start Page *UCD Sistem ATM

1 Default...

Project Browser

- Model
- 1 Use Case Diagram
- UCD Sistem ATM
- Nasabah

Actor:Nasabah Left: 90 x Top: 130 - Width: 50 x Height: 100

The screenshot displays the Enterprise Architect interface. The main workspace shows a Use Case Diagram with a single actor named 'Nasabah'. The actor is represented by a stick figure icon with a yellow circle for a head. The actor's name 'Nasabah' is written below the figure. The actor is selected, and a tooltip 'Use case and Actor elements for Usecase diagrams' is visible. The Project Browser on the right shows the project structure: 'Model' > '1 Use Case Diagram' > 'UCD Sistem ATM' > 'Nasabah'. The bottom status bar shows the actor's properties: 'Actor:Nasabah Left: 90 x Top: 130 - Width: 50 x Height: 100'.

SistemATM - Enterprise Architect

FILE EDIT VIEW PROJECT PACKAGE DIAGRAM ELEMENT TOOLS ANALYZER EXTENSIONS WINDOW HELP

Toolbox

- Use Case
 - Actor
 - Use Case
 - Test Case
 - Collaboration
 - Collaboration Use
 - Boundary
 - Package
- Use Case Relationships
- Use Case Patterns
- Common
- Artifacts

Use Case Diagram: "UCD Sistem ATM"

Start Page *UCD Sistem ATM

Default...

Sistem ATM

Nasabah

Project Browser

- Model
 - 1 Use Case Diagram
 - UCD Sistem ATM
 - Nasabah

Property... Notes Toolbox

Use Case Diagram:UCD Sistem ATM: created: 16-Oct-2017 17:23:00 modified: 16-Oct-2017 17:24:51 100% 827 x 1169

17:25 16-Oct-2017

SistemATM - Enterprise Architect

FILE EDIT VIEW PROJECT PACKAGE DIAGRAM ELEMENT TOOLS ANALYZER EXTENSIONS WINDOW HELP

Toolbox

- Use Case
 - Actor
 - Use Case
 - Test Case
 - Collaboration
 - Collaboration Use
 - Boundary
 - Package
- Use Case Relationships
- Use Case Patterns
- Common
- Artifacts

Project Browser

- Model
 - 1 Use Case Diagram
 - UCD Sistem ATM
 - Nasabah

Start Page *UCD Sistem ATM

Diagram: UCD Sistem ATM

Actor: Nasabah

Use Case: System ATM

Association: Use Case

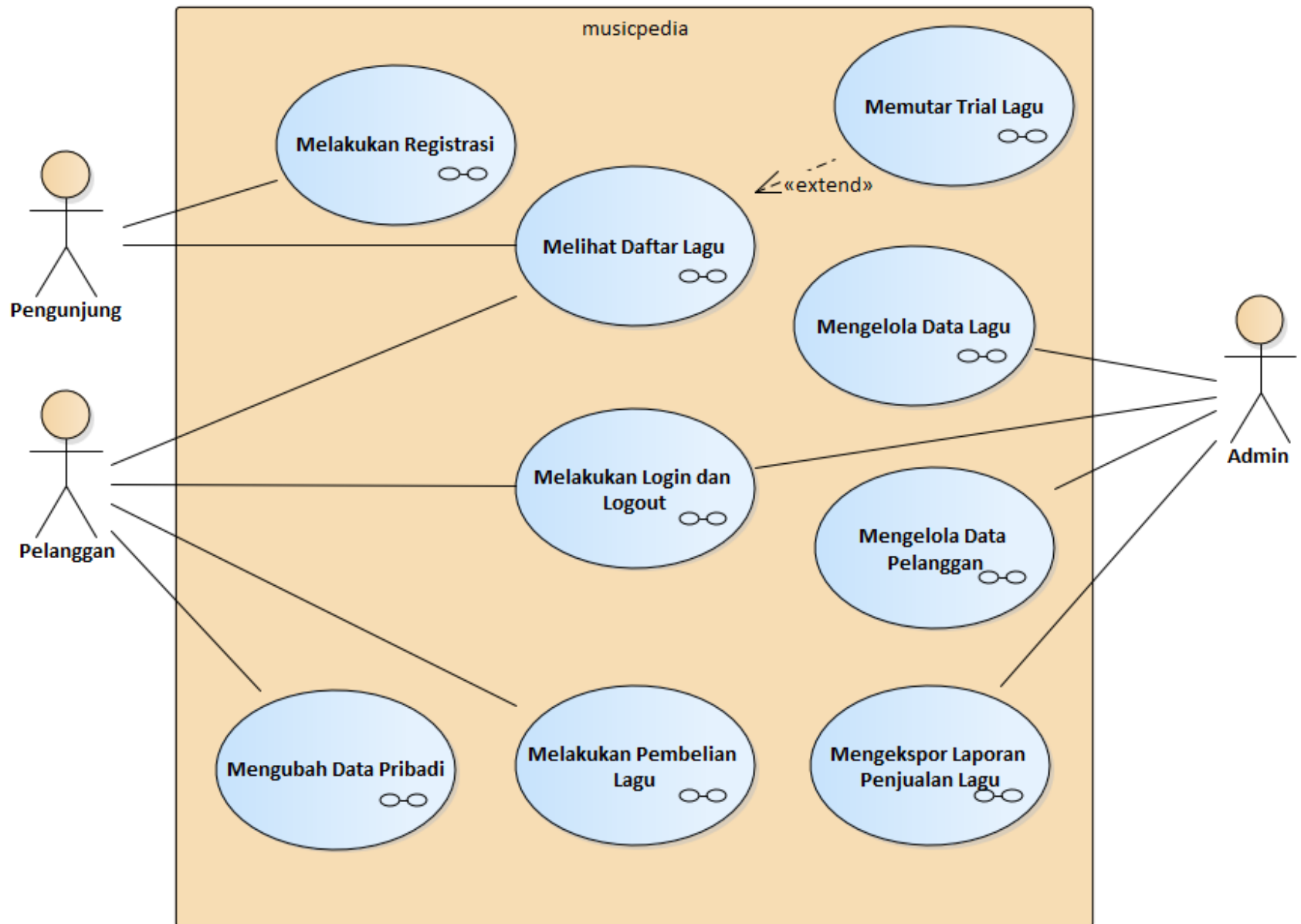
Association: Use

Association: Association

Release the mouse over a node to create a new path.

17:25 16-Oct-2017

Use Case Diagram MusicPedia





2.3 Pemodelan Proses Bisnis dengan AD atau BPMN

2.3.1 Activity Diagram

2.3.2 Business Process Model and Notation (BPMN)

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

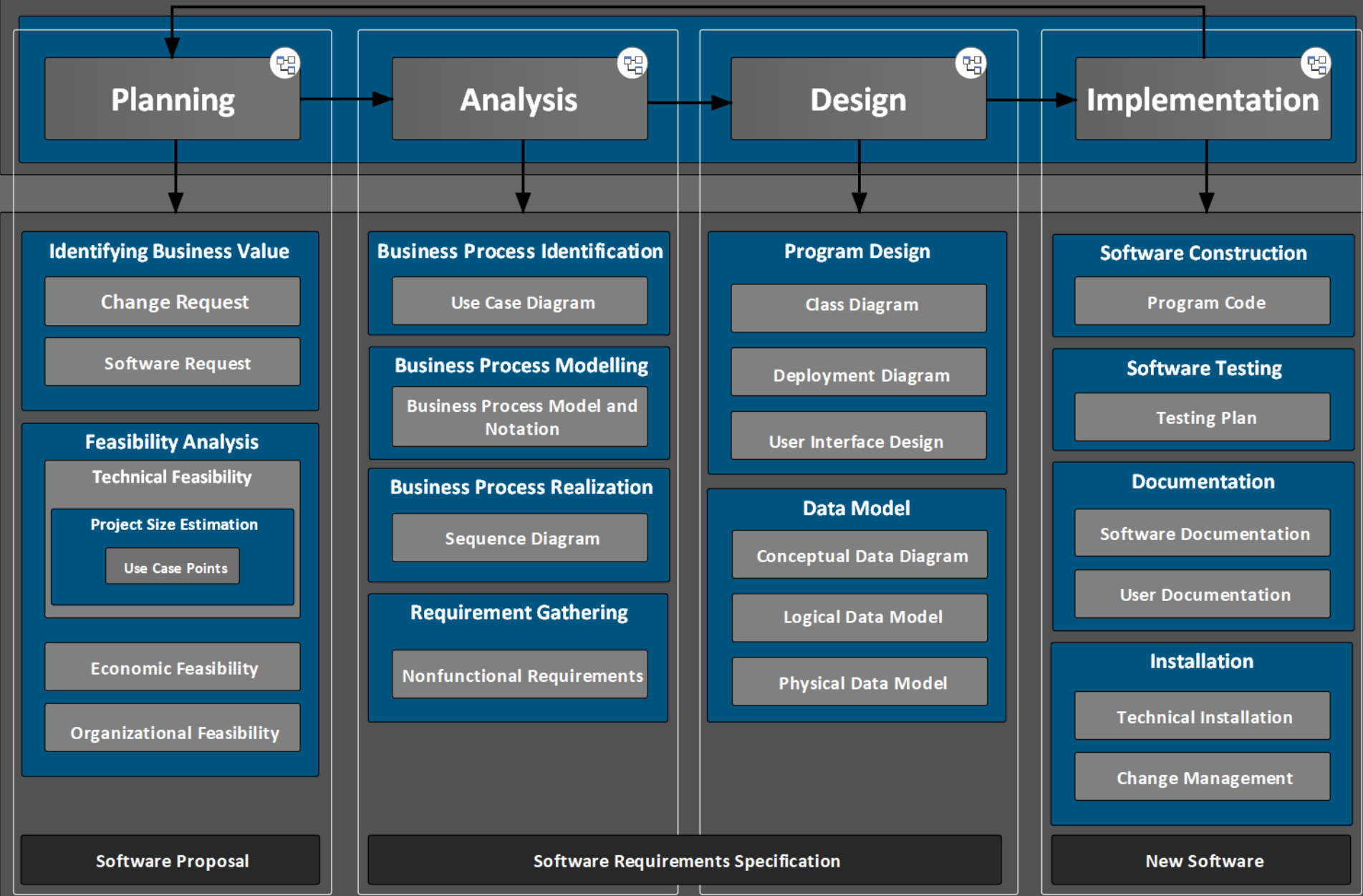
2.2 Pemodelan **User Interface Design**

2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

Application Development Governance

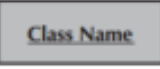
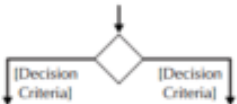
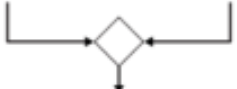
Software Development Life Cycle





2.3.1 Activity Diagram

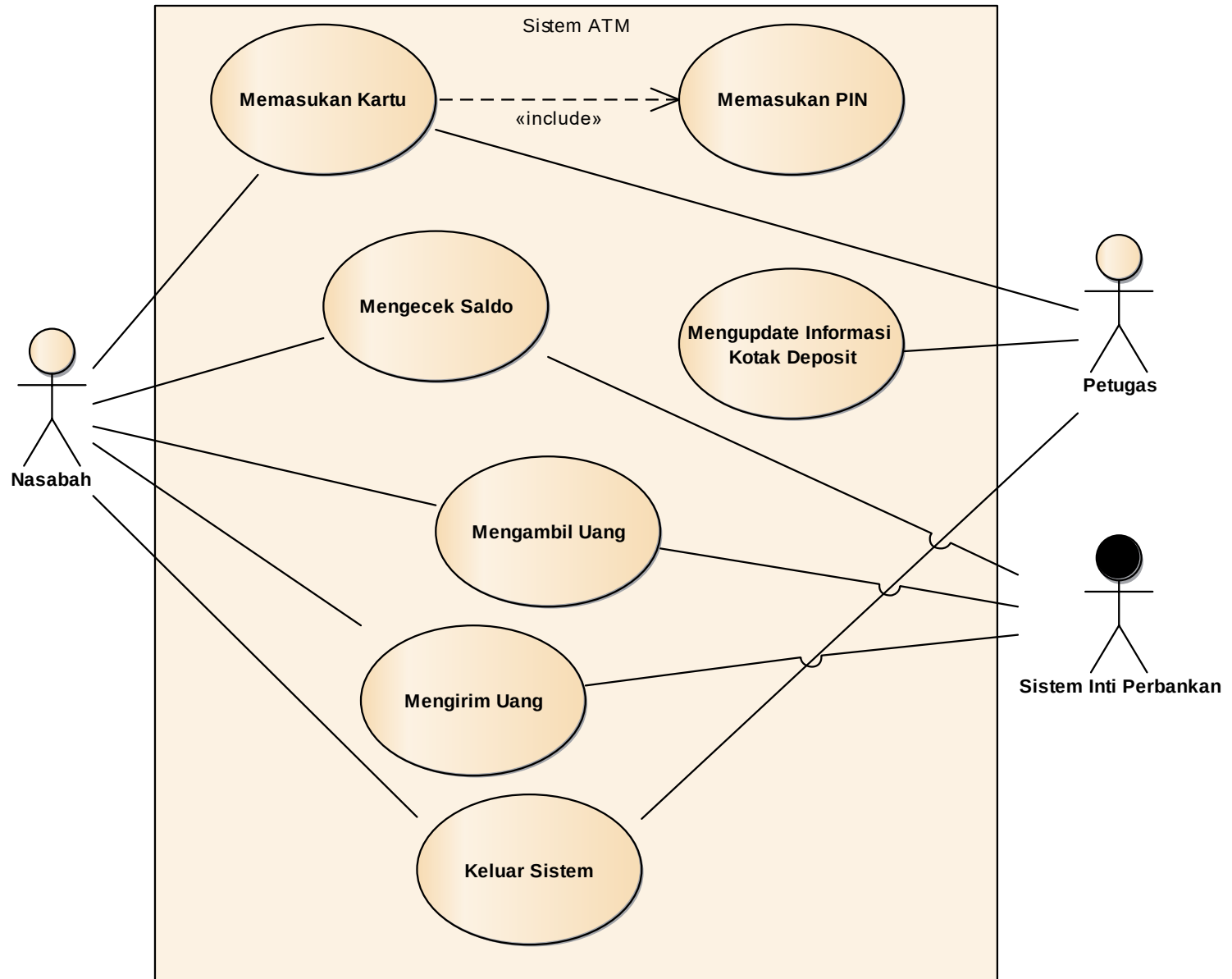
Activity Diagram Syntax

An action: ■ Is a simple, nondecomposable piece of behavior. ■ Is labeled by its name.	
An activity: ■ Is used to represent a set of actions. ■ Is labeled by its name.	
An object node: ■ Is used to represent an object that is connected to a set of object flows. ■ Is labeled by its class name.	
A control flow: ■ Shows the sequence of execution.	
An object flow: ■ Shows the flow of an object from one activity (or action) to another activity (or action).	
An initial node: ■ Portrays the beginning of a set of actions or activities.	
A final-activity node: ■ Is used to stop all control flows and object flows in an activity (or action).	
A final-flow node: ■ Is used to stop a specific control flow or object flow.	
A decision node: ■ Is used to represent a test condition to ensure that the control flow or object flow only goes down one path. ■ Is labeled with the decision criteria to continue down the specific path.	
A merge node: ■ Is used to bring back together different decision paths that were created using a decision node.	
A fork node: Is used to split behavior into a set of parallel or concurrent flows of activities (or actions)	
A join node: Is used to bring back together a set of parallel or concurrent flows of activities (or actions)	
A swimlane: - Is used to break up an activity diagram into rows and columns to assign the individual activities (or actions) to the individuals or objects that are responsible for executing the activity (or action) - Is labeled with the name of the individual or object responsible	

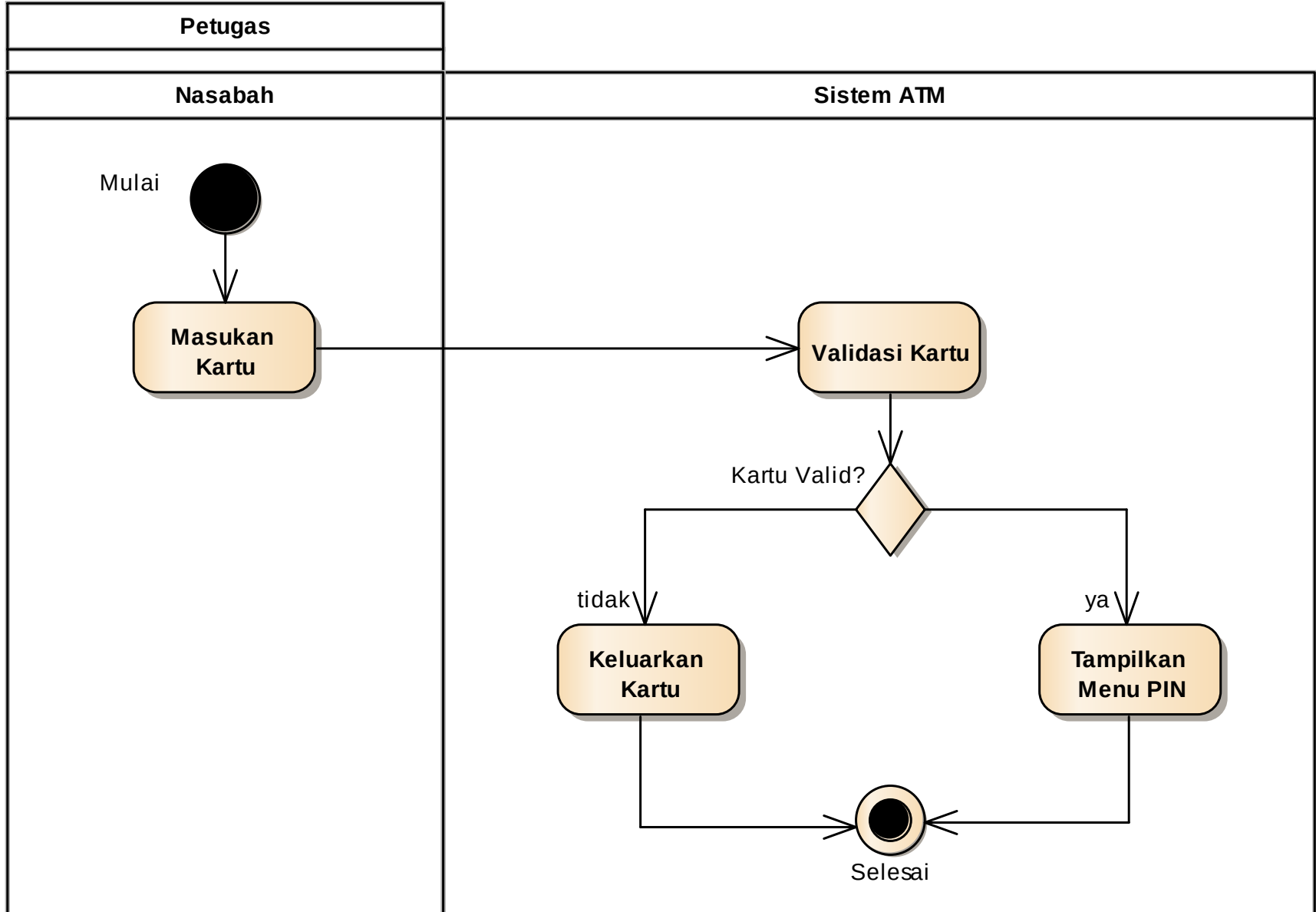


Studi Kasus: Activity Diagram Sistem ATM

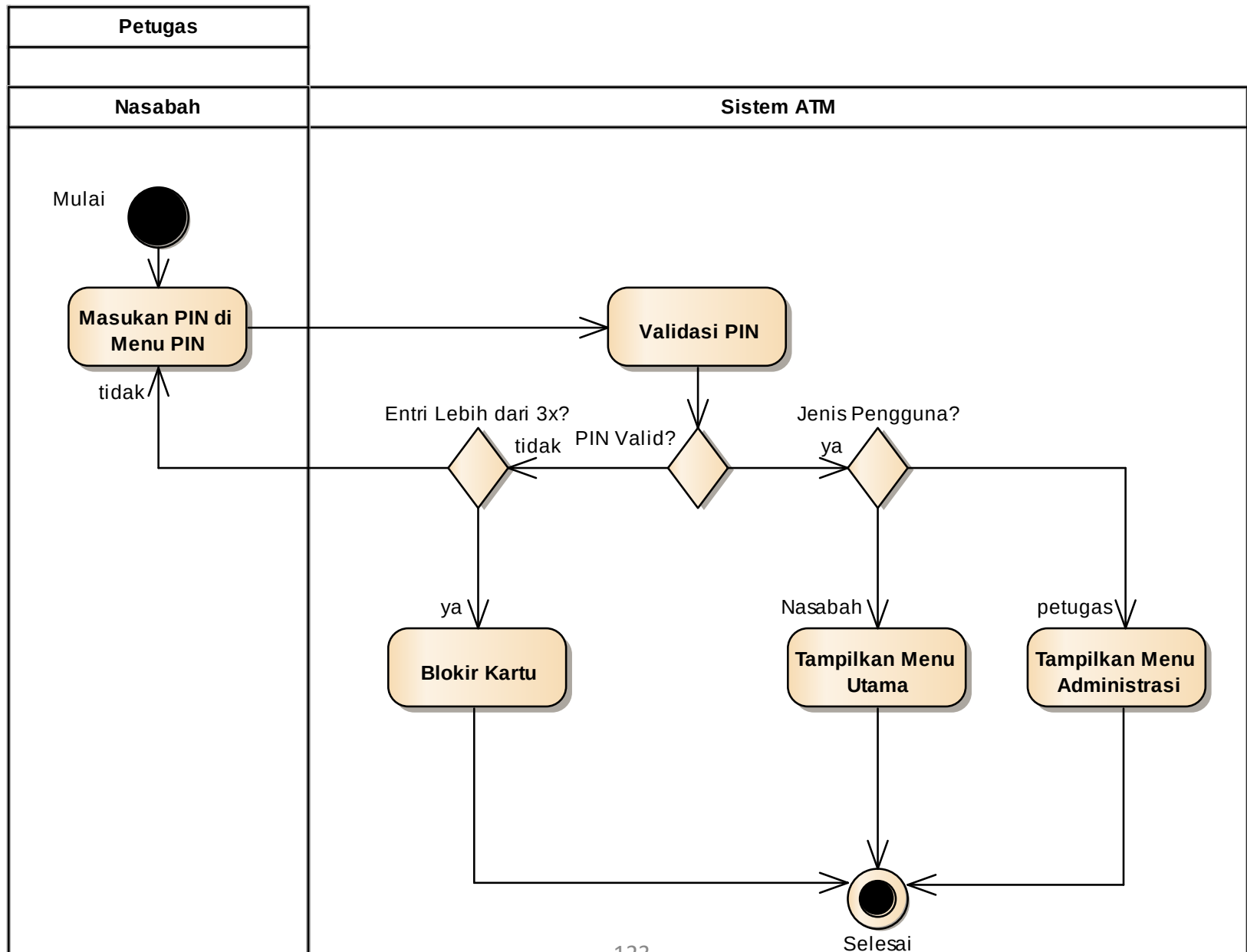
Use Case Diagram Sistem ATM



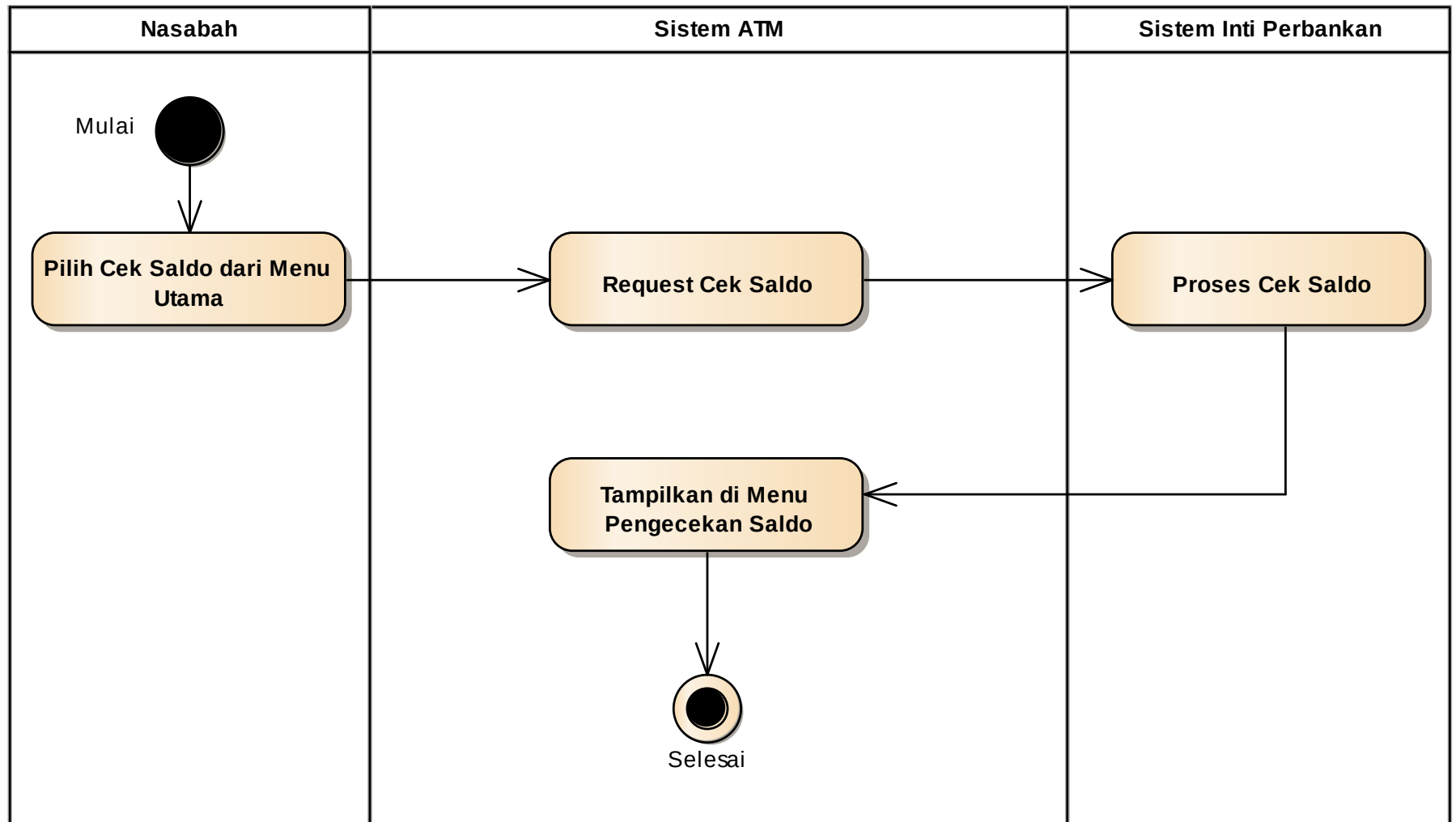
Activity Diagram: Memasukkan Kartu



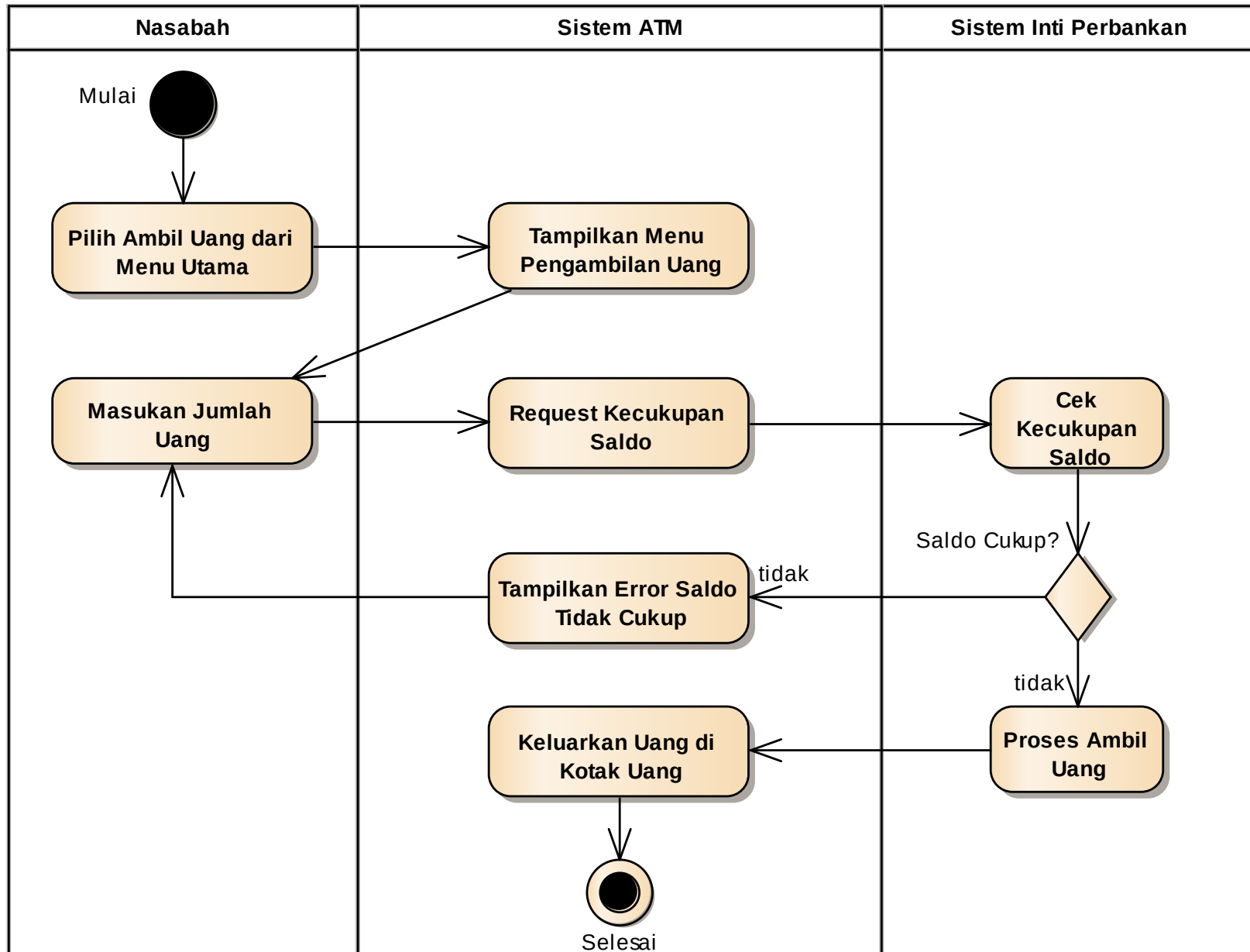
Activity Diagram: Memasukkan PIN



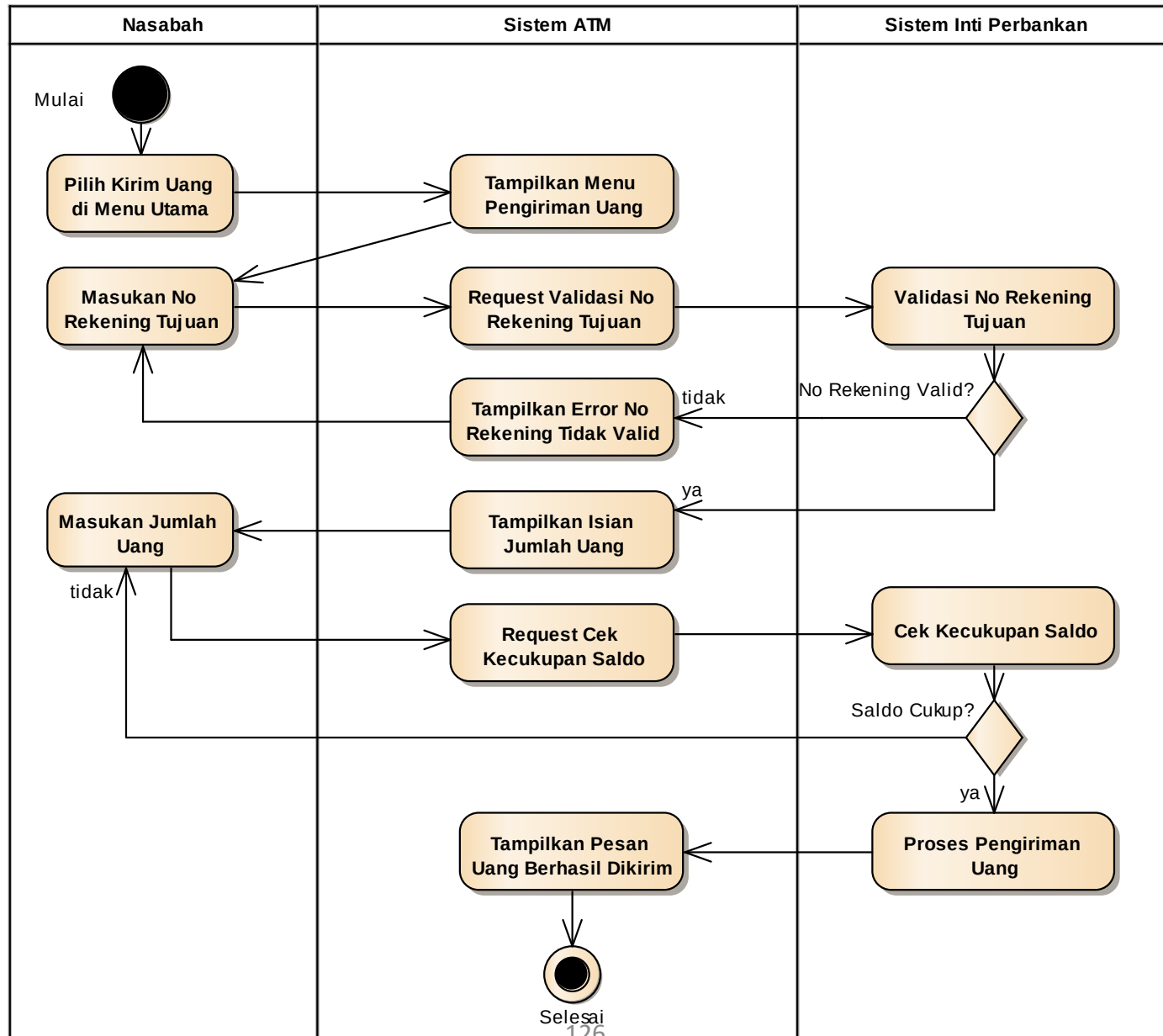
Activity Diagram: Mengecek Saldo



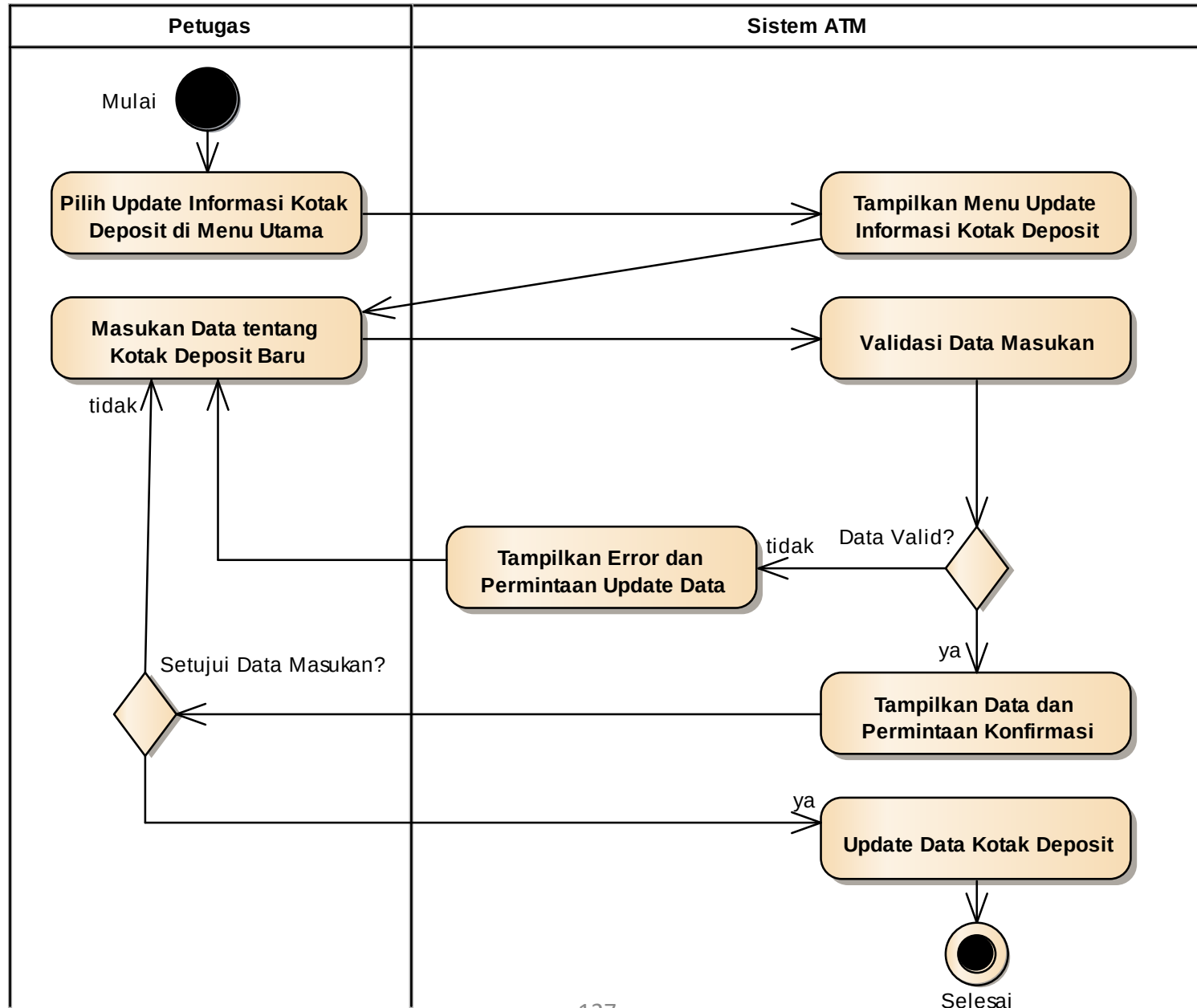
Activity Diagram: Mengambil Uang



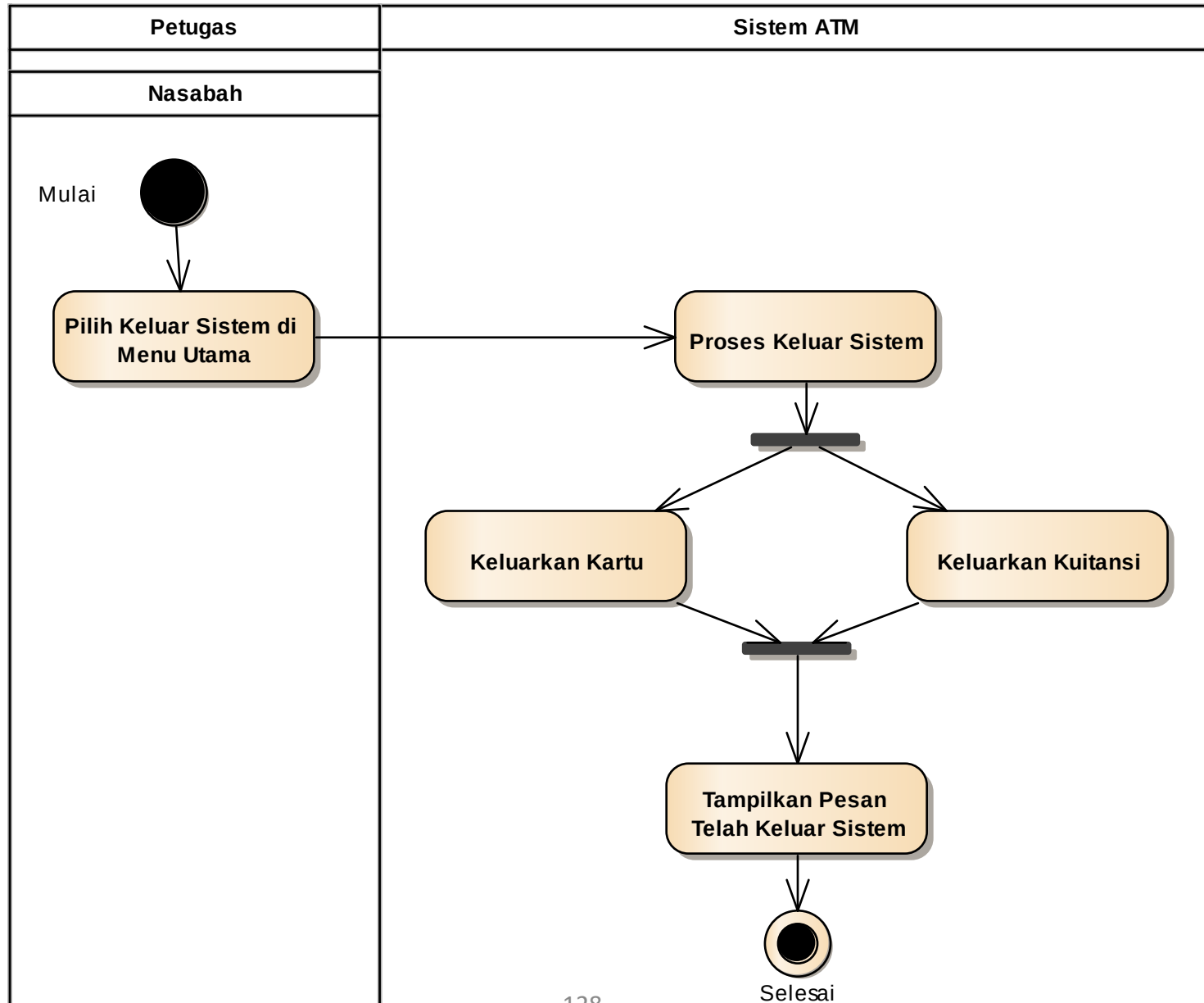
Activity Diagram: Mengirim Uang



Activity Diagram: Mengupdate Informasi Kotak Deposit



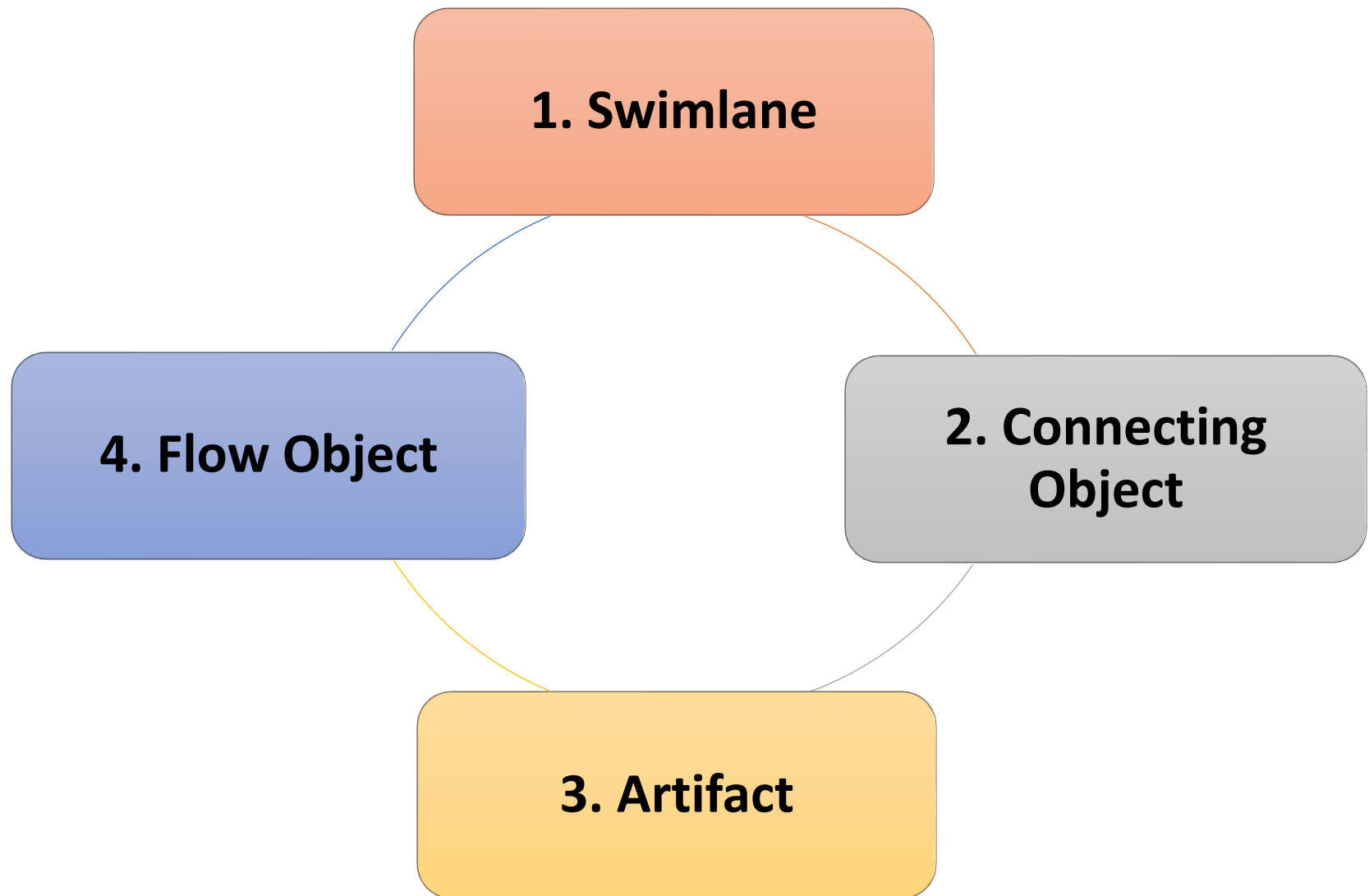
Activity Diagram: Keluar Sistem





2.3.2 Business Process Model and Notation (BPMN)

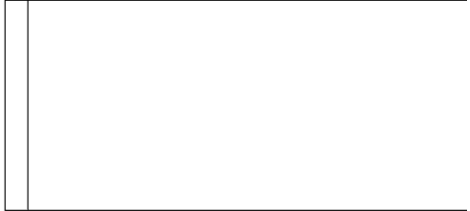
Elemen BPMN



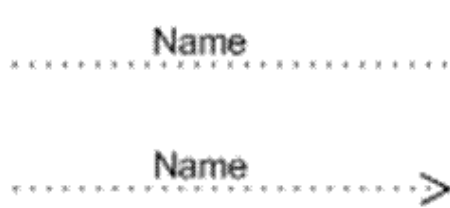
Elemen dan Notasi BPMN

ELEMEN	DESKRIPSI	NAMA NOTASI
Swimlane	Mekanisme untuk mengatur dan memisahkan peran atau penanggungjawab dari suatu proses	Pool
		Lane
Connecting Object	Konektor dari obyek yang mengalir pada suatu proses	Sequence Flow
		Message Flow
		Association
Artifact	Informasi tambahan dalam suatu proses	Annotation
		Group
		Data Object
		Data Store
Flow Object	Obyek yang mengalir pada suatu proses	Event
		Activity
		Gateway

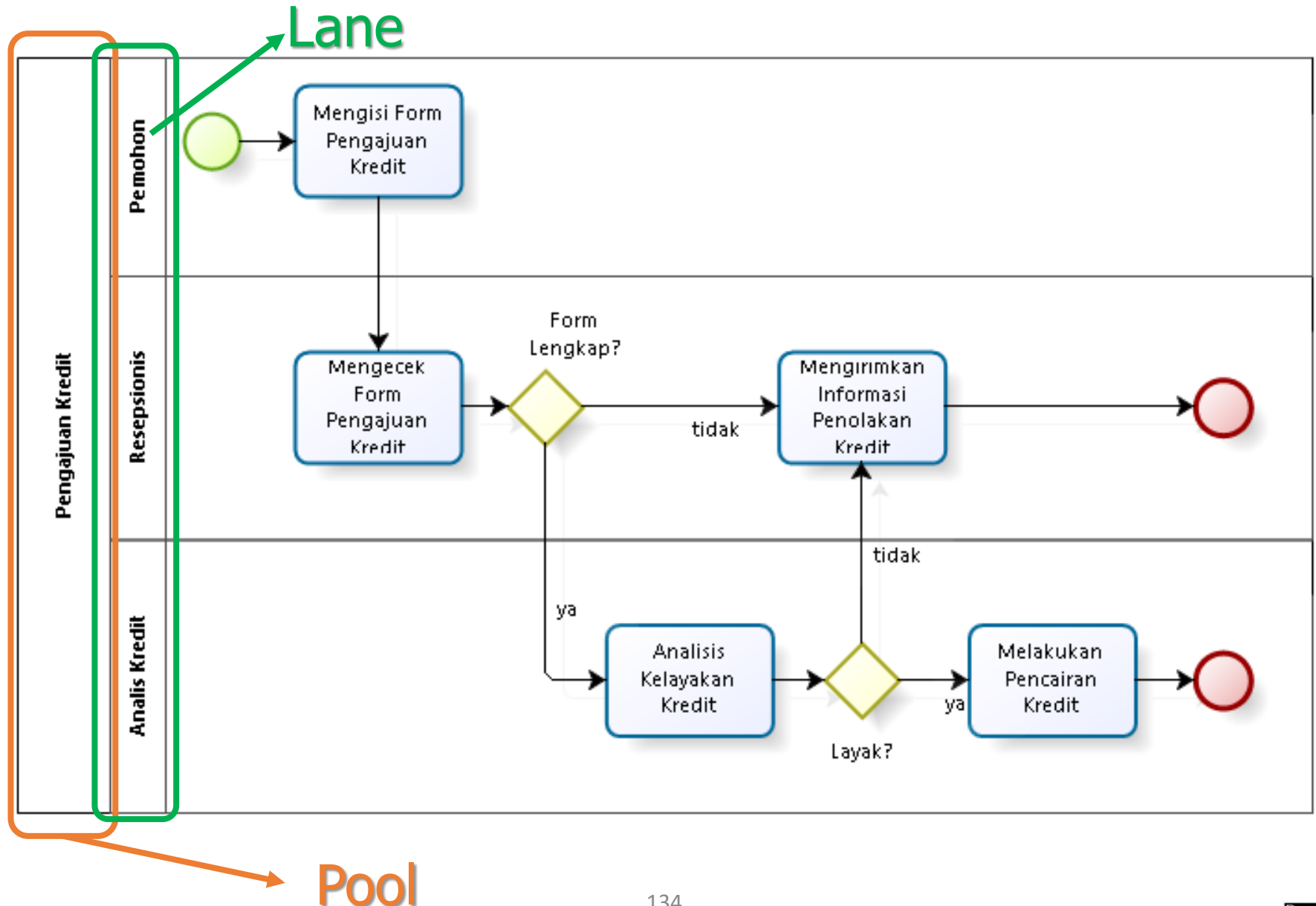
Swimlane

Nama Notasi	Deskripsi	Notasi
Pool	Kontainer dari satu proses	
Lane	Partisi dari suatu proses, yang menunjukkan sub organisasi , jabatan, peran atau penanggungjawab	

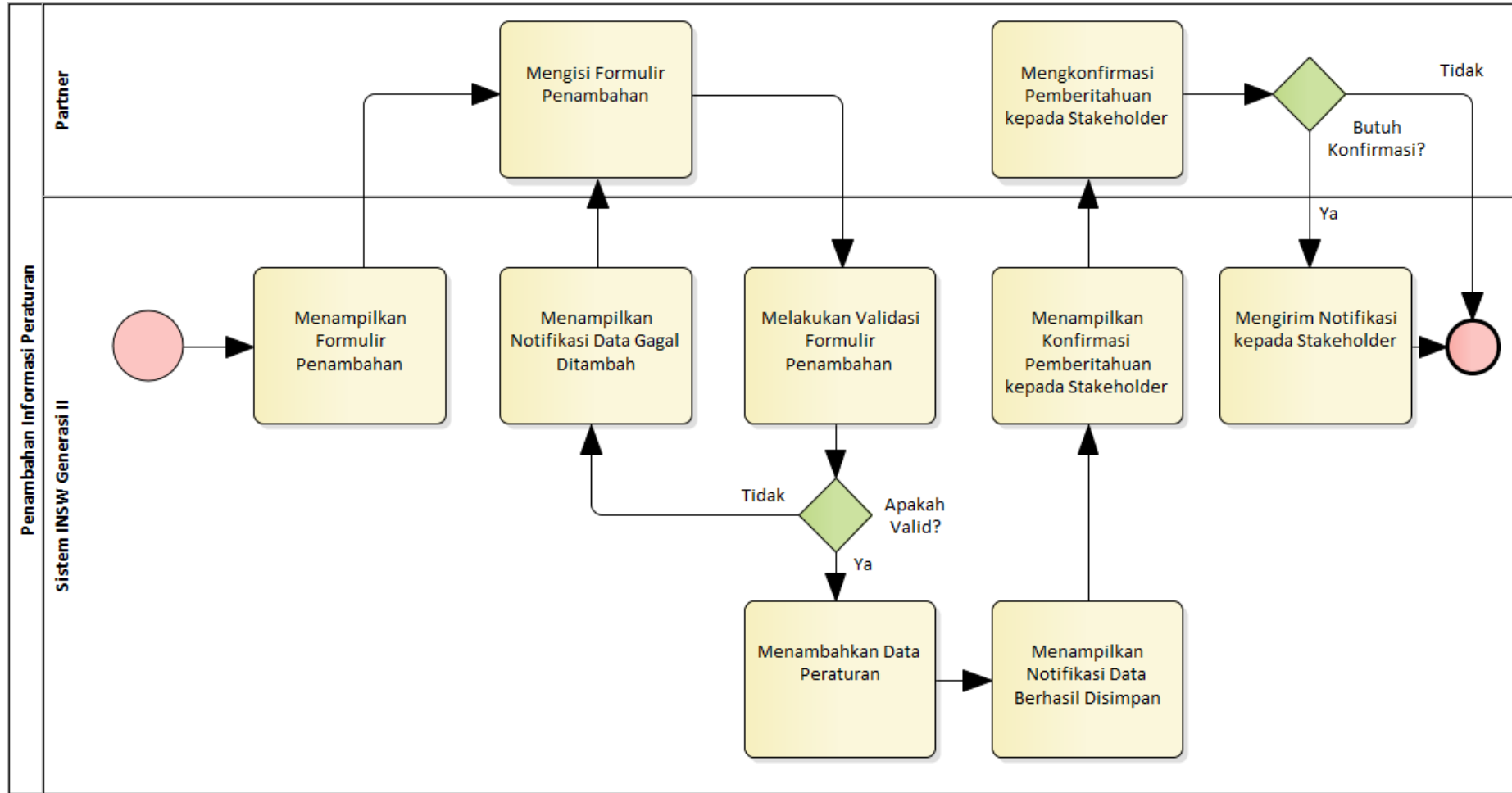
Connecting Object

NAMA NOTASI	DESKRIPSI	NOTASI
Sequence Flow	Konektor yang menghubungkan antar obyek yang mengalir dalam satu proses (satu pool)	
Message Flow	Konektor yang menghubungkan antar obyek yang mengalir antar proses (beda pool)	
Association	Konektor yang menghubungkan obyek yang mengalir ke artifact	

Swimlane - Proses Bisnis Organisasi



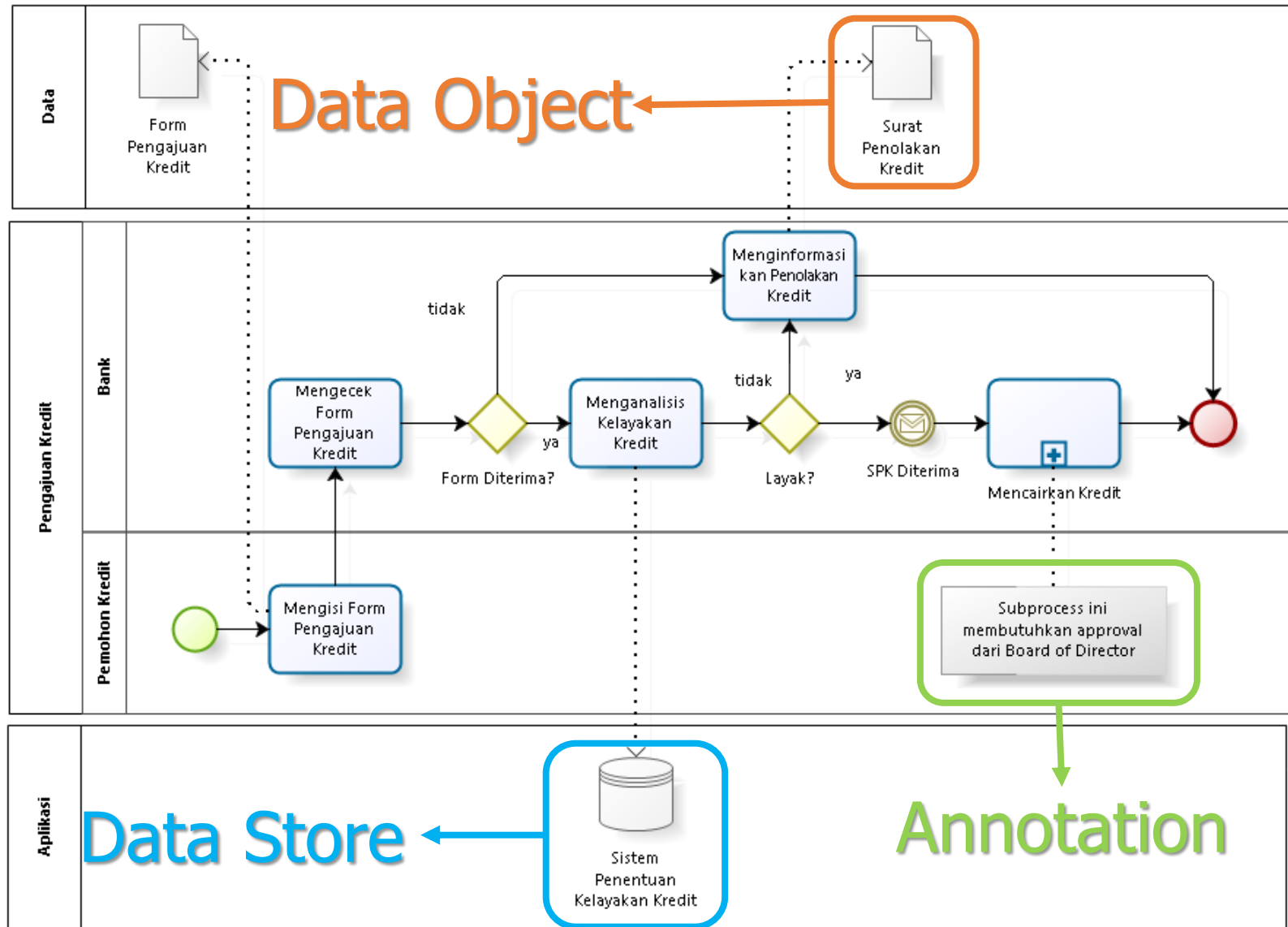
Swimlane - Proses Bisnis Sistem



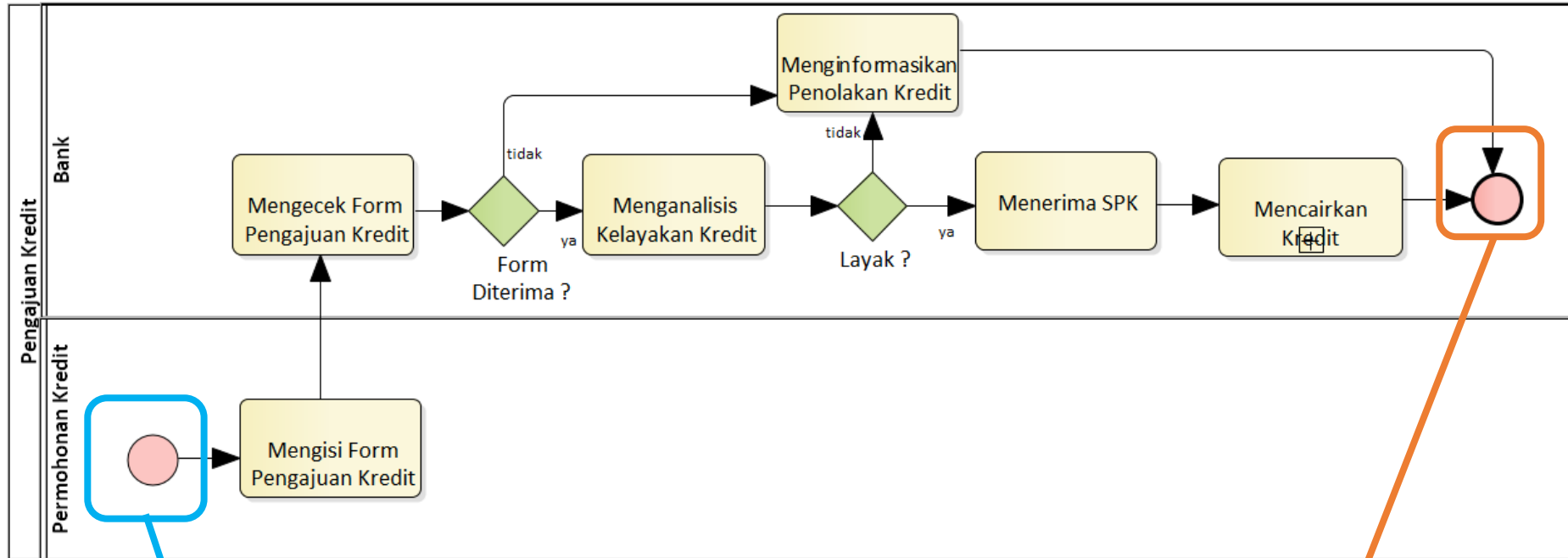
Artifact

NAMA NOTASI	DESKRIPSI	NOTASI
Annotation	Penjelasan dari suatu obyek yang mengalir	
Group	Pengelompokan dari beberapa obyek yang mengalir	
Data Object	File dan dokumen yang digunakan dan dihasilkan oleh suatu aktifitas	
Data Store	Sistem dan aplikasi yang digunakan dan dihasilkan oleh suatu aktifitas	

Annotation, Data Object dan Data Store



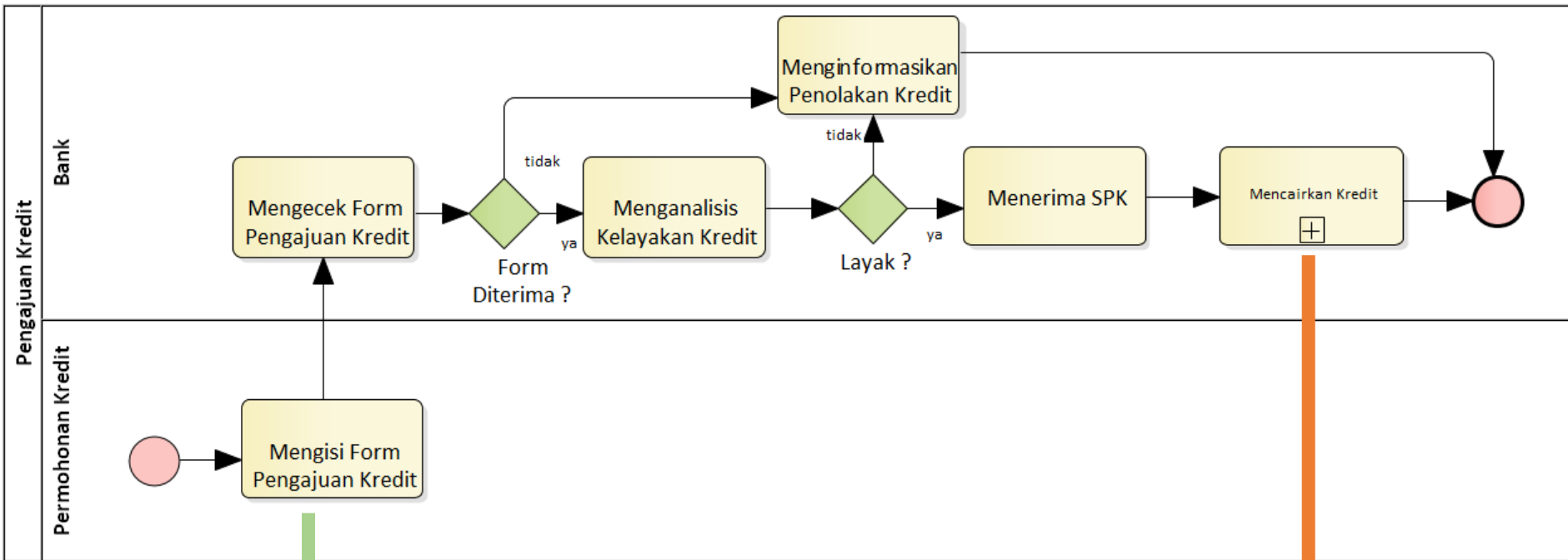
Event



Start
Event

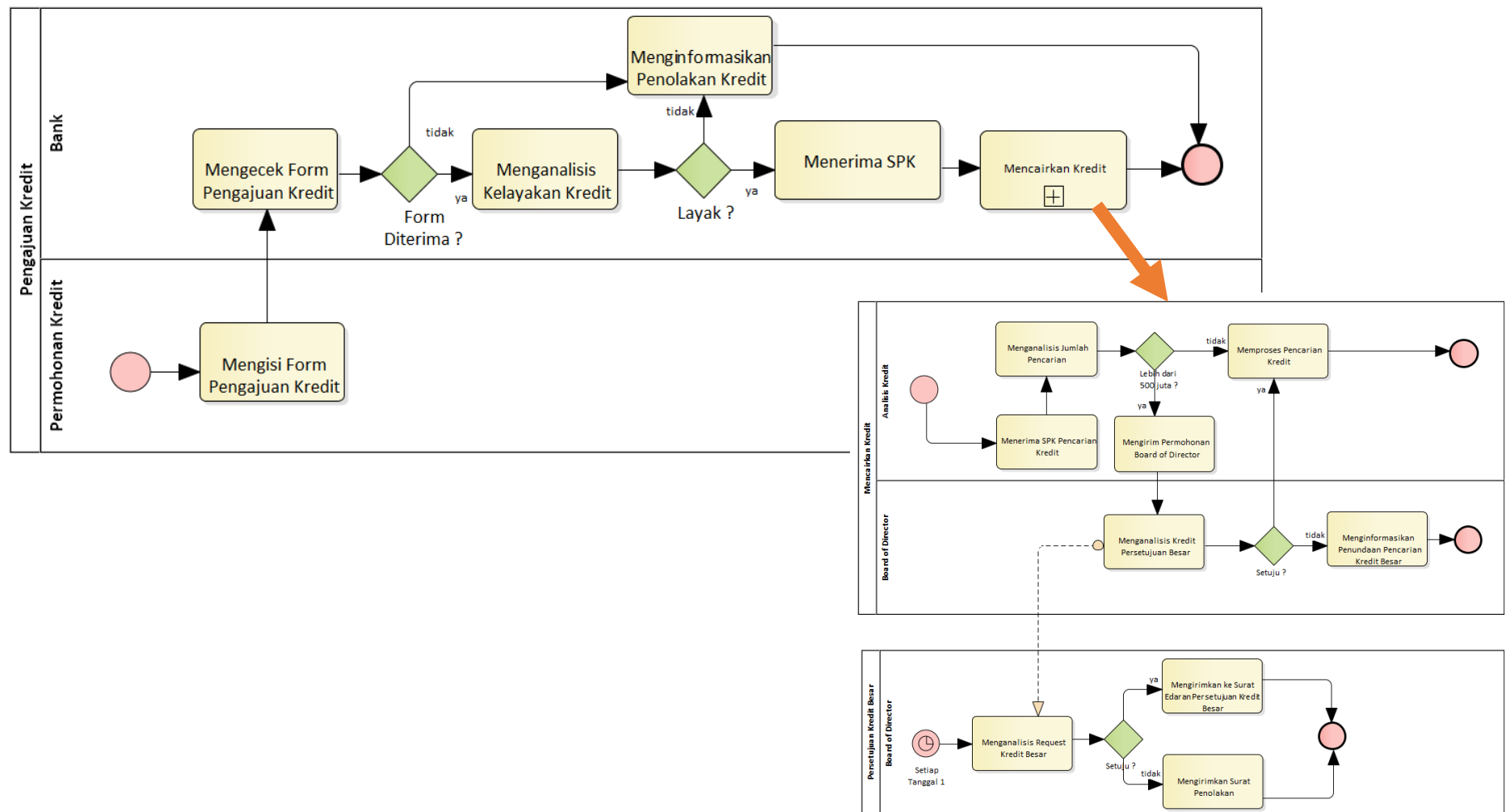
End Event

Activity



Subprocess

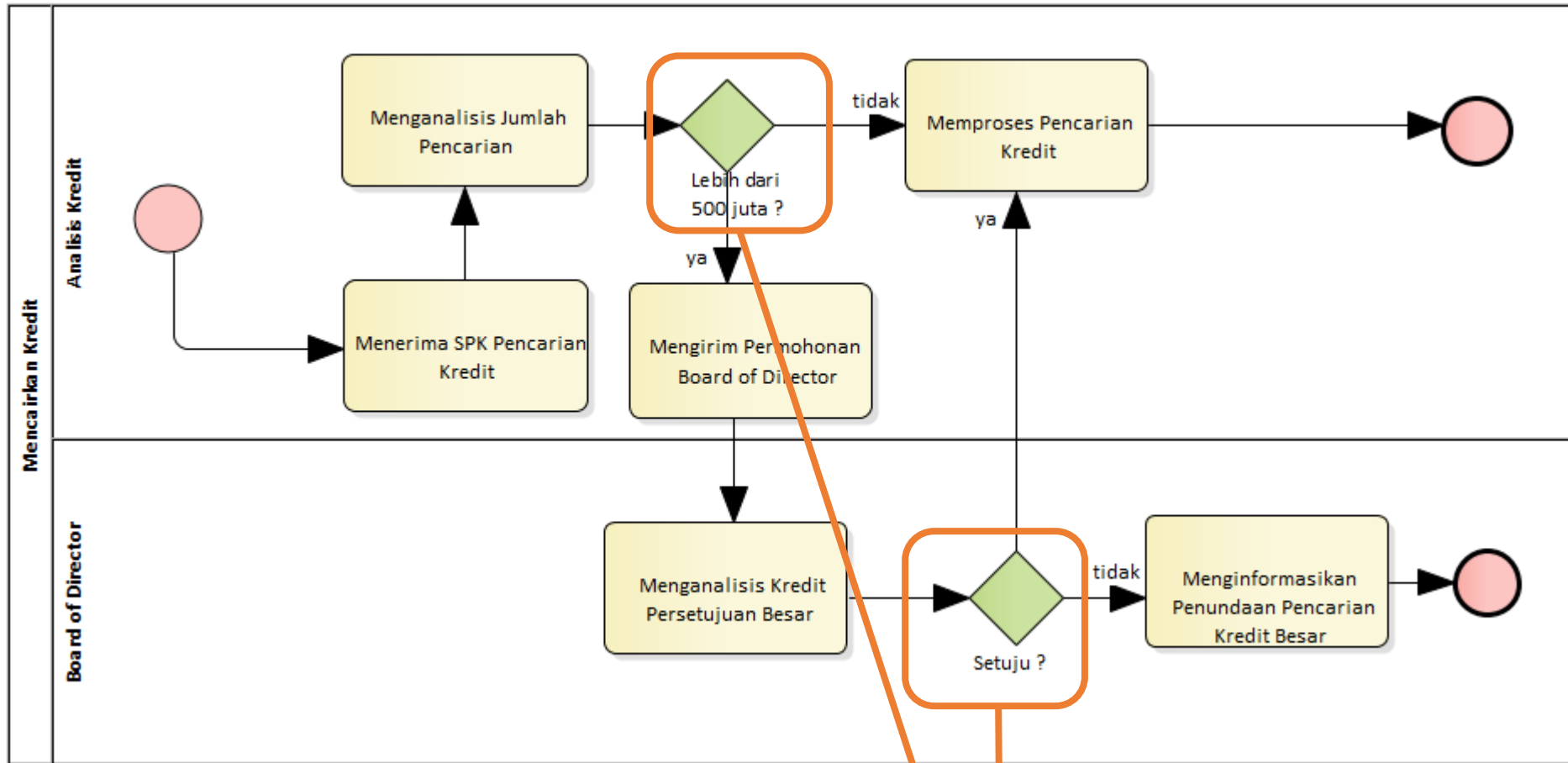
Leveling pada proses bisnis menggunakan BPMN menggunakan **subprocess**



Gateway

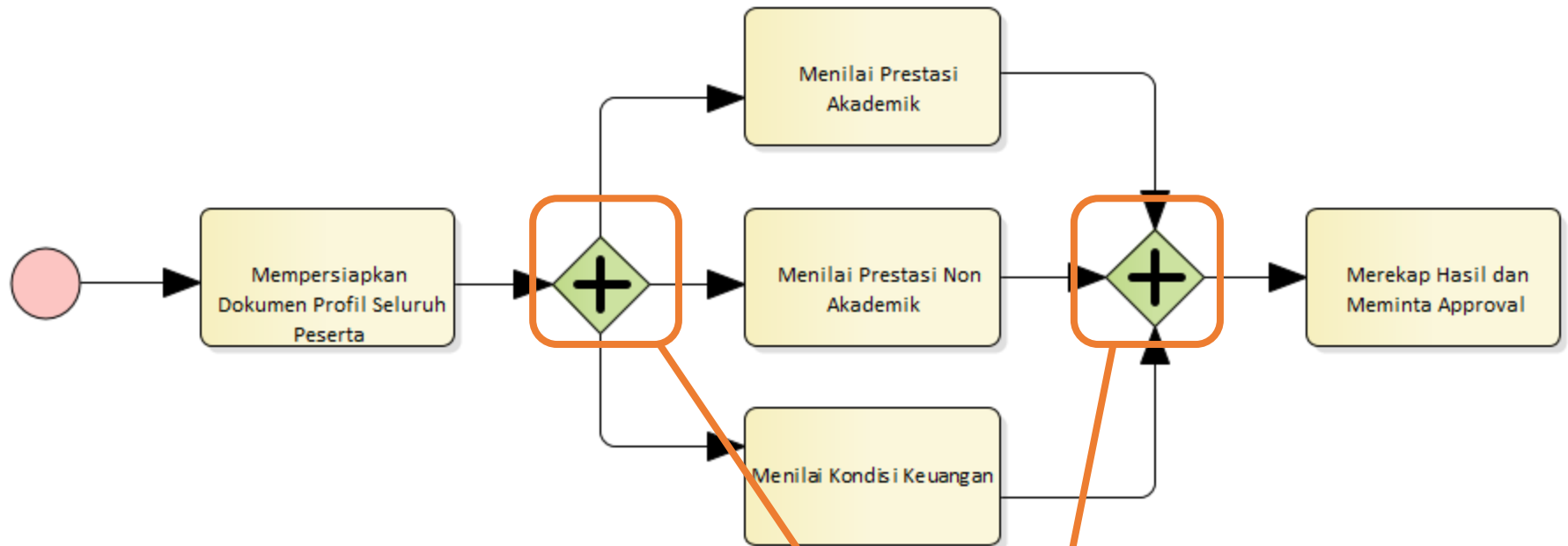
Nama Notasi	Deskripsi	Notasi
Exclusive Gateway	Pilih salah satu	
Parallel Gateway	Kegiatan bersamaan (paralel) dalam satu waktu	

Exclusive Gateway



Exclusive Gateway

Parallel Gateway



Parallel Gateway

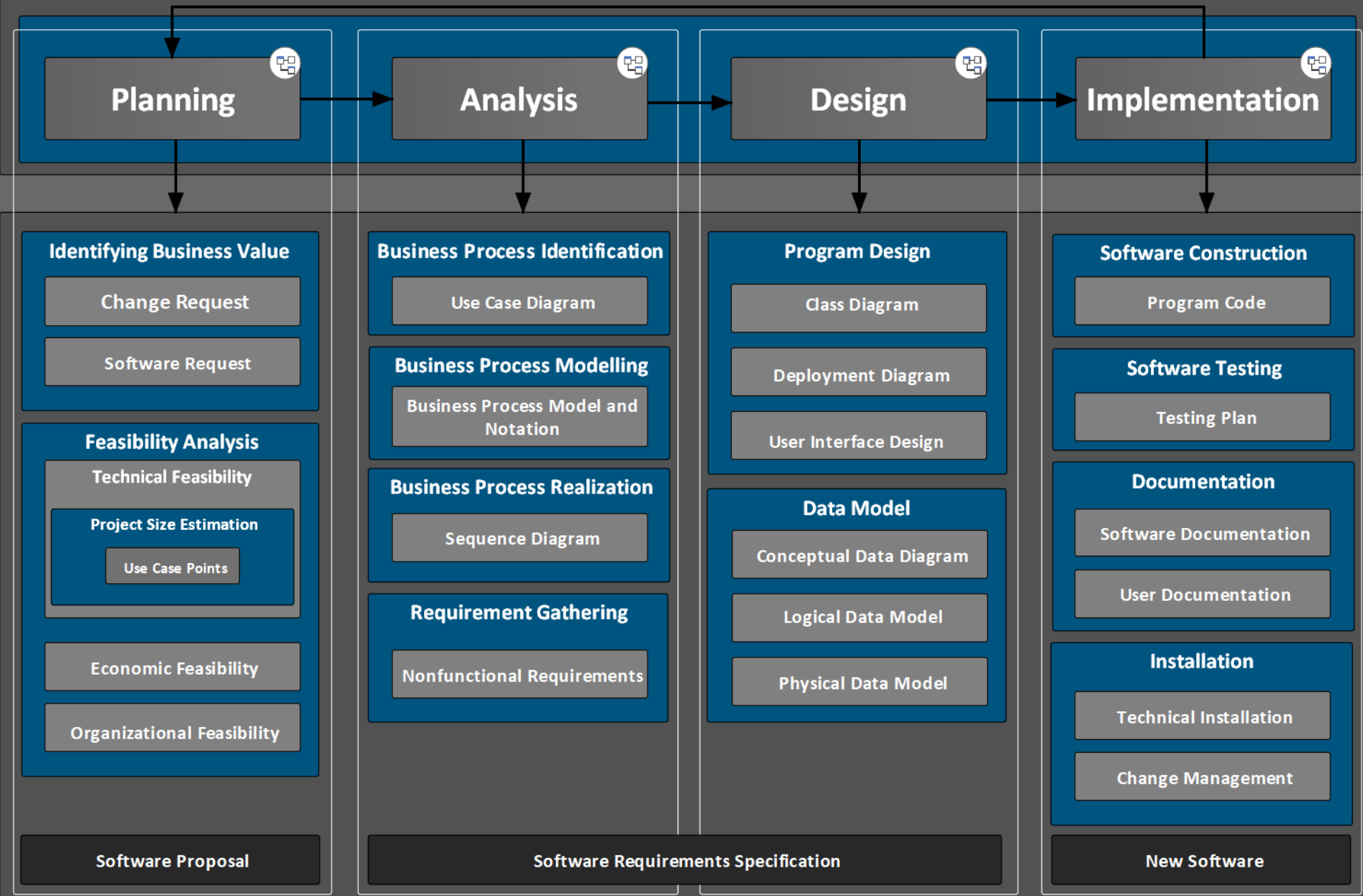


Studi Kasus: BPMN MusicPedia

User Interface Diagram langsung digambarkan (**prototyping**) untuk mempermudah pengguna memahami konteks dari alur proses berjalannya software

Application Development Governance

Software Development Life Cycle



Software Request

musicpedia				
Date	26 Oktober 2018			
Description	Musicpedia adalah aplikasi layanan download musik & audio dimana saja dan kapan saja, menawarkan akses lengkap ke jutaan lagu dari semua artis papan atas di industri musik barat maupun musik lokal, mendengarkan musik baru dan top musik dunia dalam bentuk audio mp3 secara offline			
Project Sponsor	Wahyu Utomo, VP Business Development, PT Musika Indonesia			
Business Need	1. Tidak Setuju	2. Ragu-Ragu	3. Setuju	4. Sangat Setuju
Aplikasi yang dikembangkan mampu meningkatkan pendapatan perusahaan?	☐	☐	☐	☒
Aplikasi yang dikembangkan mampu mengurangi biaya operasional perusahaan?	☐	☐	☒	☐
Aplikasi yang dikembangkan mampu meningkatkan produktifitas kerja pegawai?	☐	☒	☐	☐
Aplikasi yang dikembangkan mampu meningkatkan nilai tambah perusahaan yang bersifat intangible?	☐	☐	☒	☐
Business Value	Intangible Value: a. Meningkatkan brand recognition perusahaan di dunia internet b. Meningkatkan produktivitas kerja pegawai dan mengurangi kuantitas pegawai Tangible Value: a. Mengurangi biaya operasional perusahaan: - Sewa ruangan: Rp120.000.000,- - Biaya komunikasi: Rp6.000.000,- b. Meningkatkan penjualan musik: Rp400.000.000,-			

Technical Feasibility

musicpedia

Date: 26 Oktober 2018

Penjelasan isian	1. Sangat Kurang	2. Kurang	3. Baik	4. Sangat Baik
Kefamiliaran dengan Aplikasi				
	1	2	3	4
Pengguna familiar terhadap pengoperasian aplikasi ini.	☐	☐	☒	☐
Pengembang familiar terhadap pengembangan aplikasi ini.	☐	☐	☐	☒
Kefamiliaran dengan Teknologi				
	1	2	3	4
Pengguna familiar dengan teknologi pendukung aplikasi.	☐	☒	☐	☐
Pengembang familiar mengembangkan aplikasi dengan platform, bahasa pemrograman dan tool IDE yang dipilih.	☐	☐	☐	☒
Ukuran Proyek				
Jumlah pengembang yang dibutuhkan.	7 Man/Month			
Waktu yang dibutuhkan dalam mengembangkan aplikasi ini.	6 Month			
Kompatibilitas				
	1	2	3	4
Kebutuhan pengguna terhadap kompatibilitas aplikasi untuk terintegrasi dengan aplikasi lain.	☐	☐	☒	☐
Kompatibilitas aplikasi terhadap teknologi yang ada pada organisasi.	☐	☐	☐	☒
Secara analisi kelayakan teknis, apakah aplikasi layak dikembangkan sesuai kriteria di atas? ☒ Layak ☐ Tidak Layak				

Technical Feasibility

Use Case Points

Tahap 1 - Menghitung Person Hours (PH)				
Use Case Points (UCP)		Person Hours Multiplier (PHM)	Person Hours (PH)	
51		20	1020	
51		28	1428	
Tahap 2 - Menghitung Person Month (PM)				
PHM	Person Hours (PH)	Lama Bekerja Perhari	Jumlah Bekerja Sebulan	Person Months (PM)
20	1020	8	22	5.80
	1020	10	26	3.92
28	1428	8	22	8.11
	1428	10	26	5.49
Tahap 3 - Menghitung Time (Month)				
PHM	Formula Penghitung Waktu		Jumlah Bekerja Sebulan	Waktu dalam Bulan (M)
20	3 * PM^(1/3)		22	5.39
			26	4.73
28			22	6.03
			26	5.29

Economic Feasibility

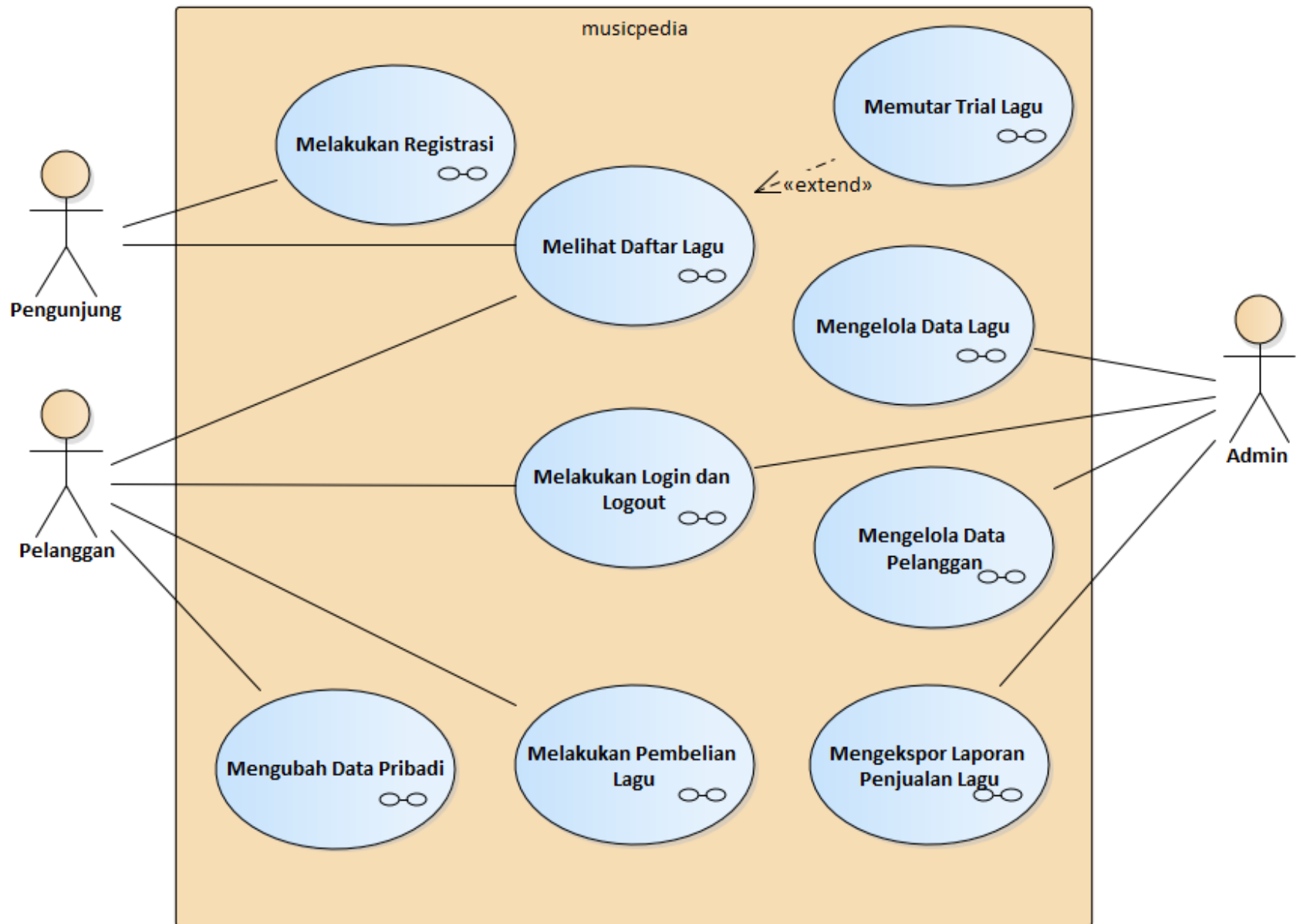
Cost-Benefit Analysis

Tahun	2019	2020	2021	2022
Peningkatan Pendapatan Penjualan Lagu		400,000,000	400,000,000	400,000,000
Pengurangan Biaya Sewa Ruangan		120,000,000	120,000,000	120,000,000
Pengurangan Biaya Komunikasi		6,000,000	6,000,000	6,000,000
Total Benefits	0	526,000,000	526,000,000	526,000,000
PV of Benefits	0	468,138,127	441,639,743	416,641,267
PV of All Benefits	0	468,138,127	909,777,870	884,779,394
Honor Tim (Analysis, Design and Implementation)	250,000,000	120,000,000	120,000,000	120,000,000
Total Development Costs	250,000,000	120,000,000	120,000,000	120,000,000
Honor Pengelola Web	72,000,000	72,000,000	72,000,000	72,000,000
Biaya Lisensi Software	10,000,000	10,000,000	10,000,000	10,000,000
Hardware upgrades	50,000,000	50,000,000	50,000,000	50,000,000
Biaya Komunikasi	1,000,000	1,000,000	1,000,000	1,000,000
Biaya Marketing	50,000,000	50,000,000	50,000,000	50,000,000
Total Operational Costs	183,000,000	183,000,000	183,000,000	183,000,000
Total Costs	433,000,000	303,000,000	303,000,000	303,000,000
PV of Costs	408,490,566	269,668,921	153,650,329	144,953,140
PV of all Costs	408,490,566	678,159,487	831,809,816	976,762,957
Total Project Costs Less Benefits	-433,000,000	223,000,000	223,000,000	223,000,000
Yearly NPV	-408,490,566	198,469,206	187,235,100	176,636,887
Cumulative NPV	-408,490,566	-210,021,360	-22,786,260	153,850,627
Return on Investment (ROI)	-100.00%	-0.309693168	-0.027393593	0.15751071
Break-even Point (BEP)				3.129000574

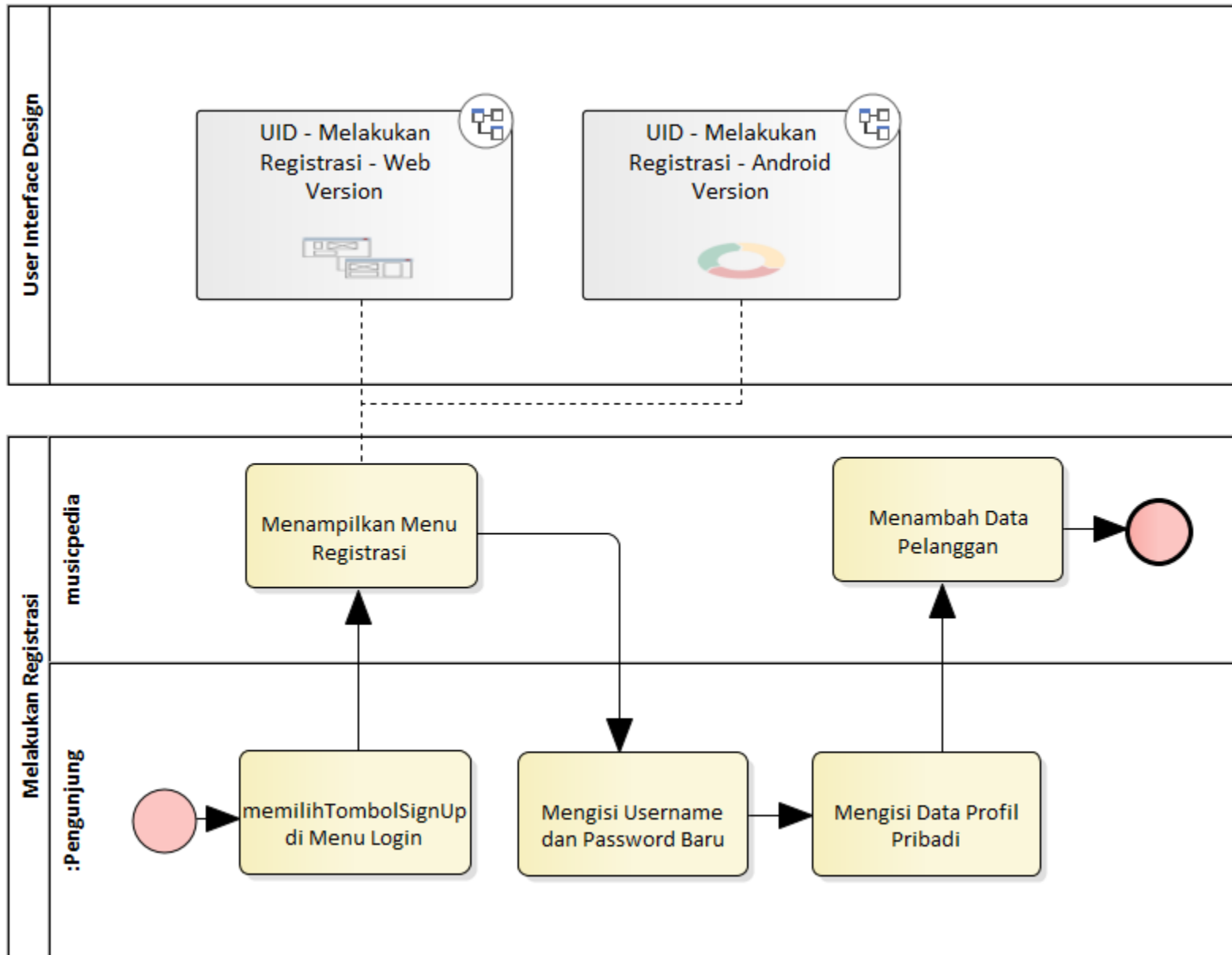
Organizational Feasibility

musicpedia		
Date	26 Oktober 2018	
Anggota Tim		
User/Product Owner	Wahyu Utomo	
Project Manager	Haris Dermawan	
System Analyst	Risa Dhani Horasman Purba	
Business Analyst	Mulyana	
Programmer	Achmad Fatkarrofiqi	
Tester	Januar Sapareza	
Apakah aplikasi ini mendukung visi dan misi organisasi?		
Ya		
Apakah aplikasi ini sesuai dengan tugas, fungsi dan KPI unit kerja anda?		
Ya		
Apakah aplikasi ini selaras dengan proses bisnis unit kerja anda?		
Ya		
Secara analisis kelayakan organisasi, apakah aplikasi layak dikembangkan sesuai kriteria di atas?	☒ Layak	☐ Tidak Layak

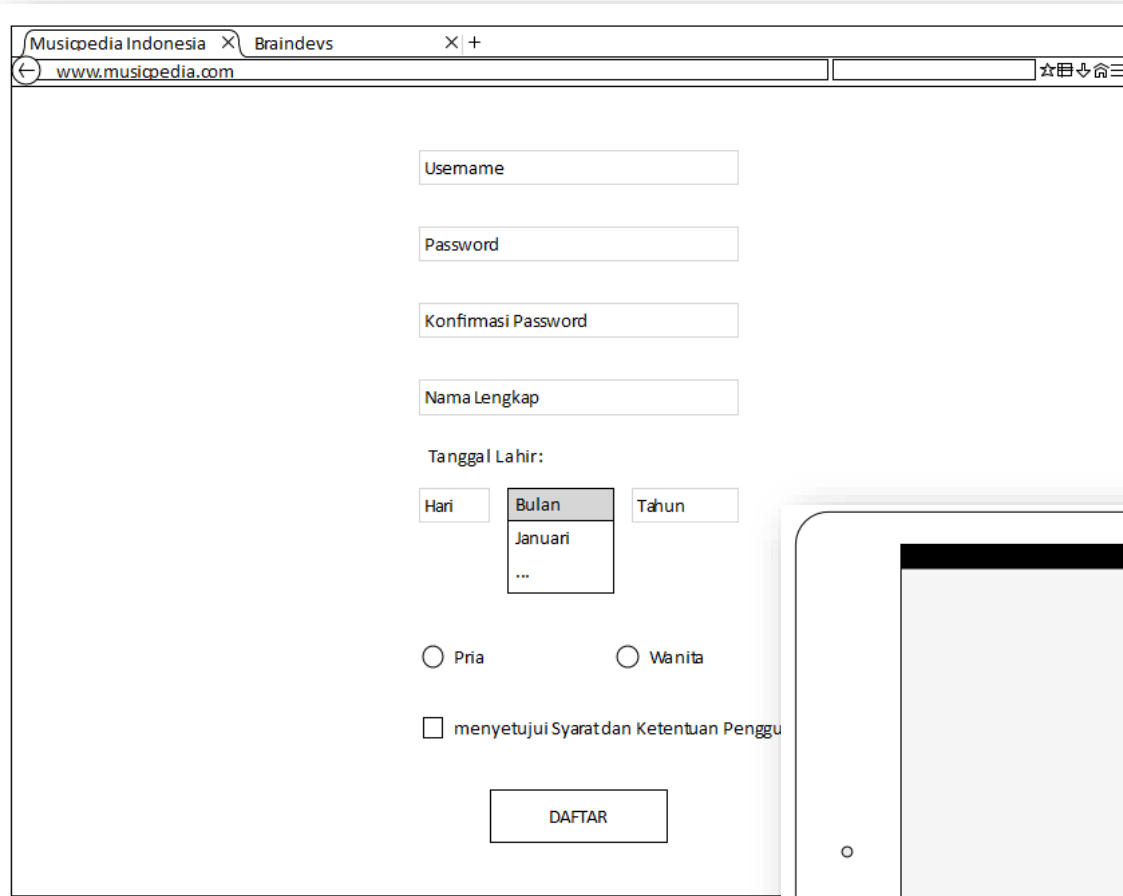
Use Case Diagram MusicPedia



BPMN Melakukan Registrasi



User Interface Design Melakukan Registrasi (versi Web dan versi Android)



A screenshot of a web browser showing a registration form. The browser's address bar displays 'www.musicpedia.com'. The form includes input fields for 'Ustream', 'Password', 'Konfirmasi Password', and 'Nama Lengkap'. Below these is a 'Tanggal Lahir:' section with three dropdown menus for 'Hari', 'Bulan', and 'Tahun'. The 'Bulan' dropdown is open, showing 'Januari' and '...'. At the bottom, there are radio buttons for 'Pria' and 'Wanita', a checkbox for 'menyetujui Syarat dan Ketentuan Pengguna', and a 'DAFTAR' button.

Ustream

Password

Konfirmasi Password

Nama Lengkap

Tanggal Lahir:

Hari Bulan Tahun

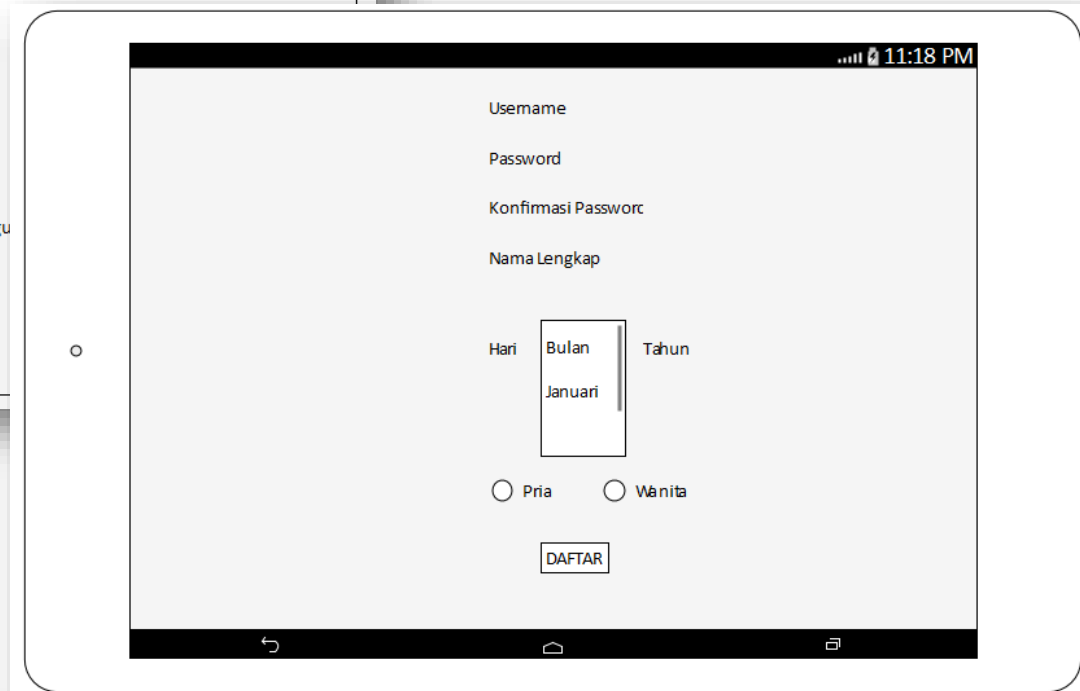
Januari

...

☐ Pria ☐ Wanita

☐ menyetujui Syarat dan Ketentuan Pengguna

DAFTAR



A screenshot of an Android mobile interface showing a registration form. The status bar at the top shows the time as 11:18 PM. The form layout is similar to the web version, with input fields for 'Ustream', 'Password', 'Konfirmasi Password', and 'Nama Lengkap'. The 'Tanggal Lahir:' section has three dropdown menus for 'Hari', 'Bulan', and 'Tahun'. The 'Bulan' dropdown is open, showing 'Januari' and '...'. At the bottom, there are radio buttons for 'Pria' and 'Wanita', and a 'DAFTAR' button.

Ustream

Password

Konfirmasi Password

Nama Lengkap

Hari Bulan Tahun

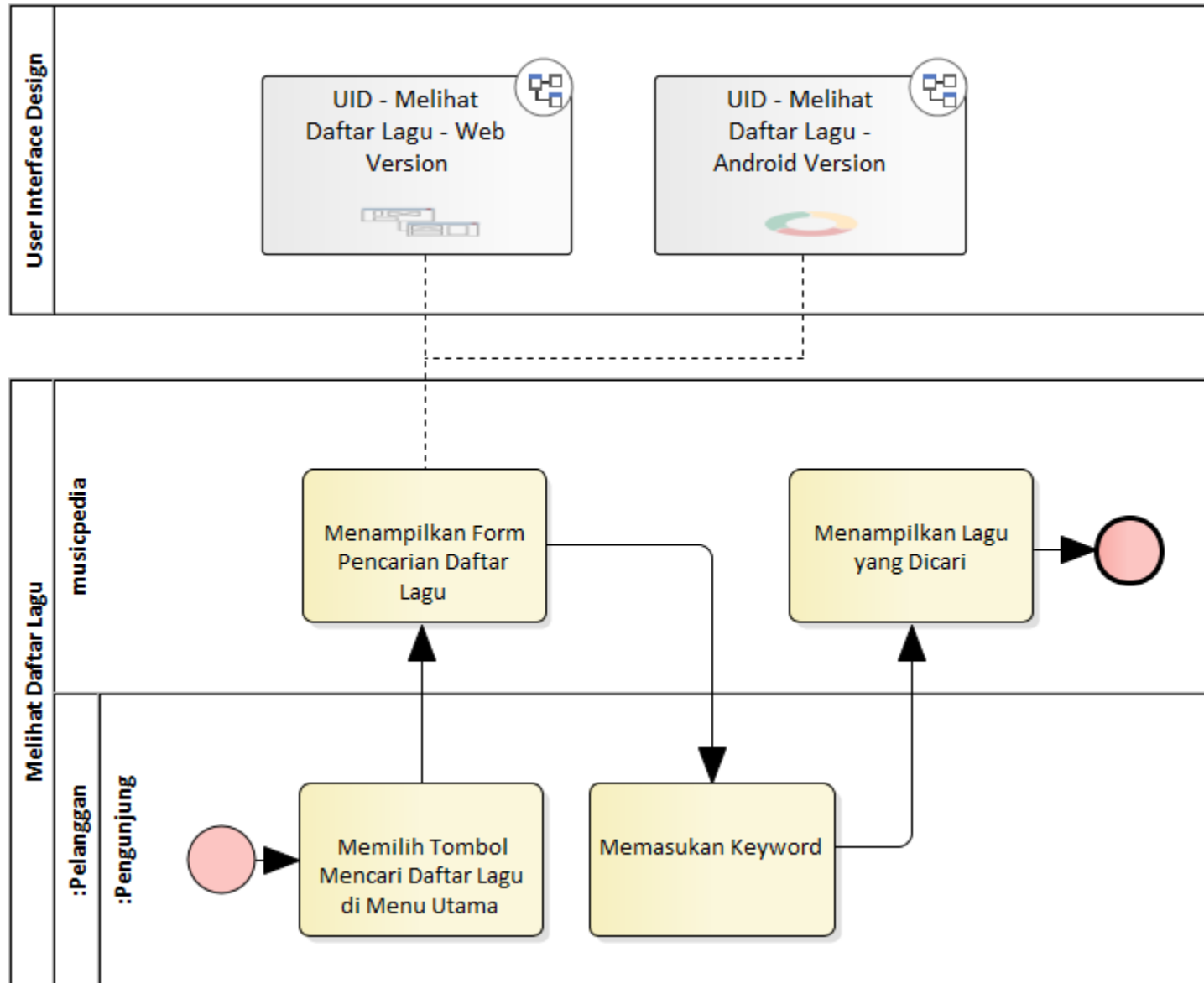
Januari

...

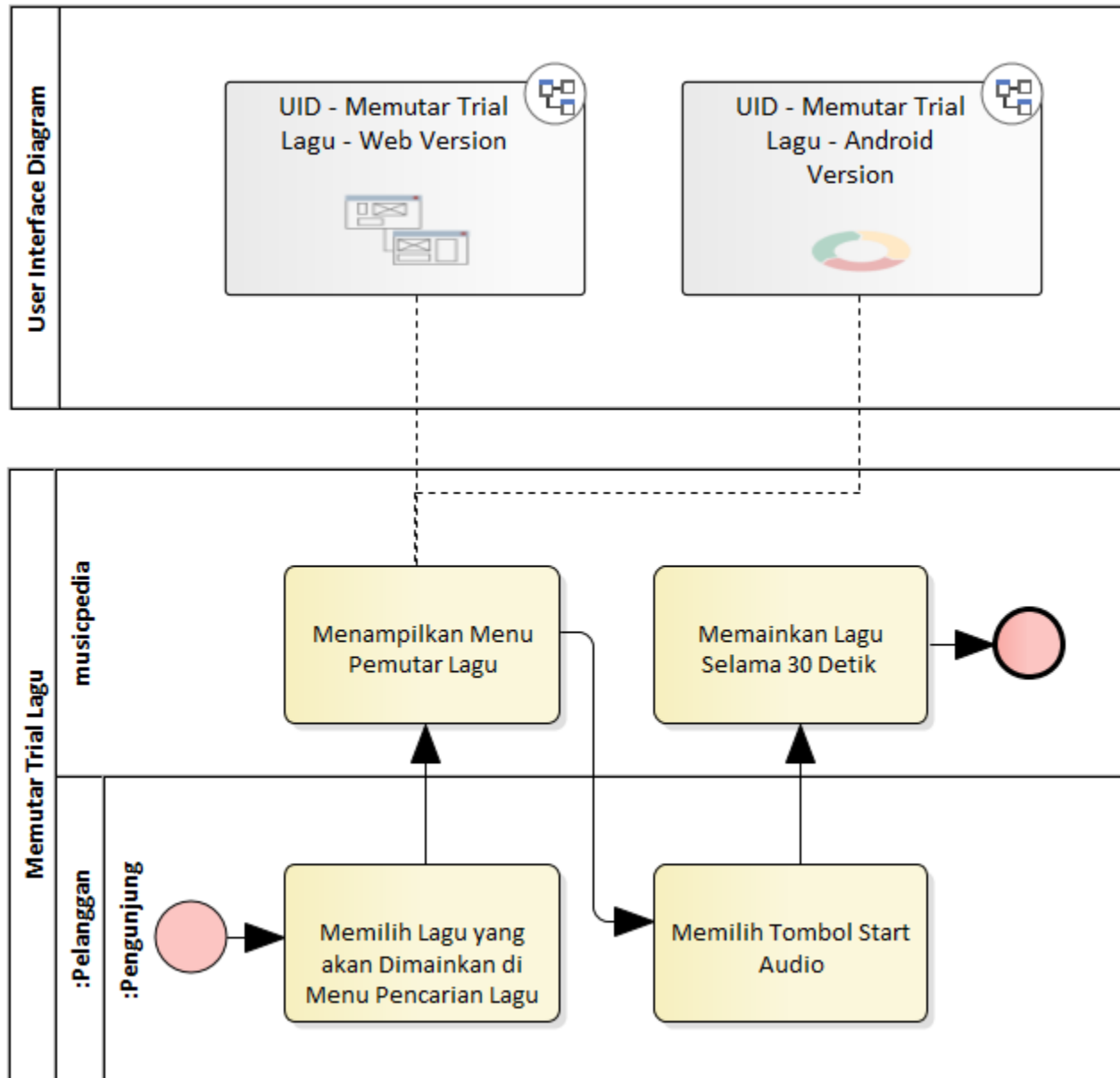
☐ Pria ☐ Wanita

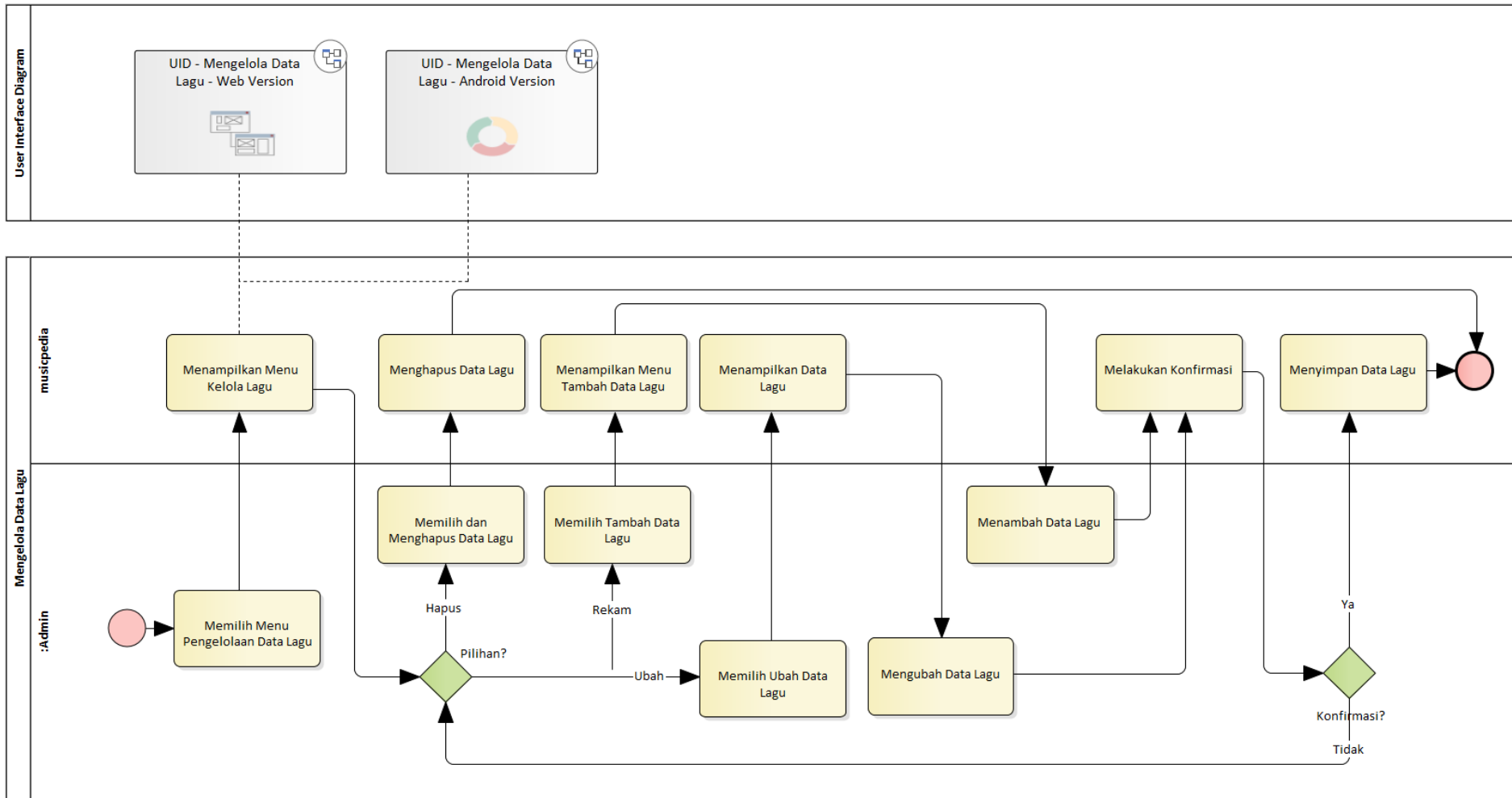
DAFTAR

BPMN Melihat Daftar Lagu

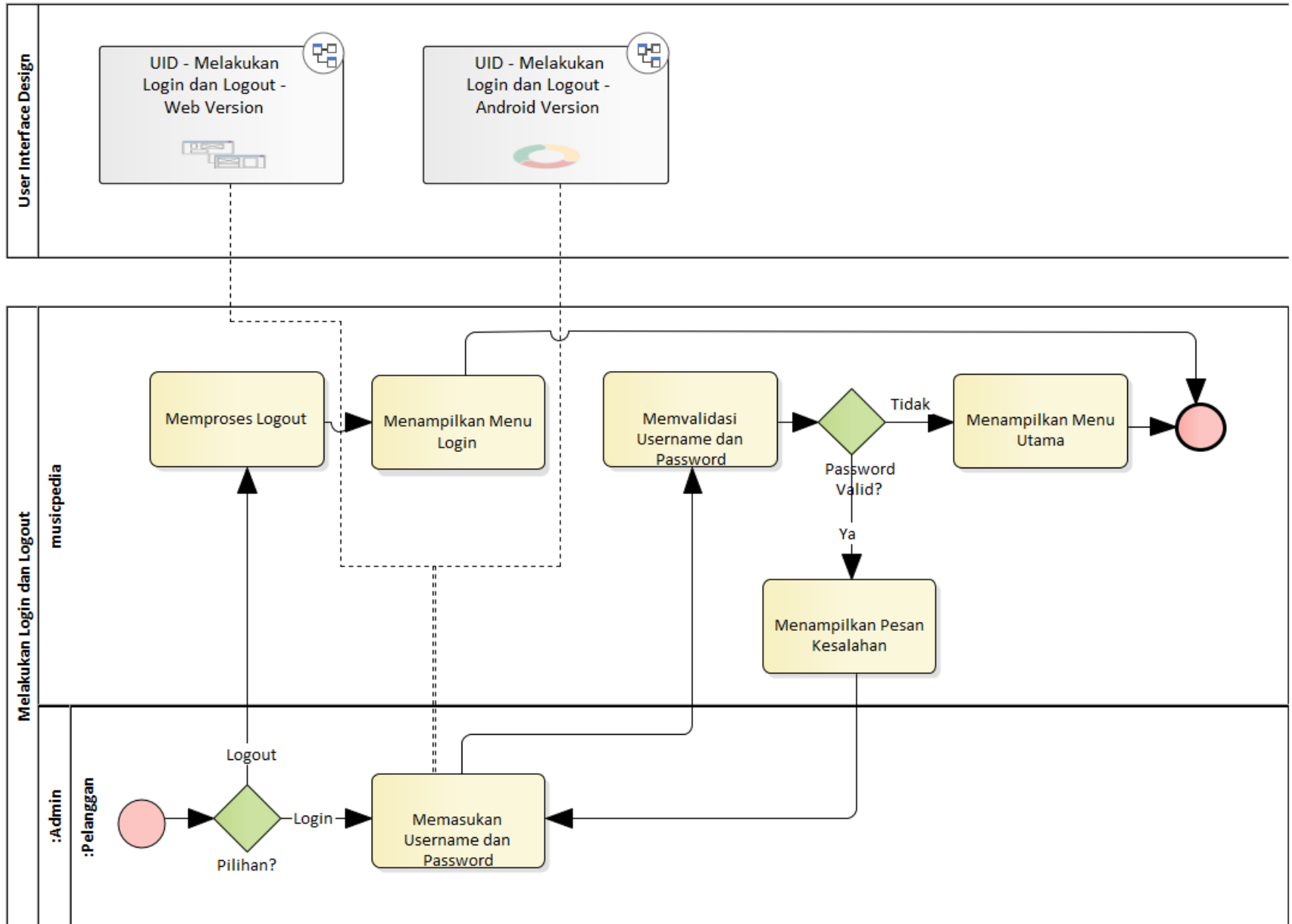


BPMN Memutar Trial Lagu

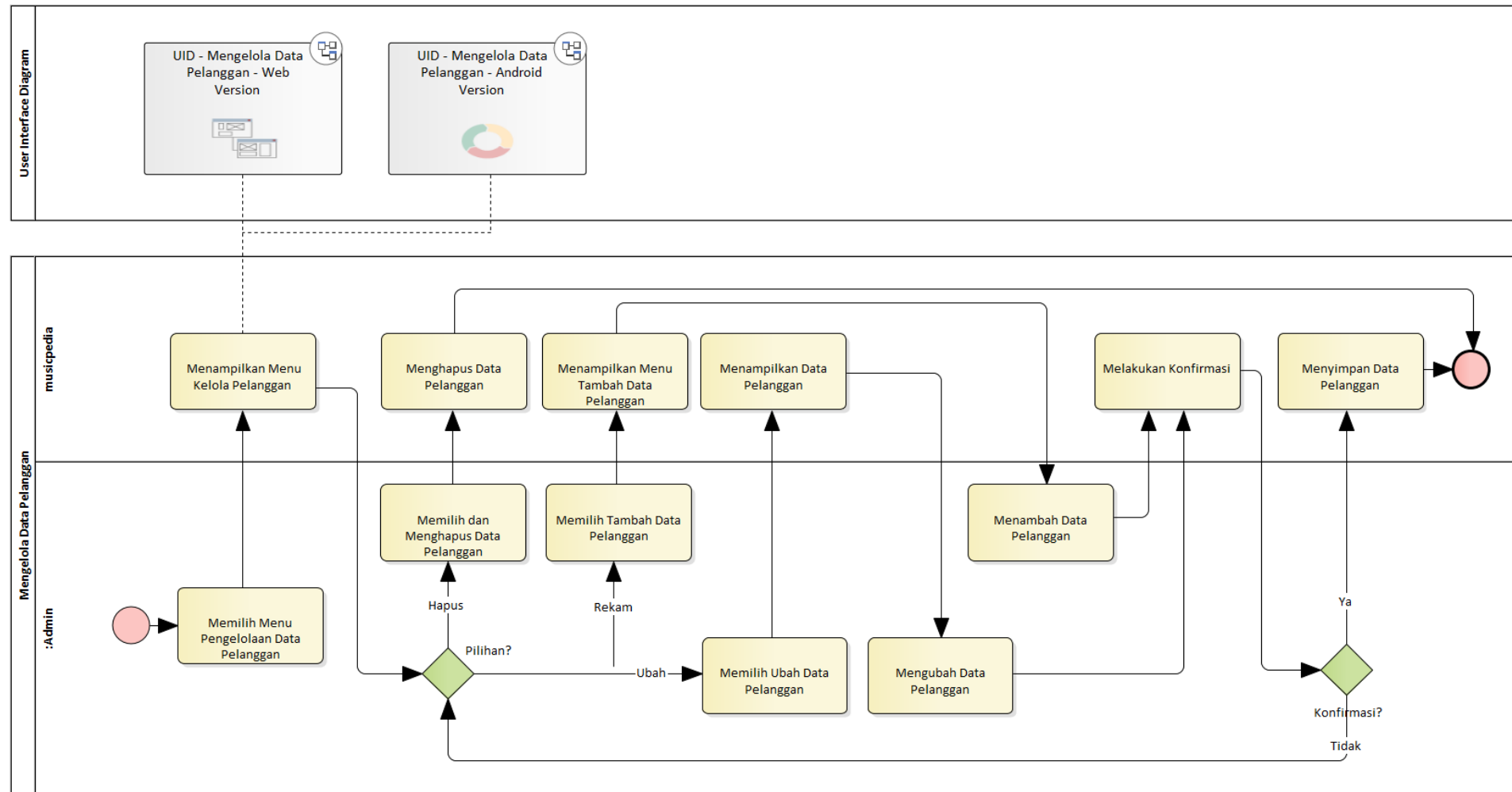




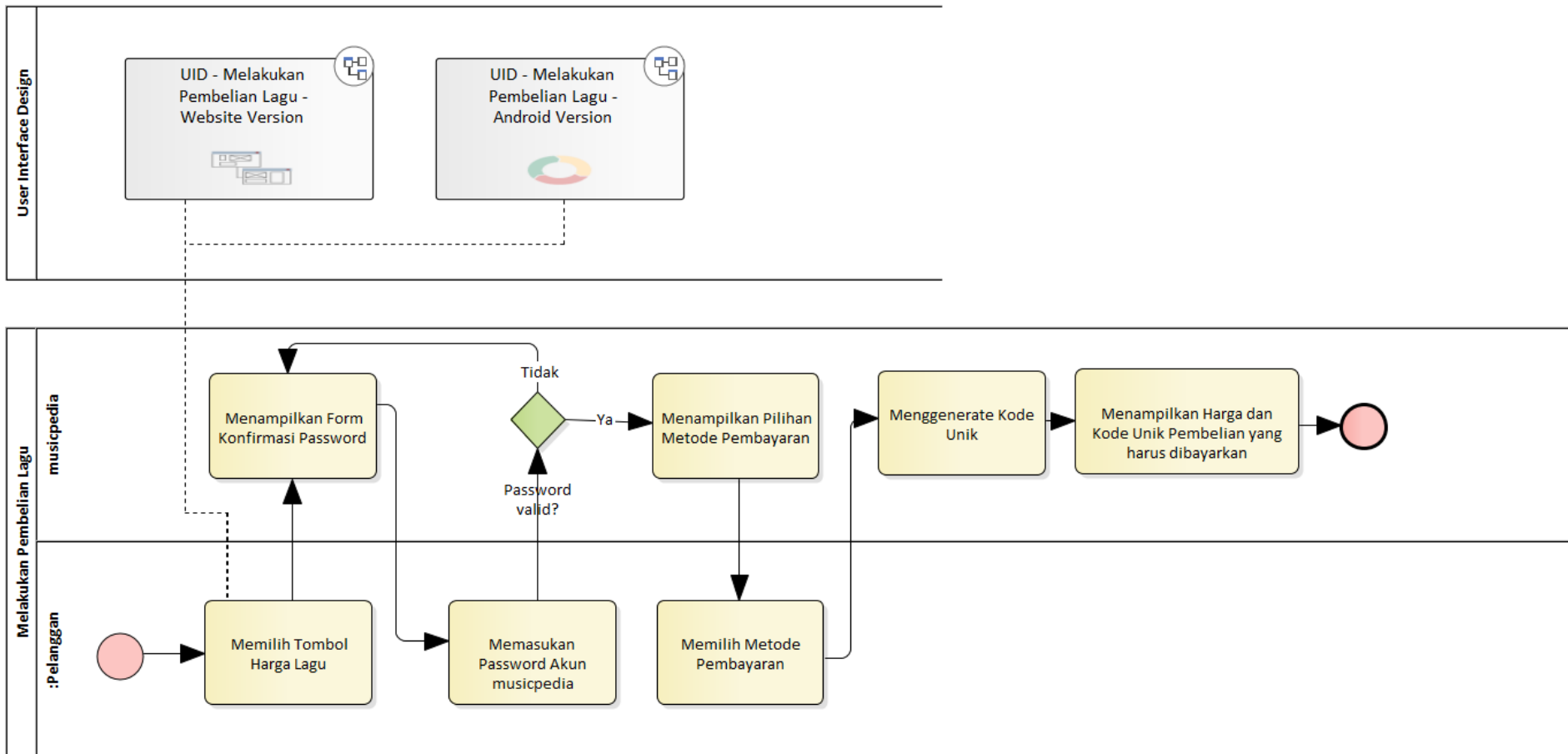
BPMN Melakukan Login dan Logout



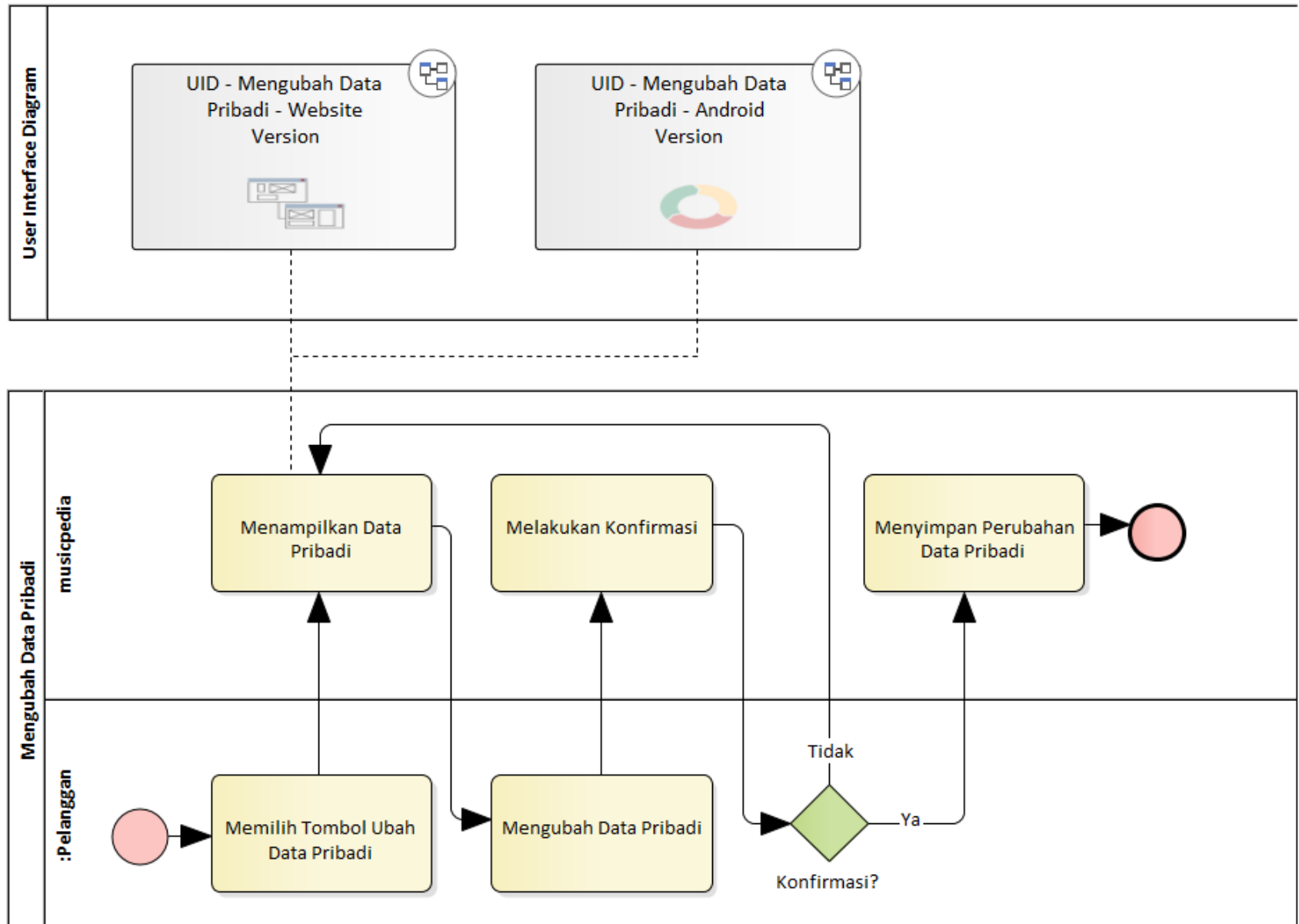
BPMN Mengelola Data Pelanggan



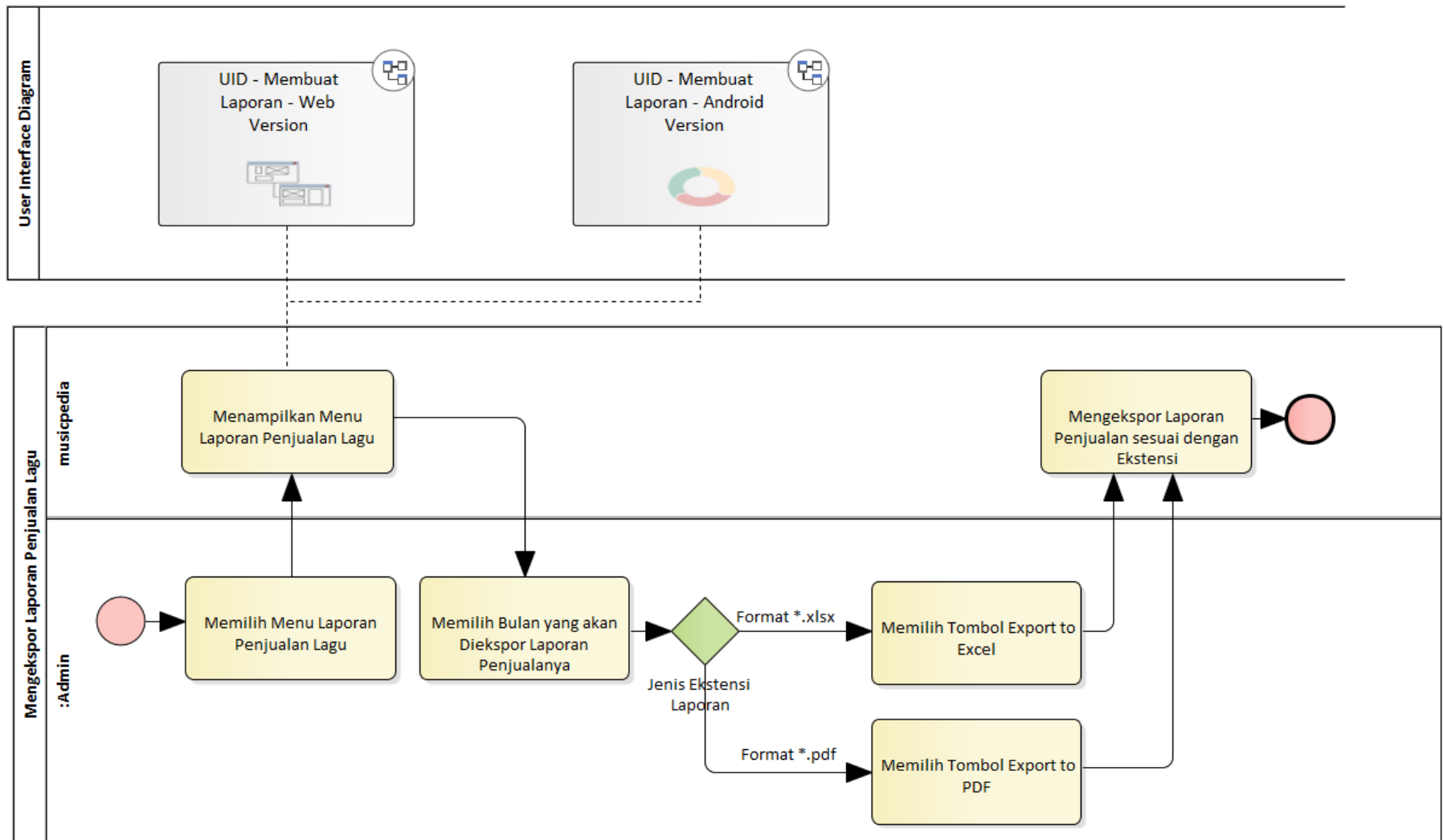
BPMN Melakukan Pembelian lagu



BPMN Mengubah Data Pribadi



BPMN Mengekspor Laporan Penjualan Lagu





2.4 Realisasi Proses Bisnis dengan Sequence Diagram

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

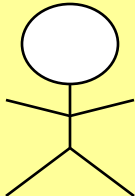
2.3 Pemodelan **Data Model**

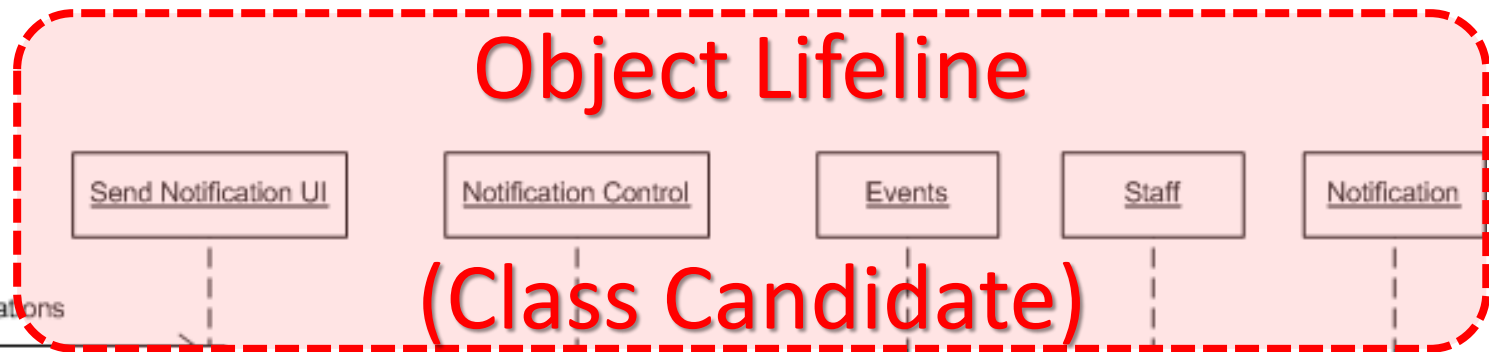
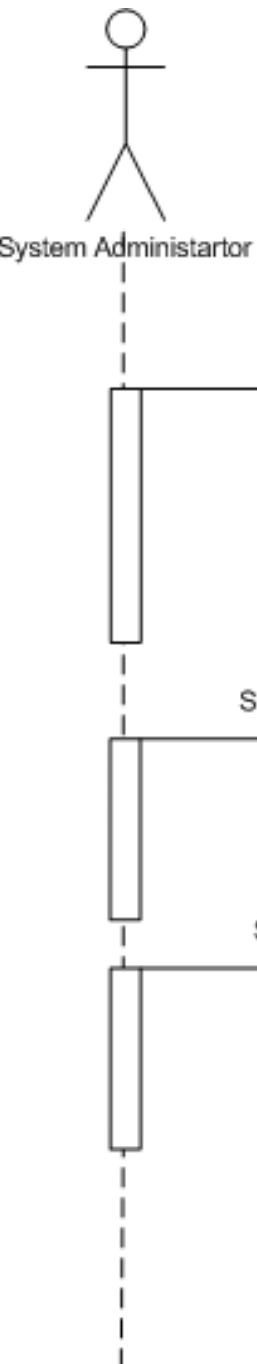
2.4 Pemodelan **Deployment Diagram**

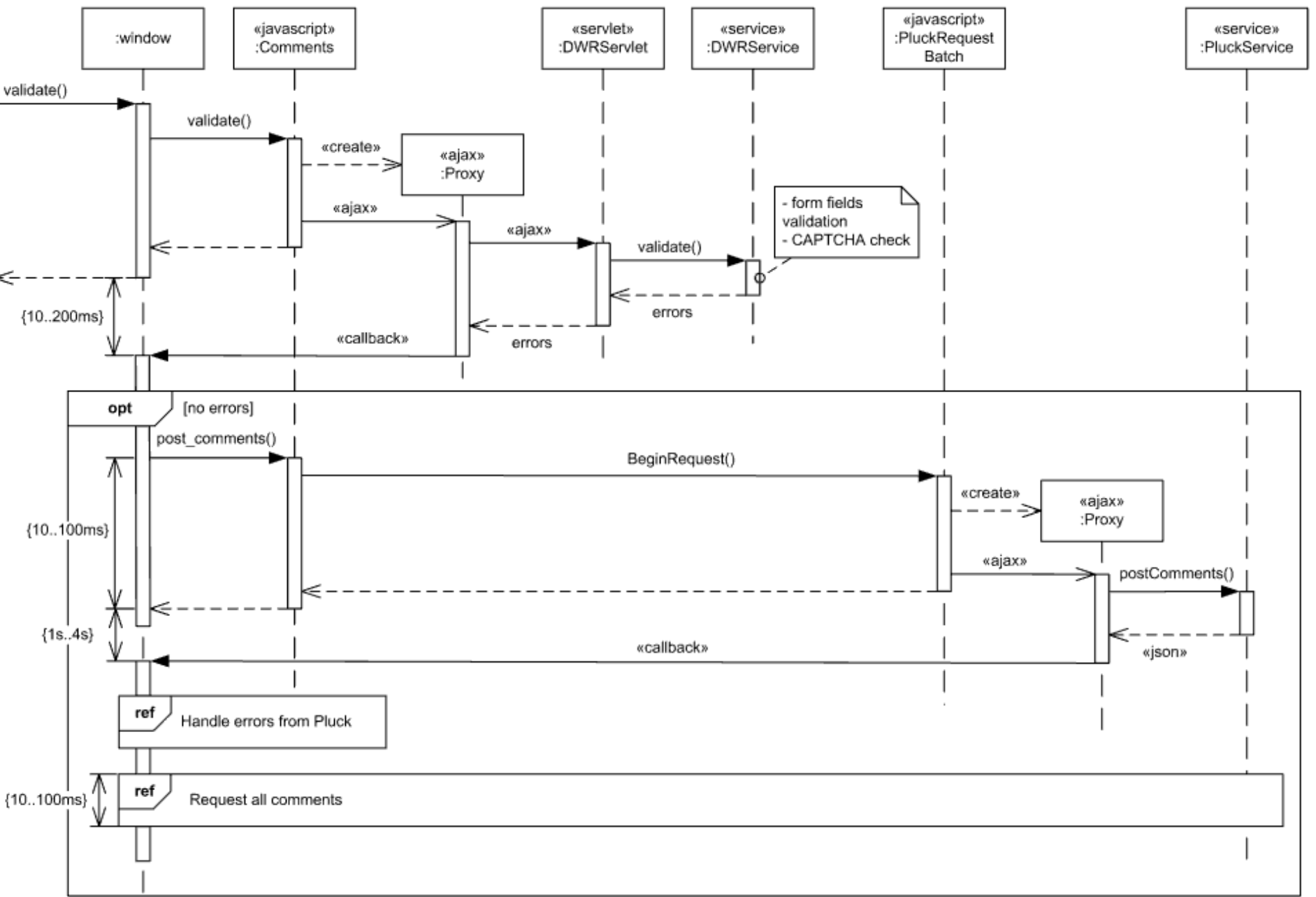
Sequence Diagram

- Sequence diagram menggambarkan **interaksi antar object (class) dalam software** pada suatu **sekuensial waktu**
 - Interaksi object (class) berdasarkan **alur proses bekerjanya software**, dimana interaksi tersebut menggambarkan **pesan yang dikirimkan secara sekuensial antara object (class)**
 - Garis **vertikal (lifeline)** menunjukkan **object (class)**, garis **horizontal** menunjukkan **pesan yang mengalir** antara object (class) tersebut
- Sequence diagram adalah **alat komunikasi System Analyst dengan Programmer**, menggambarkan alur proses bekerjanya software sekaligus dengan komposisi software akan seperti apa

Sequence Diagram Syntax

AN ACTOR	
AN OBJECT	anObject:aClass
A LIFELINE	
A FOCUS OF CONTROL	
A MESSAGE	
OBJECT DESTRUCTION	X



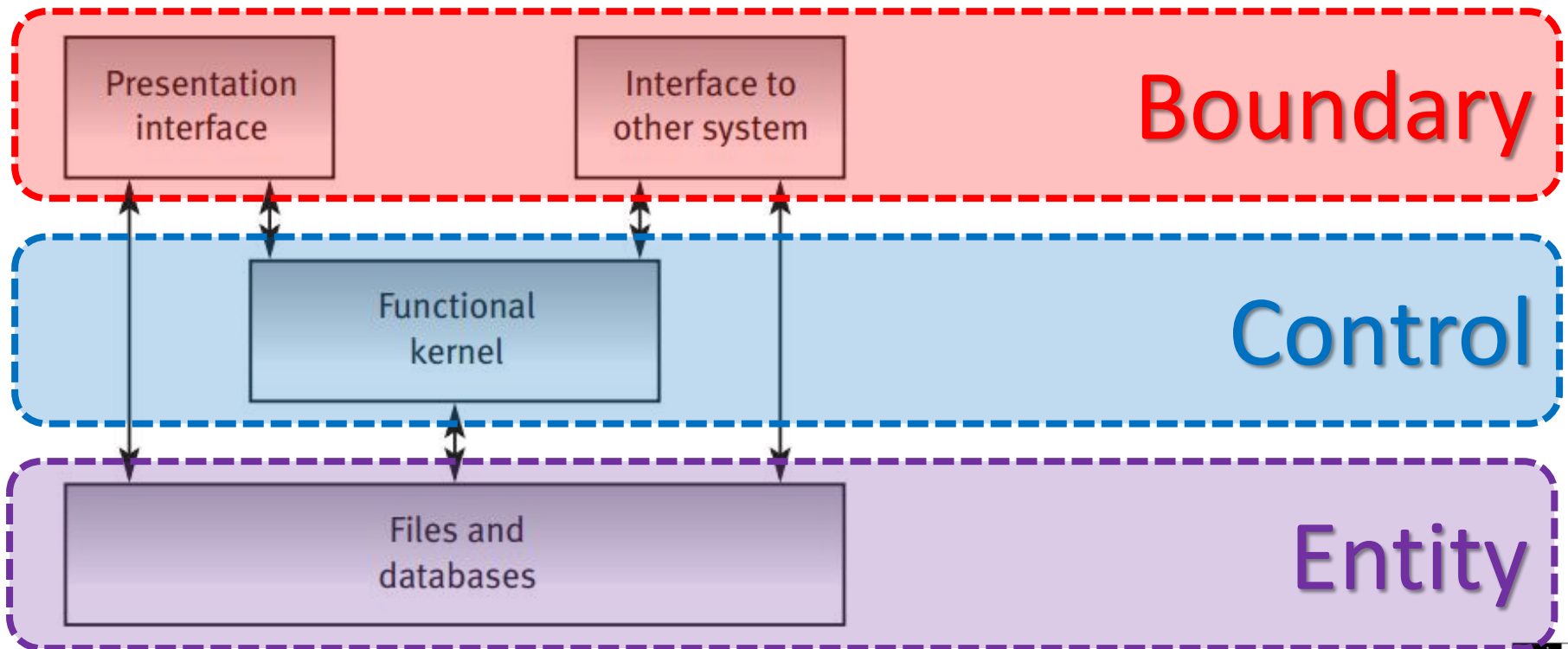


Denert's Law (1991)

Separation of concerns leads to standard architectures

(Endres, 2003)

[L9]



Sequence Diagram berbasis Arsitektur

Boundary – Control – Entity

1. Boundary Class:

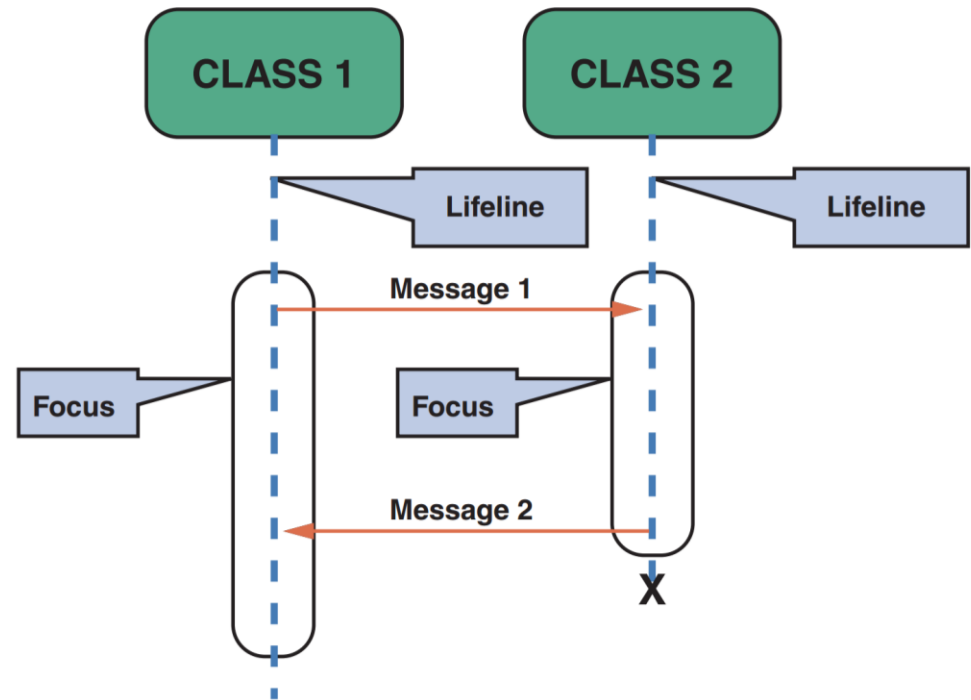
- Class yang berinteraksi dengan aktor langsung (user interface)
- Form, input, UI, dsb

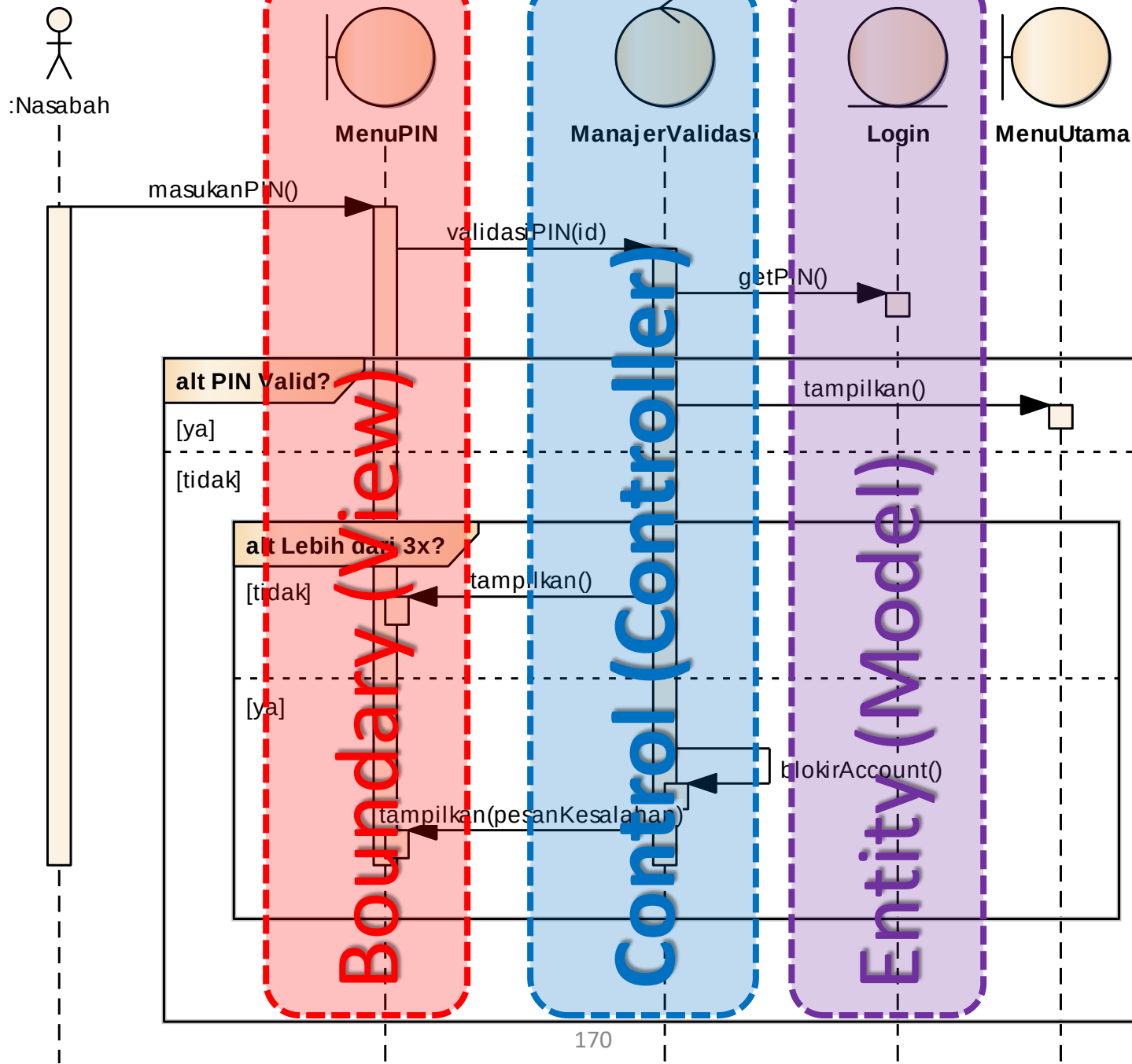
2. Control Class:

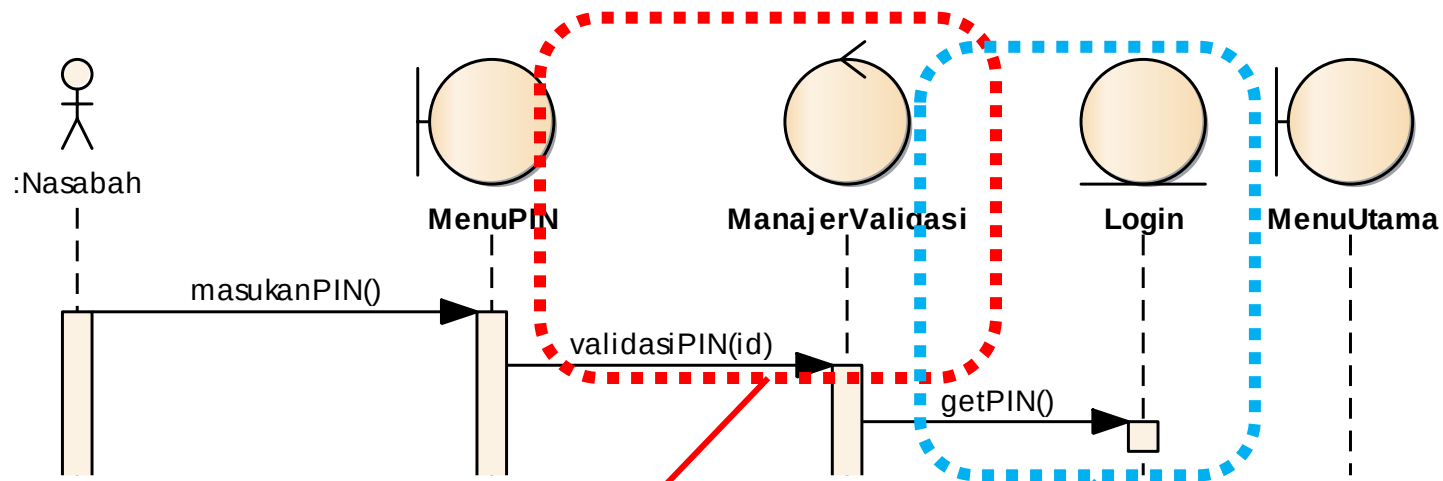
- Class yang berhubungan dengan pemrosesan, penghitungan, kalkulasi, komputasi, query, dst

3. Entity Class:

- Class yang berhubungan dengan data, penyimpanan data/file







```

public class ManajerValidasi {
    private Login m_Login;
    public int validasiKartu(){
    }

    public int validasiPIN(){
        int pin = m_Login.getPIN();
        if(pin.equals(pinuser)){
            MenuUtama.show();
        }

        public void blokirKartu(){
        }
    }
}
  
```

```

public class Login {

    public int getPIN(){

    }

}
  
```

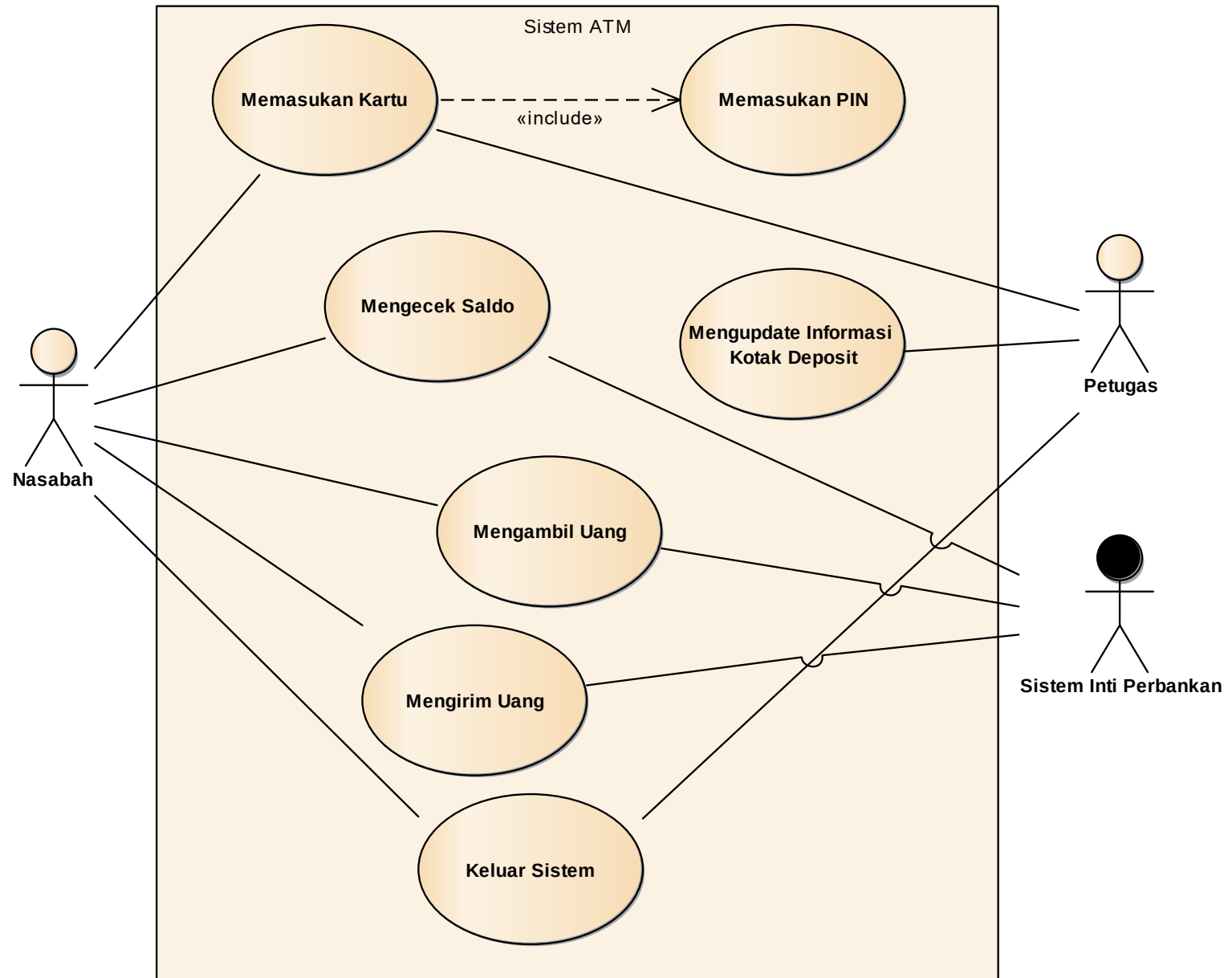
Sequence Diagram

- Sequence Diagram dibuat untuk setiap Use Case yang dibuat
- Dimulai dari menarik Actor yang ada di Use Case Diagram, dilanjutkan dengan membuat sequence detail dari alur proses berjalannya Use Case dengan message yang mengalir didalamnya
- Catatan: Objek dari Lifeline di Sequence Diagram akan menjadi kandidat Class, karena itulah harus mengikuti arsitektur Boundary – Control – Class

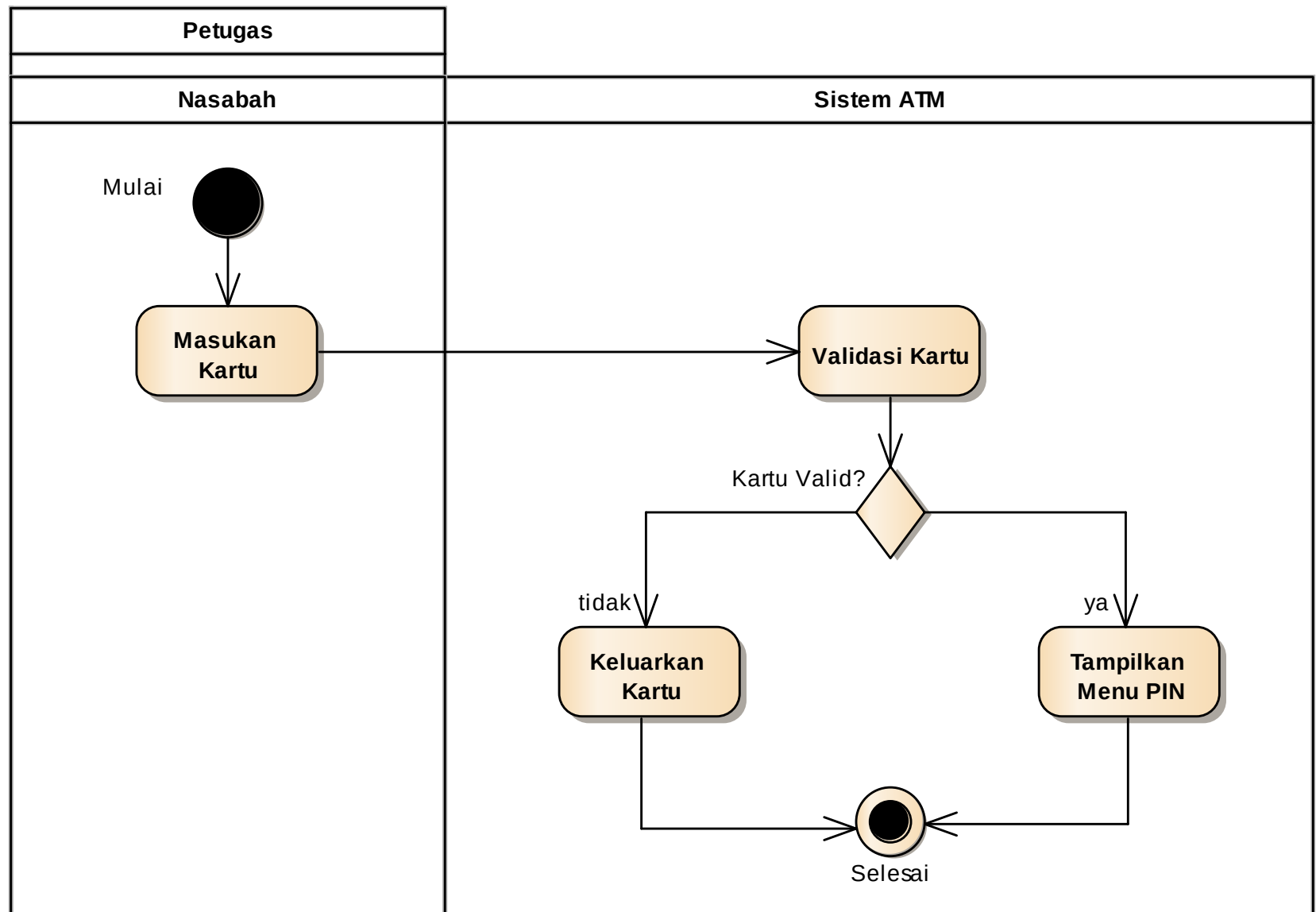


Studi Kasus: Sequence Diagram Sistem ATM

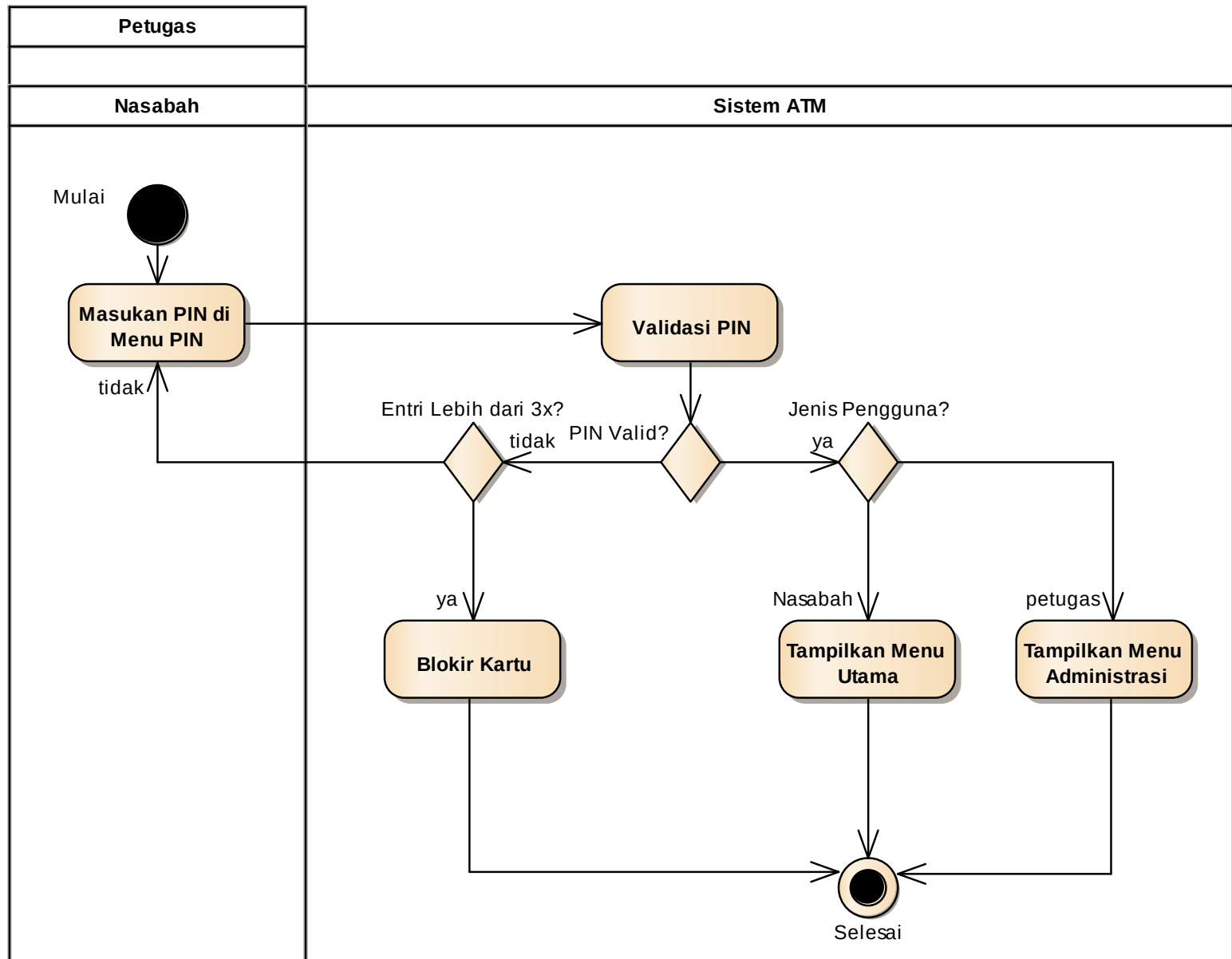
Use Case Diagram Sistem ATM



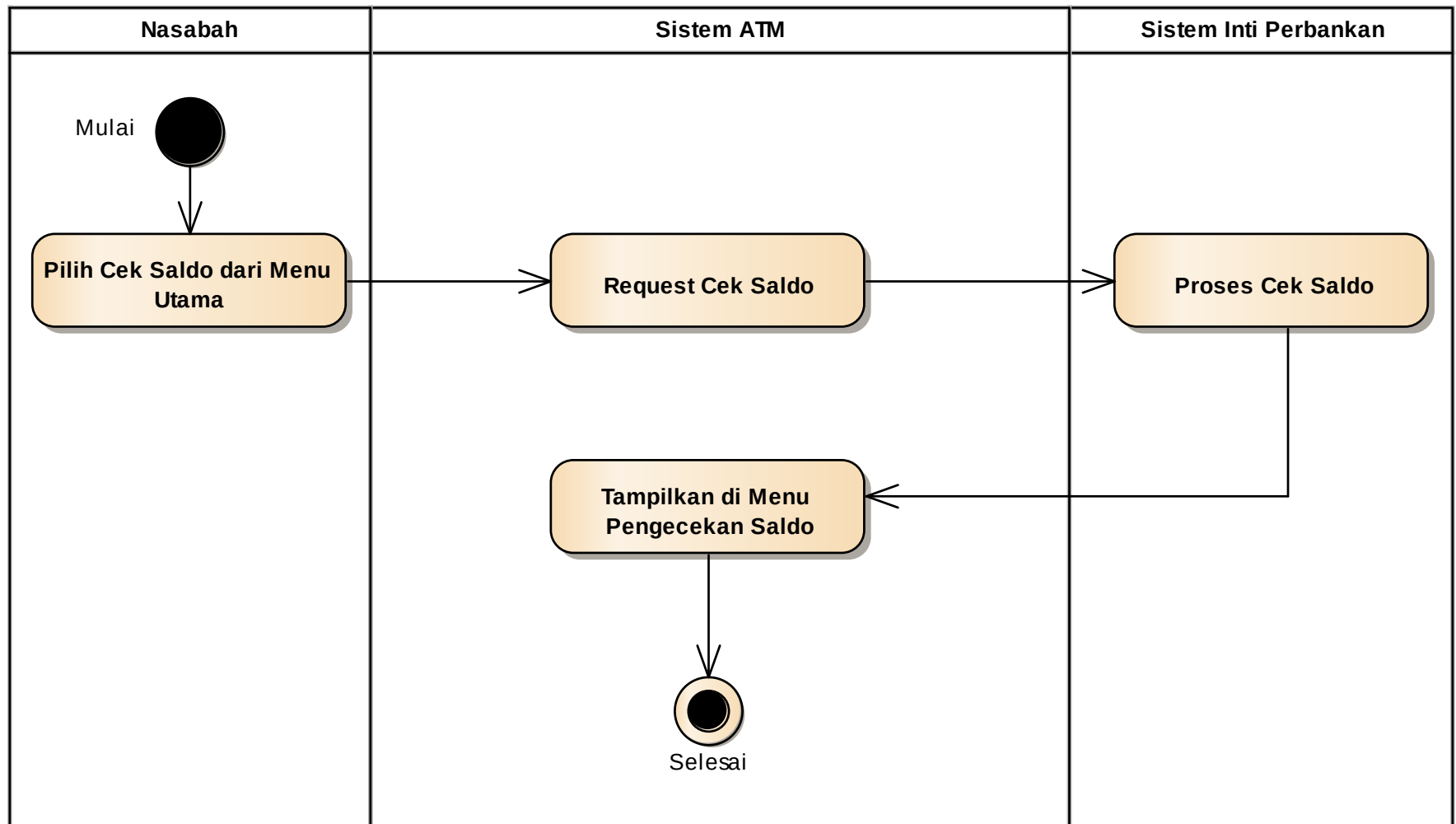
Activity Diagram: Memasukkan Kartu



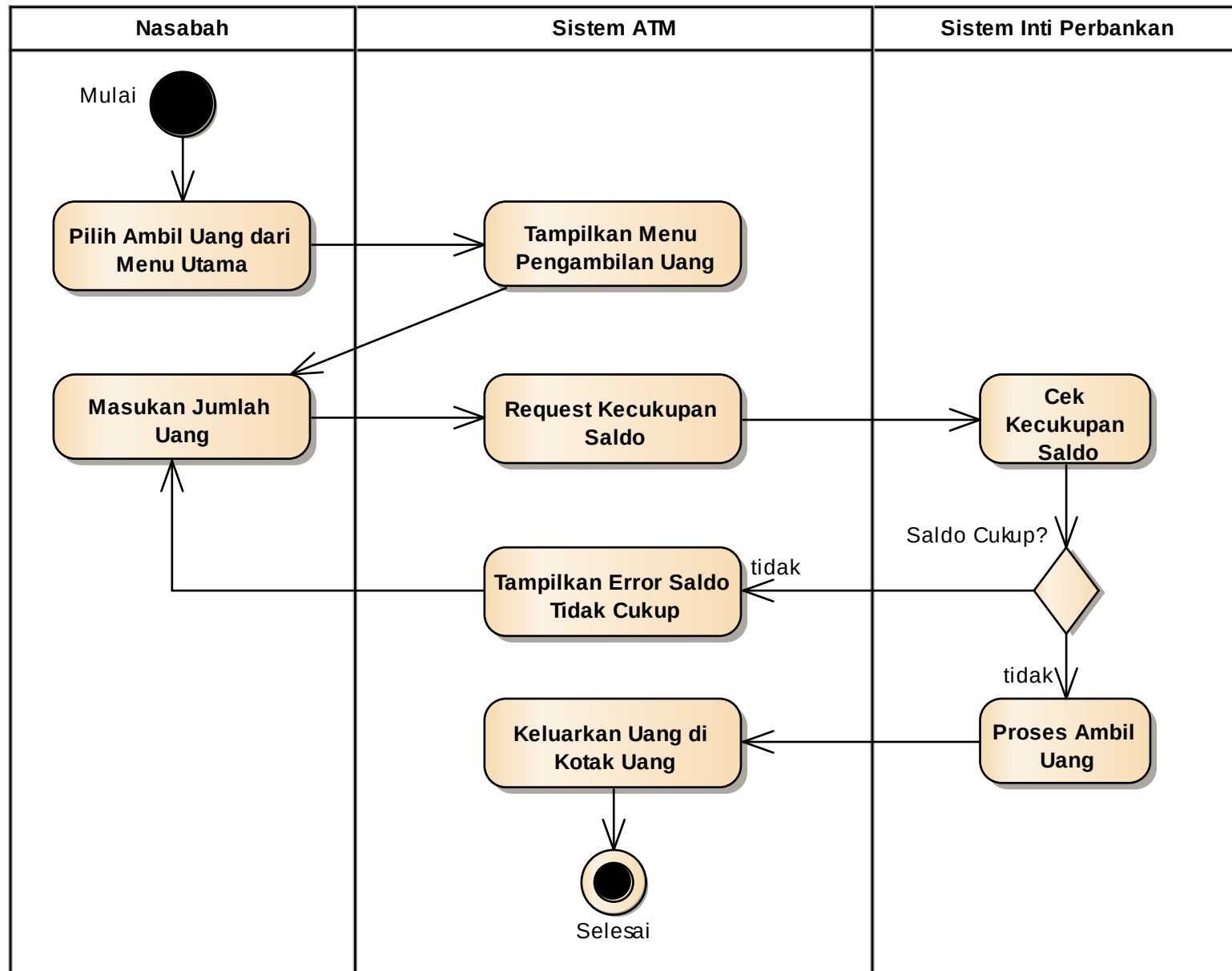
Activity Diagram: Memasukkan PIN



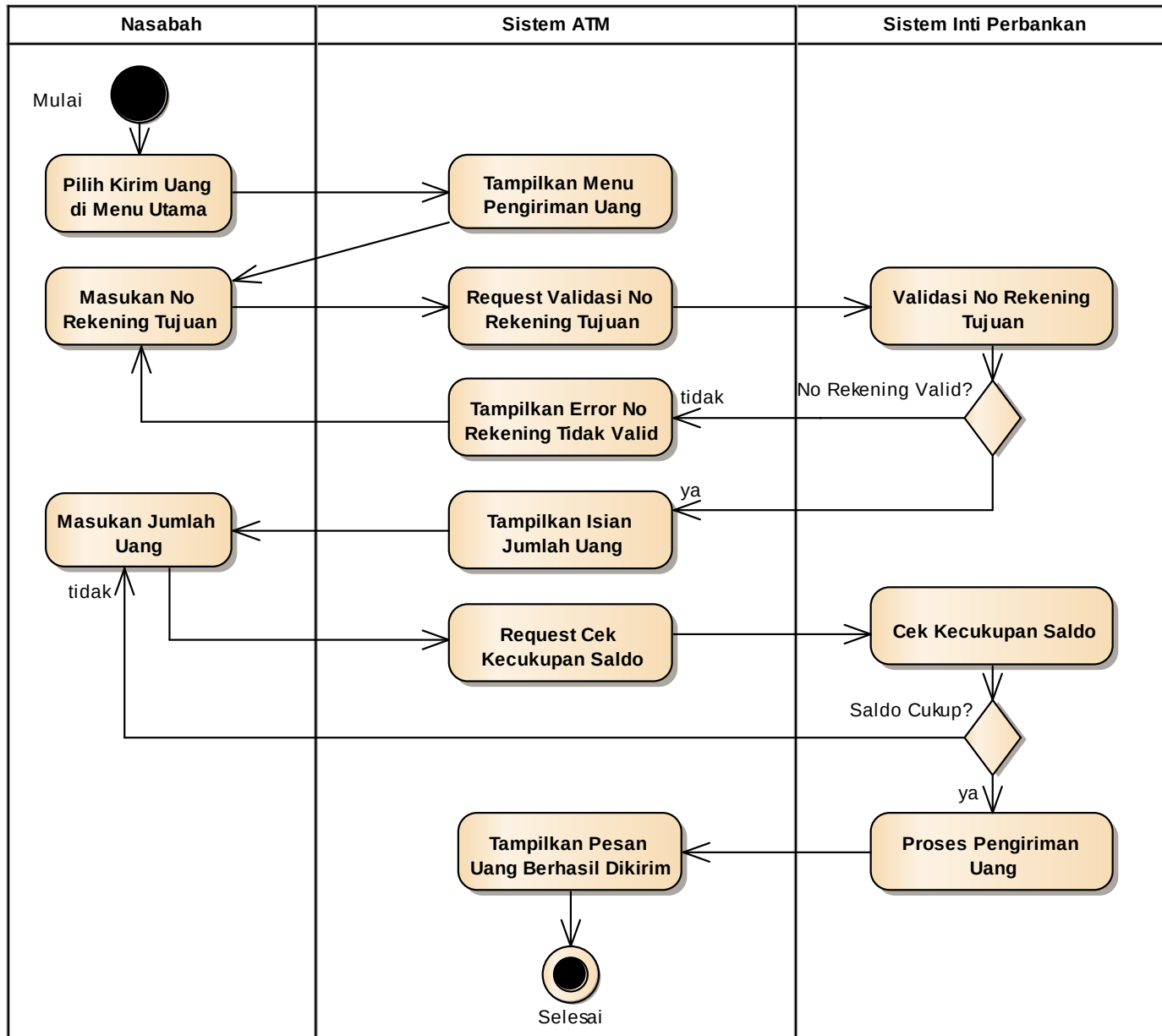
Activity Diagram: Mengecek Saldo



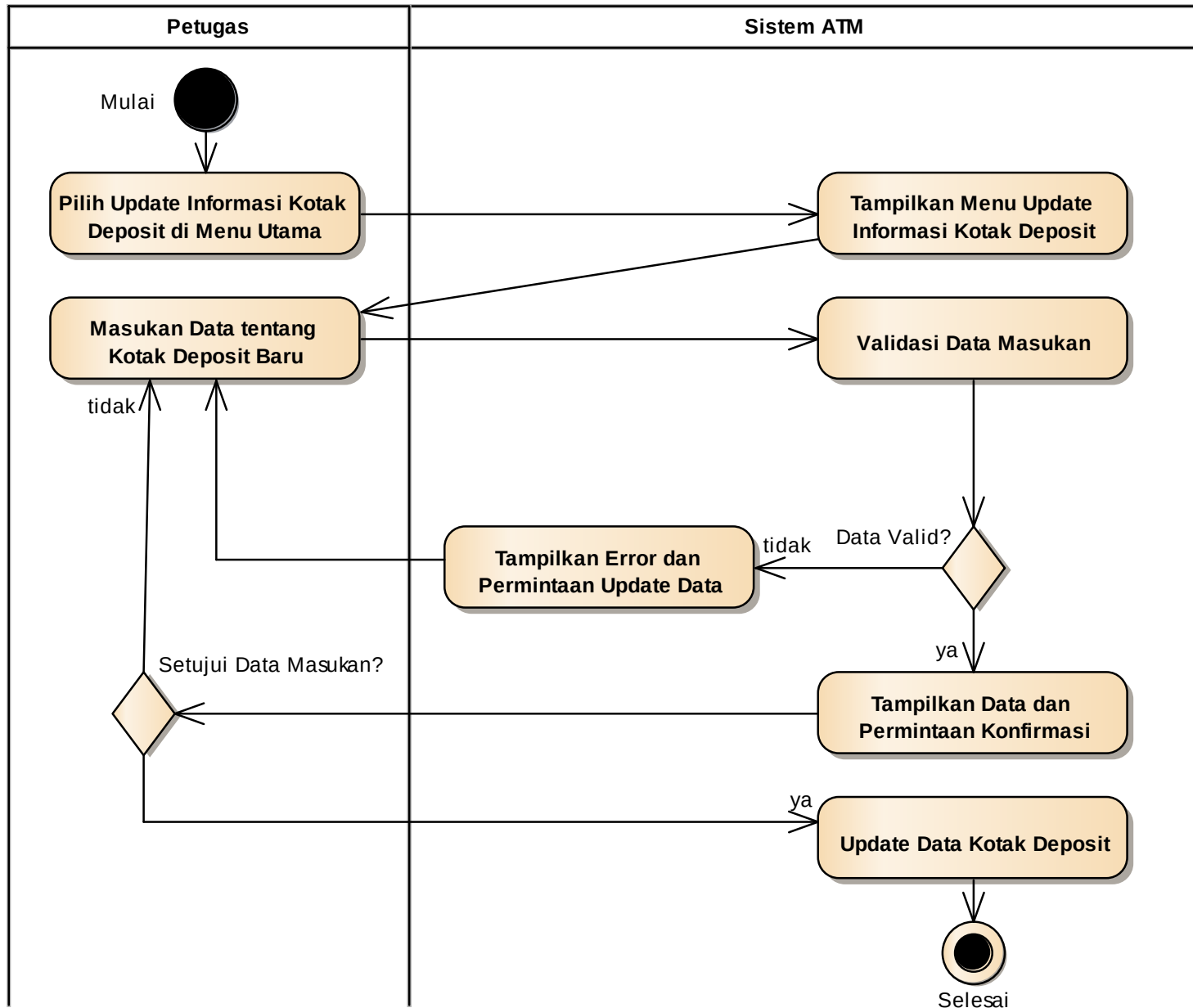
Activity Diagram: Mengambil Uang



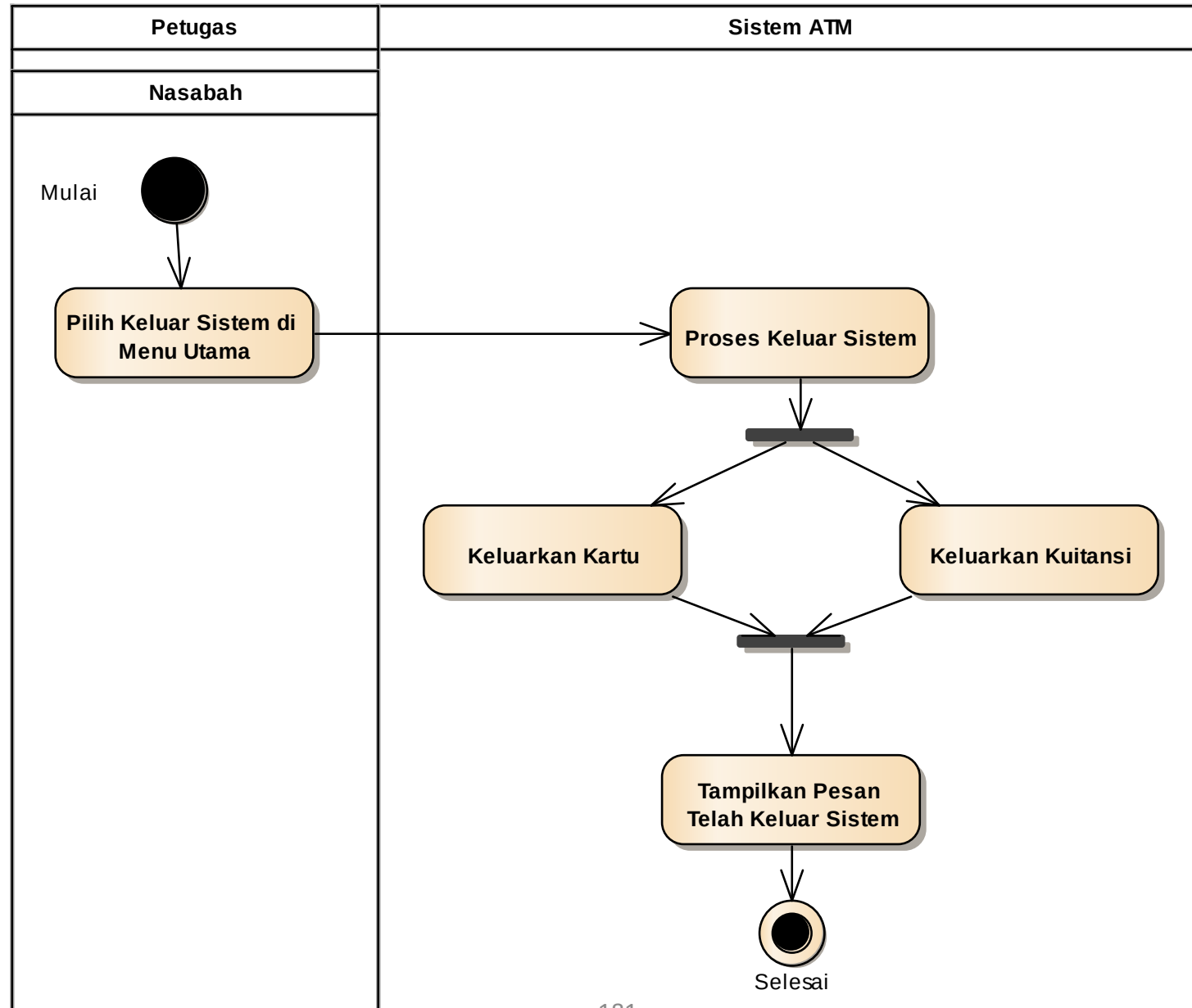
Activity Diagram: Mengirim Uang



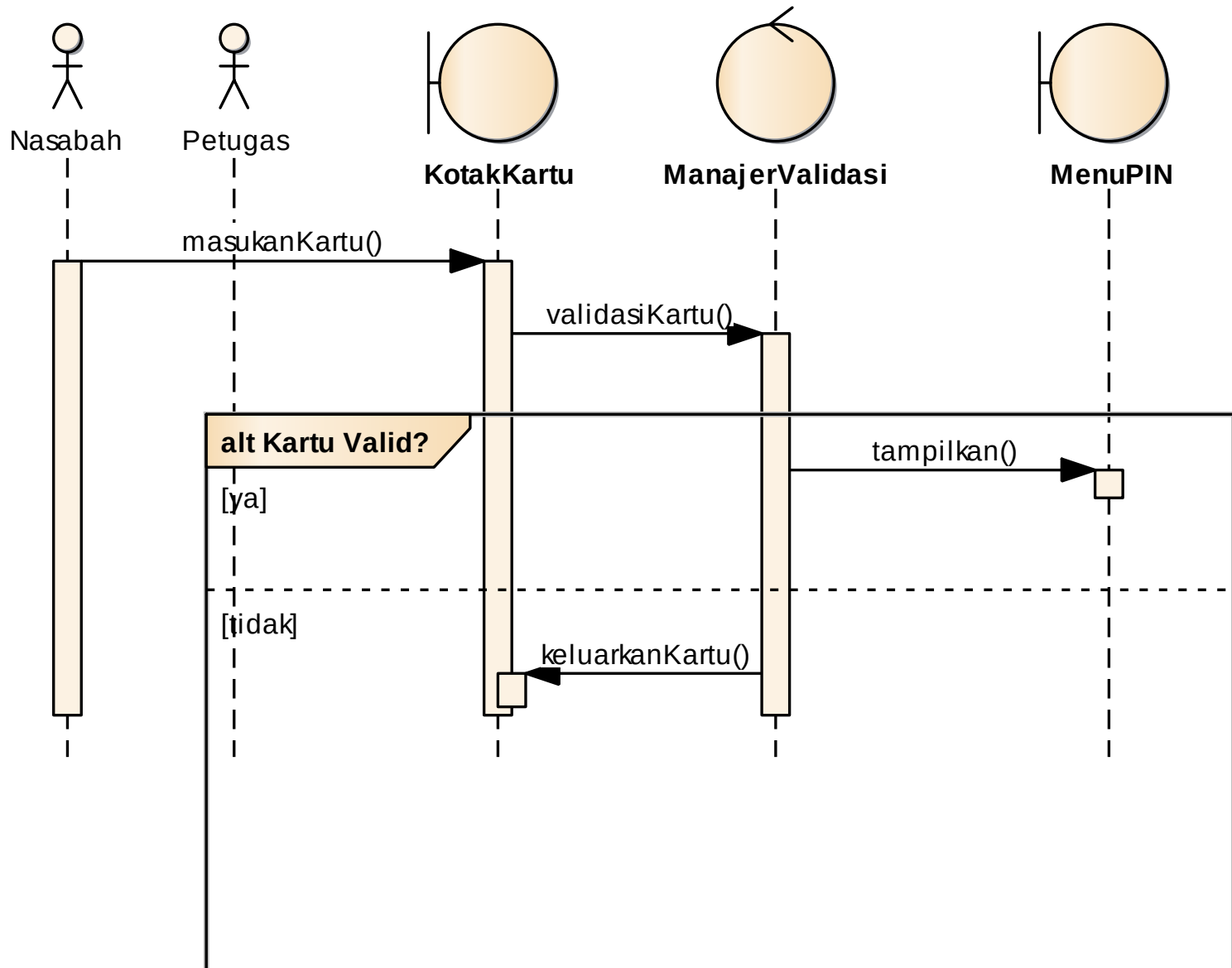
Activity Diagram: Mengupdate Informasi Kotak Deposit



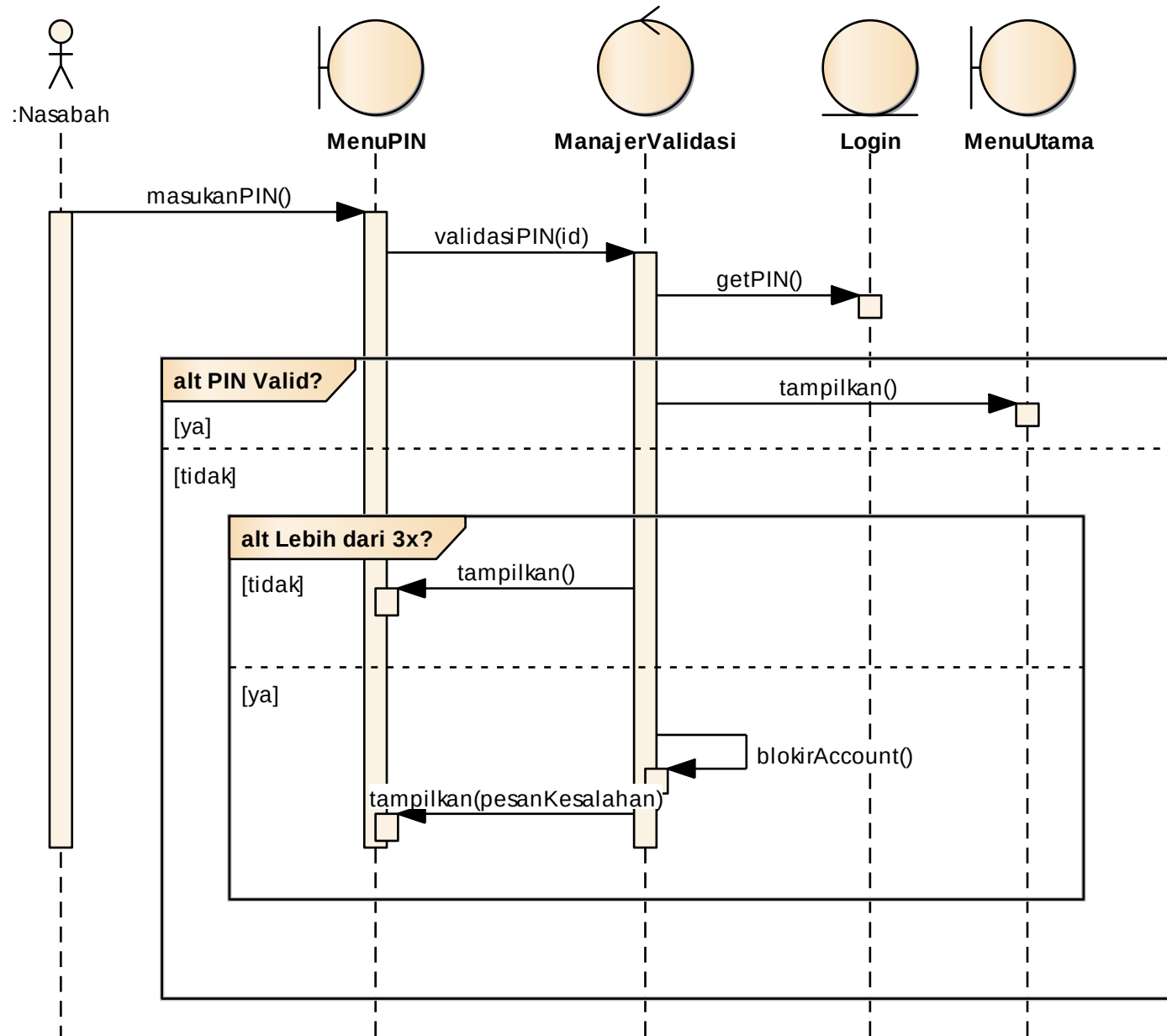
Activity Diagram: Keluar Sistem



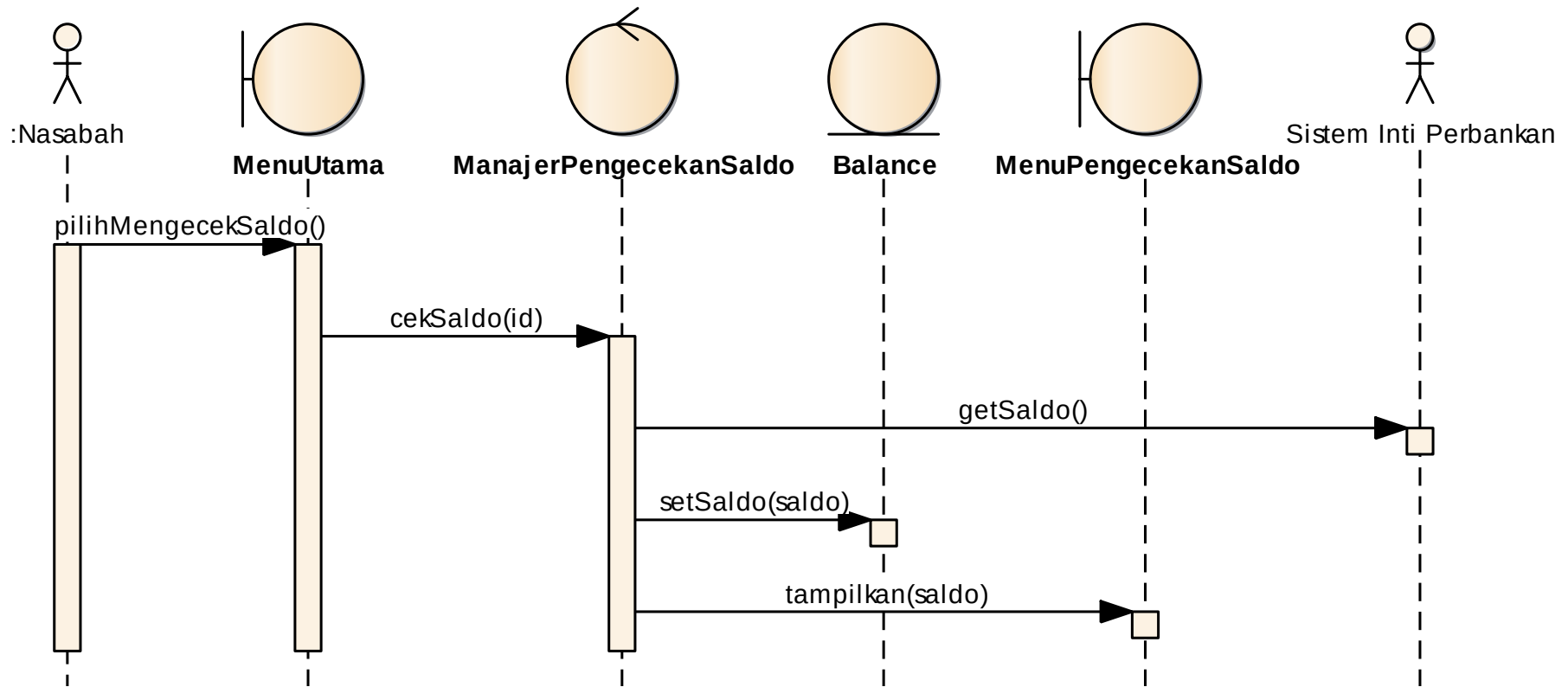
Sequence Diagram: Memasukkan Kartu



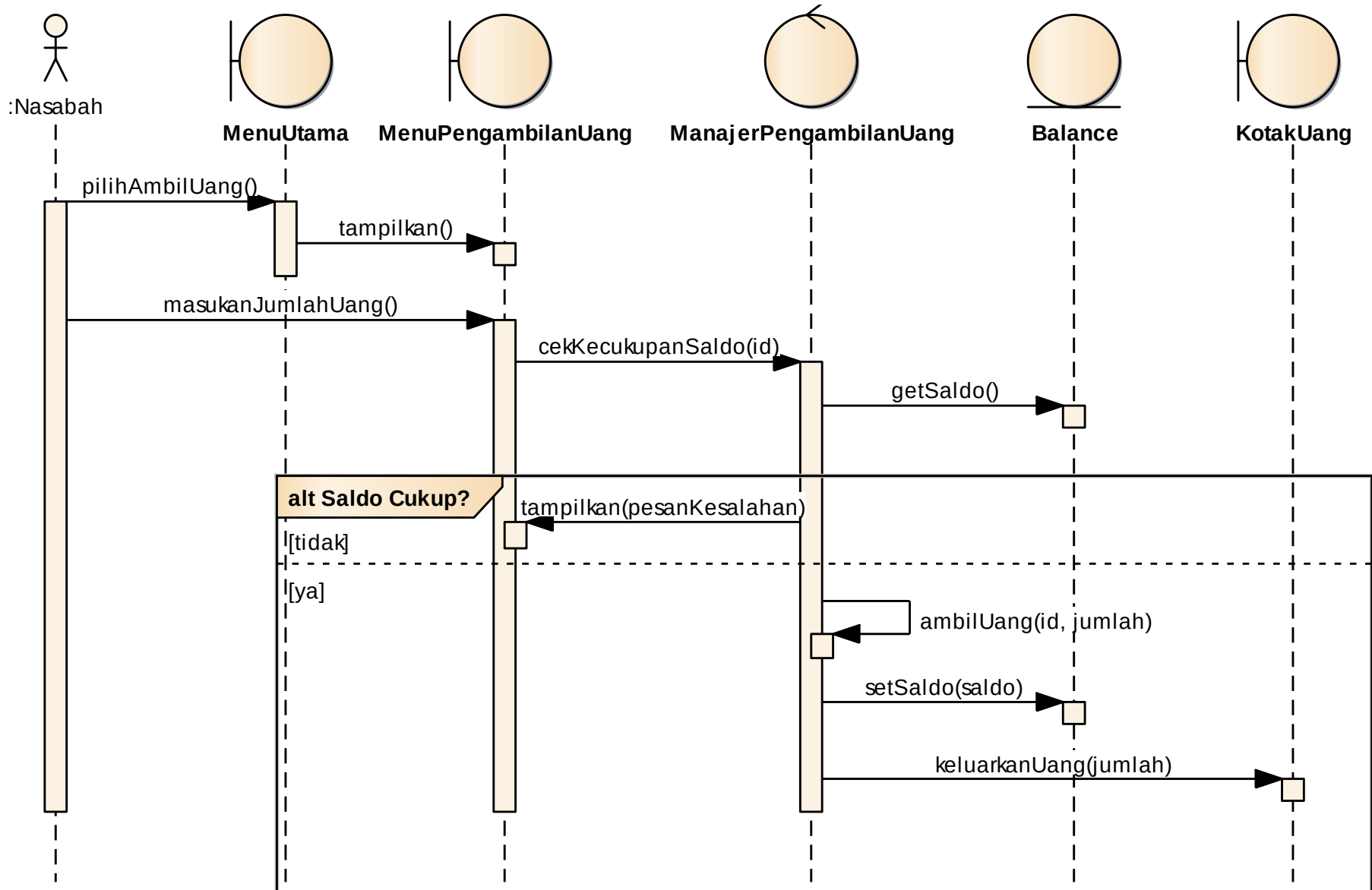
Sequence Diagram: Memasukkan PIN



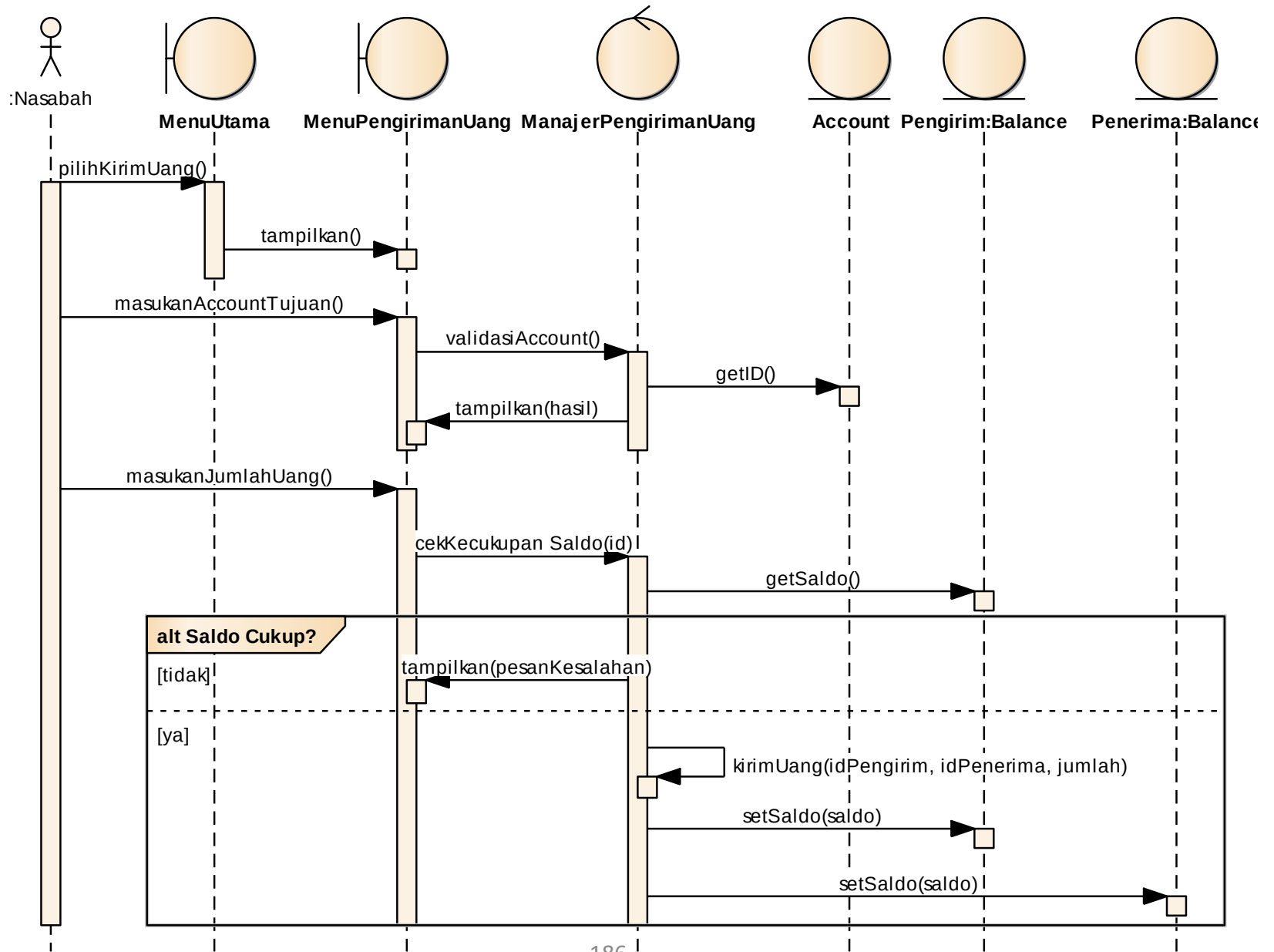
Sequence Diagram: Mengecek Saldo



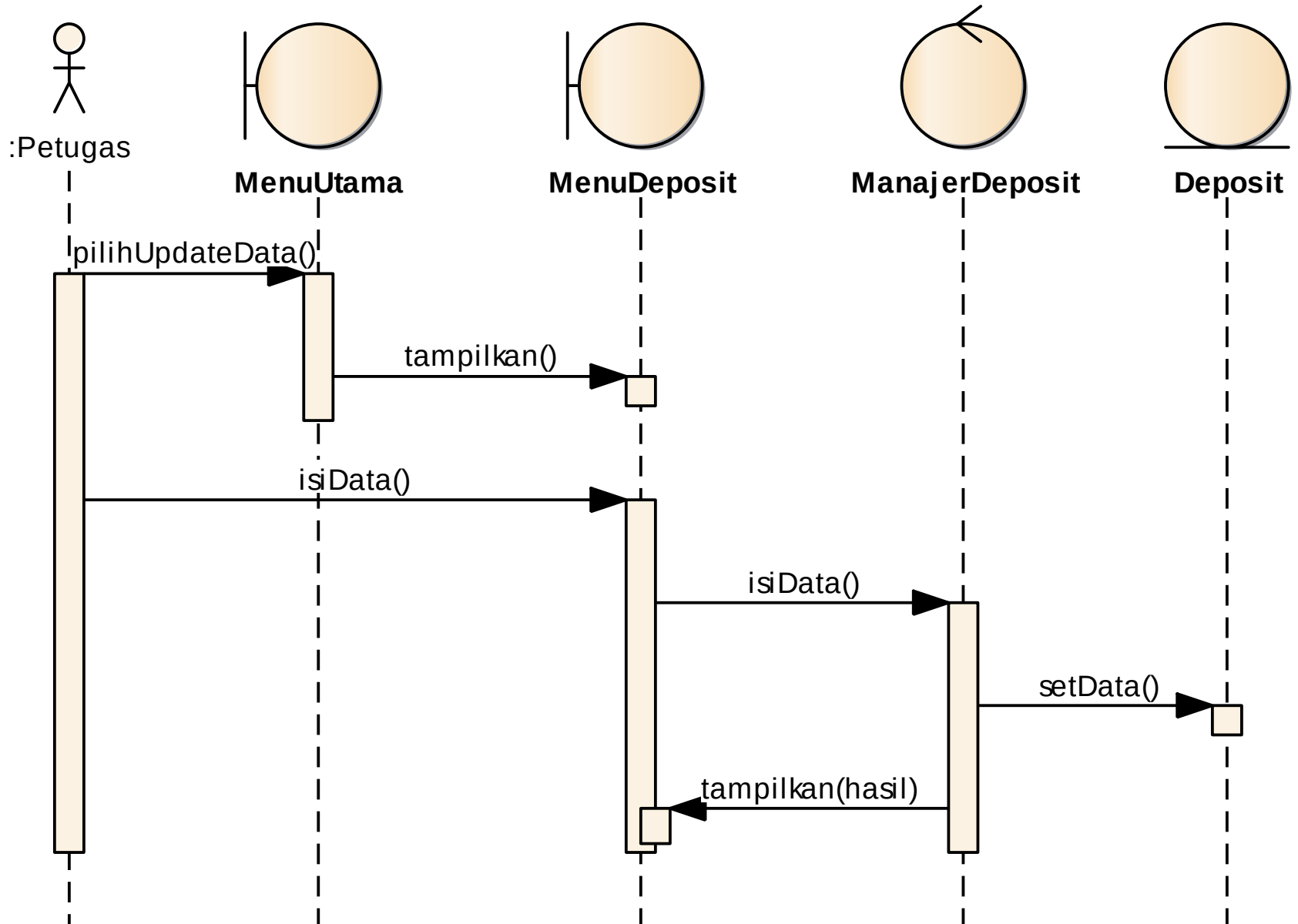
Sequence Diagram: Mengambil Uang



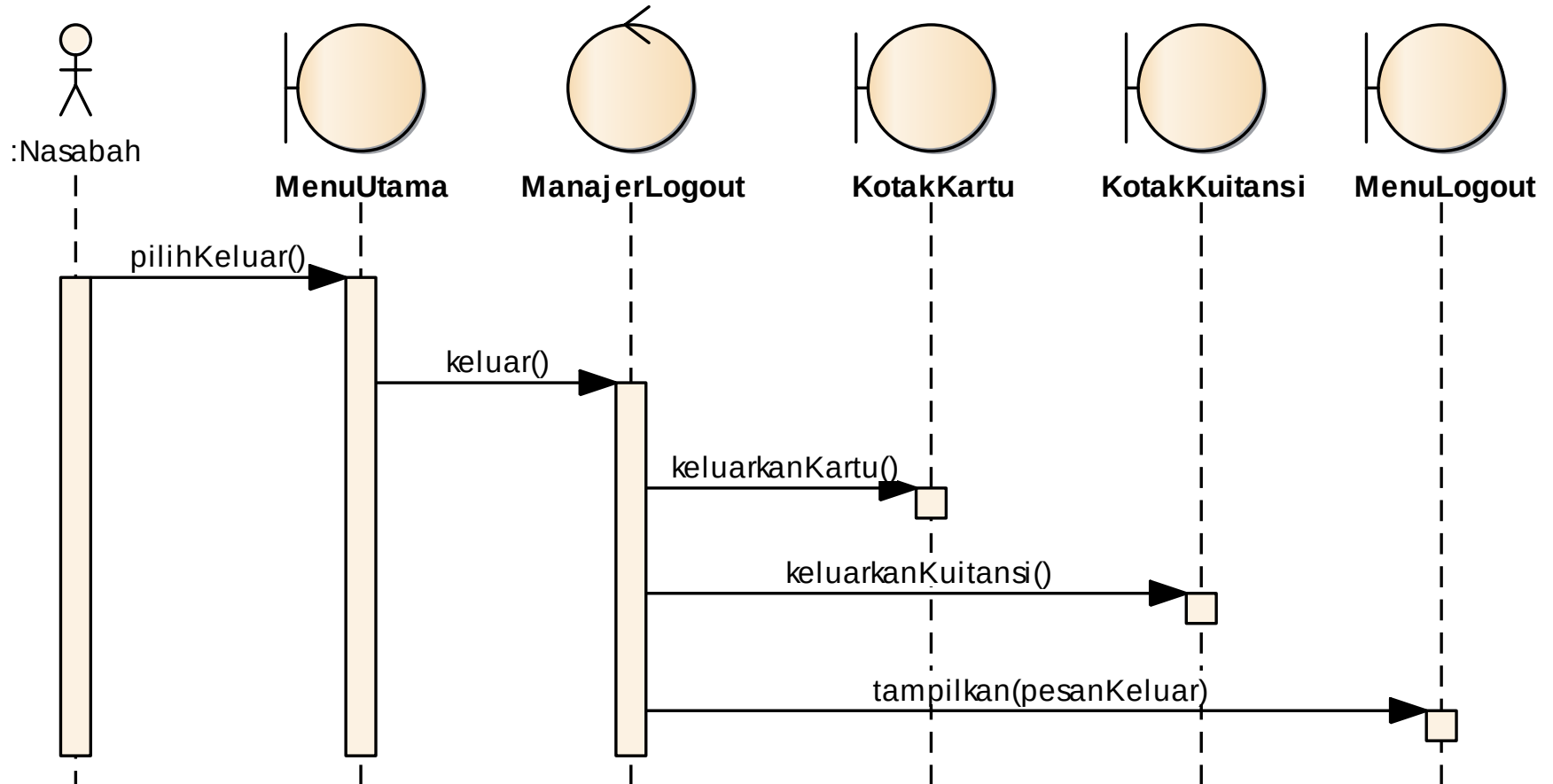
Sequence Diagram: Mengirim Uang



Sequence Diagram: Mengupdate Informasi Kotak Deposit

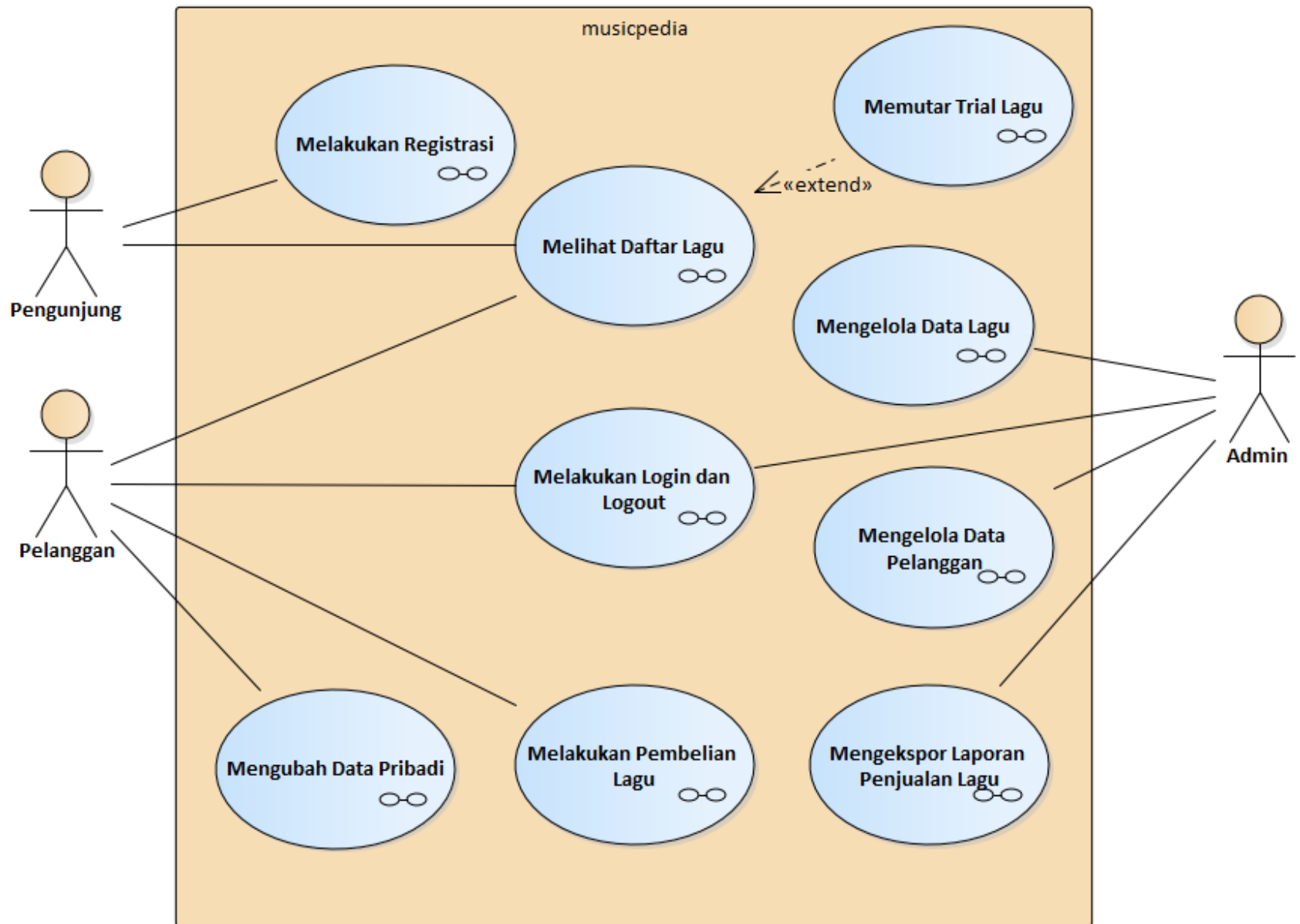


Sequence Diagram: Keluar Sistem

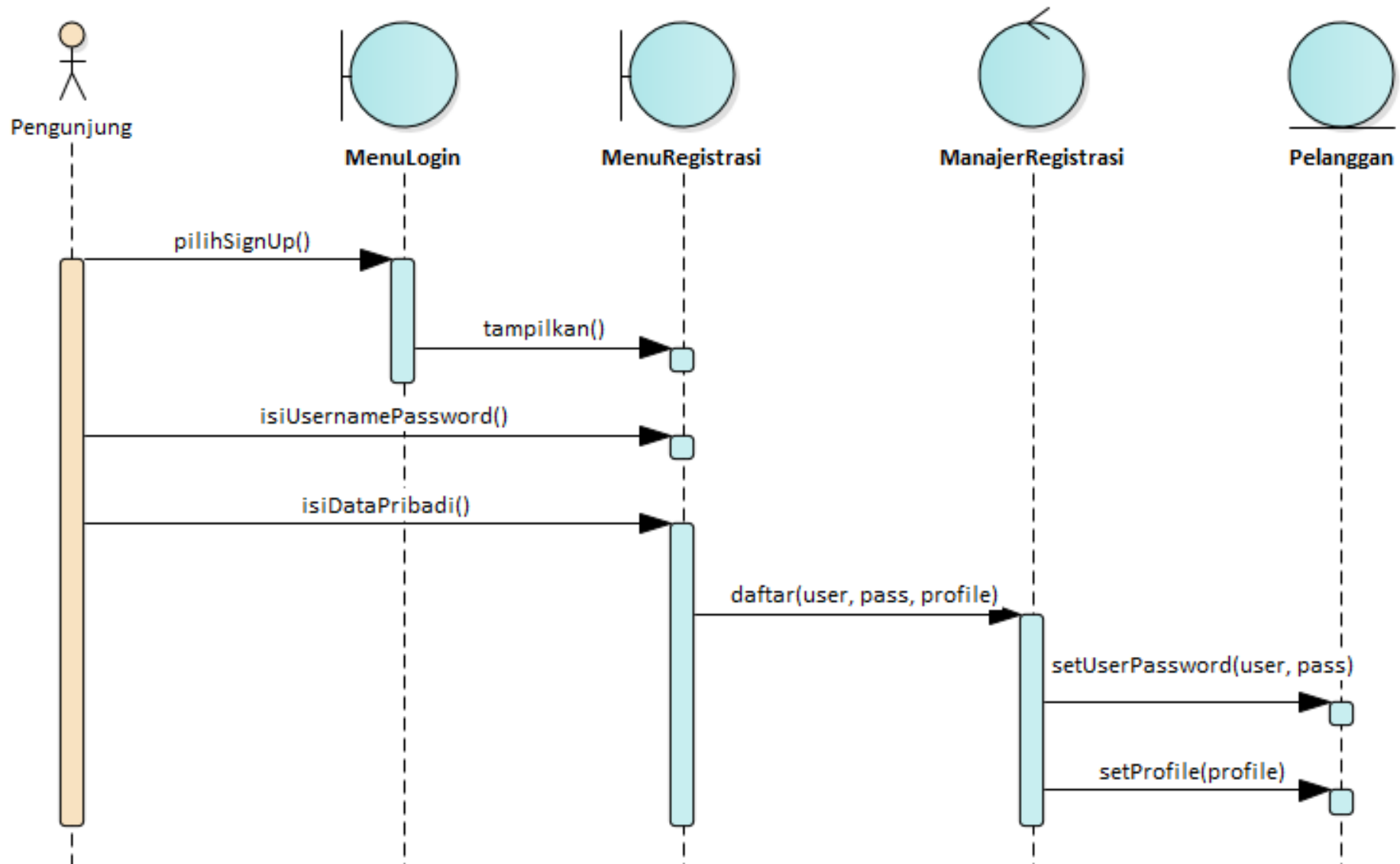


Studi Kasus: Sequence Diagram MusicPedia

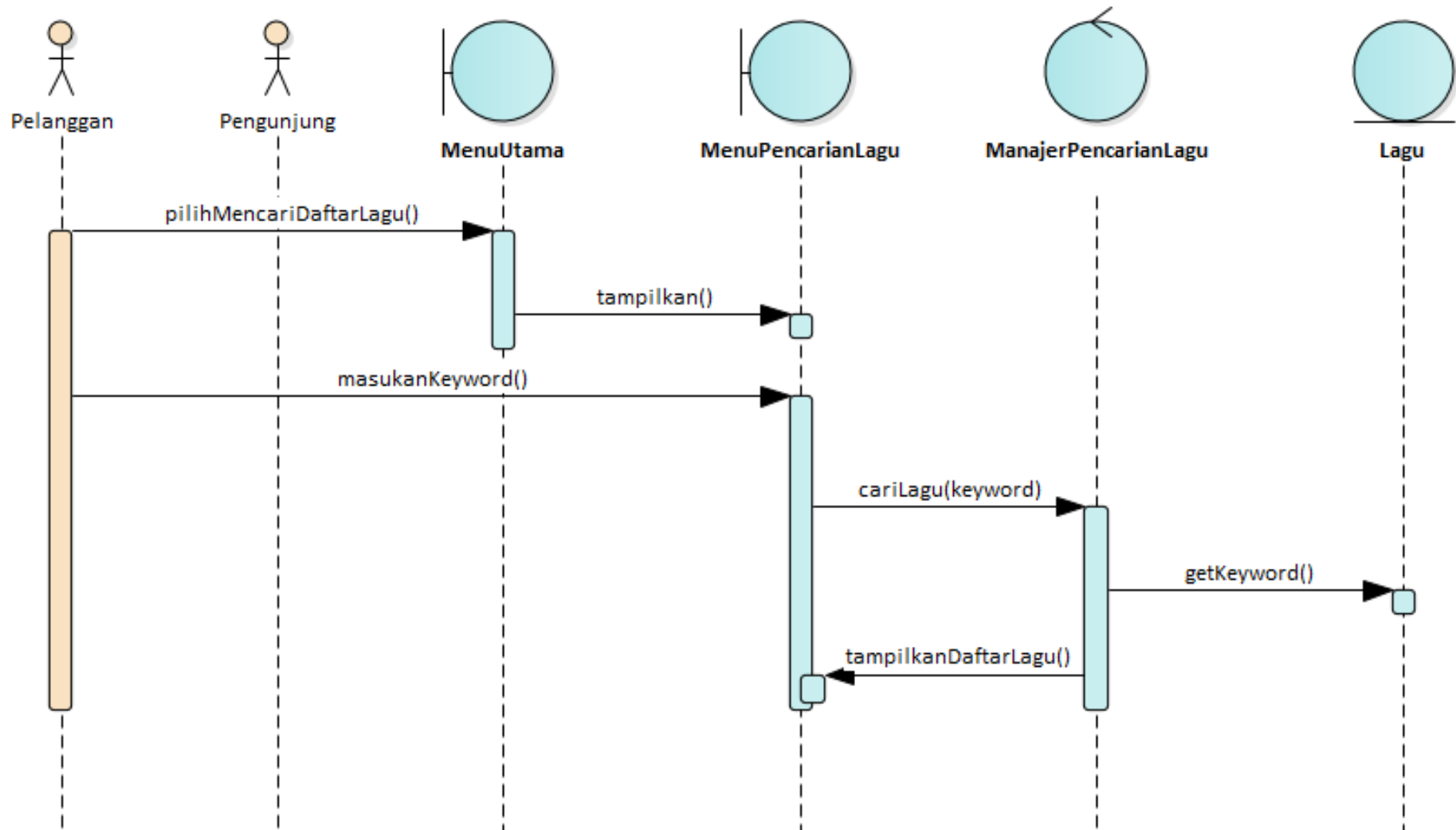
Use Case Diagram MusicPedia



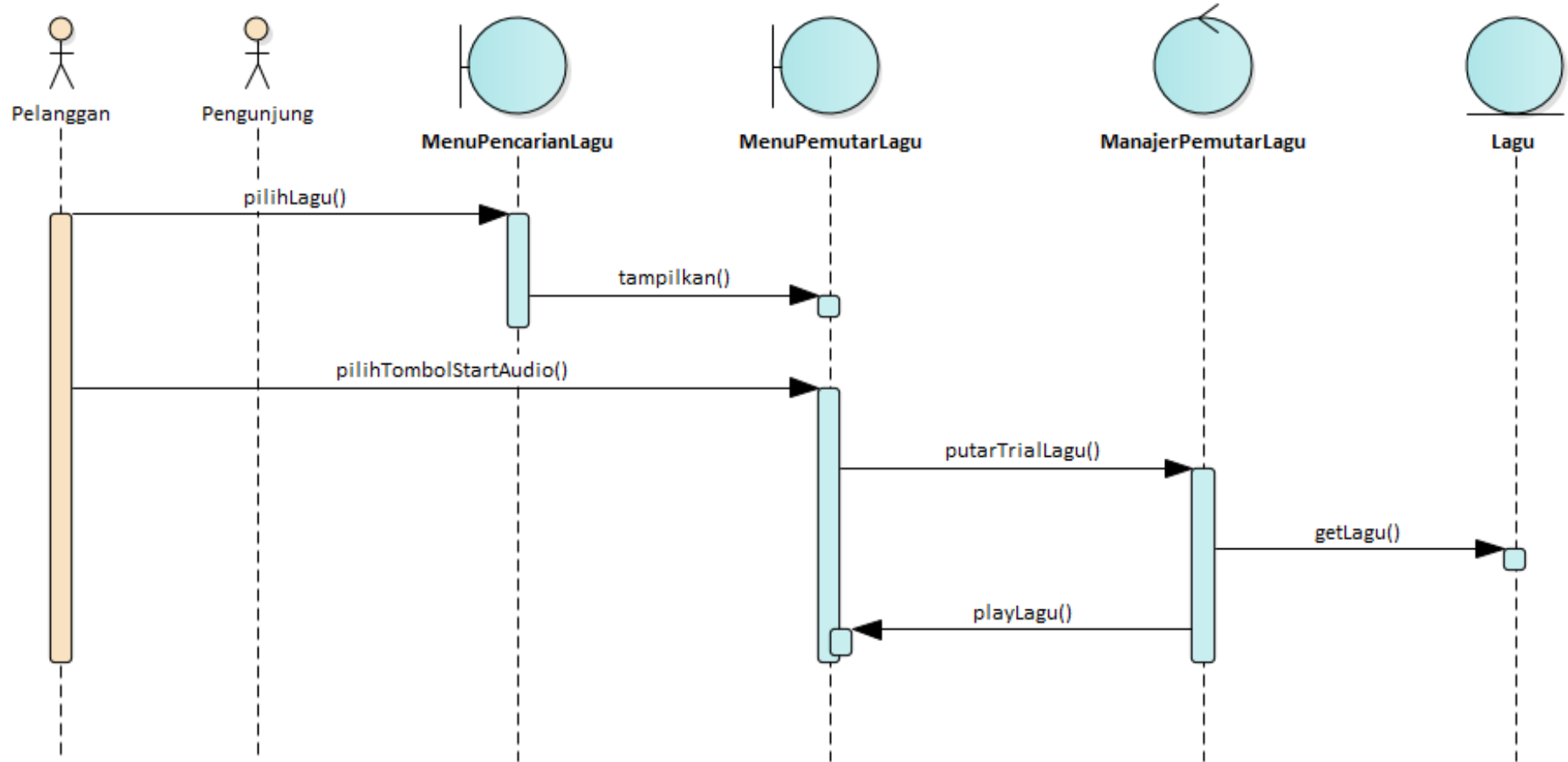
Sequence Diagram: Melakukan Registrasi



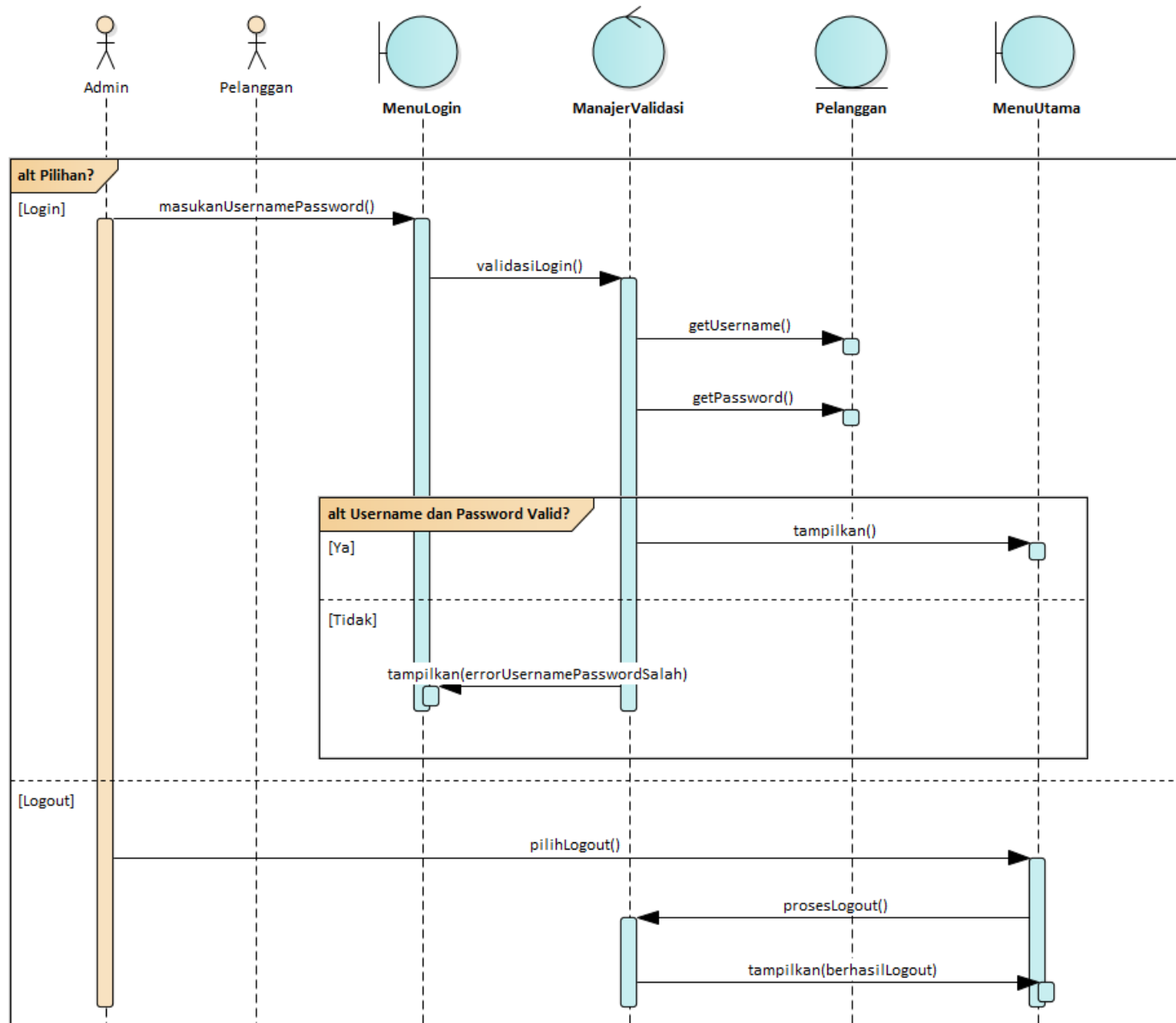
Sequence Diagram: Melihat Daftar Lagu



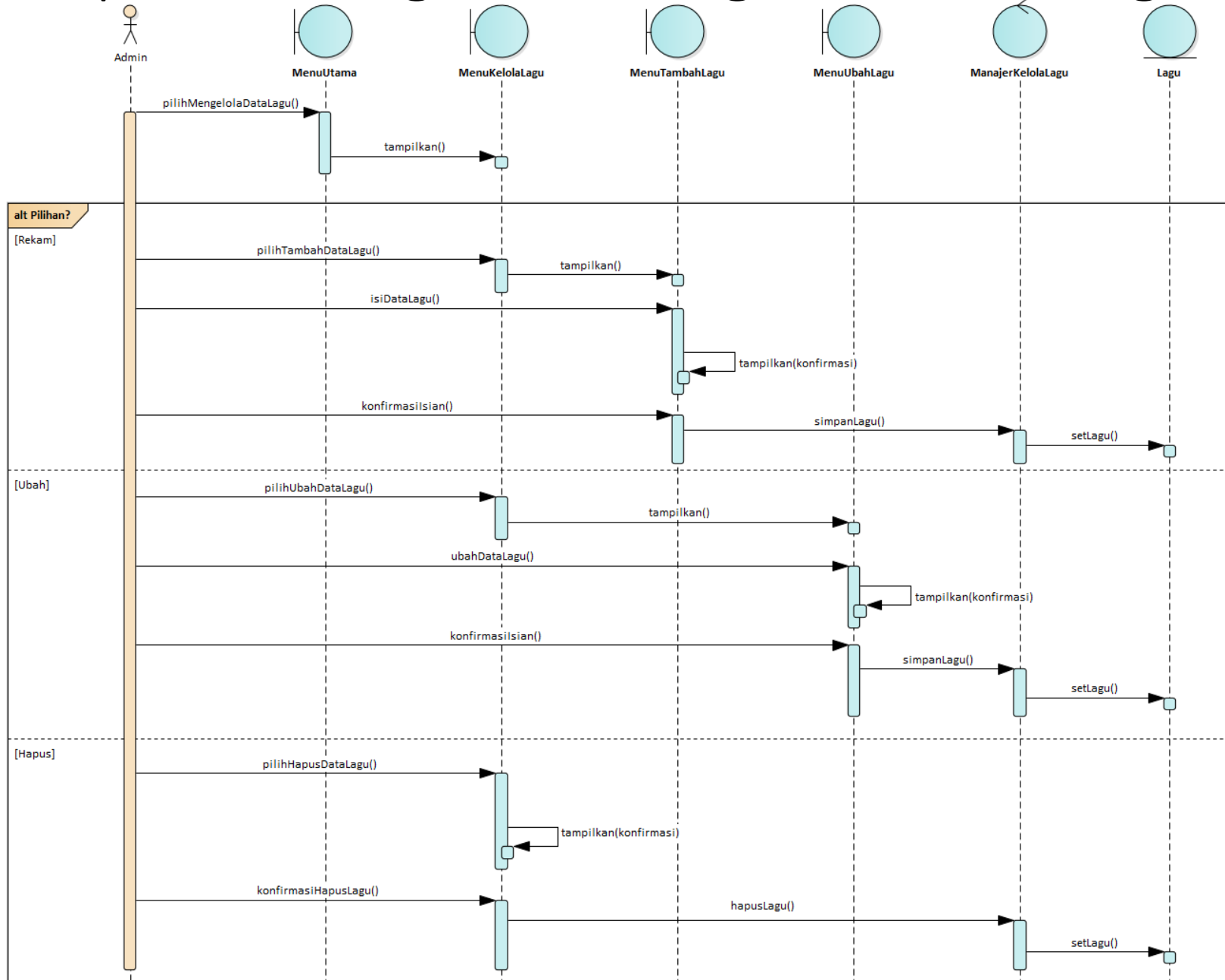
Sequence Diagram: Memutar Trial Lagu



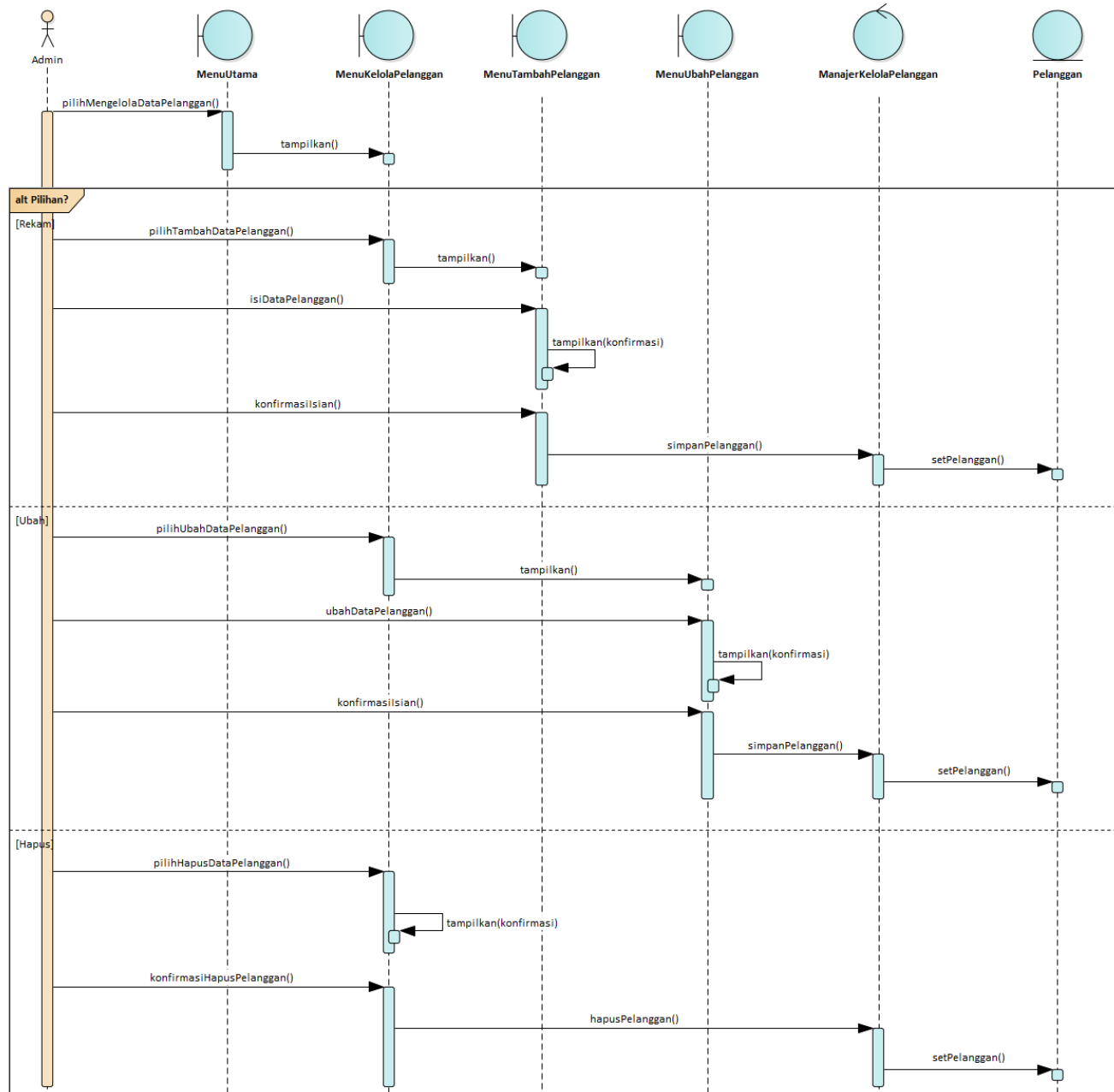
Sequence Diagram: Melakukan Login dan Logout



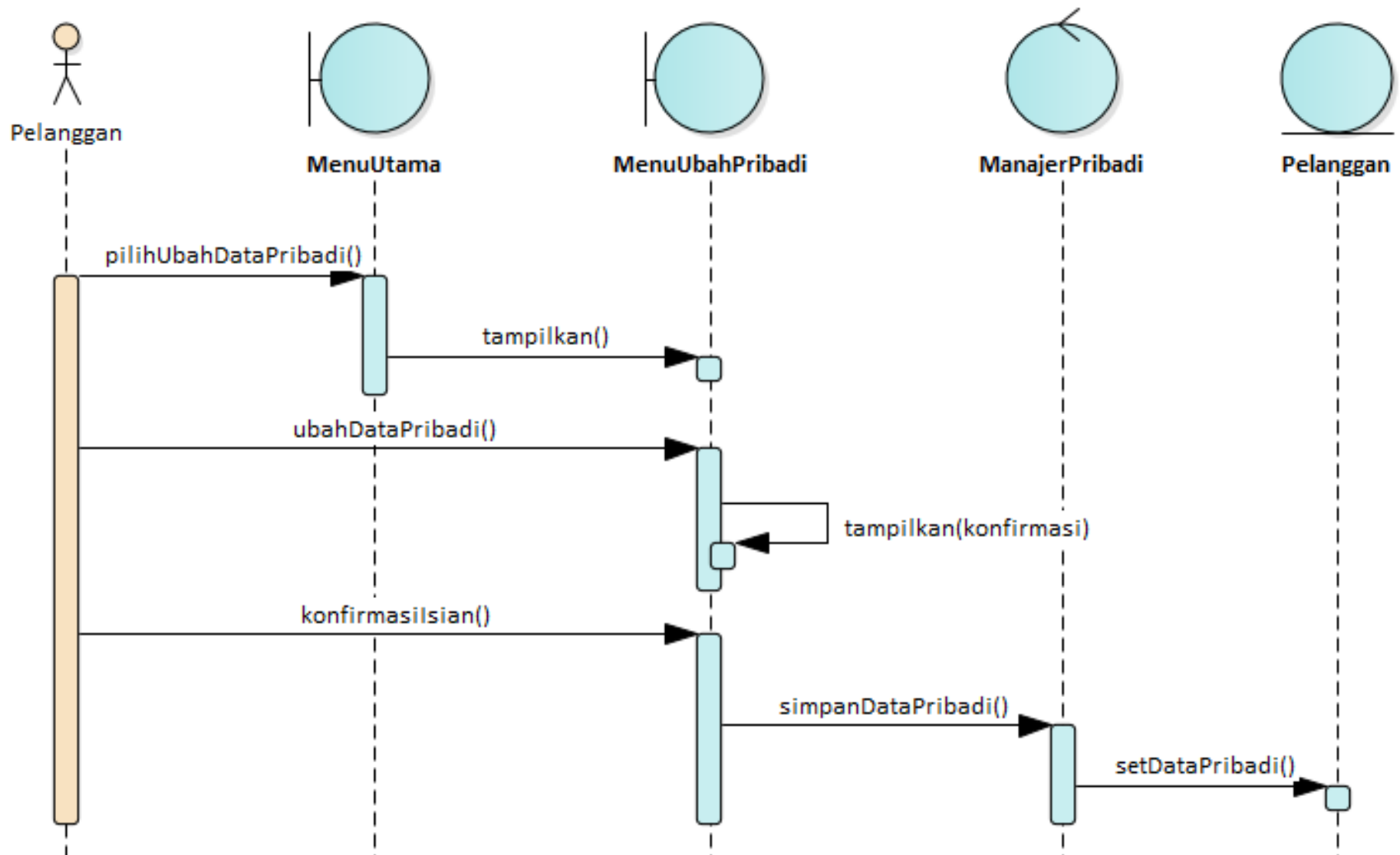
Sequence Diagram: Mengelola Data Lagu



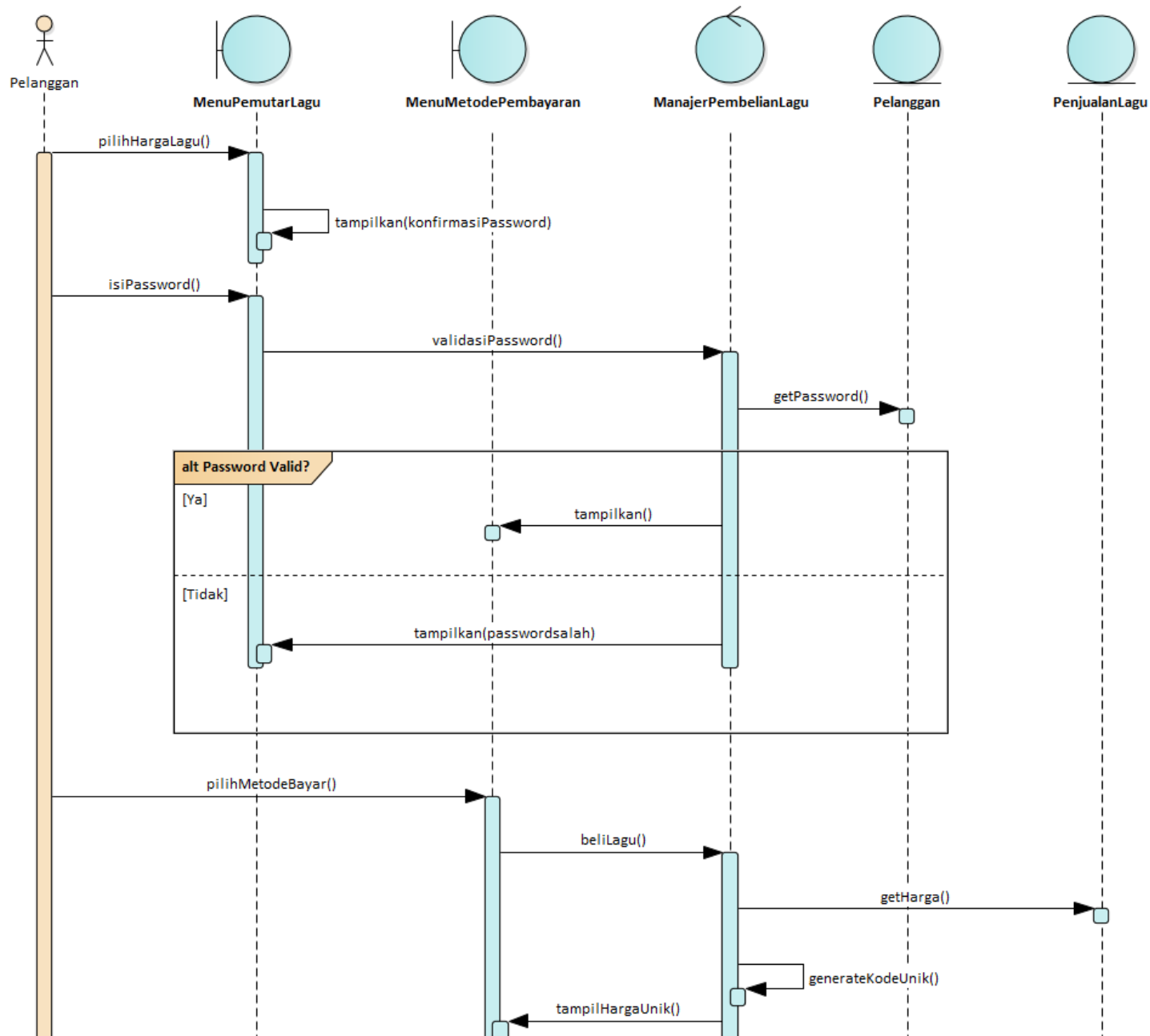
Sequence Diagram: Mengelola Data Pelanggan



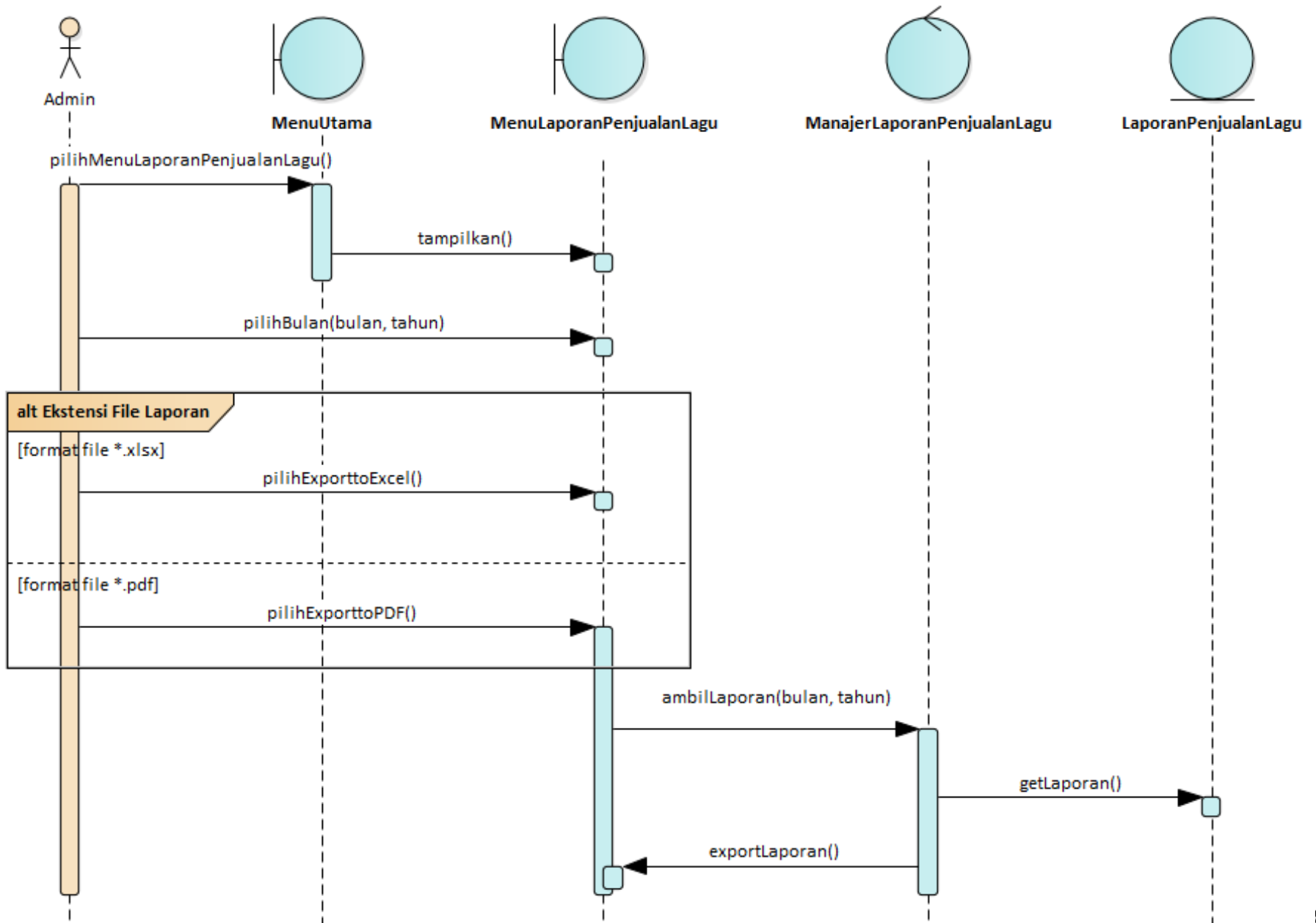
Sequence Diagram: Mengubah Data Pribadi



Sequence Diagram: Melakukan Pembelian Lagu



Sequence Diagram: Mengekspor Laporan Penjualan



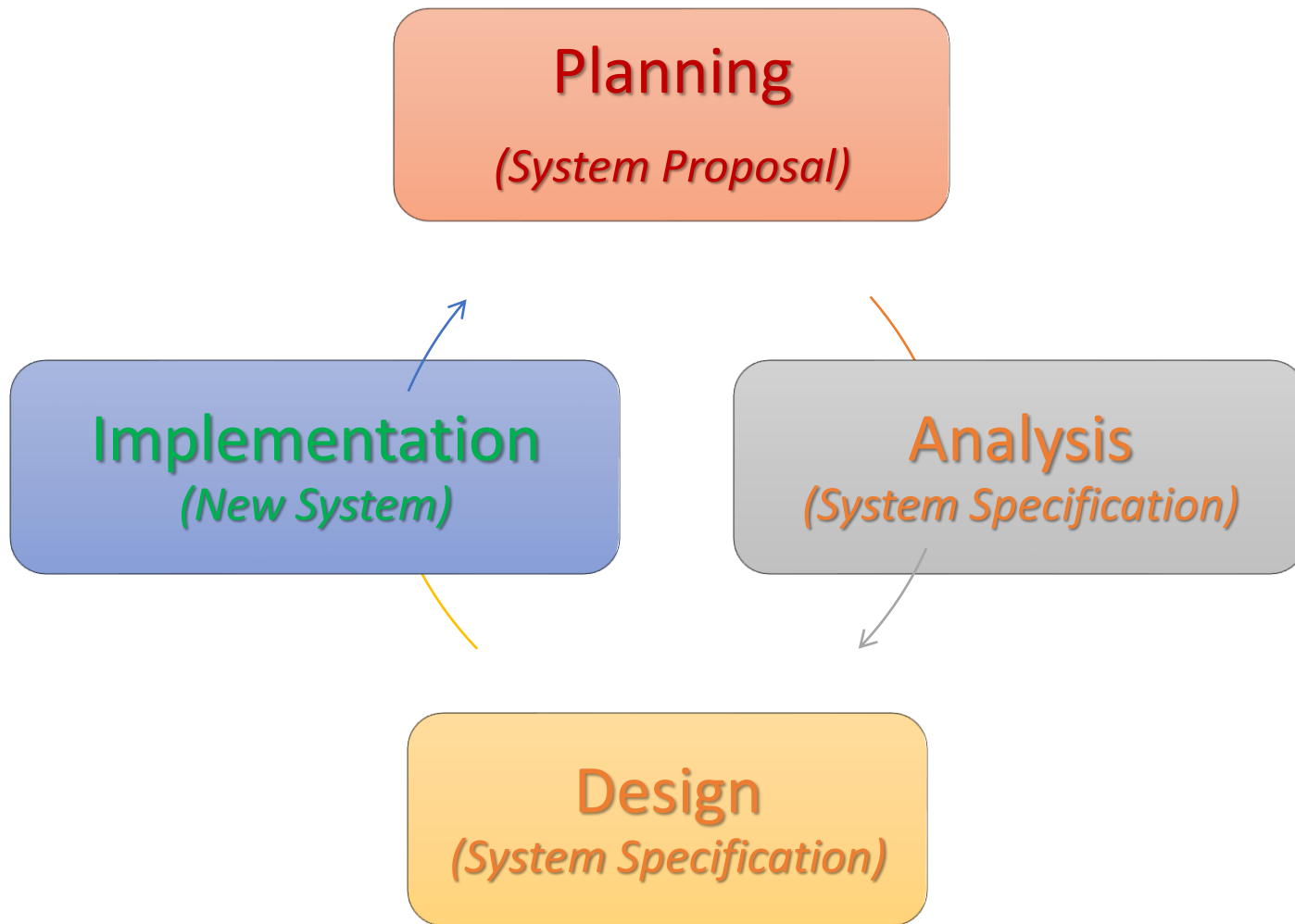
Catatan Pola Kesalahan

- Gunakan pola Subject-Verb-Object (**S-V-O**) untuk use case diagram (Actor – Use Case) dan Activity Diagram (Partition – Action)
- Use Case Diagram adalah **apa yang dilakukan Actor di sistem**, bukan apa yang dilakukan oleh sistem
- Pada Sequence Diagram **perhatikan transaksi yang harusnya datang dari actor**, dan bukan dari object lain
- Naming untuk **object dan class adalah kata benda** (noun), untuk message di Sequence Diagram (method) adalah kata kerja
- Object lifeline dan message di Sequence Diagram dan nama class di Class Diagram, **tidak boleh menggunakan spasi**, karena akan jadi Class dan Method di kode program
- **Actor (manusia)** akan mengirim **message hanya ke Boundary Class**, tidak ke Control atau Entity Class. Sedangkan **Actor (System)** akan mengirim message ke **Control**
- **Entity Class** bukan consumer, jadi **tidak pernah mengirim message** ke Boundary atau Control Class
- Boundary class akan menjadi UI Design, entity class akan menjadi Data Model
- Class diagram tidak menunjukkan alur, tapi menunjukkan **struktur dan komposisi dari sistem** yg kita bangun



3. Systems Design

Siklus Pengembangan Software



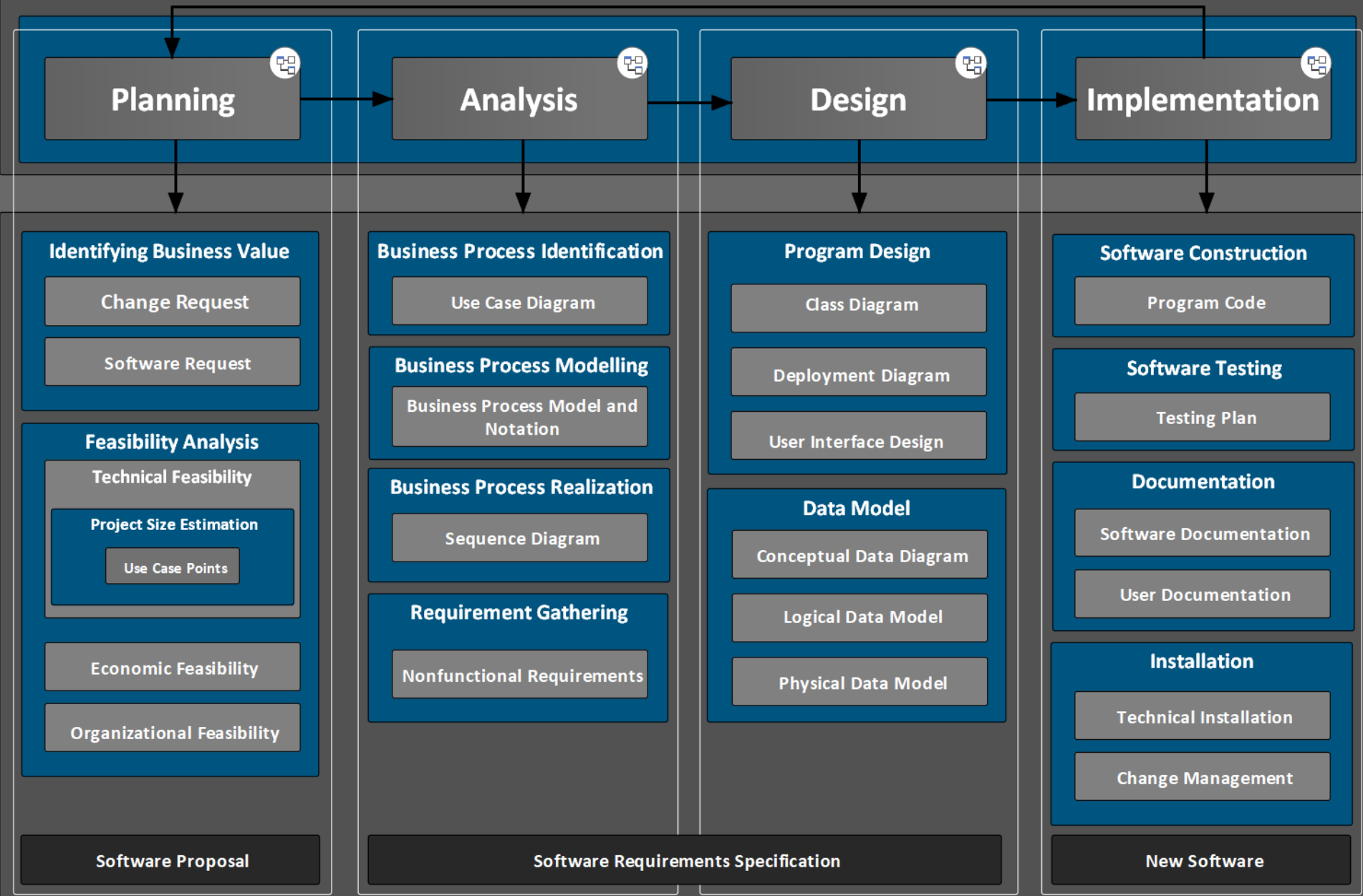
(Tilley, 2012)

(Dennis, 2016)

(Valacich, 2017)

Application Development Governance

Software Development Life Cycle





3.1 Perancangan Class Diagram

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

Class

MenuPIN

```
public class MenuPIN{
```

```
.....
```

```
}
```

Attributes

- **Visibility** of attributes
 - **+** **Public**: not hidden from any object
 - **#** **Protected**: hidden from all but immediate subclasses
 - **-** **Private**: hidden from all other classes
- **Default is private**

Class with Attribute and Method

ManajerValidasi	
-	m_Login: Login
+	blokirKartu(): void
+	validasiKartu(): int
+	validasiPIN(): int

```
public class ManajerValidasi {  
  
    private Login m_Login;  
  
    public int validasiKartu(){  
        return 0;  
    }  
  
    public int validasiPIN(){  
        return 0;  
    }  
  
    public void blokirKartu(){  
  
    }  
  
}
```

Relationships

1. Generalization

- “Is-A” relationship
- Enables inheritance of attributes & operations
- Subclasses and superclasses

2. Aggregation

- “Has-A” relationship
- Relates parts to wholes
- Uses decomposition

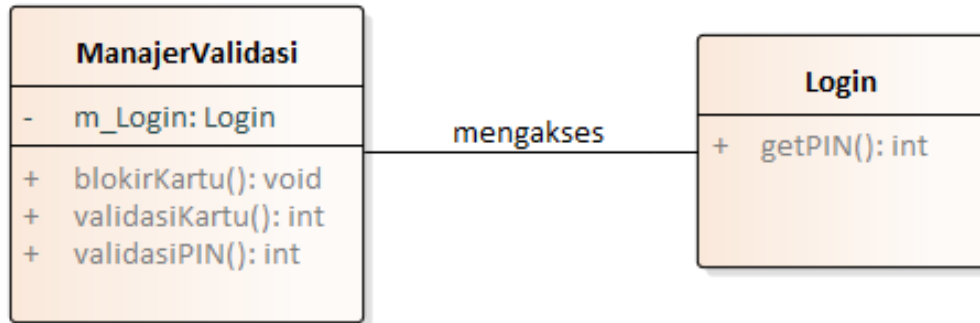
3. Composition

- “Has-A” relationship
- Similar but stronger than Aggregation

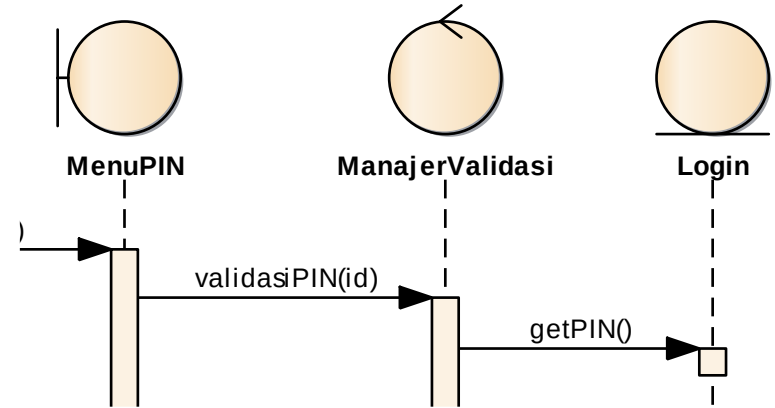
4. Association

- Relationships that don't fit “Is-A” or “Has-A”
- Often a weaker form of “Has-A”
- Miscellaneous relationships between classes

Class Association

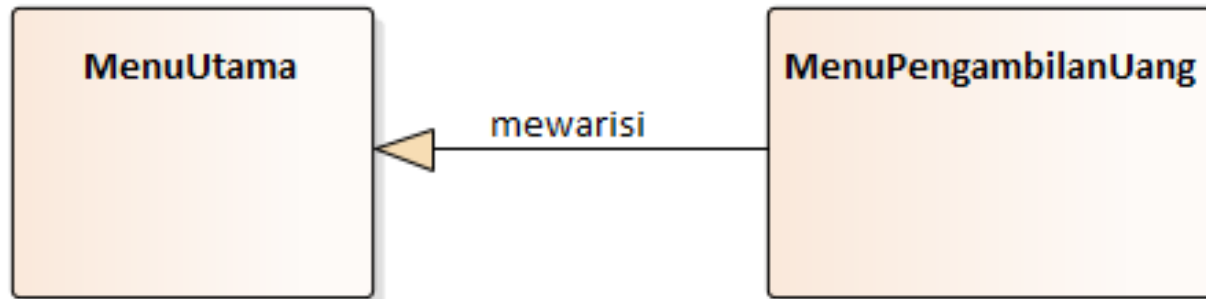


```
public class ManajerValidasi {  
    private Login m_Login;  
  
    public int validasiKartu(){  
        return 0;  
    }  
  
    public int validasiPIN(){  
        int pin = m_login.getPIN();  
        return 0;  
    }  
  
    public void blokirKartu(){  
  
    }  
}
```



```
public class Login {  
  
    public int getPIN(){  
        return 0;  
    }  
  
}
```

Class Inheritance



```
public class MenuUtama {
```

```
.....
```

```
}
```

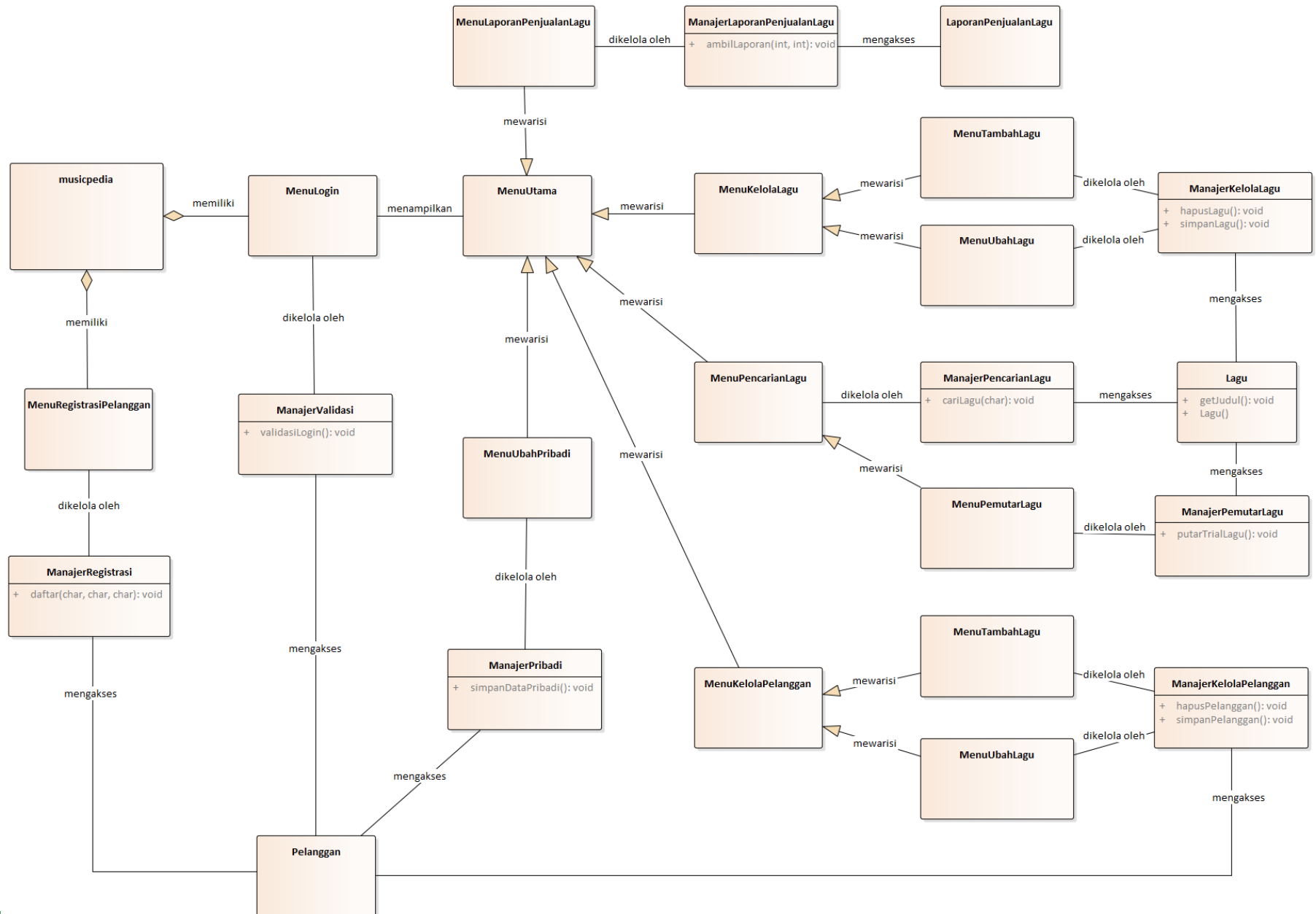
```
public class MenuPengambilanUang
    extends MenuUtama {
```

```
.....
```

```
}
```



Class Diagram: MusicPedia





3.2 Perancangan User Interface Design

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

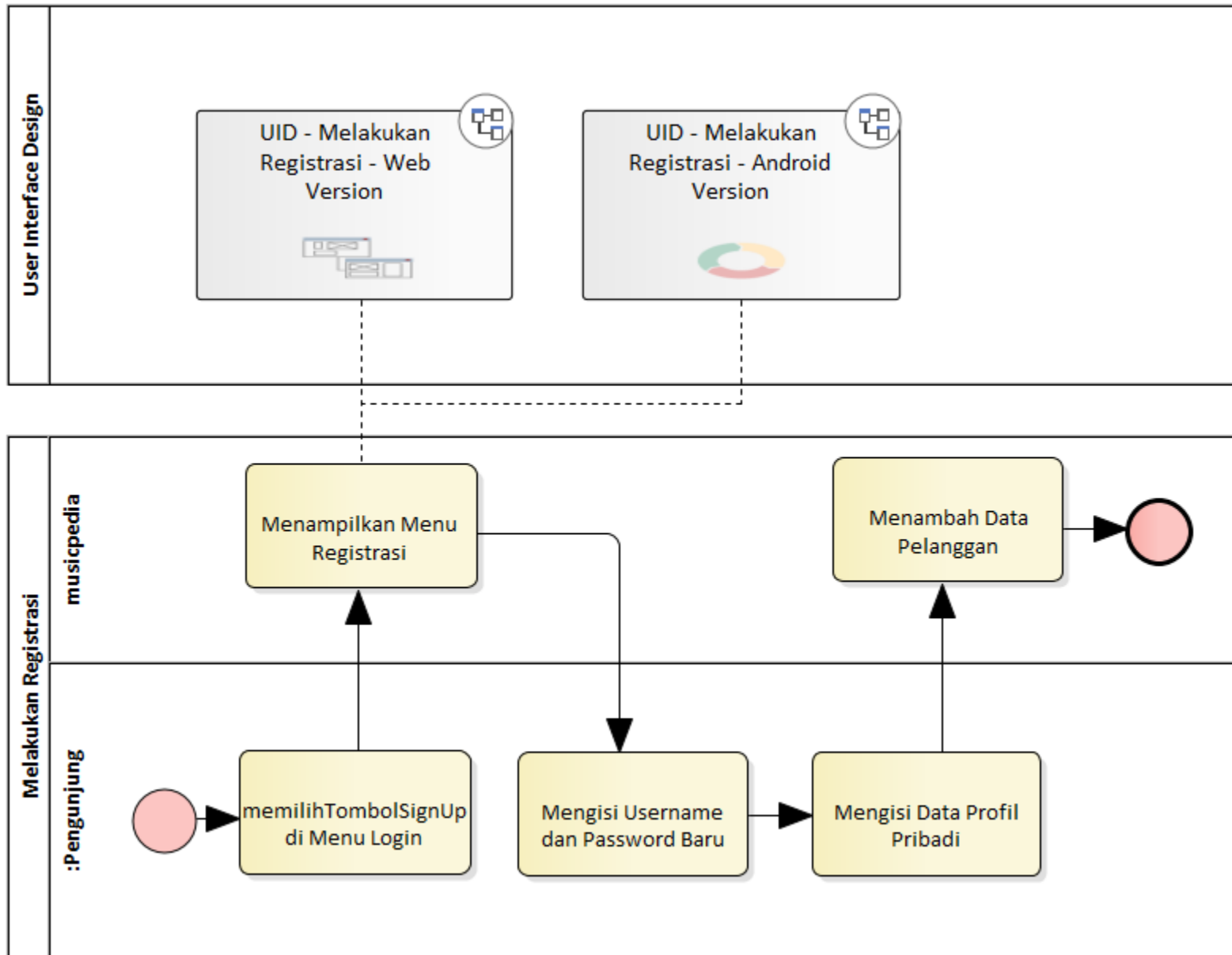
2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

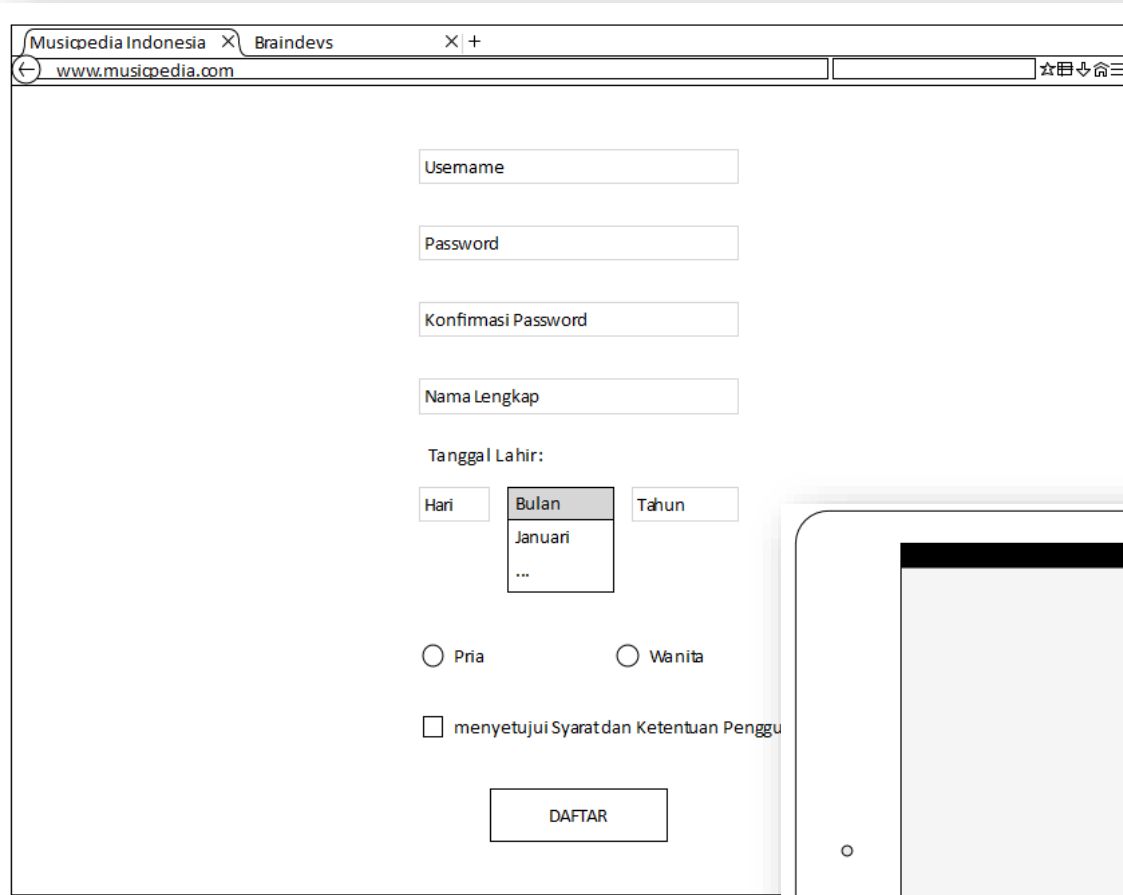
Philosophy and Principles

- Interface design is an **art**
- **Balance** between
 - Making the **interface useful** and
 - Presenting **too much information**
- **Principles** for User Interface Design:
 1. **Layout:** **Consistent** use of screen area
 2. **Content awareness:** Users **know where** they are
 3. **Aesthetics:** White space vs. functionality
 4. **User experience:** Ease of use vs. learning curve
 5. **Consistency:** User can predict **what will happen** for each action
 6. **Minimal user effort:** **Simple to use**, three click rule

BPMN Melakukan Registrasi



User Interface Design Melakukan Registrasi (versi Web dan versi Android)



A screenshot of a web browser showing a registration form on the website 'www.musicpedia.com'. The browser's address bar shows the URL. The form includes input fields for 'Ustream', 'Password', 'Konfirmasi Password', and 'Nama Lengkap'. Below these is a 'Tanggal Lahir:' section with three dropdown menus for 'Hari', 'Bulan', and 'Tahun'. The 'Bulan' dropdown is open, showing 'Januari' and '...'. There are radio buttons for 'Pria' and 'Wanita', and a checkbox for 'menyetujui Syarat dan Ketentuan Pengguna'. A 'DAFTAR' button is at the bottom.

Ustream

Password

Konfirmasi Password

Nama Lengkap

Tanggal Lahir:

Hari Bulan Tahun

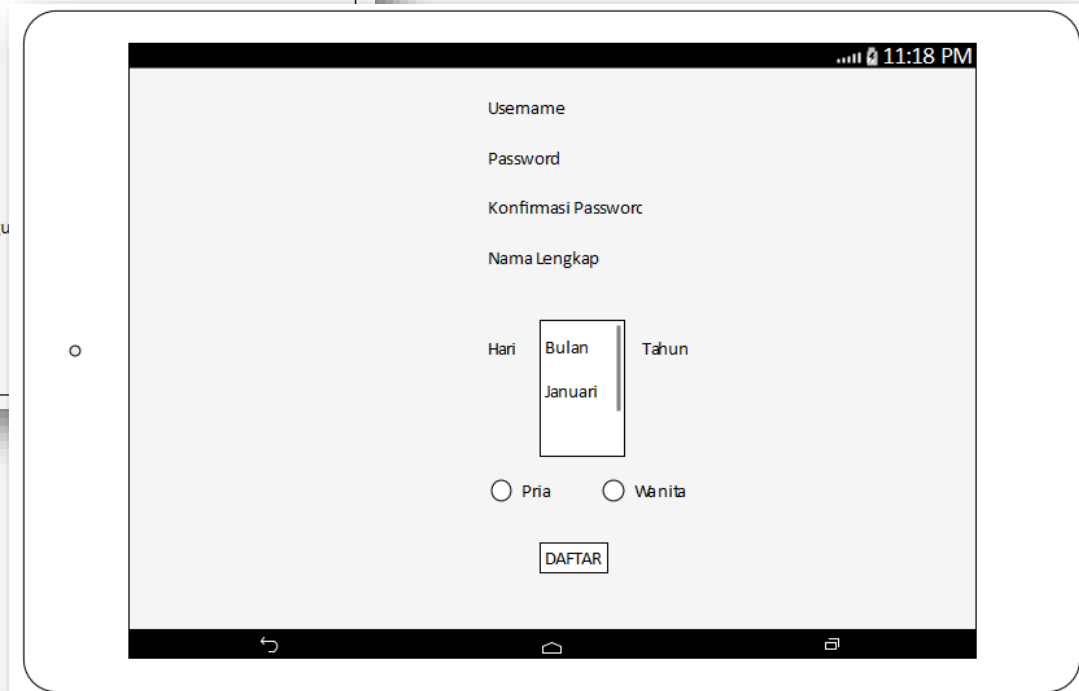
Januari

...

☐ Pria ☐ Wanita

☐ menyetujui Syarat dan Ketentuan Pengguna

DAFTAR



A screenshot of an Android mobile app showing a registration form. The status bar at the top shows the time as 11:18 PM. The form layout is similar to the web version, with input fields for 'Ustream', 'Password', 'Konfirmasi Password', and 'Nama Lengkap'. The 'Tanggal Lahir:' section has three dropdown menus for 'Hari', 'Bulan', and 'Tahun'. The 'Bulan' dropdown is open, showing 'Januari' and '...'. There are radio buttons for 'Pria' and 'Wanita', and a 'DAFTAR' button at the bottom. The Android navigation bar is visible at the very bottom.

Ustream

Password

Konfirmasi Password

Nama Lengkap

Hari Bulan Tahun

Januari

...

☐ Pria ☐ Wanita

DAFTAR



3.3 Perancangan Data Model

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

1.1 Identifikasi Proses Bisnis dengan **Use Case Diagram**

1.2 Pemodelan Proses Bisnis dengan **Activity Diagram** atau **BPMN**

1.3 Realisasi Proses Bisnis dengan **Sequence Diagram**

(**Boundary** - **Control** - **Entity**)

2. Systems Design

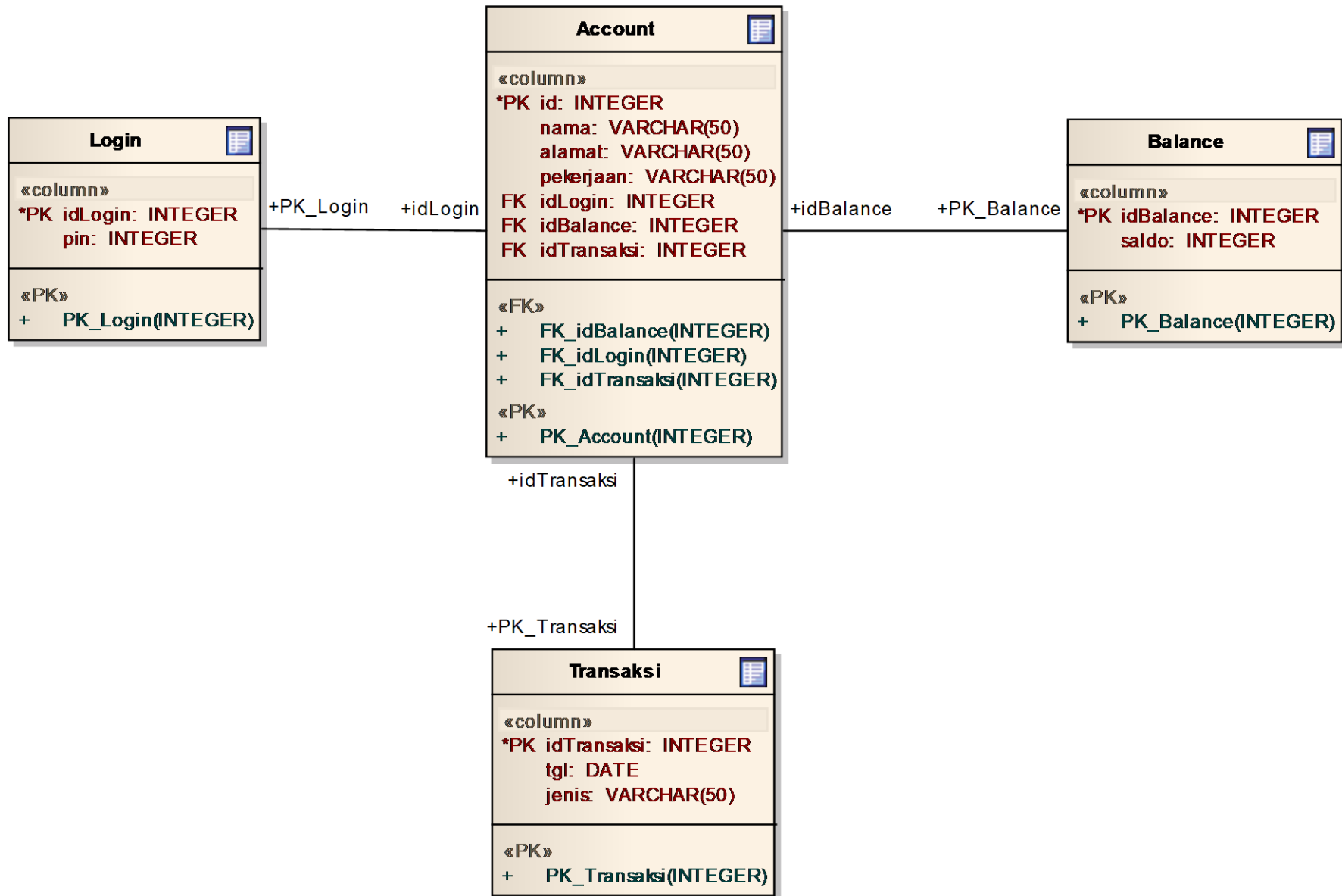
2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

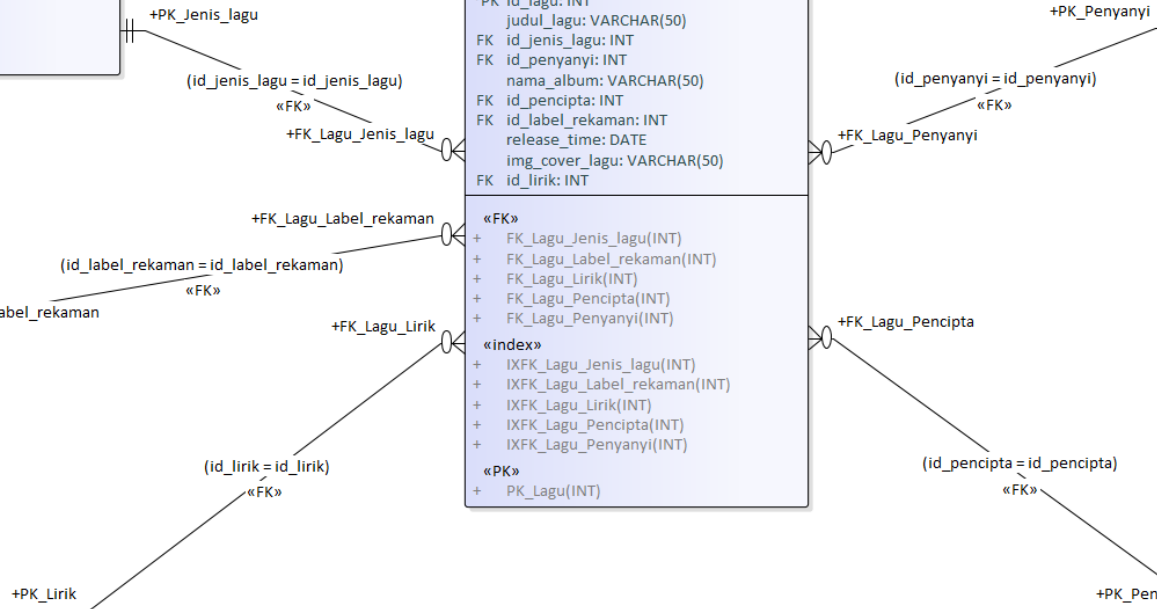
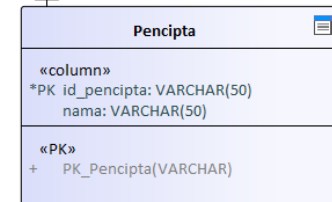
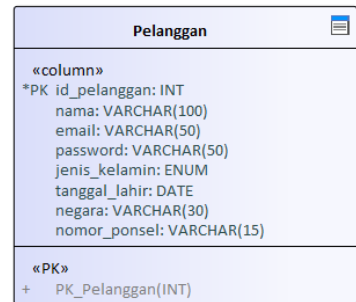
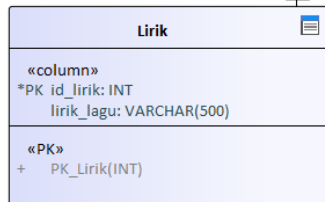
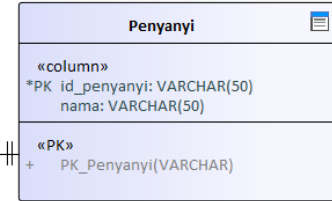
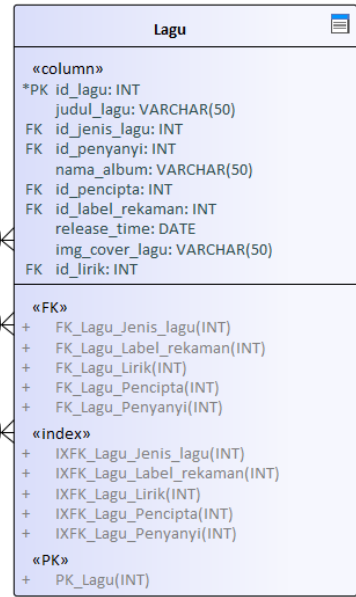
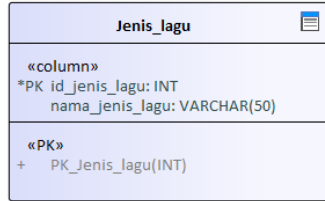
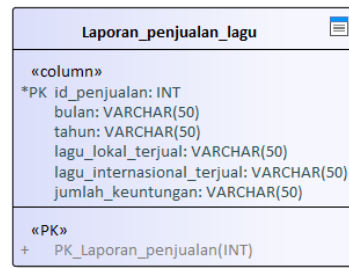
2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

Data Model Sistem ATM



Data Model MusicPedia





3.4 Perancangan Deployment Diagram

Deployment Diagram

- **Servers**
 - Mainframes, Minis, Micros
- **Clients**
 - Input/Output HW used by users
 - Terminals, PCs, special purpose HW
- **Network**
 - HW and SW to connect clients to servers
- **Nodes**
 - Any piece of hardware in the model
 - A computational resource
 - Labeled by its name
 - Stereotype to label the type of node
- **Artifacts**
 - Piece of the information system, such as software or a database table
- **Node with Deployed Artifact**
 - Shows artifact placed on a physical node
 - Good for showing distribution data or software
- **Communication paths**
 - Links between nodes of the network

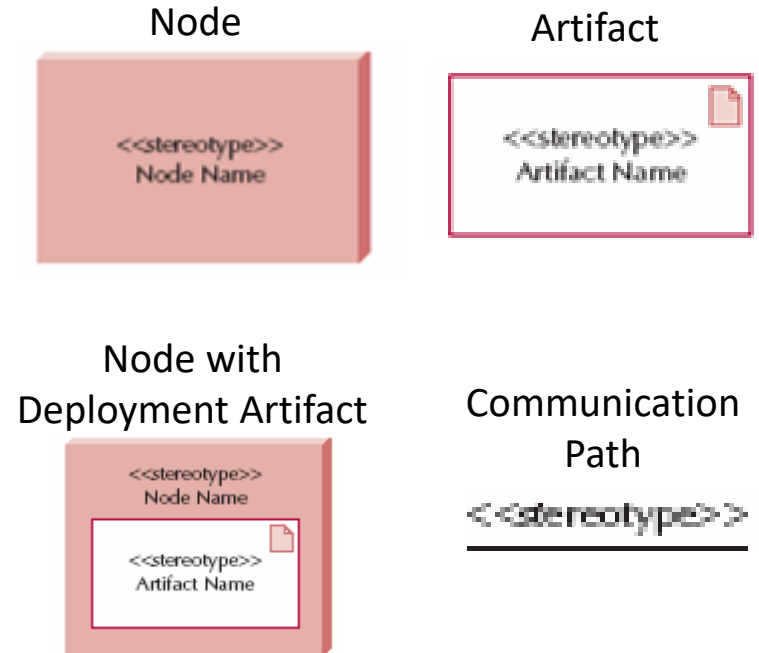
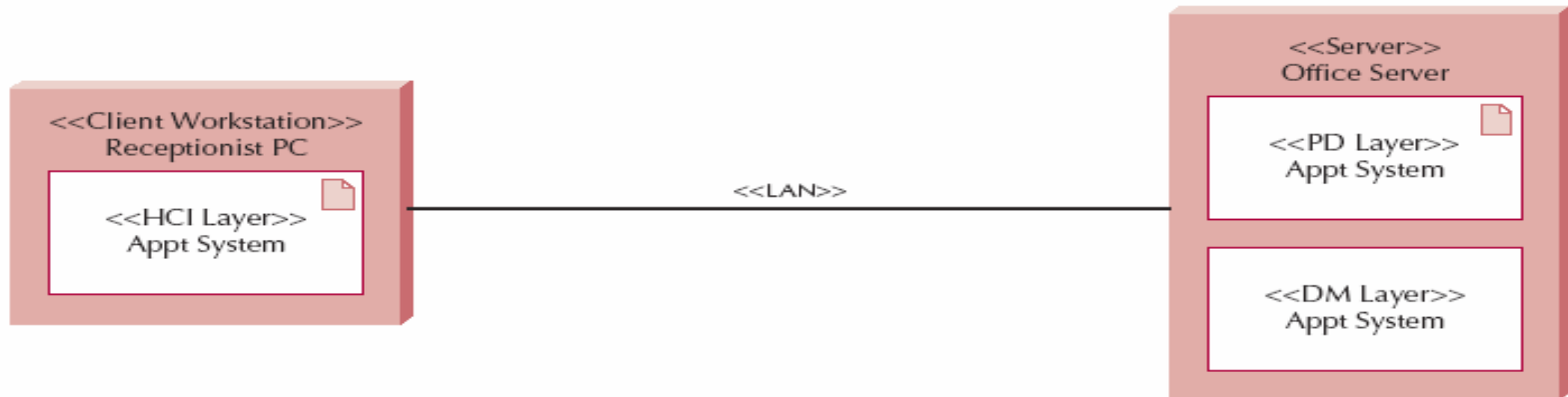


Diagram Examples



(A)

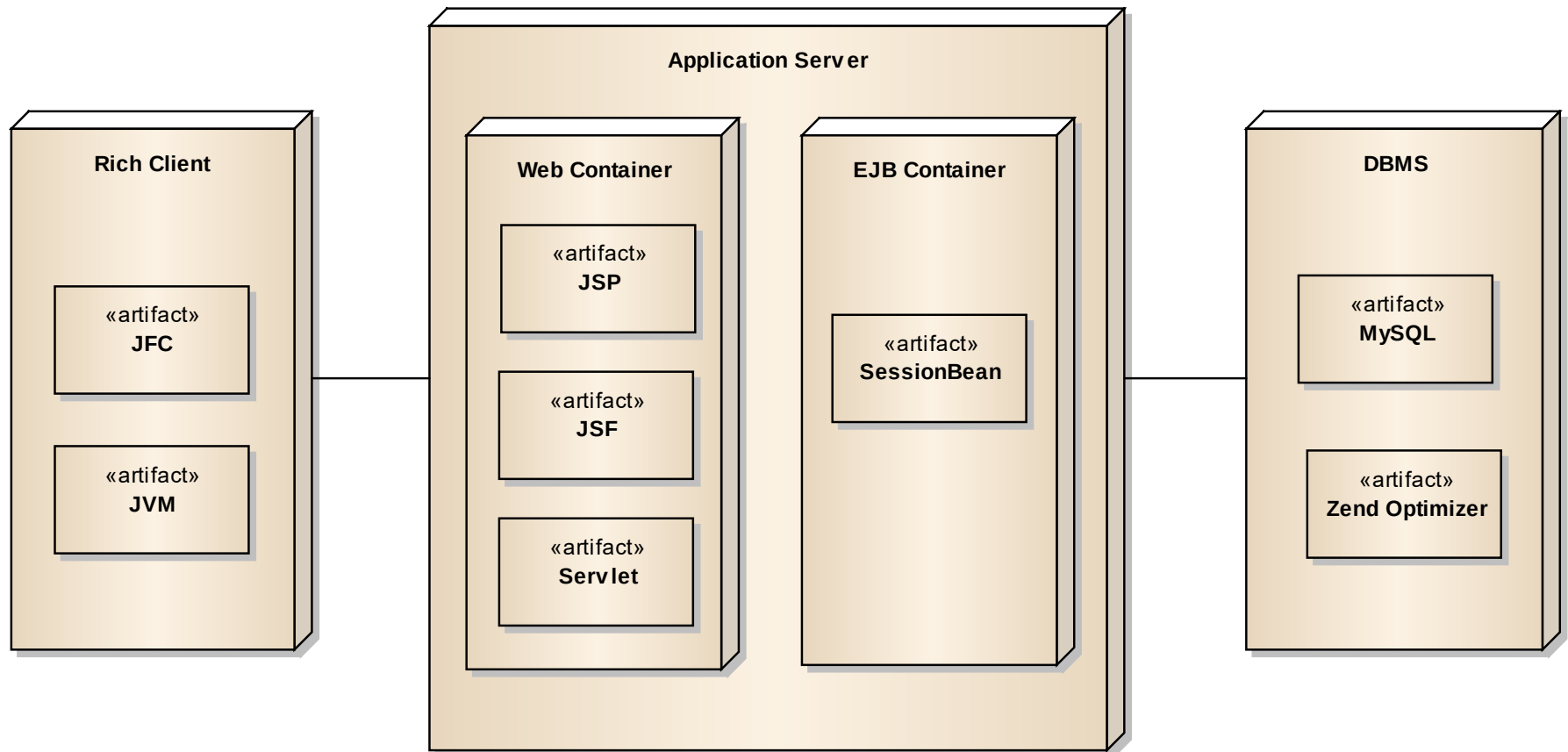


(B)



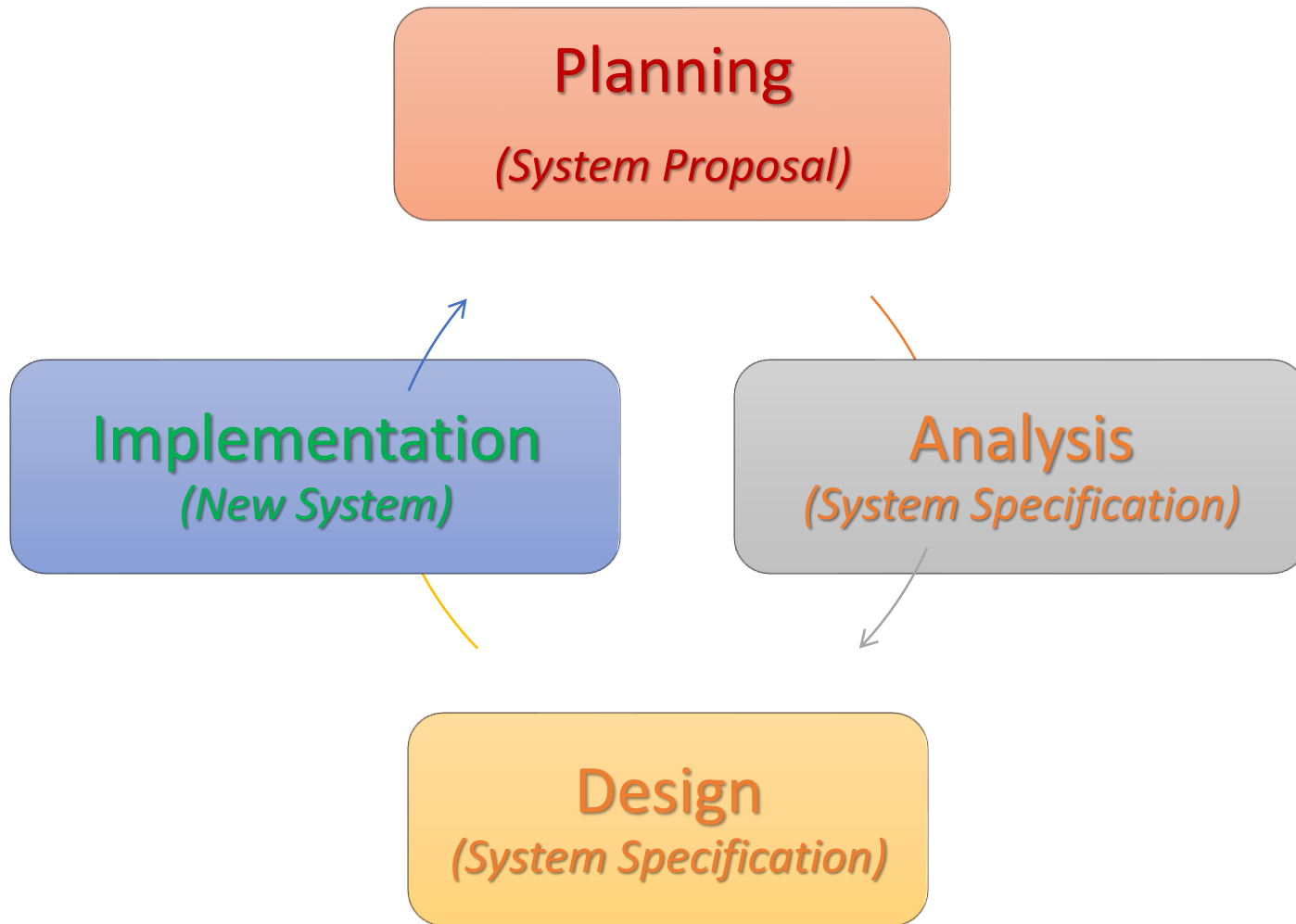
(C)

Deployment Diagram (3 Tier)



Rangkuman Studi Kasus Sistem ATM

Siklus Pengembangan Software



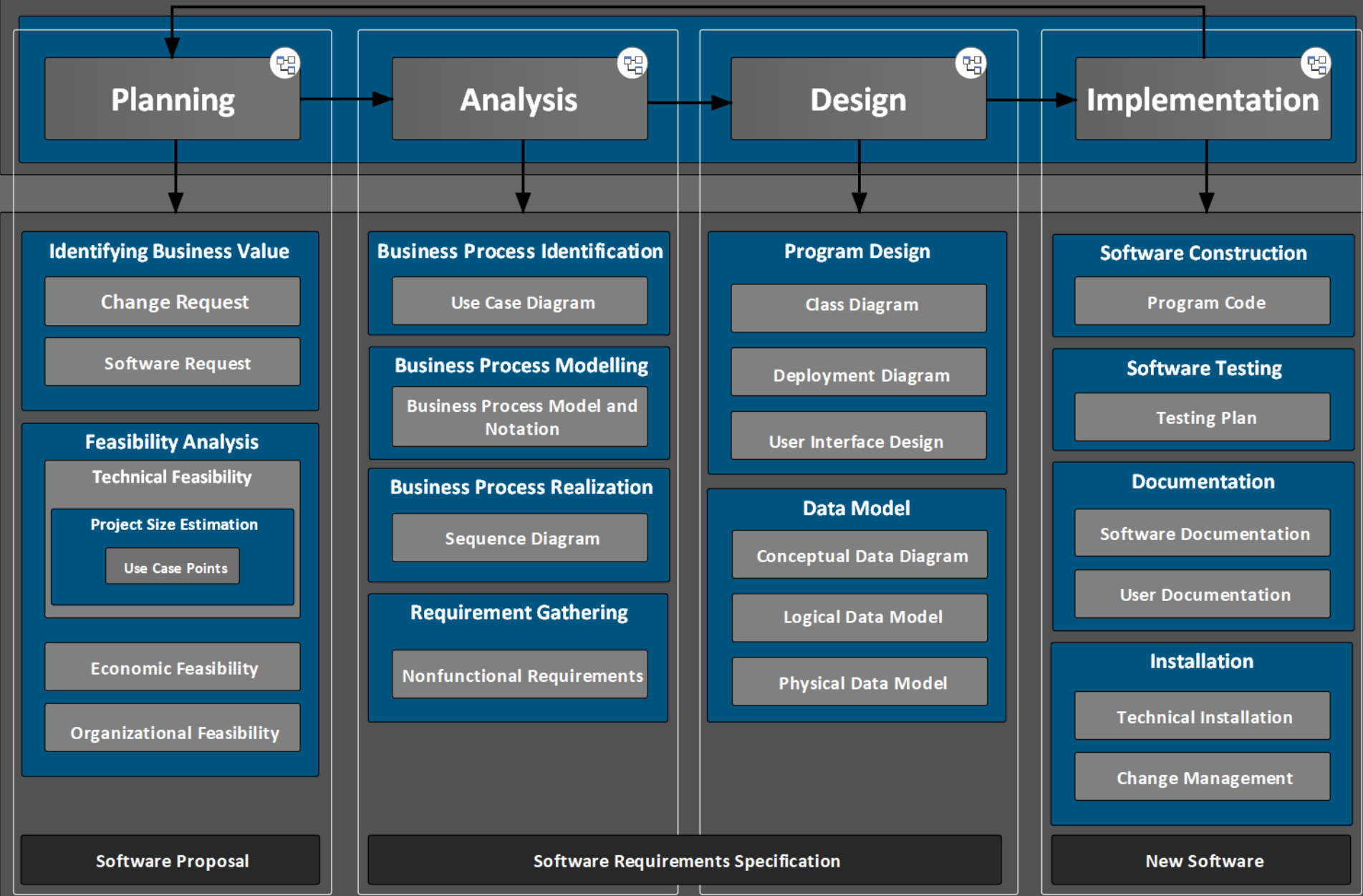
(Tilley, 2012)

(Dennis, 2016)

(Valacich, 2017)

Application Development Governance

Software Development Life Cycle

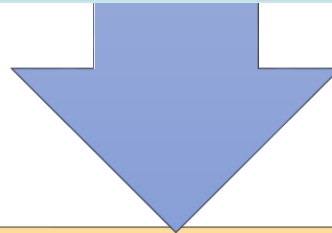


System Request (Business Value Identification)

Lower Cost

*Increase
Productivity*

Increase Profit



Feasibility Analysis

*Technical
(Capabilities)*

*Economic
(ROI, BEP)*

*Organizational
(Goals, Core Business)*

System Request: Sistem Penjualan Musik Online

Project Sponsor: Margaret Mooney, Vice President of Marketing

Business Needs: Project ini dibangun untuk:

1. Mendapatkan pelanggan baru lewat Internet

2. Memberikan layanan pendukung dengan menggunakan internet

Business Requirements:

Sistem yang mendukung penjualan musik secara online. Fitur-fitur yang harus ada:

1. Fitur Pencarian Produk
2. Fitur Pencarian Toko yang Menyediakan Stok Produk
3. Fitur Pemesanan Produk Melalui Toko yang Menyediakan
4. Fitur Pembayaran dengan Berbagai Pilihan Pembayaran

Business Value:

Intangible Value:

- Meningkatkan kenyamanan dan **kepuasan pelanggan**
- Meningkatkan **brand recognition** tentang perusahaan di dunia Internet

Tangible Value:

1. Meningkatkan penjualan dari pelanggan baru lewat Internet:

- Rp 400 juta **peningkatan penjualan** dari pelanggan baru dan Rp 600 juta dari pelanggan lama

2. Mengurangi biaya operasional untuk menangani komplain dari pelanggan

- Rp 100 juta **pengurangan** tahunan biaya telepon untuk menangani pelanggan

Studi Kelayakan Sistem Penjualan Musik Online

Margaret Mooney dan Alec Adams membuat studi kelayakan untuk pengembangan Sistem Penjualan Musik Online

Kelayakan Teknis

Sistem penjualan musik online layak secara teknis, meskipun memiliki beberapa risiko.

Risiko Berhubungan dengan **Kefamiliaran dengan Aplikasi**: Risiko **Tinggi**

- Divisi Marketing **tidak memiliki pengalaman** menggunakan sistem penjualan online
- Divisi IT memiliki pemahaman yang baik tentang sistem penjualan offline, akan tetapi **tidak berpengalaman** mengembangkan sistem penjualan musik online

Risiko Berhubungan dengan **Kefamiliaran dengan Teknologi**: Risiko **Sedang**

- Divisi IT tidak menguasai masalah infrastruktur dan ISP, tetapi akan menyewa konsultan
- Divisi IT cukup familier dengan framework dan IDE yang akan digunakan
- Divisi Marketing tidak memiliki pengalaman menggunakan teknologi Web

Risiko berhubungan dengan **Ukuran Project**: Risiko **Rendah**

- Perusahaan memiliki total **30 orang pengembang**
- Project dikerjakan oleh **5 orang pengembang** dengan estimasi waktu **6 bulan**

Kompatibilitas dengan sistem dan infrastruktur yang ada: Risiko **Rendah**

- Sistem pemesanan yang ada sekarang menggunakan *open standard*, jadi sangat **kompatibel** dengan sistem penjualan berbasis web yang akan dibangun

Kelayakan Ekonomi

Cost benefit analysis telah dilakukan. Sistem Penjualan musik online memiliki peluang yang baik untuk bisa **meningkatkan pendapatan perusahaan**.

- Return on Investment (ROI) setelah 3 tahun: **31%**
- Break-even point (BEP): **2.25 tahun**
- Total keuntungan setelah 3 tahun: **Rp. 503.559.986,-**

Keuntungan **Intangible**

- Meningkatkan **kepuasaan pelanggan**
- Meningkatkan **branding perusahaan**

Kelayakan Organisasi

- Secara organisasi, **resikonya rendah**. Tujuan dari pengembangan sistem penjualan musik online adalah meningkatkan penjualan perusahaan. Dan ini selaras dengan KPI marketing yang ke arah peningkatan kuantitas penjualan
- Project champion dari pengembangan sistem penjualan musik online ini adalah Margaret Mooney, Vice President of Marketing

	2019	2020	2021
Peningkatan penjualan dari pelanggan baru	0	400,000,000	500,000,000
Peningkatan penjualan dari pelanggan lama	0	600,000,000	700,000,000
Pengurangan biaya operasional dan telepon	0	100,000,000	100,000,000
Total Benefits:	0	1,100,000,000	1,300,000,000
PV of Benefits:	0	978,996,084	1,091,505,068
PV of All Benefits:	0	978,996,084	2,070,501,152
Honor Tim (Analysis, Design and Implementation)	360,000,000	0	0
Honor Konsultan	90,000,000	0	0
Total Development Costs:	450,000,000	0	0
Honor Pengelola Web	60,000,000	70,000,000	80,000,000
Biaya Lisensi Software	50,000,000	60,000,000	70,000,000
Hardware upgrades	100,000,000	100,000,000	100,000,000
Biaya Komunikasi	20,000,000	30,000,000	40,000,000
Biaya Marketing	100,000,000	200,000,000	300,000,000
Total Operational Costs:	330,000,000	460,000,000	590,000,000
Total Costs:	780,000,000	460,000,000	590,000,000
PV of Costs:	735,849,057	409,398,362	495,375,377
PV of all Costs:	735,849,057	1,145,247,419	1,640,622,796
Total Project Costs Less Benefits:	-780,000,000	640,000,000	710,000,000
Yearly NPV:	-735,849,057	569,597,722	669,811,321
Cumulative NPV:	-735,849,057	-166,251,335	503,559,986
Return on Investment (ROI) di Tahun 3: 30.70%	-100.00%	-0.145166304	0.306932213
Break-even Point (BEP): 2.25 tahun	234		2.248206218

UML based Software Analysis and Design

(Wahono, 2009)

1. Systems Analysis

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2. Systems Design

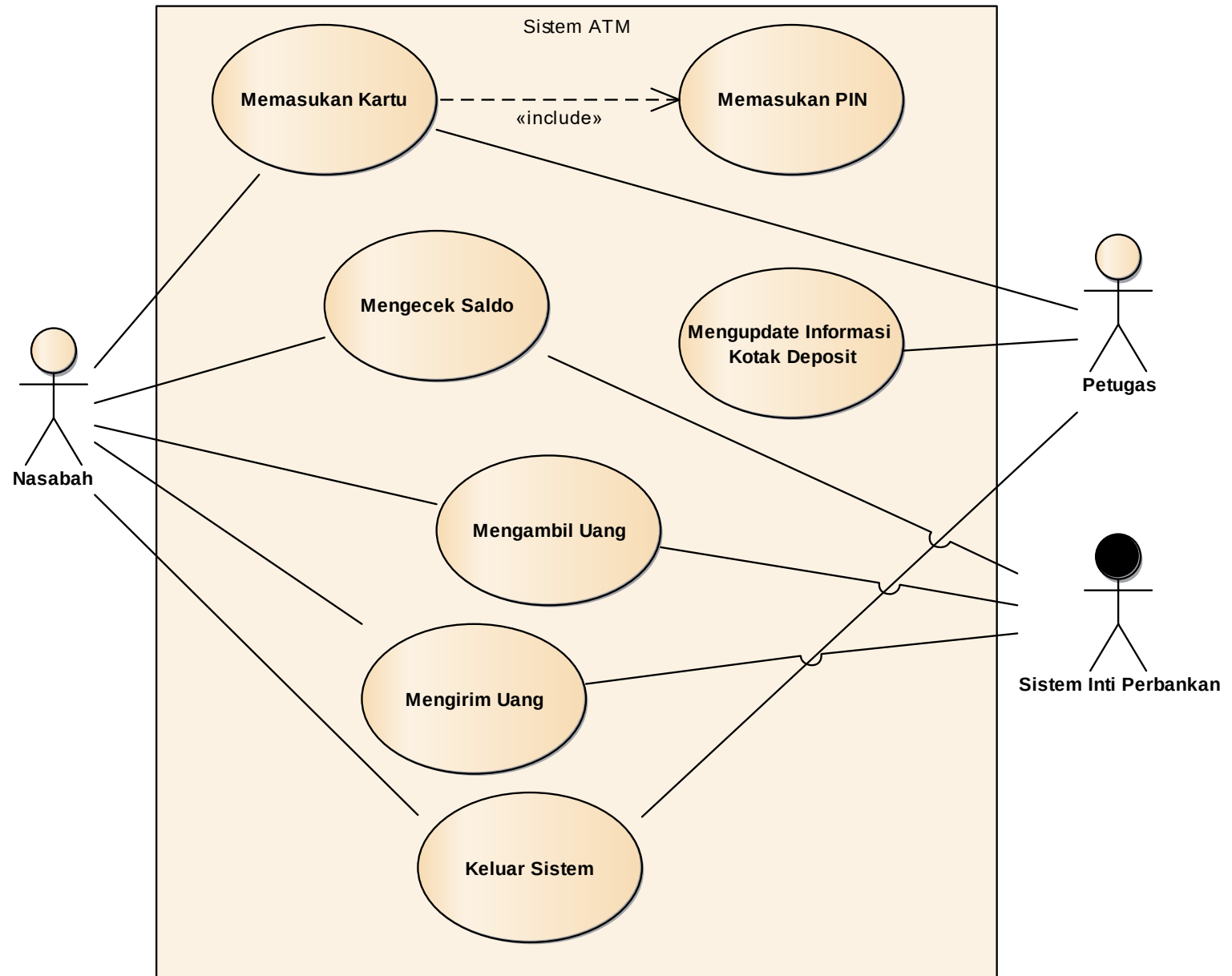
2.1 Pemodelan **Class Diagram**

2.2 Pemodelan **User Interface Design**

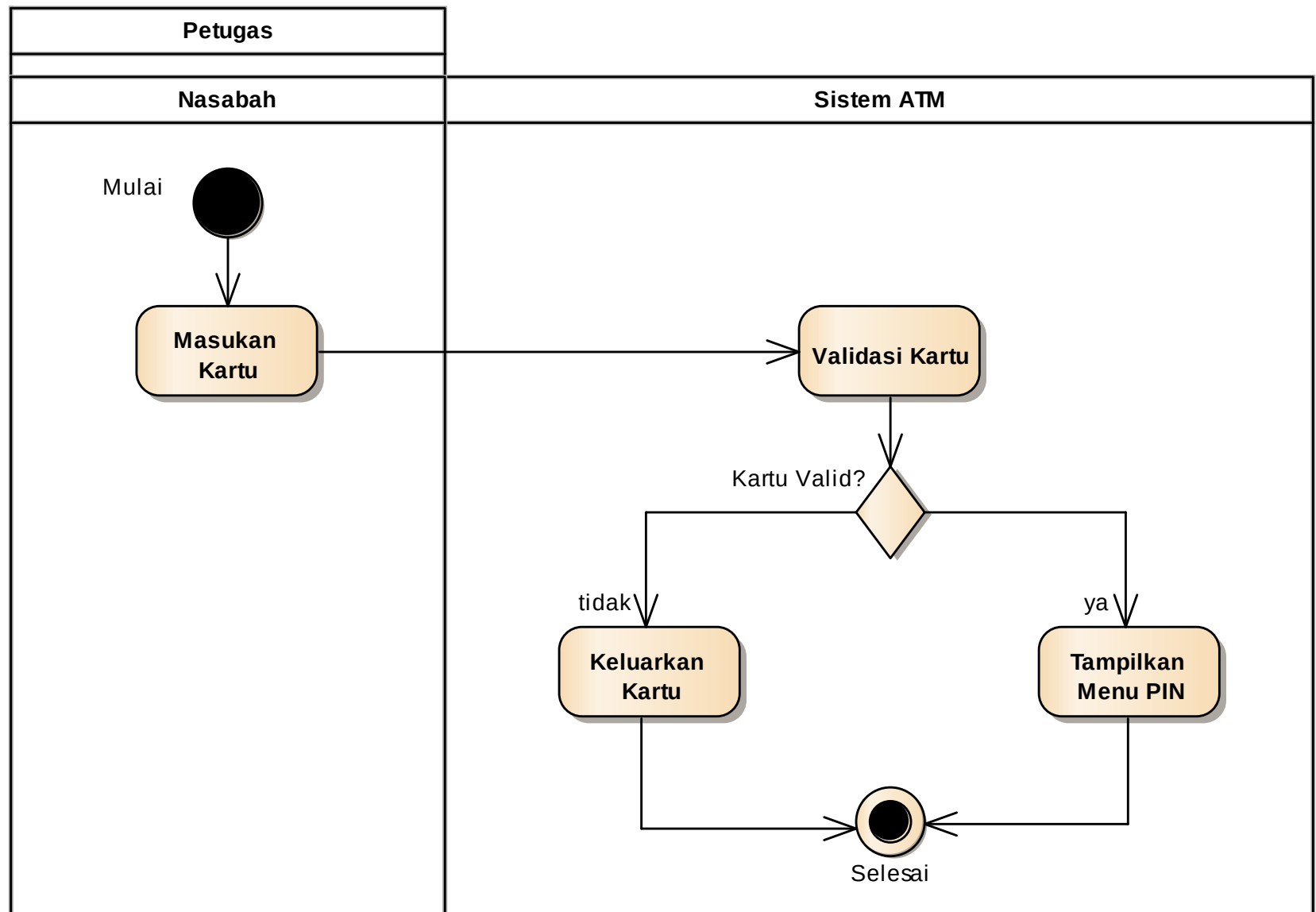
2.3 Pemodelan **Data Model**

2.4 Pemodelan **Deployment Diagram**

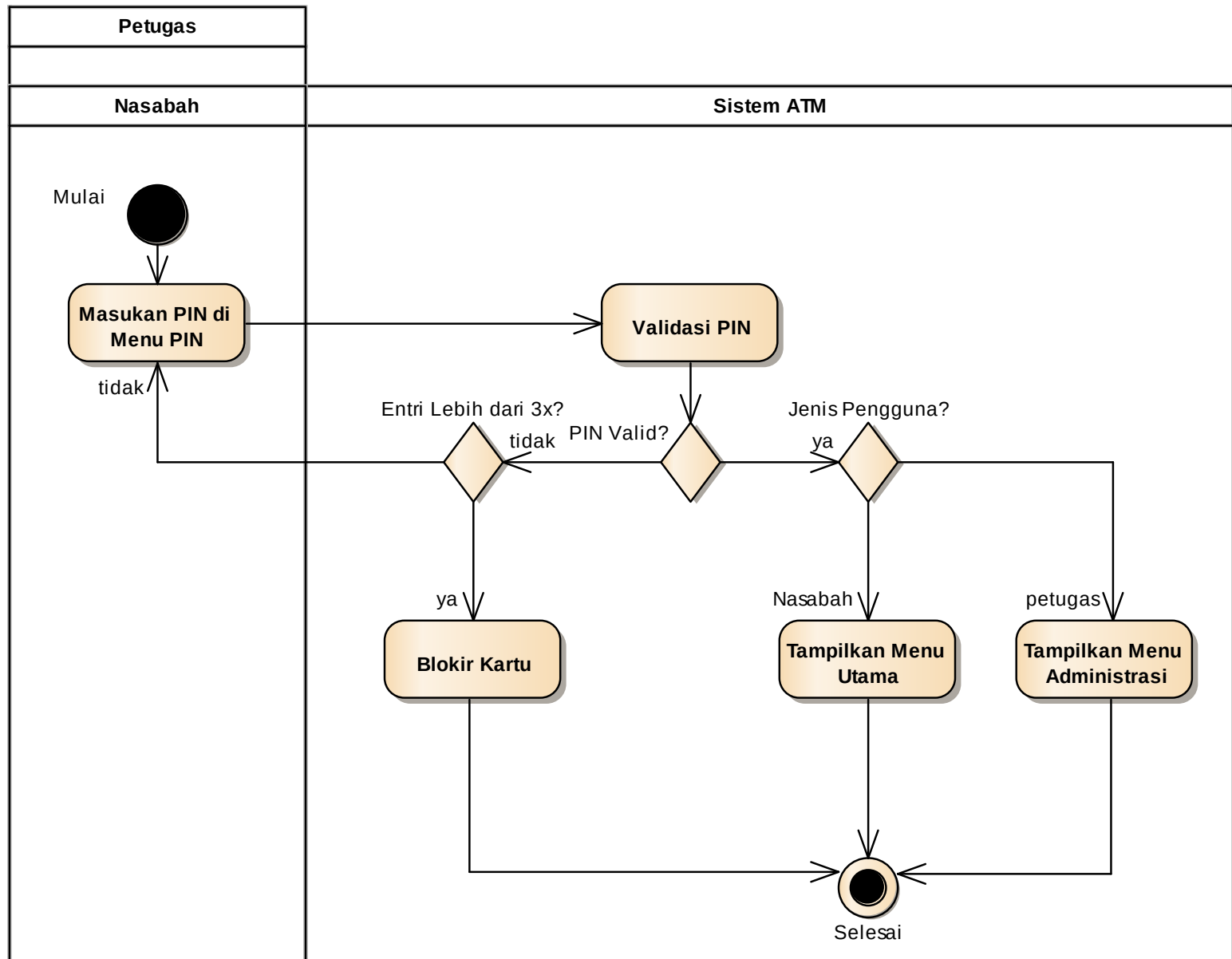
Use Case Diagram



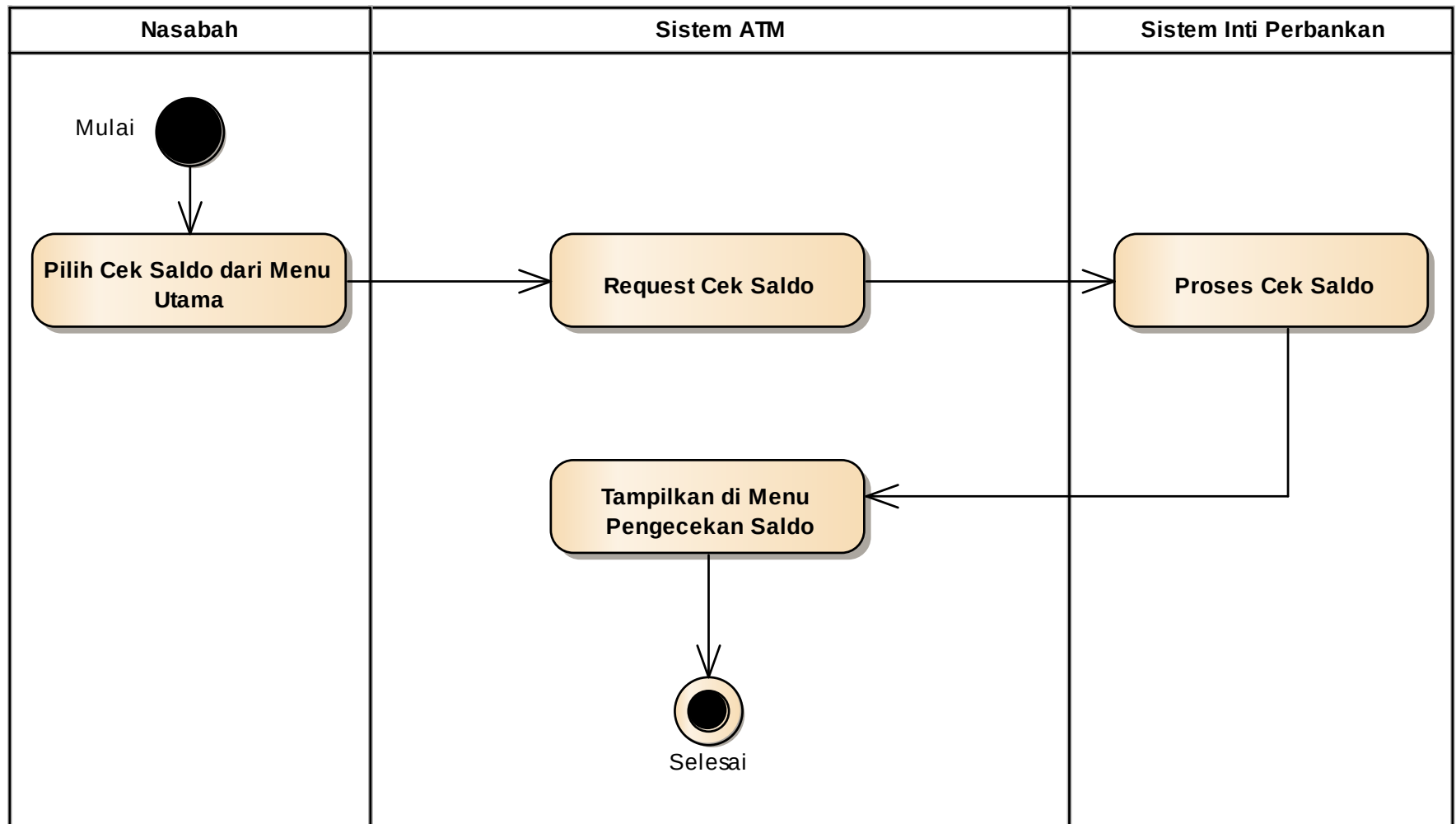
Activity Diagram: Memasukkan Kartu



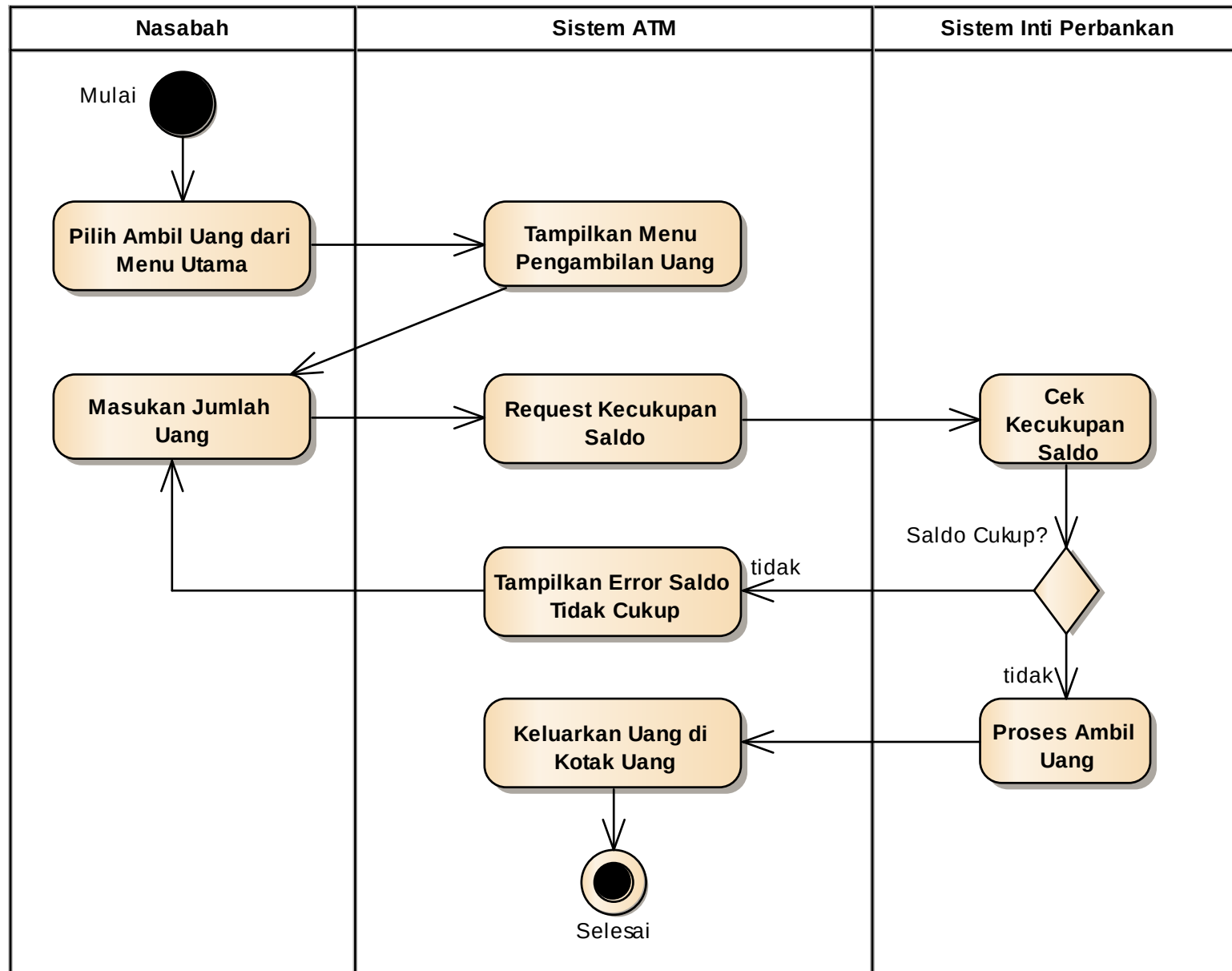
Activity Diagram: Memasukkan PIN



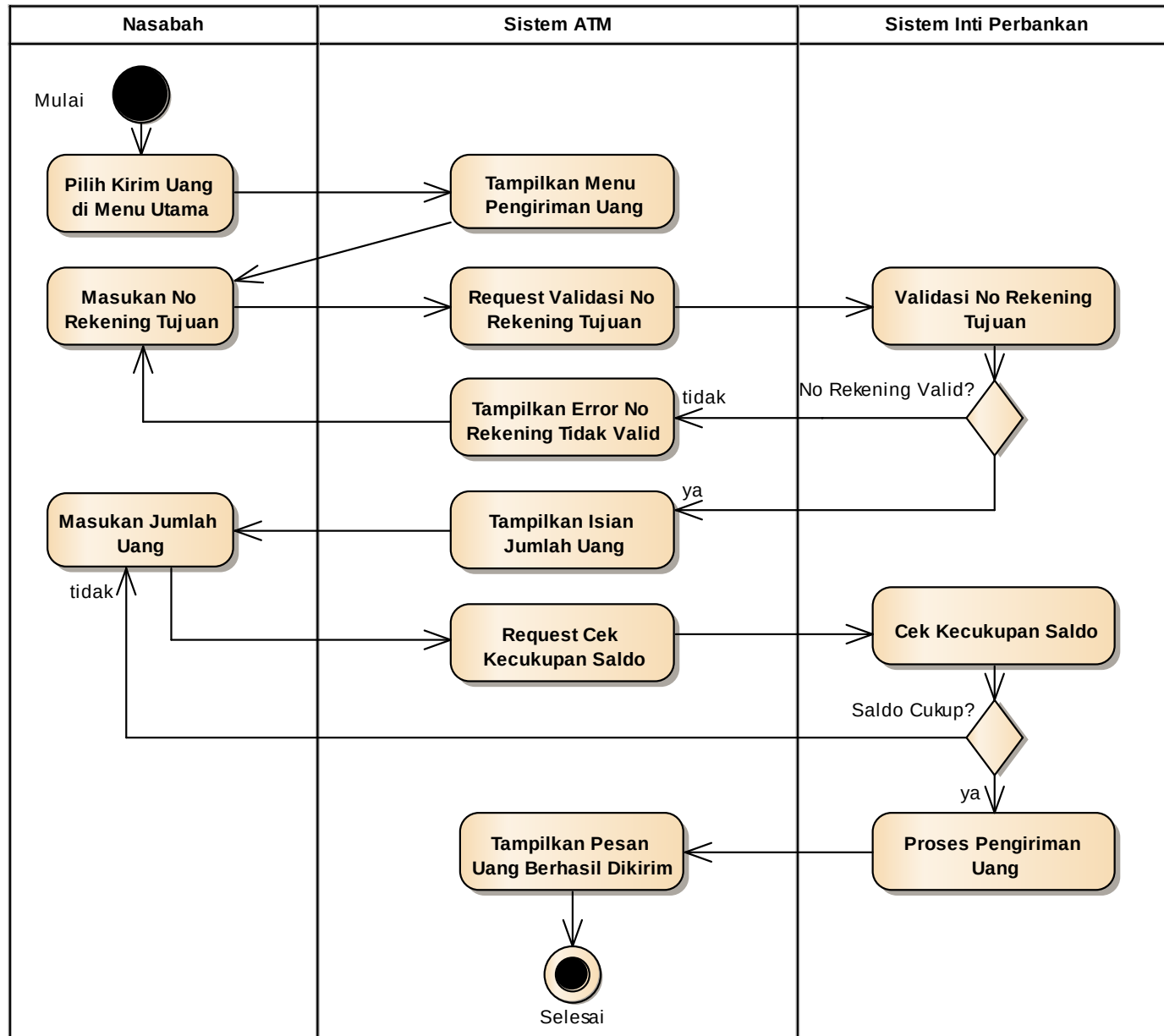
Activity Diagram: Mengecek Saldo



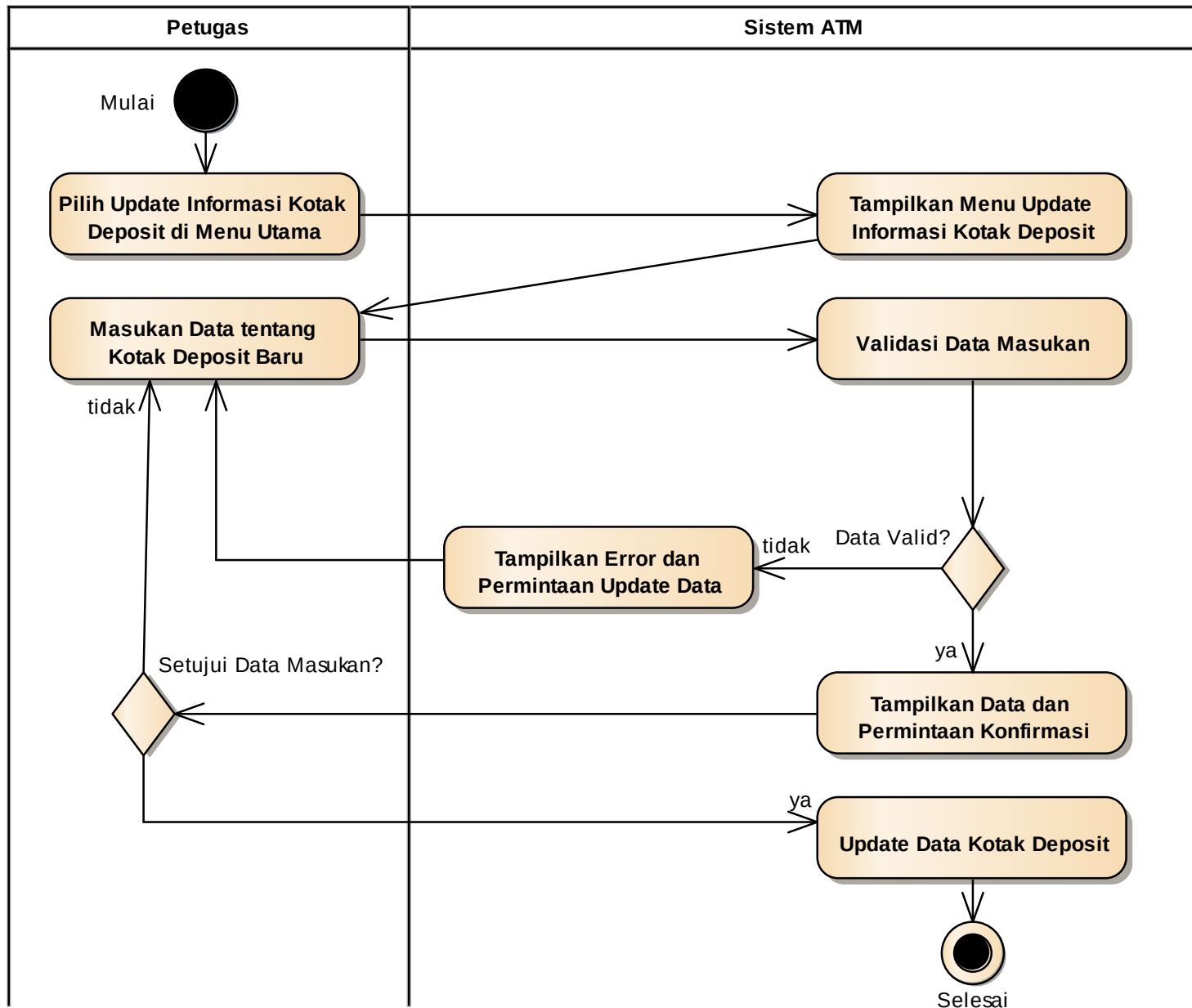
Activity Diagram: Mengambil Uang



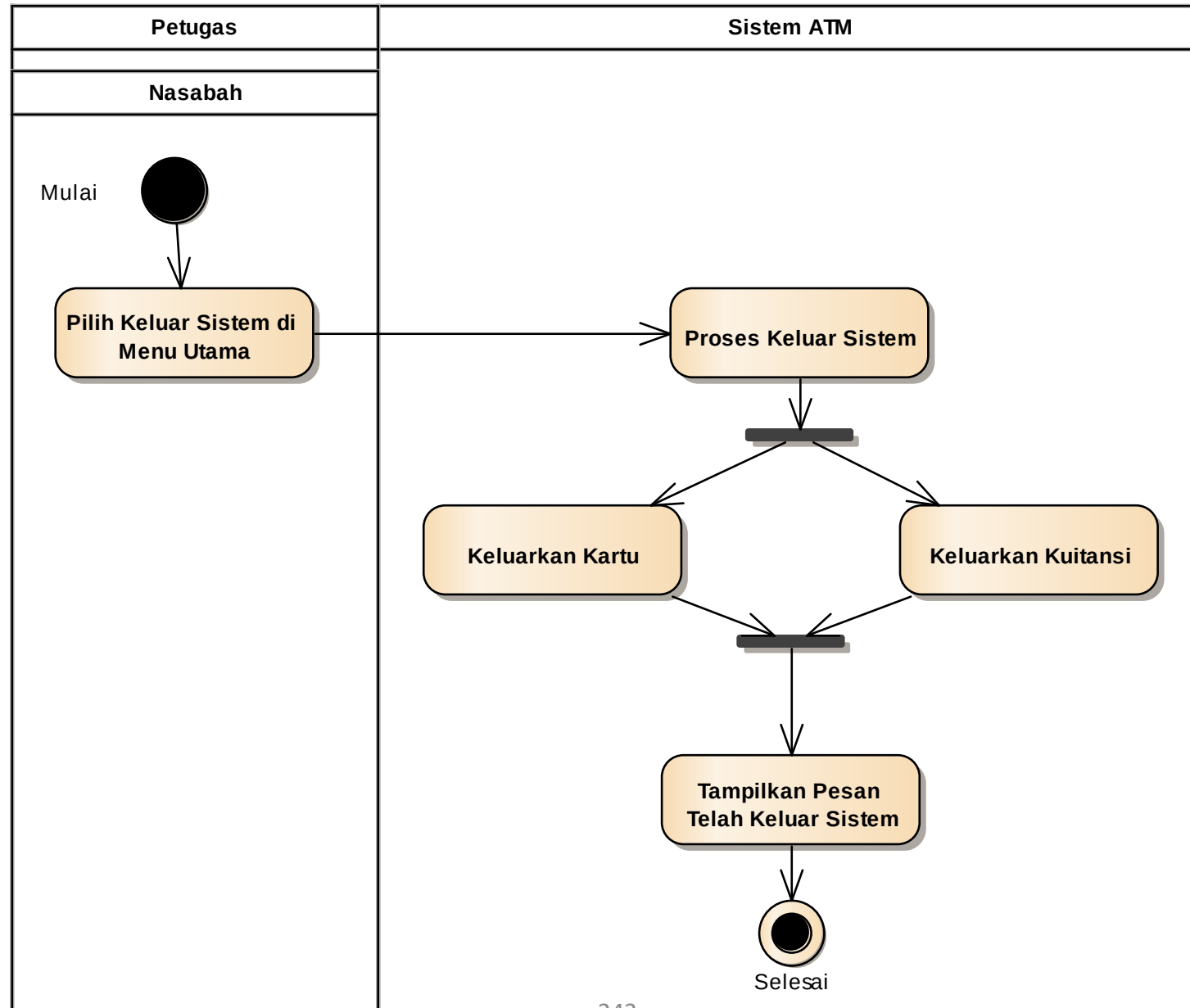
Activity Diagram: Mengirim Uang



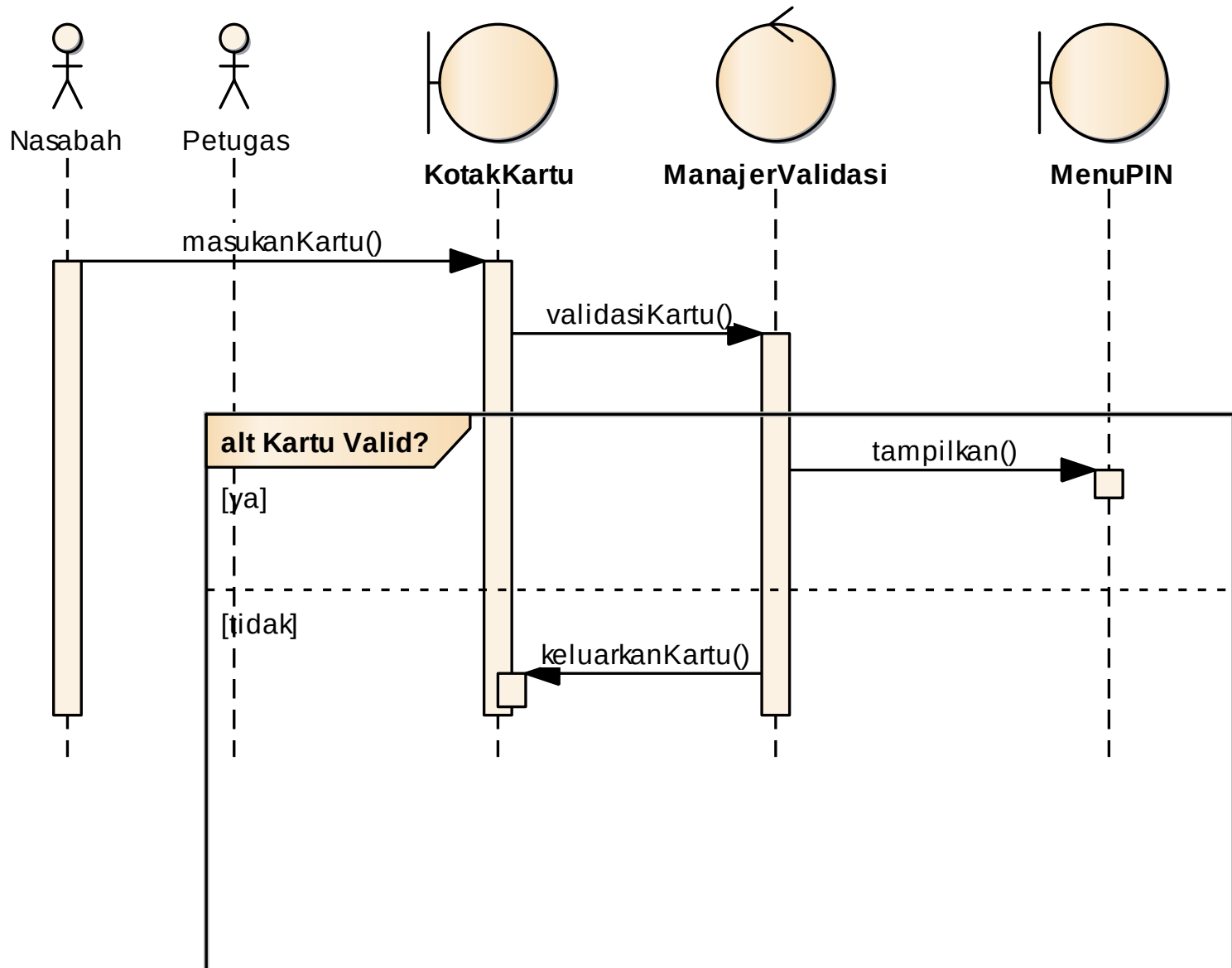
Activity Diagram: Mengupdate Informasi Kotak Deposit



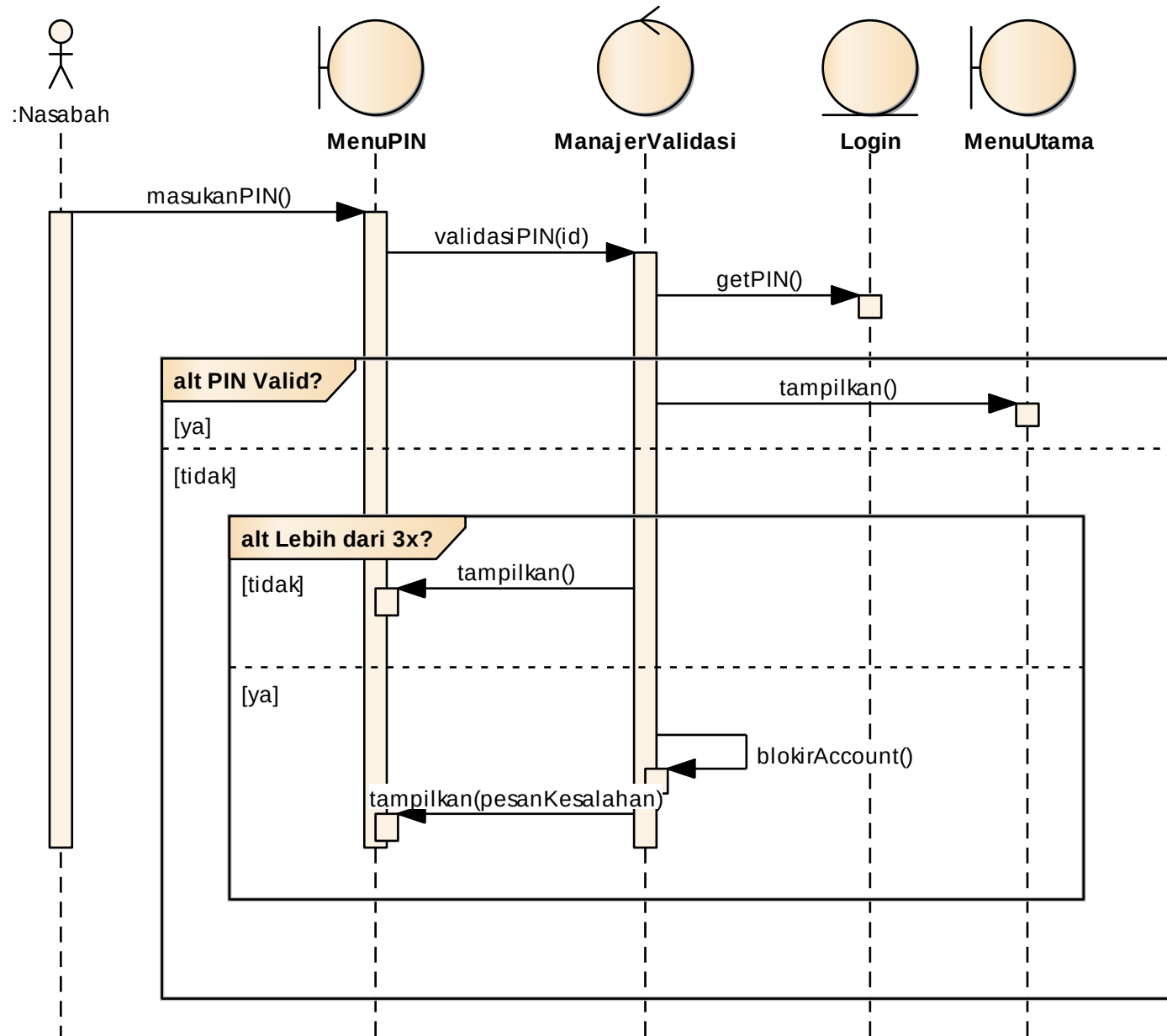
Activity Diagram: Keluar Sistem



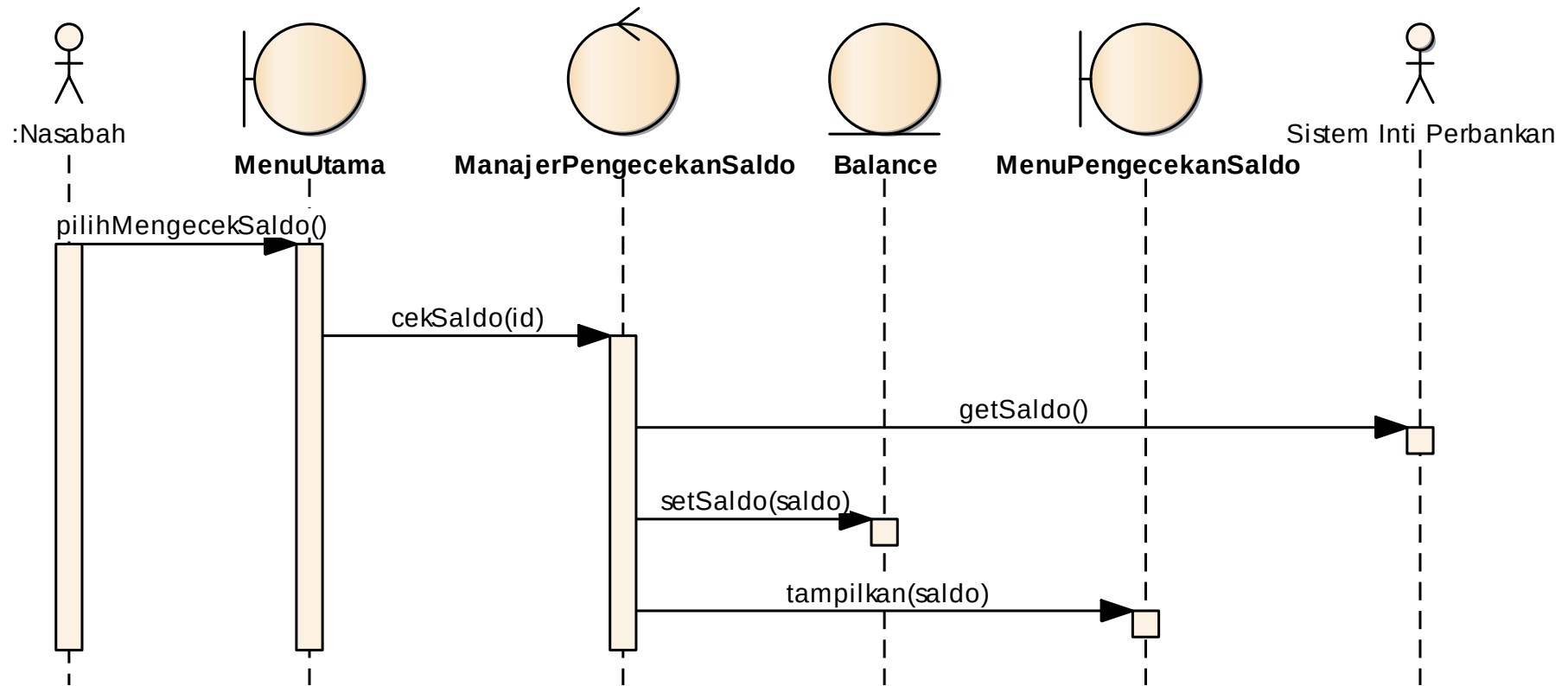
Sequence Diagram: Memasukkan Kartu



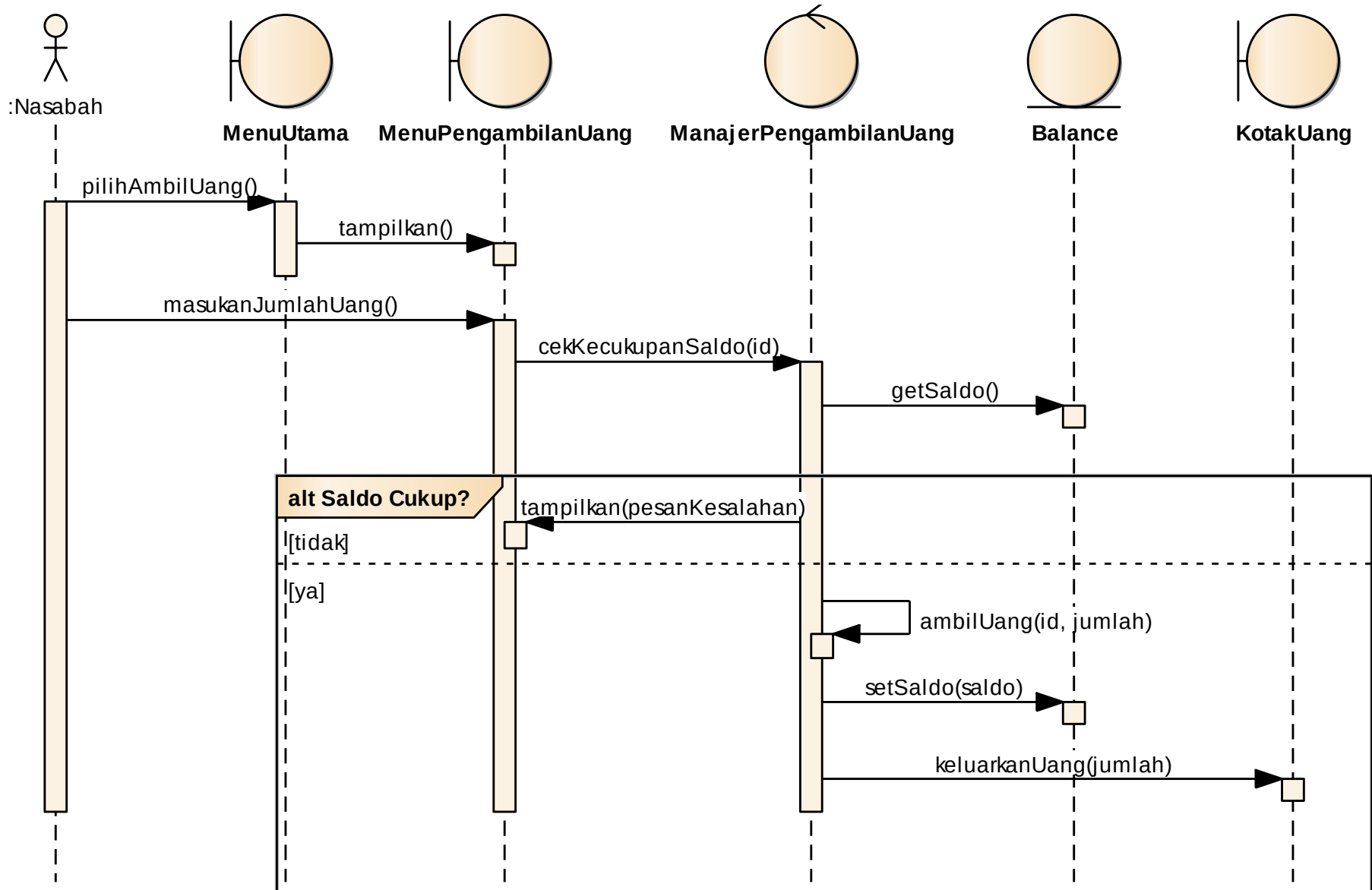
Sequence Diagram: Memasukkan PIN



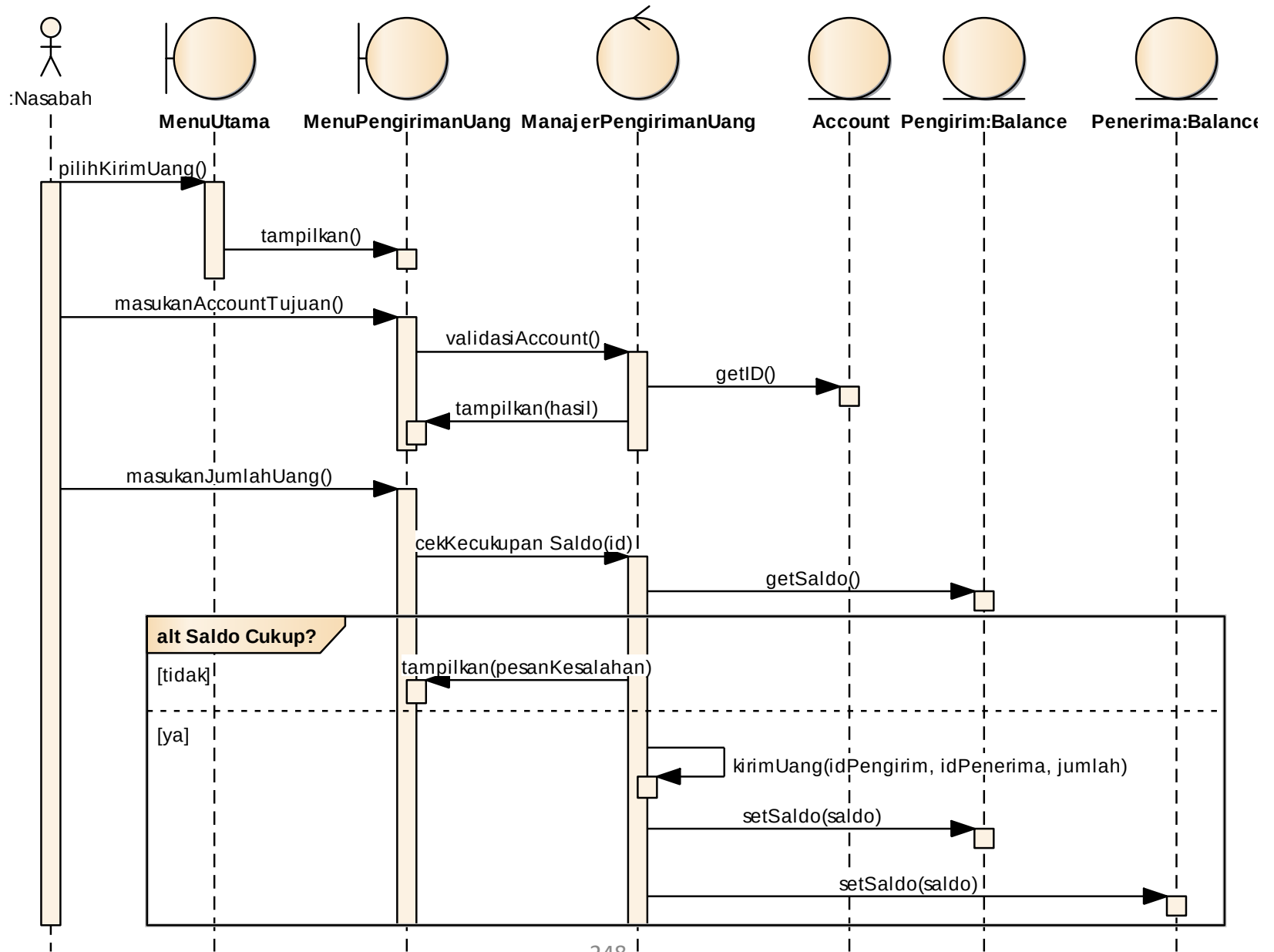
Sequence Diagram: Mengecek Saldo



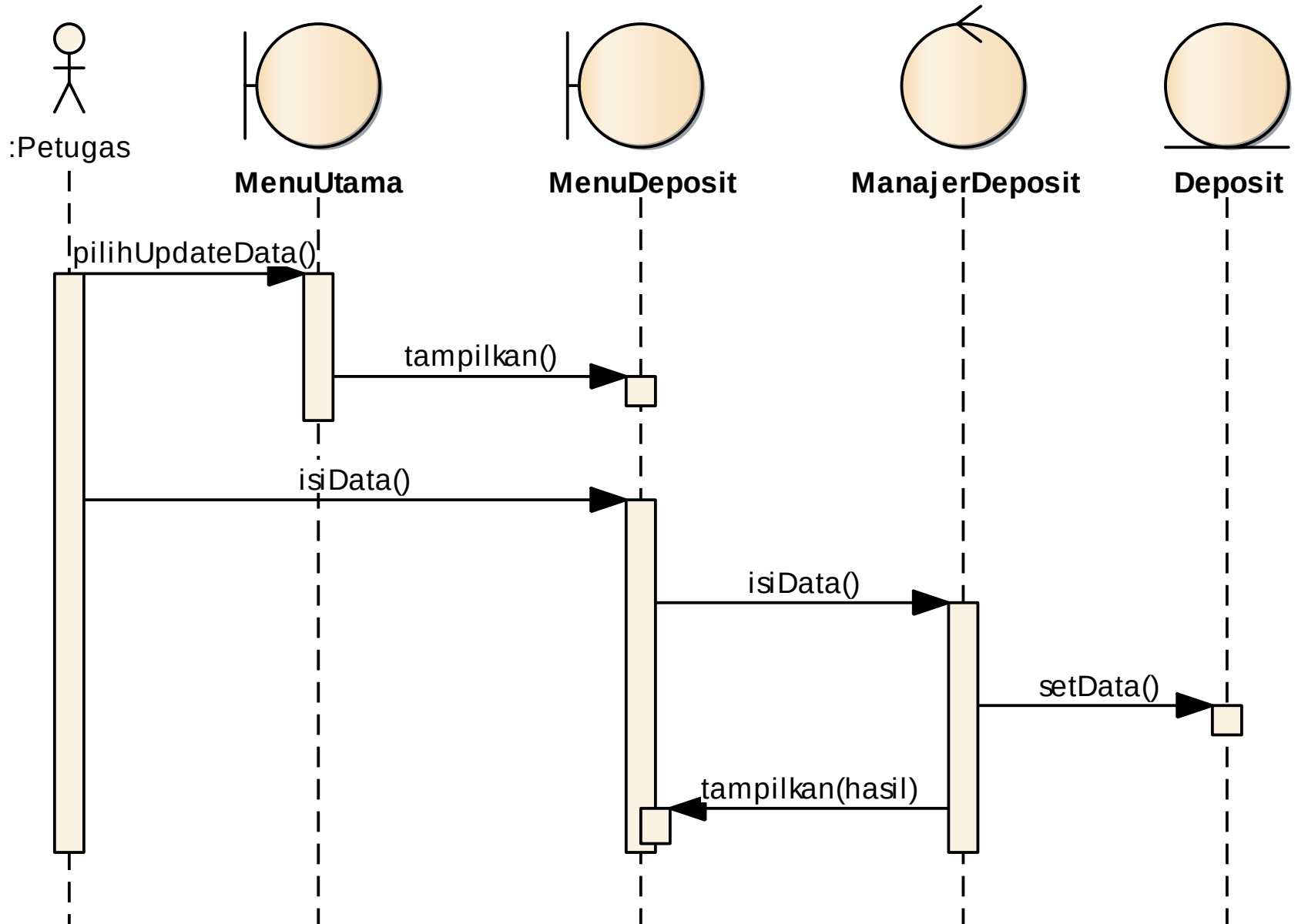
Sequence Diagram: Mengambil Uang



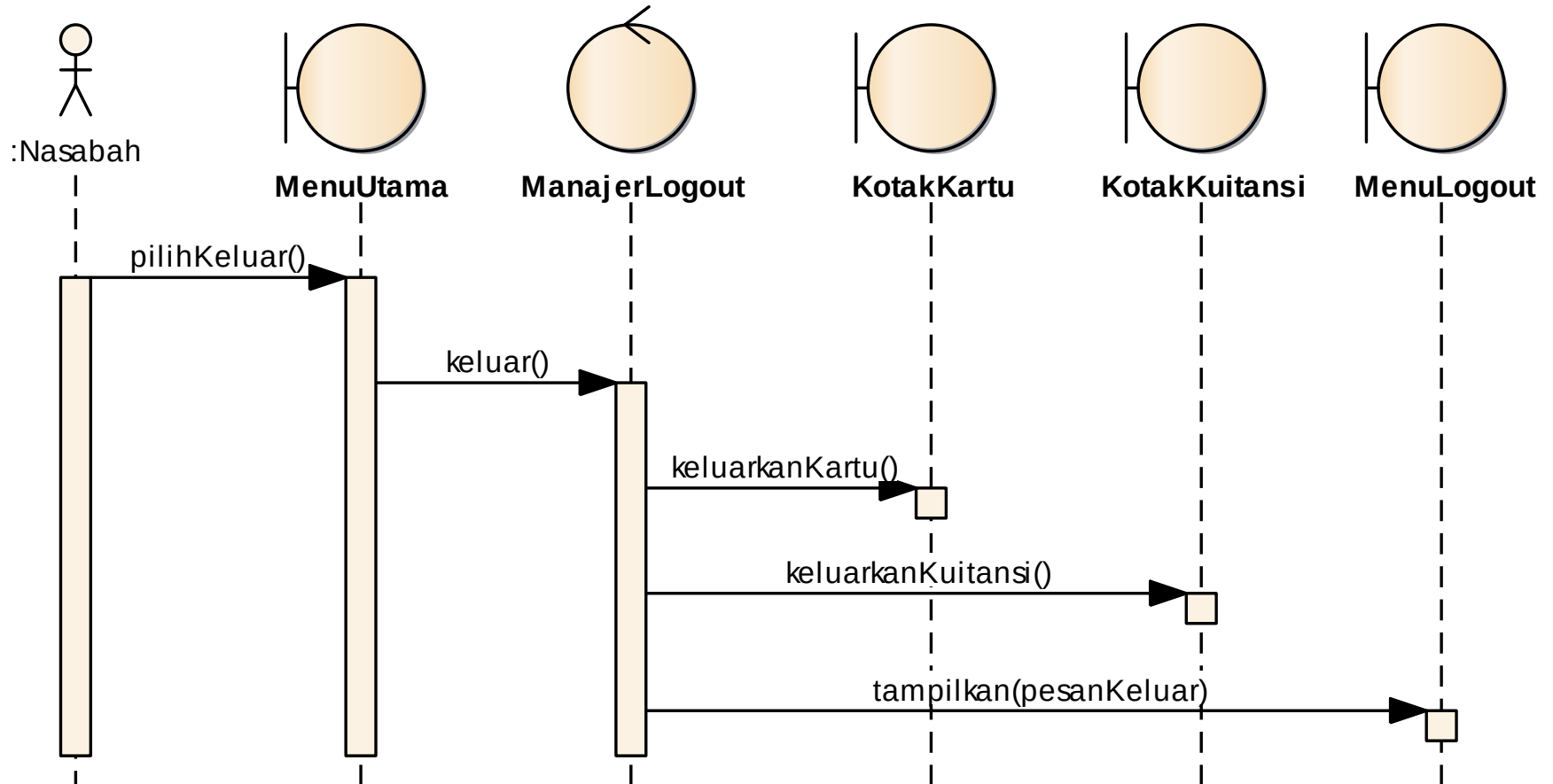
Sequence Diagram: Mengirim Uang



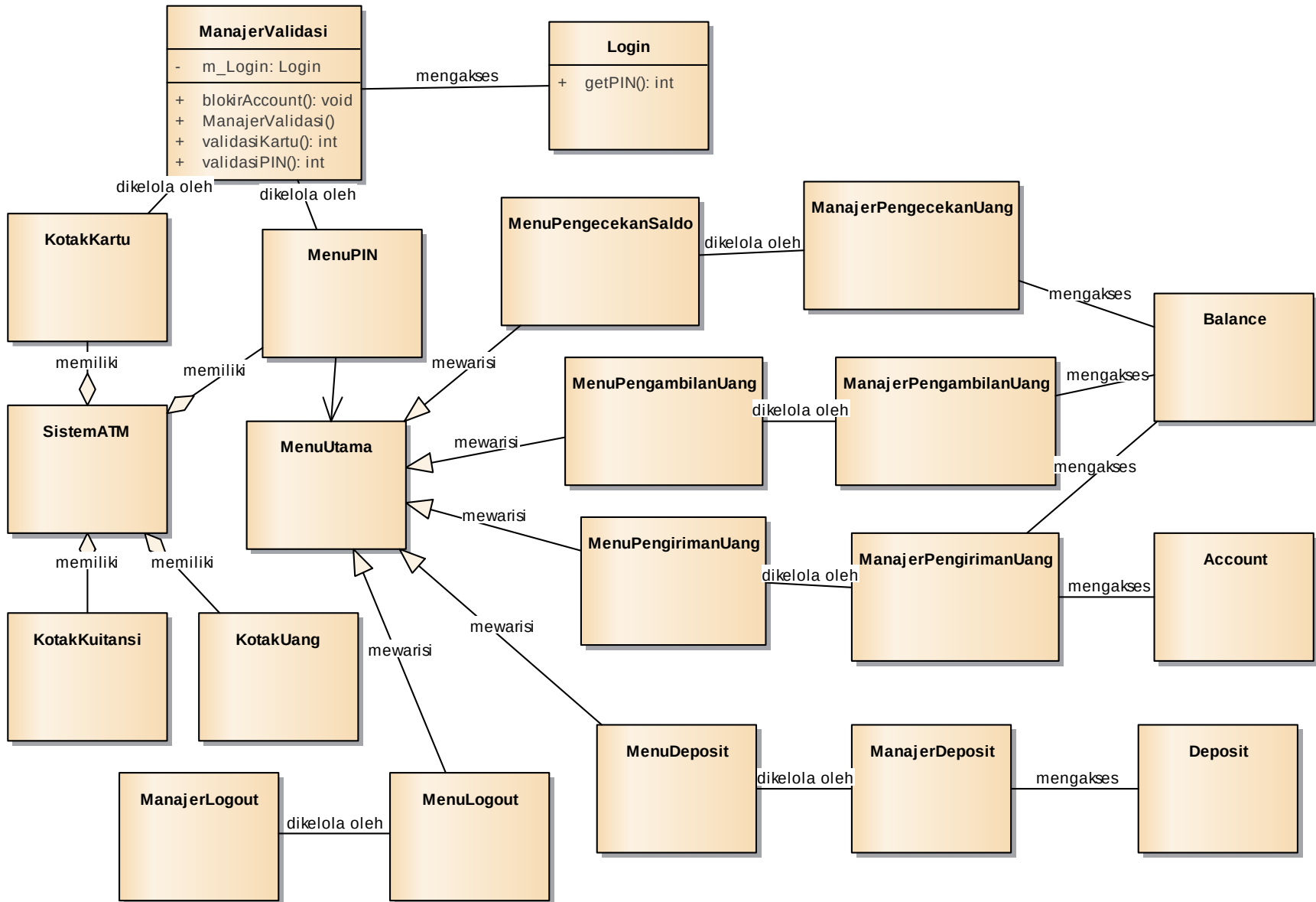
Sequence Diagram: Mengupdate Informasi Kotak Deposit



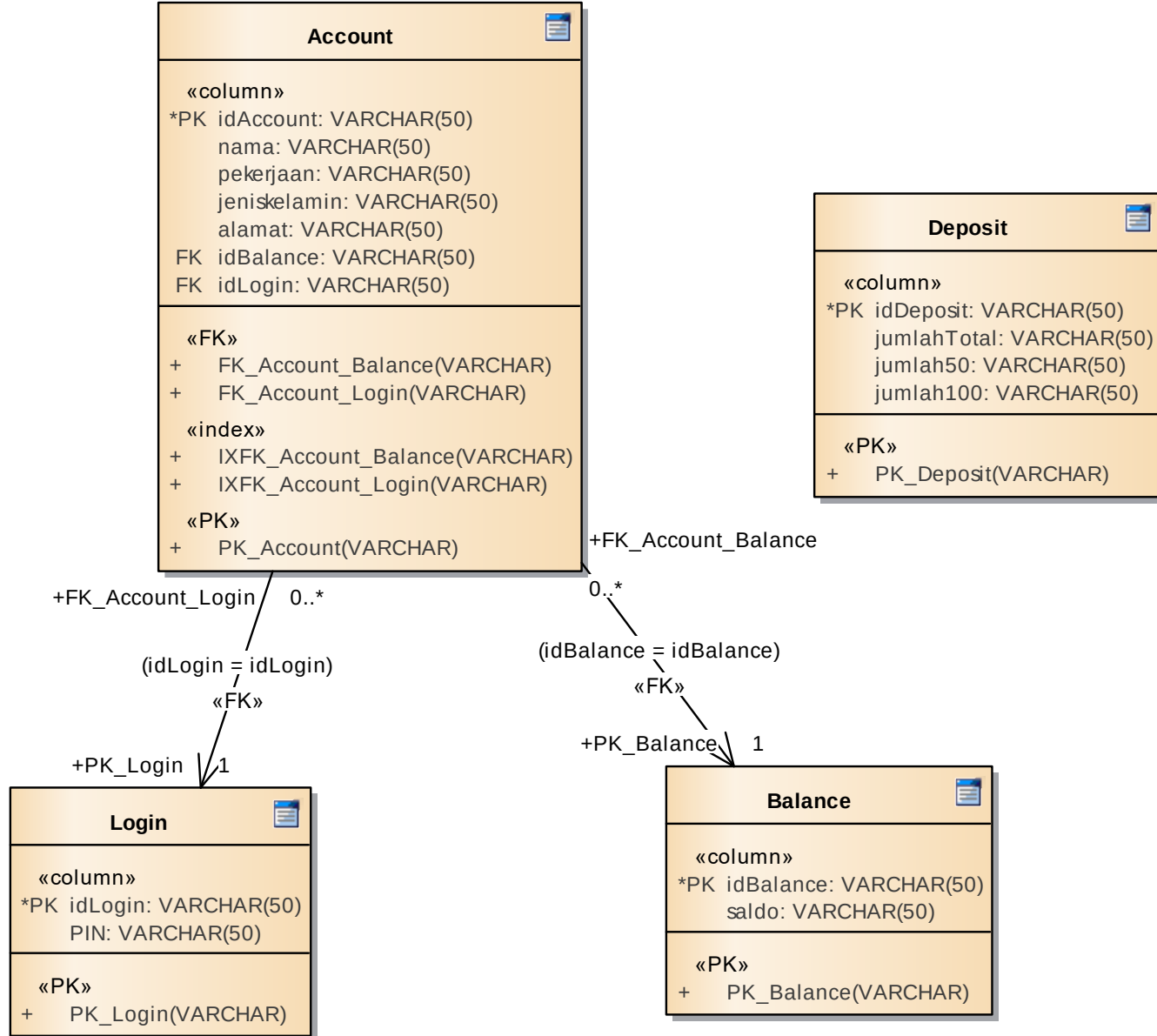
Sequence Diagram: Keluar Sistem



Class Diagram



Data Model



User Interface Design

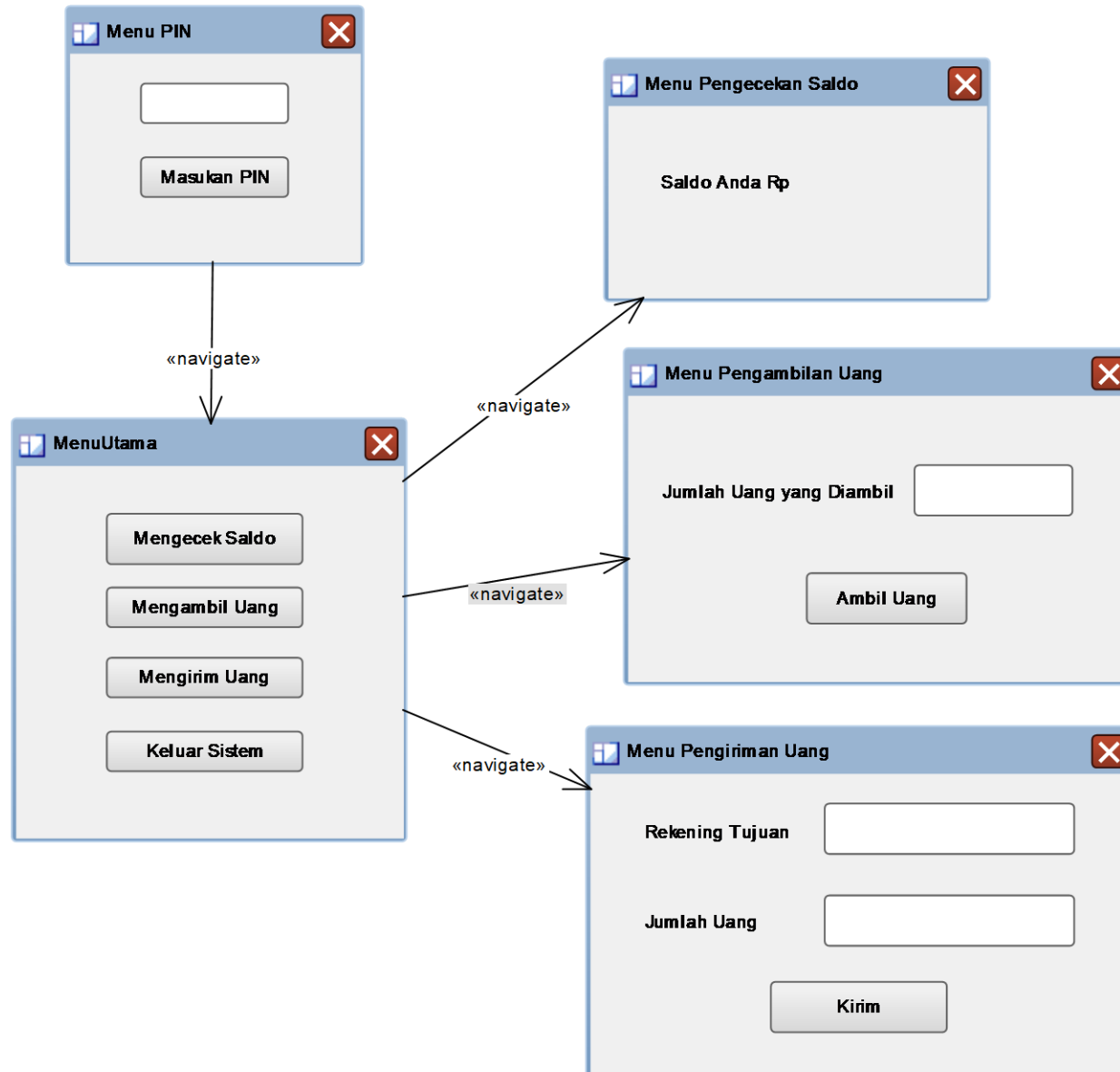
Menu Utama

7	8	9
4	5	6
1	2	3
0	Enter	

Kotak Kartu

Kotak Uang

User Interface Design (Sparx EA)



User Interface Design (Netbeans)

Menu Login

Masukkan PIN

Login

Menu Transaksi

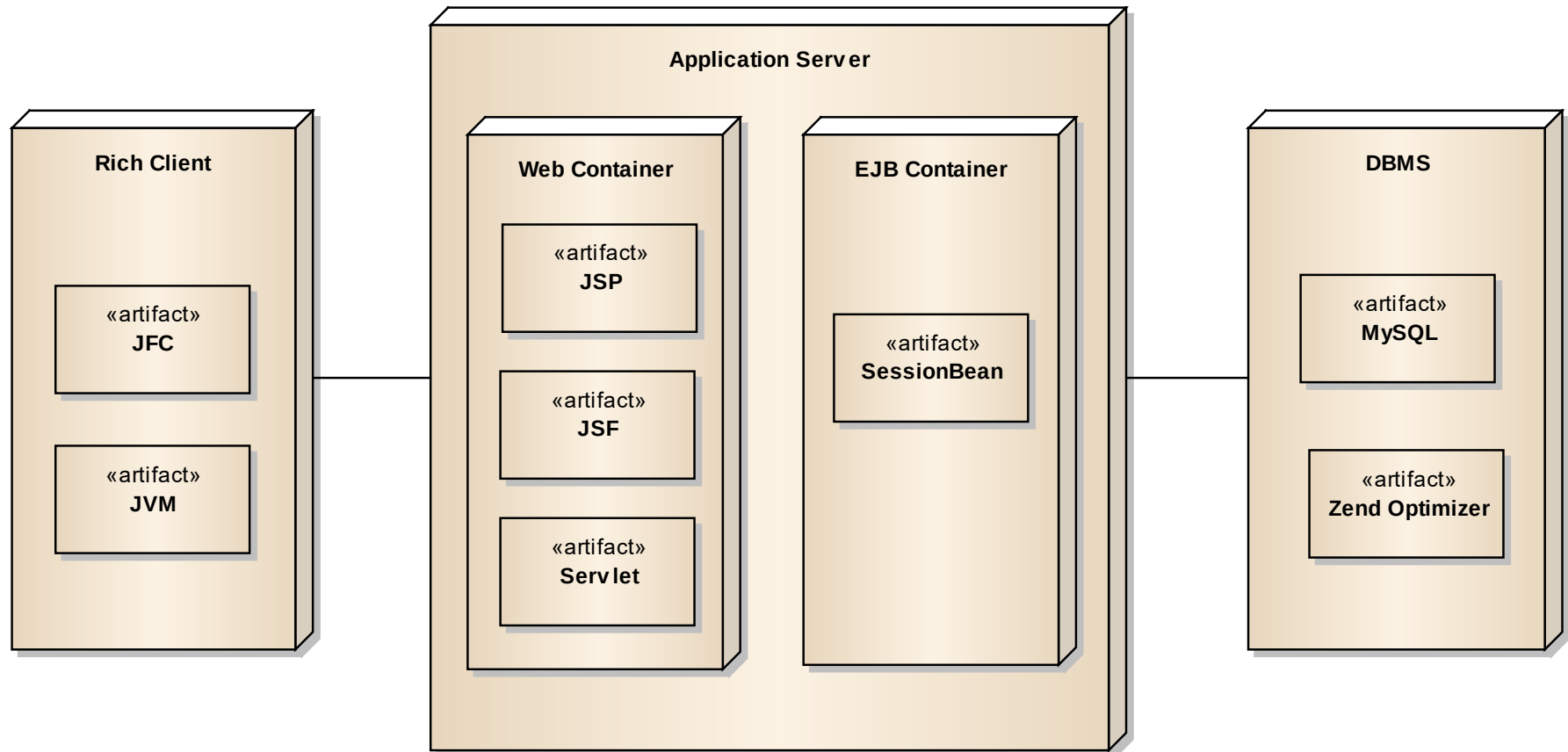
Melihat Saldo

Mengirim Uang

Mengambil Uang

Keluar

Deployment Diagram (3 Tier)



Contoh Template System Specification (SRS, BRD, FSD)

1. System Planning

- 1.1 Project Scope
- 1.2 Project Schedule
- 1.3 Project Team

2. System Design

2.1 Functional Requirements

- 2.1.1 Actor
- 2.1.2 Use Case Diagram
- 2.1.3 Activity Diagram (BPMN)
- 2.1.4 Sequence Diagram
- 2.1.5 Class Diagram
- 2.1.6 Data Model
- 2.1.7 User Interface Design
- 2.1.8 Deployment Diagram
- 2.1.9 Relational Matrices
 - 2.1.9.1 Actor – Activity Diagram
 - 2.1.9.2 Actor – Sequence Diagram

2.2 Nonfunctional Requirements

- 2.2.1 Operational
- 2.2.2 Performance
- 2.2.3 Security
- 2.2.4 Hardware
- 2.2.5 Development Platform
- 2.2.6 Deadline

3. System Implementation

- 3.1 Testing Strategy
- 3.2 Installation Strategy
- 3.3 Change Management Strategy

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